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Statistical Models of Cell Populations

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Statistical models for the description of microbial population growth have been reviewed with emphasis on their features that make them useful for applications. Evidence is shown that the integrodifferential equations of population balance are solvable using approximate methods. Simulative techniques have been shown to be useful in dealing with growth situations for which the equations are not easily solved.

The statistical foundation of segregated models has been presented identifying situations, where the deterministic segregated models would be adequate. The mathematical framework required for dealing with small populations in which random behavior becomes important is developed in detail.

An age distribution model is presented, which accounts for the correlation of life spans of sister cells in a population. This model contains the machinery required to incorporate correlated behavior of sister cells in general. It is shown that the future of more realistic segregated models, which can describe growth situations more general than repetitive growth, lies in the development of models similar to the age distribution model mentioned above.

1 Introduction

In recent years the modeling of microbial cell populations has been of particular interest to engineers, bioscientists and applied mathematicians. Consequently, the literature has grown considerably, although with somewhat varied motivations behind modeling. The focus of this article is, however, specific to the engineer's interest in the industrial role of microorganisms, which throws a different perspective in regard to the coverage of pertinent material in the literature. Thus we do not undertake a review of the mathematical work, for example in the probabilists' area of branching processes¹⁾ in spite of its relevance to microbial population growth, because the primary concern therein is the discovery of interesting results of a mathematical nature. Further, we will limit ourselves to populations of unicellular organisms, which reproduce by binary fission and unlike cells in a tissue have no direct means of communication between them.

Although, our specific interest is in *statistical models*, it would be in the interest of a proper perspective to examine the various efforts that have been made in the past to model microbial populations and to identify the special role of statistical models. Tsuchiya, Fredrickson and Aris²⁾ have provided a classification of mathematical models of microbial populations, the essence of which is retained here in Figure 1 with rather minor revisions.

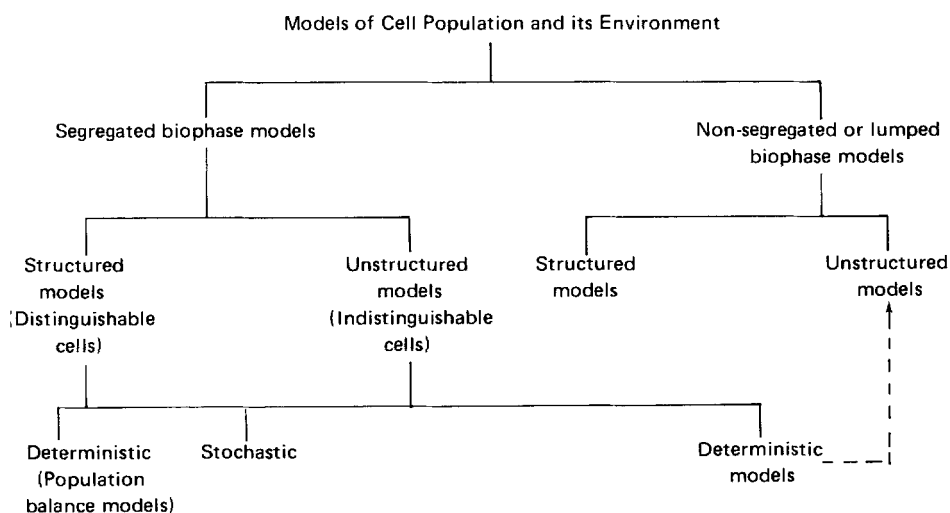


Fig. 1. A classification of mathematical models of microbial populations

The role of the environment in population growth has now been popularly recognized and accounted for except in situations where a conscious omission has been made of environmental effects for special purposes. The primary basis of the classification in Figure 1 is recognition of the integrity of the individual cell; thus segregated models view

the population as *segregated* into individual cells, that are different from one another with respect to some *distinguishable* traits. The nonsegregated models, on the other hand, treat the population as a 'lumped biophase' interacting with constituents of the environment. The commonly employed Monod model is an example of the nonsegregated model. The mathematical simplicity of such models permits a deeper analysis of their implications, although insofar as their premise is questionable their omnipotence is in doubt. Introduction of 'chemical structure' into the biophase greatly enhances the capabilities of nonsegregated models because it is an attempt to overcome the effects of straitjacketing a complex *multicomponent* reaction mixture into a *single* entity.

Segregated models derive their appeal from their very premise, viz., they recognize the obvious fact that a culture of microorganisms consists of distinct individuals. Of course, what is required to "identify" a single cell is an open question, which leads to the classification based on *structure* of a single cell, determined by one or more quantities; without structure, the cell's identity is established merely by its existence, each cell being indistinguishable from its fellows. The segregated model naturally accommodates the possibility that single cell behavior could be random, although this does not necessarily imply that the population will also behave randomly. Random population behavior is the target of *stochastic*, segregated models. Until recently, engineers had recognized the stochastic formulation only of the unstructured variety of segregated models. The so-called "population balance" models are *deterministic* formulations of the structured, segregated models^a; their stochastic formulations will also be dealt with here. Since the recognition which the segregated models accord to the individual cell is, of necessity, that under a *statistical* framework, we will refer to them as *statistical models*. The processes of growth and reproduction are a manifestation of the *physiological* activity of the cell. The extent of this activity may be said to depend on the *physiological state* of the cell and the constitution of the cell's environment both in a qualitative and quantitative sense. Fredrickson, Ramkrishna and Tsuchiya³ have developed a statistical framework for characterizing the dynamic behavior of cell populations based on a *vector* description of the physiological state; i. e., the physiological state can be determined by the quantitative amounts of all the cellular constituents. The approach was justified for cells of the procaryotic type, in which the subcellular organization is apparently negligible. It will not be our objective to elaborate on matters of this kind, for little can be added to what has already been said³. Thus we preserve the assumption that the physiological state of a cell can be represented by a vector in finite dimensional space.

To outline the objectives of this article in specific terms, let us consider the minimum attributes of "useful" modeling. First, a model must be built around *concepts* that are experimentally *measurable*, i. e., observables. Second, the mathematical formulation should produce equations that are not entirely intractable from the point of view of obtaining solutions. Third, the model must be identifiable; i. e., models contain

^a Unfortunately, the term "structure" here has a connotation different from that used in connection with nonsegregated models, where it implies a subdivision of biomass. With segregated models, this implication arises only for vector indices of the physiological state. It is not certain that this distinction is more irksome than a more elaborate classification.

unknown phenomenological quantities, which are either constants or functions, and *identification* of the model consists in being able to adapt experimental data to the model by a proper choice of the phenomenological quantities. Clearly, this third quality of a “useful” model is intimately tied up with the previous two. We gauge “usefulness” of the model specifically by its ability to *correlate* satisfactorily experimental data on the observables over the range of interest^a. The stated requirements of usefulness can now be interpreted more specifically. The choice of the index of physiological state must be such that one can measure the *distribution* of the chosen property or properties among the population. It is known that the model equations are integrodifferential in nature and unless methods are available for their solution, the model cannot be considered useful. In regard to the requirement of identifiability, we are concerned, for example, with the determination of the growth rate expression, cell division probability, probability distribution for the physiological states of daughter cells formed by division, etc. There is at present no statistical model that may be judged “useful” in the light of the foregoing remarks. It will be the first objective of this article, however, to show that there is *promise* for the future development of useful statistical models in that *some* of the stated requirements for usefulness can be met satisfactorily. It was pointed out earlier that stochastic, segregated models of the structured variety had not been recognized before. Such models would be essential when dealing with *small* populations of organisms, for which random fluctuations may exist about expected behavior. As a second objective, this article will examine the statistical foundation of segregated models of cell populations in terms of which the ramifications of deterministic and stochastic versions would be elucidated. Another important aspect of cell populations is that the daughter cells belonging to the same parent (i. e., sister cells) may display *correlated* behavior. The existing segregated models have no machinery to account for this effect. Of course if the physiological state has been completely accounted for, there would in fact be no special need to take explicit account of such correlated behavior; such would be the case, for example, with the framework presented by Fredrickson et al.³⁾. However, a model based on an elaborate physiological state is likely to violate the requirement that it contains only observables. One is therefore forced to account explicitly for such correlated behavior for simple indices of the physiological state. The third and final objective of this article is to present the machinery for such models.

In attempting to meet the foregoing objectives, we will adhere to the following scheme. The unstructured, segregated models will be omitted altogether, since these have been discussed elsewhere^{3,4)}. Besides their usefulness is quite limited, because growth processes cannot be described by these models. Thus we begin our discussion with structured, segregated models, which are *deterministic*. These models are perhaps more important since they are applicable to *large* populations, which occur more

^a One may take issue with the criterion of “usefulness” here, because frequently it is possible to learn a great deal about natural phenomena from models without extensive quantitative correlation (Failure to concede this would indeed mean an abuse of the great equations of mathematical physics!). The present stricture on usefulness is motivated by the focus on the role of models in the *industrial* sphere.

frequently. Further, the reader uninterested in stochastic features will have been spared from the more elaborate machinery required for them. It may be borne in mind, however, that Sections 2.2 and 2.3.3 which appear under the deterministic, segregated models also apply to stochastic models.

2 Structured, Segregated Models

We are concerned here with the *deterministic* formulation of structured, segregated models. The determinism pertains to the *number* of cells of any given range of physiological states at any instant of time. The models are still *statistical*, because the physiological state of an *arbitrarily selected* member from the cell population is *statistically* distributed. Such models may be satisfactory for large populations since fluctuations of the actual number of cells about its mean value N are generally of $O(\sqrt{N})$ implying that relative fluctuations are of $O(1/\sqrt{N})^a$. The more common situations involve large populations so that the deterministic models of this section are particularly important.

The framework developed by Fredrickson et al.³⁾, using the vectorial physiological state seems an ideal starting point for the discussion of structured, segregated models.^b Thus we assume that the cell's physiological state is described by a finite dimensional vector $\mathbf{z} \equiv (z_1, z_2, \dots, z_n)$, which can be anywhere in the region $\mathfrak{B}$ of admissible states in n -dimensional space; z_j represents the amount of the j^{th} cellular constituent in the cell. The environment of the population, which consists of the nutrient medium, is assumed to be specified by the concentration vector $\mathbf{c} \equiv (c_1, c_2, \dots, c_m)$, where c_i is the concentration of the i^{th} component in the medium. Note that in a growing culture $\mathbf{c}$ would be dependent on time.^c The state of the population is described by a time-dependent number density function $f_1(\mathbf{z}, t)$ such that $f_1(\mathbf{z}, t)d$ represents the number of cells in the population at time t with their physiological states in d located at $\mathbf{z}$.^d The total number of cells in the population, $N(t)$ is then given by

$$N = \int_{\mathfrak{B}} f_1(\mathbf{z}, t) d\mathbf{z} \quad (1)$$

The number of cells in any specific "volume of the physiological state space" is obtained by replacing the region of integration in Eq. (1) by the "volume" of interest.

The *growth rate* vector for any cell, denoted $\dot{\mathbf{Z}} \equiv (\dot{Z}_1, \dot{Z}_2, \dots, \dot{Z}_n)$ represents the rates of increase in the amounts of the various cellular constituents. These growth rates

^a $O(\sqrt{N})$ must be read as 'of the order of $\sqrt{N}$ '. Mathematically this implies that $\lim_{N \rightarrow \infty} \frac{1}{\sqrt{N}} O(\sqrt{N}) < \infty$.

^b Conceivably, a more general starting point is to assume that the physiological state, in view of the uncertainty associated with its proper characterization, belongs to an abstract "sample" space; such a treatment would be quite irrelevant to the present article.

^c For a careful consideration of all the assumptions involved in this framework, the reader is referred to Fredrickson et al.³⁾

^d The subscript 1 on the number-density function defies an immediate explanation. It is used for conformity with a future notation.

would obviously depend on the prevailing physiological state and the environmental concentration vector. Although Fredrickson et al.³⁾ assumed that the growth rate is random, the final model equations contained only the *expected* or *average* growth rate vector $\bar{Z}$.^a We take the growth rate here to be deterministic and given by $\bar{Z}(z, c)$.

Cell reproduction is characterized by two functions. The first is the transition probability function $\sigma(z, c)$ such that $\sigma(z, c)dt$ represents the probability that during the time interval t to $t + dt$ a cell of physiological state z , placed in an environment of concentration c will divide (into two daughter cells when division is binary). Clearly, division has been interpreted in a purely probabilistic manner without regard to any specific circumstances that may lead to it. Of course, it is within the scope of the theory to be able to find the proper expression for $\sigma(z, c)$ if the actual circumstances ensuring fission were known. The second function relates to the distribution of the biochemical constituents between the two daughter cells. This is given by the probability density function $p(z, z', c)$ such that $p(z, z', c)dv$ represents the probability that the division of a mother cell of physiological state z' in an environment of state c will result in a daughter cell of physiological state somewhere in dv located at z ; $p(z, z', c)$ has been called the *partitioning function* and satisfies the normalization condition

$$\int_{\mathbb{R}^n} p(z, z', c) dv = 1 \quad (2)$$

Further it satisfies the constraints imposed by conservation of each biochemical entity, viz., that it is zero for any $z_i > z'_i$ and

$$p(z, z', c) = p(z' - z, z', c) \quad (3)$$

Equations (2) and (3) together can be shown to yield

$$\int_{\mathbb{R}^n} z p(z, z', c) dv = \frac{1}{2} z' \quad (4)$$

which implies that the *expected* amount of any biochemical component in the daughter cell is one half of that present in the mother cell at the instant of fission. However, this does not necessarily mean that this is the most probable distribution of the biochemical components.^b

^a This is because they postulated the existence of a probability density for the growth rate vector *conditional* on a given physiological state and environmental concentration vector, which makes the covariance terms such as $(\bar{Z}_i - \bar{Z}_j)dt(\bar{Z}_j - \bar{Z}_i)dt$ of order dt^2 . However, if it is assumed that random cell growth is such that it is describing "Brownian motion" in the physiological state space, i.e., the state of the cell undergoes infinitely rapid changes about a mean during the time interval dt , then the foregoing covariance terms will be of order dt and the model equation will show "diffusion" terms. We do not consider this generalization here.

^b For example, experiments by Collins⁴²⁾ on the distribution of penicillinase in a population of *Bacillus licheniformis* seems to indicate very uneven partitioning of the enzyme between sister cells.

2.1 Model Equations

Since the state of the population is described by $f_1(z, t)$ and that of the environment is described by $c(t)$, the model equations would be in terms of these variables.

The derivation of the number density equation becomes particularly simple if the population is assumed to be embedded in a hypothetical n -dimensional continuum in the physiological state space, which deforms in accordance with the “kinematic” vector field $\bar{Z}(z, c)$.^a Consider an arbitrary *material volume* $\Delta\mathfrak{B}$ of the continuum in which the total number of embedded cells is given by

$$\int_{\Delta\mathfrak{B}} f_1(z, t) d\mathfrak{v} \quad (5)$$

Except for those cells, which disappear from $\Delta\mathfrak{B}$ at time t , all others will be constrained to remain on it. As in continuum mechanics we denote the material derivative by

$\frac{D}{Dt}$ so that the *rate of change* of the cell population embedded to $\Delta\mathfrak{B}$ is given by

$$\frac{D}{Dt} \int_{\Delta\mathfrak{B}} f_1(z, t) d\mathfrak{v} \quad (6)$$

The *rate of disappearance* of cells by division from $\Delta\mathfrak{B}$ is given by

$$\int_{\Delta\mathfrak{B}} \sigma(z, c) f_1(z, t) d\mathfrak{v} \quad (7)$$

Of course some of the daughter cells from these may again be in $\Delta\mathfrak{B}$ but we account for them separately in the “source” term. Thus cells may be born into $\Delta\mathfrak{B}$ by division of other cells at the rate

$$2 \int_{\Delta\mathfrak{B}} d\mathfrak{v} \int_{\mathfrak{B}} p(z, z', c) \sigma(z', c) f_1(z', t) d\mathfrak{v}' \quad (8)$$

Thus a number balance on the cells in $\Delta\mathfrak{B}$ leads to

$$\begin{aligned} \frac{D}{Dt} \int_{\Delta\mathfrak{B}} f_1(z, t) d\mathfrak{v} &= 2 \int_{\Delta\mathfrak{B}} d\mathfrak{v} \int_{\mathfrak{B}} p(z, z', c) \sigma(z', c) f_1(z', t) d\mathfrak{v}' \\ &\quad - \int_{\Delta\mathfrak{B}} \sigma(z, c) f_1(z, t) d\mathfrak{v} \end{aligned} \quad (9)$$

We may now use the Reynolds Transport theorem⁵⁾ for the left hand side of Eq. (9), which yields

$$\frac{D}{Dt} \int_{\Delta\mathfrak{B}} f_1(z, t) d\mathfrak{v} = \int_{\Delta\mathfrak{B}} \left[\frac{\partial}{\partial t} f_1(z, t) + \nabla \cdot \bar{X}(z, c) f_1(z, t) \right] d\mathfrak{v} \quad (10)$$

^a An organism of physiological state z in an environment c , will change its physiological state at a rate given by $\bar{Z}(z, c)$. This is interpreted as “motion” of the cell *with* the continuum, which deforms with “velocity” $\bar{Z}(z, c)$ at the point z . If cell growth were random, we could construe this as “Brownian motion” of the cell in the physiological state space *relative* to this deforming continuum. Thus “diffusion” terms would arise in the model equation.

where the gradient operator ∇ belongs to the n -dimensional physiological state space. Collecting all the terms in Eq. (9) into a single volume integral one obtains

$$\Delta \mathfrak{V} \left[\frac{\partial}{\partial t} f_1(\mathbf{z}, t) + \nabla \cdot \bar{\mathbf{Z}}(\mathbf{z}, \mathbf{c}) f_1(\mathbf{z}, t) + \sigma(\mathbf{z}, \mathbf{c}) f_1(\mathbf{z}, t) - 2 \int_{\mathfrak{V}} \sigma(\mathbf{z}', \mathbf{c}) p(\mathbf{z}, \mathbf{z}', \mathbf{c}) f_1(\mathbf{z}', t) d\mathbf{v}' \right] d\mathbf{v} = 0 \quad (11)$$

The arbitrariness of $\Delta \mathfrak{V}$ together with the continuity of the integrand then imply that the integrand must be zero, i. e.,

$$\frac{\partial}{\partial t} f_1(\mathbf{z}, t) + \nabla \cdot \bar{\mathbf{Z}}(\mathbf{z}, \mathbf{c}) f_1(\mathbf{z}, t) = -\sigma(\mathbf{z}, \mathbf{c}) f_1(\mathbf{z}, t) + 2 \int_{\mathfrak{V}} \sigma(\mathbf{z}', \mathbf{c}) p(\mathbf{z}, \mathbf{z}', \mathbf{c}) f_1(\mathbf{z}', t) d\mathbf{v}' \quad (12)$$

Thus the number density function $f_1(\mathbf{z}, t)$ must satisfy Eq. (12), which is popularly known as a population balance equation in the chemical engineering literature⁶⁾. The equation as written holds for a perfectly stirred batch culture. (It is easily modified for a continuous culture by adding the term $-\frac{1}{\theta} f_1(\mathbf{z}, t)$ to the right hand side of Eq. (12). The solution of Eq. (12) must be considered in conjunction with a material balance equation for the environment. The latter equation is readily derived using the intrinsic reaction rate vector $\bar{\mathbf{R}}$, whose dimensionality is the number of independent chemical reactions within the cell involving cellular constituents and environmental substances. The actual rate of *consumption* of environmental substances would be obtained by multiplying the expected reaction rate vector $\bar{\mathbf{R}}$ by a stoichiometric matrix $(-\gamma)$ whose i, j^{th} coefficient γ_{ij} is the stoichiometric coefficient of i^{th} substance in the environment in the j^{th} reaction^a. For a batch culture, we then write

$$\frac{d\mathbf{c}}{dt} = \gamma \cdot \int_{\mathfrak{V}} \bar{\mathbf{R}}(\mathbf{z}, \mathbf{c}) f_1(\mathbf{z}, t) d\mathbf{v} \quad (13)$$

Again the modification for a continuous culture involves adding to the right hand side of (13) the term $\frac{1}{\theta} (\mathbf{c}_f - \mathbf{c})$, where $\mathbf{c}_f$ comprises concentrations of the environmental substances in the feed. There is also a stoichiometric matrix β associated with the cellular constituents participating in the reactions such that

$$\bar{\mathbf{Z}}(\mathbf{z}, \mathbf{c}) = \beta \cdot \bar{\mathbf{R}}(\mathbf{z}, \mathbf{c}) \quad (14)$$

which, when substituted into (12), yields the number density equation of Fredrickson et al.³⁾.

$$\frac{\partial}{\partial t} f_1(\mathbf{z}, t) + \nabla \cdot [\beta \cdot \bar{\mathbf{R}}(\mathbf{z}, \mathbf{c}) f_1(\mathbf{z}, t)] = -\sigma(\mathbf{z}, \mathbf{c}) f_1(\mathbf{z}, t) + 2 \int_{\mathfrak{V}} \sigma(\mathbf{z}', \mathbf{c}) p(\mathbf{z}, \mathbf{z}', \mathbf{c}) f_1(\mathbf{z}', t) d\mathbf{v}' \quad (15)$$

^a Here γ_{ij} is positive for a product of reaction and negative for a reactant.

The specification of a segregated model lies in the identification of the functions $\bar{R}(z, c)$, $\sigma(z, c)$ and $p(z, z', c)$ without which Equation (15) signifies no more than a straightforward number balance. It is unlikely however that such a general framework could be followed experimentally. Indeed the role of this general framework is to provide a base from which can be deduced the conditions, under which simpler descriptions of the physiological state may be “satisfactorily” used. Since the counterpart of Eq. (15) for a simpler choice of the physiological state is also a number balance, whose propriety is beyond question, what we mean by its “satisfactoriness” needs further explanation. Consider for example a single “size” variable s related to the physiological state z by

$$s = g(z) \quad (17)$$

If s is specified, $g(z)$ represents a hypersurface (in n -dimensional space), whose expanse may be denoted $\mathfrak{S}_s$. Then the conditional probability density^a $f_{z|s}(z, s, t)$ is given by

$$f_{z|s}(z, s, t) = \frac{f_1(z, t)}{\int_{\mathfrak{S}_s} f_1(z, t) d\mathfrak{f}} \quad (18)$$

where we have used d to denote an infinitesimal surface area on $\mathfrak{S}_s$. If we denote the number density function in cell size by $f_1(s, t)$, then it is given by^b

$$f_1(s, t) = \int_{\mathfrak{S}_s} f_1(z, t) d\mathfrak{f} \quad (19)$$

The population balance equation for $f_1(s, t)$ may be written as

$$\begin{aligned} \frac{\partial f_1}{\partial t}(s, t) + \frac{\partial}{\partial s} [f_1(s, t) \bar{S}(s, c, t)] = & -\Gamma(s, c, t) f_1(s, t) \\ & + 2 \int_s^\infty \Gamma(s', c, t) r(s, s', c, t) f_1(s', t) ds' \end{aligned} \quad (20)$$

where $\bar{S}(s, c, t)$ is the growth rate, $\Gamma(s, c, t)$ is the transition probability of cell division, $r(s, s', c, t)$ is the partitioning function, all expressed in terms of cell size. They are related to the corresponding quantities in the physiological state as below

$$\bar{S}(s, c, t) = \int_{\mathfrak{P}} \nabla g(z) \cdot \bar{Z}(z, c) f_{z|s}(z, s, t) dv \quad (21)$$

$$\Gamma(s, c, t) = \int_{\mathfrak{P}} \sigma(z, c) f_{z|s}(z, s, t) dv \quad (22)$$

$$r(s, s', c, t) = \frac{\int_{\mathfrak{P}} dv' \sigma(z', c) f_{z|s}(z', s', t) \int_{\mathfrak{S}_s} p(z, z', c) d\mathfrak{f}}{\Gamma(s', c, t)} \quad (23)$$

^a $f_{z|s}(z, s, t)$ is the probability density of the physiological state of a cell, given its size.

^b We prefer to use the same symbol (f_1) for the number density function independently of its argument. However, to avoid confusion the argument will always be specified.

The double overbar on the growth rate for cell size signifies two statistical averagings, the first inherited from $\bar{Z}$ and the second represented by the right hand side of Eq. (21). Equations (21)–(23) are obtained by application of the fundamental rules of probability. The central point to be noted here is that the $\bar{S}$, Γ and r are explicit functions of time. Such models are hardly of any practical utility, and it is in this sense that the satisfactoriness of Eq. (20) is subject to question. The mass balance equations for the environmental variables are easily shown to be

$$\frac{dc}{dt} = \gamma \cdot \int_0^\infty \bar{r}(s', c, t) f_1(s', t) ds' \quad (24)$$

where

$$\bar{r}(s', c, t) = \int_{\mathfrak{B}} \mathbf{R}(\mathbf{z}, c) f_{Z|S}(\mathbf{z}, s', t) d\mathbf{v} \quad (25)$$

which also displays an explicit time dependence.

One may also obtain equations for the segregated model based on cell age. By cell age is normally implied the time elapsed since the cell has visibly detached from its mother. The mean population density in terms of cell age, a is given by

$$\frac{\partial}{\partial t} f_1(a, t) + \frac{\partial}{\partial a} f_1(a, t) = -\Gamma(a, c, t) f_1(a, t), a > 0 \quad (26)$$

where $\Gamma(a, c, t)$ is the age-specific transition probability function for cell fission. There are no integral terms on the right hand side of (26) analogous to those in (20) because Eq. (26) is written for $a > 0$, and newborn cells are necessarily of age zero. Thus we also have the *boundary condition*

$$f_1(0, t) = 2 \int_0^\infty \Gamma(a', c, t) f_1(a', t) da' \quad (27)$$

As in the model based on cell size, the age-specific transition probability, $\Gamma(a, c, t)$ is given by an expression similar to (22)

$$\Gamma(a, c, t) = \int_{\mathfrak{B}} \sigma(\mathbf{z}, c) f_{Z|A}(\mathbf{z}, a, t) d\mathbf{v} \quad (28)$$

The explicit time dependence in $\Gamma(a, c, t)$ is again the point to be noted, which makes the age distribution model unattractive unless some simplifying growth situations are presumed. Thus Fredrickson et al.³⁾ have postulated the concept of *repetitive growth*, a situation in which “the same sequence of cellular events (the “life cycle” of the cell) repeats itself over and over again, and at the same rate, in all cells of the population.” The mathematical definition of repetitive growth is that the conditional probability $f_{Z|A}$ is time-independent.^a This leads to time-independence of the age-specific fission probability function so that the age distribution model is relatively more useful in situations of repetitive growth. Naturally, models, which ignore detailed physiological

^a This mathematical definition of repetitive growth implies a little *more* than the continual repetition of the life cycle at identical rates (see Section 4 of this article).

structure, cannot be expected to describe general situations of growth; thus conditions such as repetitive growth are understandable constraints for the admission of models such as that of Von Foerster.

2.2 Observable States

The lesson to be learned from the foregoing discussion is that the description of situations of growth except for, say repetitive growth or *balanced*^a repetitive growth, requires a more detailed concept of the physiological state. Evidently, it cannot be so elaborate that the constraint of observability is violated. It is in this respect that some of the recent work of Bailey and co-workers^{9)–11)} becomes particularly important. Bailey⁹⁾ has used a flow microfluorometer to determine the distribution of cellular protein and nucleic acids in a bacterial population. The technique of the flow microfluorometer, as described by Bailey⁹⁾, subjects cells previously stained with fluorescent indicators to a continuous half watt argon laser (488 nm) beam. The fluorescent indicators must have the dual quality of being specific to the cellular components of interest and a high absorptivity at the available wavelength of the laser beam. As the cells in suspension pass through the laser beam, the scattered light and fluorescent signals emitted from the cells are detected by photomultiplier tubes, which store the information to be displayed and analyzed subsequently. Bailey⁹⁾ points out that the cells may flow through the instrument at rates of the order of one thousand per second, thus allowing rapid analysis. Their results on protein and nucleic acid distributions at various instants during batch growth of *Bacillus subtilis* are reproduced in Figure 2.

Indeed the above technique appears to have the potential to track quantitatively important cellular components and to permit calculation of their statistical distributions among a cell population^b, Bailey et al.¹¹⁾ have made simultaneous two-color fluorescence measurements on a bacterial growth process using a dual photomultiplier tube, the advantage of this technique being the capability for tracking multivariate distributions. Thus they have obtained the joint nucleic acid and protein distribution among a bacterial population. It is also of interest to note that relatively inexpensive light scattering and light absorption measurements on single cells leading to information on their chemical composition are available.

2.3 Solution of Equations. Some Specific Models

From Section 2.1, we have seen that the structured, segregated models give rise to integro-partial differential equations. As pointed out earlier, the practical usefulness of segregated models also depends on whether or not solutions can be obtained for such equations. Of course the solvability of the model equations cannot be separated from its dependence on the growth rate functions such as $\bar{S}$ and the probability functions

^a See ³⁾ for a definition of balanced repetitive growth or Perret⁸⁾, who refers to it as the "exponential state".

^b Recently, Eisen and Schiller³⁵⁾ have also reported a microfluorometric analysis from which the DNA distribution has been obtained. They have also attempted to obtain the DNA synthesis rate in individual cells assuming the rate to be identical and constant for all cells.

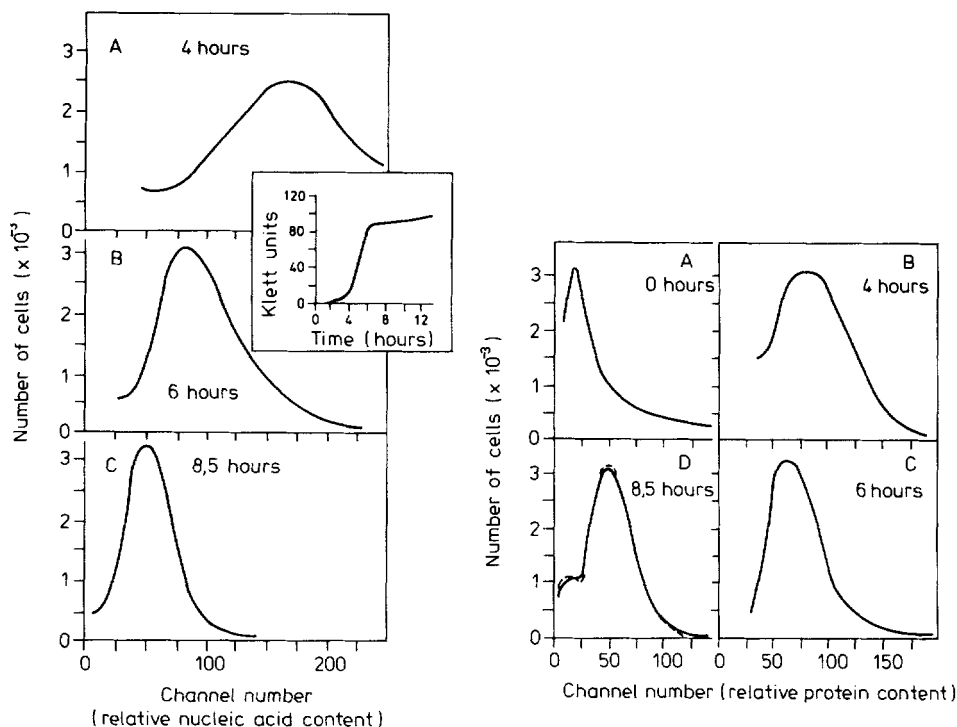


Fig. 2. Protein distributions of *Bacillus subtilis* in a batch culture at different times as determined by a microfluorometer (Bailey et al.¹⁰). Reprinted by permission of A. I. A. A. S.

such as Γ and r . Nevertheless, some general discussion is possible. Besides, in this section we will consider some of the specific models that have been proposed in the past. It is to be expected that analytical solutions are difficult except in some simplified situations. However, we begin with a brief review of analytical solutions, because they may be useful as initial approximants in a successive approximation scheme to solve more realistic problems.

2.3.1 Analytical Solutions

The age distribution model yields the most tractable equation for analytical solution. Thus Trucco^{12,13} has considered at length the solution of Von Foerster's equation (Eq. (26)) for various situations. Equation (26)^a, being a first order partial differential equation, the standard approach to its solution is via the method of characteristics (see for example Aris and Amundson¹⁴). The solution is calculated *along* the characteristics on the (t, a) plane by solving an ordinary differential equation. If the parameter *along*

^a Von Foerster's Equation is written for repetitive growth in which environmental variations, if any, have no effects on the rates of cellular processes.

any characteristic is assumed to be t itself the characteristics on the (t, a) plane may be described by

$$\frac{da}{dt} = 1 \quad (29)$$

which is instantly solved to obtain $a - t = a_0 - t_0$, so that the characteristics are straight lines originating, at, say a_0, t_0 . Since we are interested in the positive region of the (a, t) plane, for $a > t$ we may take $t_0 = 0$ and for $a < t$ we set $a_0 = 0$. The characteristic $a = t$ springs from the origin. The layout of characteristics is presented in Figure 3.

For calculating the population density, we write

$$\frac{df_1}{dt} = \frac{\partial}{\partial t} f_1(a, t) + \frac{\partial}{\partial a} f_1(a, t) \frac{da}{dt} = -\Gamma(a, c, t) f_1(a, t) \quad (30)$$

where the term on the extreme left represents the derivative of $f_1(a, t)$ along the characteristics. For analytical solutions, one must assume that the environment is virtually constant or that Γ is independent of c over the range of the latter's variation. Thus dropping the variable c in Γ , we solve (30) subject to the condition that at (a_0, t_0) , f_1 is known to be $f_{1,0}$. For any given a and t we may write

$$f_1(a, t) = f_{1,0} \exp \left[-\int_{t_0}^t \Gamma(a_0 + t' - t_0, t') dt' \right] \quad (31)$$

when $a > t$, $t_0 = 0$ and $f_{1,0}$ assumes the value of the initial age distribution, viz.

$$f_{1,0} = N_0 g(a_0) = N_0 g(a - t) \quad (32)$$

where $g(a)$ is the initial age distribution of the cell population and N_0 is the initial number of cells. When $a < t$, then $f_{1,0} = f_1(0, t_0)$, i.e., the number of newborns at time $t_0 = t - a$, which is given by Eq. (27). Thus

$$f_1(0, t - a) = 2 \int_0^\infty \Gamma(a', t - a) f_1(a', t - a) da' \quad (33)$$

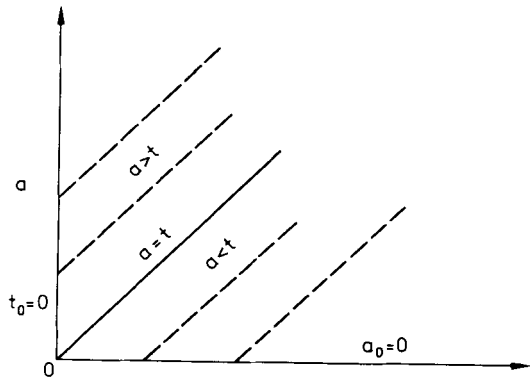


Fig. 3. Characteristic curves on the $a-t$ plane

Now (31) may be written as

$$f_1(a, t) = \begin{cases} N_0 g(a - t) \exp \left[-\int_0^t \Gamma(a - t + t', t') dt' \right] & a > t \\ f_1(0, t - a) \exp \left[-\int_{t-a}^t \Gamma(a - t + t', t') dt' \right] & a < t \end{cases} \quad (34)$$

Note that for $a > t$, the solution is already determined from (34). The solution is to be found for $a < t$. The combination of (33) and (34) produces the following Volterra integral equation

$$\psi(\tau) = \int_0^\tau 2 \Gamma(a', \tau) \exp \left[-\int_{\tau-a'}^\tau \Gamma(a' + t'' - \tau, t'') dt'' \right] \psi(\tau - a') da' + \phi(\tau) \quad (35)$$

where $\tau \equiv t - a$, $\psi(\tau) \equiv f_1(0, \tau)$ and

$$\phi(\tau) \equiv N_0 \int_\tau^\infty \Gamma(a', \tau) g(a' - \tau) \exp \left[-\int_0^\tau \Gamma(a' + t'' - \tau, t'') dt'' \right] da' \quad (36)$$

The function $\phi(\tau)$ is known on specification of N_0 and the initial age distribution of the population. This is as far as the analytical solution can carry us for the general case. Equation (35) has the property that the method of successive approximations will unconditionally converge, which can be used to advantage for numerical solutions. For the case of repetitive growth, where Γ is independent of t , Eq. (35) will transform to

$$\psi(\tau) = \int_0^\tau F(a') \psi(\tau - a') da' + \phi(\tau) \quad (37)$$

where

$$F(a') \equiv 2 \Gamma(a') \exp \left[-\int_0^{a'} \Gamma(u) du \right] \quad (38)$$

Equation (37) is amenable to solution by Laplace transform. Denoting the transform variable by a bar over it, we have

$$\bar{\psi}(s) = \frac{1}{1 - \bar{F}(s)} \bar{\phi}(s) \quad (39)$$

It is pointless to consider the inversion of (30) without a specific form for the function Γ .^a Trucco^{12, 13}) has considered various forms of Γ including one that depends on f_1 so that Eq. (26) becomes a nonlinear differential equation. The integral equation

^a However, some further interesting remarks could be made in regard to inversion of (39). $\bar{\psi}(s)$ has a singularity at $\bar{F}(s) = 1$, which may be assumed to occur at $s = \mu$, a real positive number. (This can be inferred by inspection of the Laplace transform of (38)). It can be further proved that this root is unique, from which complex roots of the equation $\bar{F}(s) = 1$ can be shown to be impossible. Interestingly enough, the residue of $\bar{\psi}(z)e^{zt}$ at $z = \mu$ leads to Powell's¹⁵) asymptotic solution for the age distribution; i.e. we arbitrarily write

$$\psi(t) = \frac{\bar{\phi}(\mu)}{-\bar{F}'(\mu)} e^{\mu t}, \quad \bar{F}'(s) \equiv \frac{d}{ds} \bar{F}(s)$$

corresponding to (37) is then also nonlinear, which may be solved by the method of successive approximations with guaranteed convergence.

Returning to Eq. (26), for the case of repetitive growth (with a time-independent Γ), Powell¹⁵ has shown that the age distribution defined by

$$f(a, t) \equiv \frac{f_1(a, t)}{N(t)} \quad (40)$$

becomes time-independent for large times. By assuming that $f_1(a, t) = N_0 e^{\mu t} f(a)$ as a trial solution one obtains (see for example²⁾ the asymptotic age distribution of Powell.

$$f(a) = \left\{ \int_0^\infty e^{\mu a} \exp \left[-\int_0^a \Gamma(u) du \right] da \right\}^{-1} \exp \left[-(\mu a + \int_0^a \Gamma(u) du) \right] \quad (41)$$

The exponential growth rate constant μ is given by

$$1 = \int_0^\infty e^{\mu a} 2 \Gamma(a') \exp \left[-\int_0^a \Gamma(u) du \right] da' \quad (42)$$

which is the same as the root of the equation $\bar{F}(s) = 1$ (see footnote in regard to the inversion of (39)).

Tsuchiya et al.²⁾ have obtained an analytical solution for a *synchronous* culture assuming that

$$\Gamma(a) = \gamma S(a - a_0), \quad \gamma > 0$$

where $S(x)$ is the step function which is zero for negative arguments and unity for positive x . This implies that the cells definitely do not divide until reaching an age a_0 after which there is a constant transition probability of a cell dividing regardless of its age. Synchrony of the culture is represented by $g(a) = \delta(a)$ where $\delta(a)$ is the Dirac delta function. The final solution for the total number of cells is given by

$$\frac{N(t)}{N_0} = 1 + \sum_{m=1}^{P(t)} 2^{m-1} f[\gamma(t - m_0), m] \quad (43)$$

where $P(t)$ is the largest integer such that $t > P(t)a_0$ and

$$f(y, m) \equiv \frac{1}{(m-1)!} \int_0^y e^{-x} x^{m-1} dx$$

where

$$\bar{\phi}(\mu) \equiv N_0 \int_0^\infty d\tau e^{\mu \tau} \int_0^\infty \Gamma(a') \exp \left[-\int_{a'-\tau}^a \Gamma(u) du \right] g(a' - \tau) da'$$

and

$$-\bar{F}'(\mu) \equiv 2 \int_0^\infty a e^{\mu a} \Gamma(a) \exp \left[-\int_0^a \Gamma(u) du \right] da$$

which is the solution arrived at by Trucco¹³⁾ using the results of Harris¹⁶⁾ on branching processes. The above formula holds for large times. From the point of view of the inversion of the Laplace transform it must be inferred that for short times the inversion integral cannot be calculated by evaluating the residue at $s = \mu$, implying that there are nontrivial contributions along suitably chosen sequences of contours enclosing the singularity.

Equation (43) predicts the progressive loss of synchrony because of randomness in the birth rate.

Analytical solutions are difficult, when for example a size variable is used to describe a cell. In most situations, it is possible to apply the method of characteristics to reduce the integro-partial differential equation to a Volterra integral equation along the characteristics. Under suitable assumptions, it may be possible to write analytical expressions for the representation of the solution by a Neumann series^a. It is safe to say, however, that analytical solutions are inaccessible for any realistic model of population growth. We therefore consider approximate methods for the solution of such equations.

2.3.2 Approximate Methods

Approximate methods cover a wide range of possibilities. We had observed that the method of successive approximations could be applied to the solution of the Volterra integral equations to which the integro-partial differential equation may be reduced. Since the upper limit of integration is infinity, the integral equation is singular and convergence of the Neumann series cannot be guaranteed. However, in most actual calculations, a finite upper limit (but suitably large) may be placed so that convergence is indeed certain.

The method of successive approximations is somewhat cumbersome computationally. Moreover, the method is even more difficult (although not impossible) to apply in situations in which the environmental variables and the cell population influence each other. In order to discuss some of the approximate methods, it will be most convenient to consider specific models that have been propounded in the past.

Eakman, Fredrickson and Tsuchiya¹⁸⁾ have investigated a statistical model using mass as the cell variable. Their model equations were given by (20) and (21) with s replaced by the variable m . They assumed a single rate-limiting substrate in the environment, whose concentration is denoted C_s and that the growth rate expression $\dot{M}(m, C_s)$ was given by

$$\dot{M}(m, C_s) = S\phi(C_s) - \mu_c m \quad (44)$$

where S is the surface area of the cell (which should depend on cell mass m), $\phi(C_s)$ is the flux of substrate across the cell surface, and μ_c is the specific mass release rate. Expression (44) was proposed by Von Bertalanffy^{19, 20)}. Eakman et al.¹⁸⁾ used a Michaelis-Menten expression for the flux, viz.,

$$\phi(C_s) = \frac{\mu C_s}{k_s + \mu C_s} \quad (45)$$

For spherical cells with $S = \left(\frac{3m}{4\rho\pi}\right)^{1/3}$, where ρ is the average density it is easily shown that (44) and (45) imply a maximum cell mass¹⁸⁾. For cylindrical cells, $S \approx \frac{2}{R\rho} m$, upon

^a See for example Courant and Hilbert¹⁷⁾ for solution of a Volterra integral equation by Neumann series.

neglecting the areas at the end; (44) and (45) then predict unlimited growth as long as nutrients do not run out^a.

The division probability $\Gamma(m, C_s)$ was assumed by Eakman et al.¹⁸⁾ to be

$$\Gamma(m, C_s) = \frac{2 e^{-\left(\frac{m-m_c}{\epsilon}\right)^2} M(m, C_s)}{e\sqrt{\pi} \operatorname{erfc}\left(\frac{m-m_c}{\epsilon}\right)} \quad (46)$$

This expression was proposed based on the assumption that cells most likely divide when their masses are over a "critical mass" m_c . The partitioning function $r(m, m', C_s)$ was assumed to be independent of C_s ,

$$r(m, m') = \frac{e^{-\left(\frac{m - \frac{1}{2}m'}{\xi}\right)^2}}{\xi\sqrt{\pi} \operatorname{erf}\left(\frac{m'}{2\xi}\right)} \quad (47)$$

implying a distribution symmetric about $\frac{1}{2}m'$ as required.

For a continuous propagator operating at steady state, we have

$$\begin{aligned} \frac{d}{dm} [\tilde{f}_1(m) \dot{M}(m, \tilde{C}_s)] = & -[\Gamma(m, \tilde{C}_s) + \frac{1}{\theta}] \tilde{f}_1(m) \\ & + 2 \int_m^\infty \Gamma(m', \tilde{C}_s) r(m, m') \tilde{f}_1(m') dm' \end{aligned} \quad (48)$$

$$0 = \frac{1}{\theta} [C_{sf} - \tilde{C}_s] - \int_0^\infty \beta S \phi(\tilde{C}_s) \tilde{f}_1(m) dm \quad (49)$$

Equation (49) is based on the assumption that β mass units of substrate are consumed per unit cell mass produced by growth and that no substrate is associated with the mass released by the cell. (The tilda over a variable is used to denote its steady state value).

Using finite differences, they solved Eq. (48) for the case

$$r(m, m') = \delta(m - \frac{1}{2}m') \quad (50)$$

which converts Eq. (48)^b to

$$\frac{d}{dm} [\tilde{f}(m) \dot{M}(m, \tilde{C}_s)] = 4\Gamma(2m, \tilde{C}_s) \tilde{f}(2m) - [\Gamma(m, \tilde{C}_s) + \frac{1}{\theta}] \tilde{f}(m) \quad (51)$$

^a If one were to solve Eq. (20) assuming, say constant C_s by the method of characteristics, the portrait of characteristics on the $(m - t)$ plane would appear significantly different for cylindrical and spherical cells.

^b Eakman et al.¹⁸⁾ show a factor of 2 (instead of 4) multiplying the first term on the right hand side of (50). Undoubtedly, this is an isolated oversight, since Eq. (51) of Eakman et al. in ¹⁸⁾ would imply that the steady state total population density $\tilde{N}$ equals zero!

Eakman et al.¹⁸⁾ have solved Eq. (51) in conjunction with (49) numerically but the solution of Eq. (48) using (47) presented considerable difficulties. Subramanian and Ramkrishna²¹⁾ solved this case by an alternative method, which will be discussed subsequently.

The segregated model equations such as Eq. (20) have been of interest to chemical engineers in the analysis of a variety of dispersed phase systems. For example, the analysis of a population of crystals in a slurry in which the crystal size distributions vary because of nucleation, growth and breakage, closely parallels that of a cell population in which cell size is considered to be distributed. Hulburt and Katz²²⁾ presented a general formulation of population balance equations for particulate systems. For the solution of equations like (20), which feature monovariate number densities, they proposed the evaluation of moments of the number density function defined by

$$\mu_n(t) = \int_0^\infty s^n f_1(s, t) ds \quad (52)$$

Frequently, a few of the leading moments themselves provide adequate engineering information. Thus for example, μ_0 represents the total number of particles in the system, $\mu_0^{-1}\mu_1$ is the average particle size, $(\mu_0\mu_1^{-2}\mu_2 - 1)^{1/2}$ is the coefficient of variation about the mean, and so on. Equations for the moments may be directly obtained from the population balance equation in some cases although such situations are more the exception than the rule. The procedure, which consists in multiplying Eq. (20) by s^n and integrating from 0 to ∞ , leads to terms that cannot be directly expressed in terms of the moments. A possible means to overcome this difficulty lies in the suggestion of Hulburt and Katz²²⁾ to expand the number density function in terms of *Laguerre functions*^a. Thus one may write

$$f_1(s, t) = e^{-s} \sum_{n=0}^{\infty} \alpha_n(t) L_n(s) \quad (53)$$

where $L_n(s)$ are the *Laguerre polynomials* given by

$$L_n(s) = e^s \frac{d^n}{ds^n} [e^{-s} s^n] \quad (54)$$

The laguerre polynomials satisfy the orthogonality relations

$$\int_0^\infty e^{-s} L_n(s) ds = \begin{cases} 0 & n \neq m \\ (n!)^2 & n = m \end{cases} \quad (55)$$

The coefficients $\{\alpha_n(t)\}$ are expressible as known linear combinations of the moment $\{\mu_n(t)\}$ ²²⁾. Obviously, in actual calculations one is forced to truncate (53), retaining only a small number of terms. Thus a finite number of moment equations can always be identified from the population balance equation by introducing the finite expansion

$$f_1(s, t) = e^{-s} \sum_{n=0}^N \alpha_n(t) L_n(s) \quad (56)$$

in the 'troublesome' spots of the equation.

^a See for example Courant and Hilbert¹⁷⁾, p. 94.

Ramkrishna²³⁾ has shown that this procedure is equivalent to a special application of the method of weighted residuals, which affords a wider repertoire of techniques. Finlayson²⁴⁾ has covered a comprehensive collection of these techniques. Subramanian and Ramkrishna²¹⁾ employed the method to solve the transient batch and continuous culture equations of the mass distribution model due to Eakman et al.¹⁸⁾ with minor variations. They also accounted for a rate limiting substrate, whose concentration diminished with growth of the population. Thus Eq. (20) and Eq. (21) (with s replaced by m) were solved simultaneously by expanding $f_1(m, t)$ in terms of a finite number of Laguerre polynomials. The residual obtained by substituting the trial solution into (20) was orthogonalized by using various choices of weighting functions. The convergence of the trial solution to the correct solution was inferred by its insensitivity to increasing the number of Laguerre polynomials. About ten Laguerre polynomials were found to be sufficient in most cases. The computation times were practically insignificant for both the batch and continuous culture calculations.

It is opportune at this point to discuss some of the results obtained by Subramanian et al.²⁵⁾ because it brings out some of the potential features of segregated models. Their calculations were based on the mass distribution model due to Eakman et al.¹⁸⁾, using essentially the same expressions for growth and the cell fission probability but the partitioning function (47) was replaced by

$$r(m, m') = \frac{30}{m} \left(\frac{m'}{m} \right)^2 \left[1 - \left(\frac{m'}{m} \right) \right]^2 \quad (57)$$

In Figure 4 are reproduced calculations of Subramanian et al.²⁵⁾ which show the evolution of the size distribution from the initial distribution to the steady state value. Figure 5 shows the calculations (under conditions the same as Figure 4) for the total population density, the biomass and the substrate concentrations. Of particular interest are the opposite initial trends of the number of cells, which at first decreases before eventually increasing, and the steadily increasing biomass concentration. The initial decrease in the number of cells occurs because the small cells then present are not ready to divide although they are growing at a rapid rate. A similar feature is shown in their calculations for a batch culture reproduced in Figure 6. Here a *lag* phase is predicted, during which the initial size distribution changes substantially to the size distribution characteristic of the exponential phase^a. As pointed out by Subramanian et al.²⁵⁾, this lag phase is not necessarily that observed experimentally, since the latter has been attributed to a period of adjustment, which the individual cells undergo when placed in an "unfamiliar" environment; indeed such adaptive delays have not been built into the growth model of Eakman et al.¹⁸⁾. What is of interest to note here is that for a lag phase to occur, it is *not necessary* that adaptive delays be involved and that inferences about individual cell behavior based on that of the *population* must be made with caution. It is most likely that the lag phase observed in a batch culture arises *both* because of adaptive delays on the part of single cells *and* due to the transient period in which the distribution of physiological states varies from its

^a If sufficient substrate is present, the exponential phase is characterized by a time-independent size distribution.

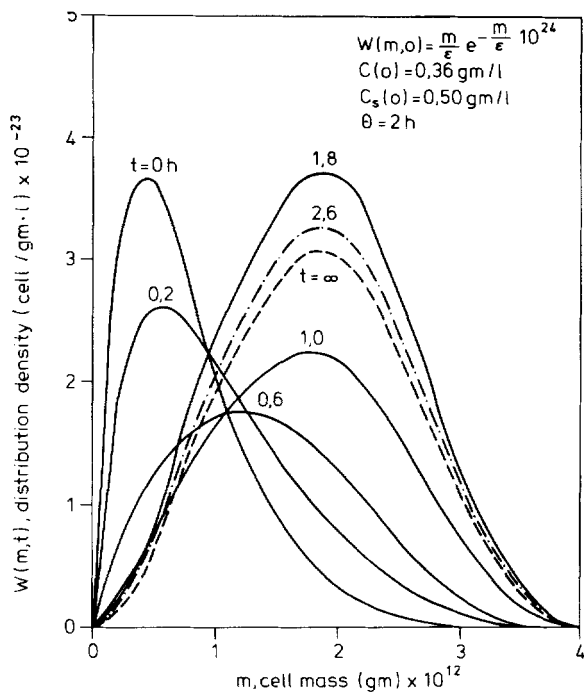


Fig. 4. Dynamics of cell mass distribution in a transient, continuous propagator from calculations of Subramanian et al.^{25). Reprinted by permission of Pergamon Press}

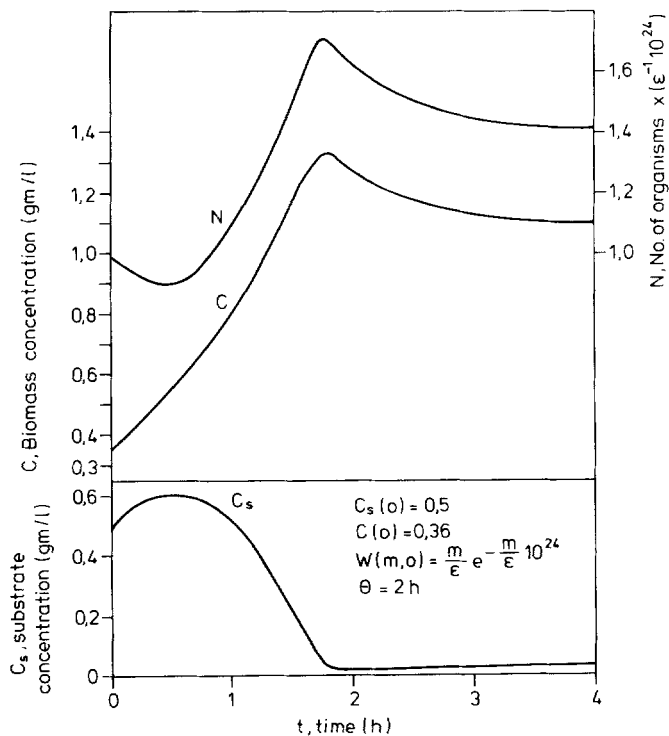


Fig. 5. Variation of population density, biomass and substrate concentration in a transient, continuous propagation from calculations of Subramanian et al.^{25). Reprinted by permission of Pergamon Press}

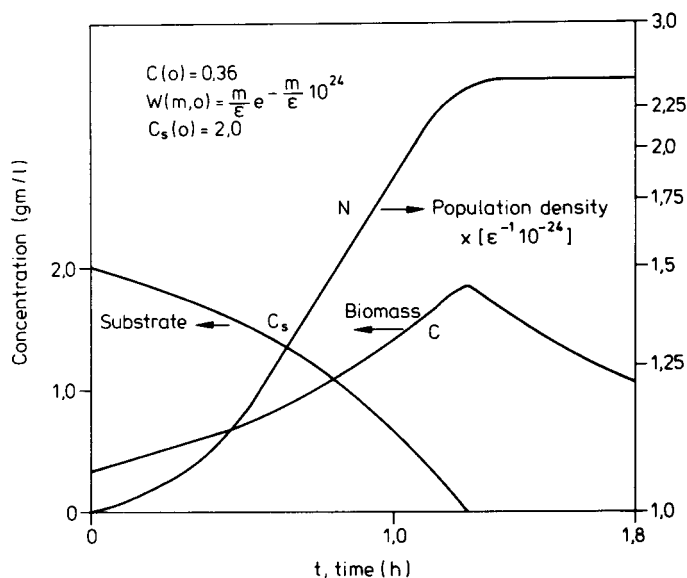


Fig. 6. Variation of population density, biomass and substrate concentrations in a batch culture from calculations of Subramanian et al.²⁵⁾ Reprinted by permission of Pergamon Press

initial value to that characteristic of the exponential phase. A particularly interesting possibility is suggested by Subramanian et al.²⁵⁾ in regard to conducting batch growth with widely varying initial distributions of physiological states. If adaptive delays associated with single cells are not important then it should be possible to produce experimentally situations in which the population initially multiplies even more rapidly than in the exponential phase. Such experiments do not appear to have been performed as yet.

We now return to the use of the method of weighted residuals for solution of the segregated model equations, which was central to the contents of this section.

The success of the method of weighted residual crucially hinges on the trial functions used in the expansion. Hulburt and Akiyama²⁶⁾ have employed generalized Laguerre polynomials in the solution of population balance equations connected with the study of agglomerating particle populations. The efficacy of the generalized Laguerre polynomials lies in the presence of additional adjustable parameters. They arise through the Gram-Schmidt orthogonalization process^a on the set $\{s^n\}$ using inner products with different weighting functions.^b

^a See for example Courant and Hilbert¹⁷⁾, p. 50.

^b The Laguerre polynomials (54) are obtained through the Gram-Schmidt orthogonalization process using the inner product

$$\langle u, v \rangle = \int_0^\infty e^{-s} u(s) v(s) ds$$

The generalized Laguerre polynomials used by Hulburt and Akiyama²⁶⁾ may be obtained by replacing the weight function e^{-s} in the above inner product by $s^a e^{-bs}$, $a, b > 0$.

Ramkrishna²⁷⁾ has pointed out that convergence of expansions in terms of trial functions may be accelerated by employing *problem-specific* orthogonal polynomials, generated by the Gram-Schmidt orthogonalization process using weighted inner products. The weight functions in the inner product are so chosen that it approximately displays the trend and shape of the required solution. Singh and Ramkrishna^{28, 29)} have solved population balance equations using such problem-specific polynomials.

It appears then that the integrodifferential equations of segregated models in which the cell state is described by a single variable such as size, are amenable to solution by approximate methods. Applications have not been made of these techniques to the solution of model equations in which the physiological state is described by two or more subdivisions of the cellular mass. There had been limited motivation for the development of such detailed segregated models because of the difficulty in procuring adequate experimental information. However, with the advent of microfluorometric techniques such as those used by Bailey and coworkers¹⁰⁾, the scope for increased sophistication has been undoubtedly widened. While it may be expected that the integrodifferential equations for multivariate number densities are less tractable, a simulation technique discussed in the next section offers considerable promise for the analysis of segregated models.

2.3.3 Monte Carlo Simulations

Kendall³⁰⁾ has described an “artificial realization” of a simple birth-and-death process in the following terms. A birth-and-death process involves the random appearance of new individuals and the disappearance of existing individuals governed by respective transition probability functions. The total population changes by one addition for every birth and a deletion for every death. Kendall defines a “time interval of quiescence” between successive events (where an event may refer to a birth or death) during which the population remains the same in number. The interval of quiescence is obviously a random quantity since the birth and death events are random. Kendall shows that the interval of quiescence has an exponential distribution with a coefficient parameter, which depends on the number of individuals at the beginning of the interval. If the population size is known at the beginning of an interval, then at the end of it, the change in the population would depend on whether the quiescence was interrupted by a birth or death. Given that either a birth or death has occurred, the probability of either event is readily obtained as the ratio of the corresponding transition probability to the sum of the two transition probabilities. An artificial realization of the birth-and-death process is now made possible by successively generating the pair of random numbers^a, the first representing the quiescence interval, which satisfies an exponential distribution and the second which identifies whether the event at the end of the interval is a birth or death.

^a See for example Moshman³¹⁾ on random number generation. More recently better methods³²⁾ have appeared for generating exponential random variables.

Shah, Borwanker and Ramkrishna³³⁾ have used Kendall's concept^a of quiescence intervals to simulate the dynamic behavior of cell populations distributed according to their age. The quiescence interval, T can again be shown to have an exponential distribution. For specificity assume that the environmental variables do not affect the population. Let

$A_t \equiv$ At time t , there are N cells of ages $a_1, a_2, \dots, a_n$.

$$P(\tau|A_t) \equiv \Pr\{T > \tau|A_t\}$$

If the transition probability function for cell fission is $\Gamma(a, t)$, then it is readily shown that

$$P(\tau|A_t) = \exp\left[-\sum_{i=1}^N \int_0^{\tau} \Gamma(a_i + u, t + u) du\right] \quad (58)$$

The cumulative distribution function for T , denoted $F(\tau|A_t)$, which is the probability that $T \leq \tau$, is given by $1 - P(\tau|A_t)$. The random number T can be generated satisfying the foregoing distribution function. The probability distribution for identifying the cell, which has divided at the end of the quiescence interval is easily seen to be

$$\Pr\{i^{\text{th}} \text{ cell has divided} \mid T = \tau, A_t\} = \frac{\Gamma(a_i + \tau, t + \tau)}{\sum_{j=1}^N \Gamma(a_j + \tau, t + \tau)} \quad (59)$$

The division of the i^{th} cell leads to two new cells of age zero making a net addition of one individual to the total population. Thus the state of the population at time $(t + \tau)$ is completely known. The procedure can now be continued until the period over which the population behavior is sought has been surpassed. The result is a *sample path* of the behavior of the cell population and the average behavior is to be calculated from a suitably large number of simulations, each of which, provides a sample path. It is also possible to calculate fluctuations about average behavior, which become important in the analysis of small populations. Shah et al.³³⁾ have shown how estimates can be made of averaged quantities from the simulations.

In dealing with, for example, the mass distribution model of Eakman et al.^{b 18)}, an additional random number is to be generated to determine the masses of the daughter cells. The probability distribution for this random variable is directly obtained from the expression for $r(m, m')$ such as (47) or (55). This simulation has been handled by Shah et al.³³⁾. Figure 7 shows a selection from their results, which have been presented as the *cumulative* number distribution of cells given by

$$\beta_1(m, t) = \int_0^m f_1(m', t) dm' \quad (60)$$

^a There have been other methods of simulation but the technique of Kendall is probabilistically exact and involves no arbitrary discretization of the time interval.

^b A similar model was also presented by Koch and Schaechter³⁴⁾.

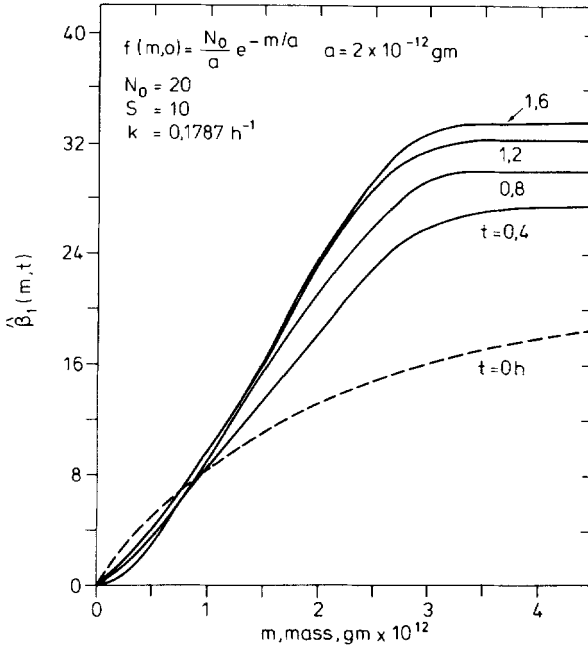


Fig. 7. Cell mass distribution in a batch culture from simulations of Shah et al.³³⁾ Reprinted by permission of Elsevier

The extension of this simulation technique to more elaborate characterizations of the physiological state is straightforward. Shah et al.³³⁾ did not account for varying environment in their simulations but the extension to this case is also straightforward. Computationally, however, the burden of generating the random quiescence interval is worsened by the more complicated probability distributions encountered. Thus, for example, the distribution function for the quiescence interval would involve the transient solution of the differential equations for the growth of *all* the cells in the population simultaneously with the equations for the environmental variables. Simplifications must therefore be introduced if such simulations have to be accomplished in reasonable time.

2.4 Identifiability

We have used the term identifiability to connote the adaptability of experimental information to recover the functions representing cellular growth rate, cell division probability and the probability distribution for the physiological states of daughter cells at the instant of birth. It is not unexpectedly that information in the literature is sparse in regard to such details. As observed earlier, only recently have become available methods for the determining the distribution of cellular components such as nucleic acids and proteins among the population. The attempt of Eisen and Schiller³⁵⁾ to determine the DNA synthesis from measurements of DNA distribution captures in spirit the process of identification of the growth rate. It would be necessary to formulate "test expressions" from more detailed modelling of growth and fission processes to make the problem of identification more tractable. To consider a specific example, Rahn³⁶⁾ postulates that cell division occurs when a certain fixed number of identical entities have been dupli-

cated. The assumption of independent replication of N entities leads to a binomial distribution (see for example ²⁾) for the number of entities replicated from which an age-specific transition probability is readily obtained under conditions of balanced growth.

Direct observations of the growth of individual cells (bacteria) date back to Ward³⁷⁾, who reported an exponential increase in length. Bayne-Jones and Adolph³⁸⁾ recorded sigmoid curves for the volumetric growth rate of yeast while the elongation rate continually decreased. Collins and Richmond³⁹⁾ provide a more complete list of such growth rate measurements. They have also observed that the foregoing growth rate measurements have been made under conditions not representative of those prevailing in a stirred liquid culture. They go on to demonstrate how elongational growth rates can be obtained from measurements of the length distribution during exponential growth. They do not derive the expression for the growth rate from the integro-differential equations but the connection has been made by Ramkrishna, Fredrickson and Tsuchiya⁴⁰⁾ ^a. It will be purposeful to present the ideas of Collins and Richmond³⁹⁾ here since it appears amenable to some useful extensions. Consider a population of bacteria distributed according to their lengths, which grow by increasing in length and multiply by binary division. It is further assumed that the population is in balanced exponential growth. If $\dot{L}(l)$ is the elongation rate of the individual cell, then it is possible to show that^{39, 40)}

$$L = k \int_0^1 [2 \psi(l') - \phi(l') - \lambda(l')] dl' / \lambda(l) \quad (61)$$

where k is the rate constant in the exponential growth phase, $\psi(l)$ is the length distribution of newly born cells at birth, $\phi(l)$ is the length distribution of dividing cells and $\lambda(l)$ is the length distribution of all cells in the population during exponential growth. The foregoing distributions have been measured by Collins and Richmond from which the elongation rate of *Bacillus cereus* were obtained using Eq. (61). Their results are reproduced in Figure 8. Ramkrishna et al.⁴⁰⁾ have shown that the transition fission probability $\Gamma(l)$ can also be calculated from the distribution functions in (61) through the equation

$$\Gamma(l) = \phi(l) \dot{L} / \int_0^1 \exp \left[k \int_1^{l'} \frac{dl''}{L} \{ 2 \psi(l'') - \phi(l'') \} \right] dl' \quad (62)$$

Equations (61) and (62) were obtained by Ramkrishna et al.⁴⁰⁾ from the number density equation. The partitioning function $p(l, l')$ could also be obtained if it is assumed that cell division is "similar", i.e.

$$p(l, l') = \frac{1}{l'} p \left(\frac{l}{l'} \right) \quad (63)$$

Equation (63) implies that the lengths of new born cells bear a constant ratio (in the

^a Harvey, Marr and Painter⁴¹⁾ have also provided a clear derivation of the results of Collins and Richmond by systematic argument.

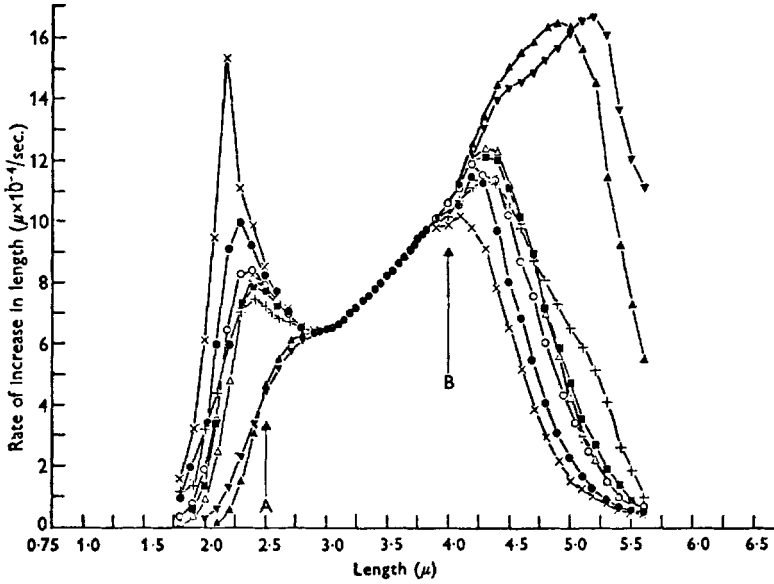


Fig. 8. Length-specific growth rate of *Bacillus cereus* between divisions from Collins and Richmond³⁹⁾
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statistical sense) to the lengths of their mothers. The function $P(x)$ is defined in the unit interval and has the properties

$$\int_0^1 P(x) dx = 1 \quad (64)$$

$$\int_0^1 xP(x) dx = \frac{1}{2} \quad (65)$$

From the number density equation, it is not difficult to show that the *moments* of $P(x)$, defined by

$$\pi_n \equiv \int_0^1 x^n P(x) dx \quad (66)$$

is given by

$$\pi_n = \frac{k \int_0^\infty l^n \psi(l) dl}{\int_0^\infty l^n \Gamma(l) \lambda(l) dl} \quad (67)$$

The right hand side of (67) is obtainable in principle although not without the hazards of substantial errors. The moments of $P(x)$ are therefore at least accessible approximately

The identification procedure just considered is of course under the constraint of balanced growth. It is not clear at this stage how one may deal with the more general situations of unbalanced population growth.

An effective way to track balanced, repetitive growth situations for identification purposes is through steady state experiments with a chemostat. Bailey and coworkers are presently engaged in such identification experiments.

Before concluding this section, we observe again that the problem of identification would be considerably simpler if specific postulates were available such as those of Koch and Schaechter³⁴⁾, which were based on extensive observations⁴³⁾. These have been the subject of considerable discussion^{43–47)}. Others, who have addressed the problem of identification are Aiba and Endo⁶⁰⁾ and Kothari et al.⁶¹⁾.

Before concluding this section, we observe again that the problem of identification would be considerably simpler if specific postulates were available such as those of Koch and Schaechter³⁴⁾, which were based on extensive observations⁴³⁾. These have been the subject of considerable discussion^{44–47)}.

3 Statistical Foundation of Segregated Models

The segregated models, discussed in the preceding sections are deterministic models because the number of cells in the population is a deterministic function of the physiological state and time. Although cell division is viewed as a random phenomenon, which should change the number of cells randomly, the assumption of a *large* population averages out this randomness. The fluctuations about the mean or expected population density $E[N]$ may be shown to be of the order of $\sqrt{E[N]}$ so that the percent fluctuation is of the order of $100/\sqrt{E[N]}$. Thus an expected population of about 10000 corresponds only to a 1% fluctuation. The normal population densities in microbial cultures are considerably higher than 10000 and a deterministic framework is generally adequate for a description of their dynamics. There are situations, however, where the population size may drop to very small values before eventually becoming extinct. For example, if a continuous culture is operated at very near the maximum dilution rate (which yields the maximum productivity of cells), a low initial population could lead to an eventual washout. When the population drops to small levels, the fluctuations about the mean number of cells may be of considerable magnitude. Whether or not an eventual washout would occur cannot also be predicted with certainty. Thus an *extinction* probability may be associated with the event of washout. The description of such features is of course the province of a stochastic framework. The deterministic, segregated models, with which we have been concerned so far, are therefore inadequate for dealing with small populations. It is well to observe at the outset that since the behavior of individual cells determine that of the population, stipulations in regard to the former, probabilistic or otherwise, should provide all the requisite information for a stochastic description of the latter. Indeed the deterministic segregated models have fed on precisely the same information, so that one is led to believe that the stochastic formulation somehow calls for a more elaborate synthesis of single cell behavior. The necessary apparatus is provided by the theory of stochastic point processes, which originally grew out of problems in the description of energy distributions of elementary particles in cascade processes⁴⁸⁾. In an abstract sense, stochastic point processes are concerned with the distribution of discrete points in a multidimensional continuum.

Ramkrishna and Borwanker^{49, 50)}, have shown that the general population balance equation is the primary descendent of an infinite hierarchy of equations in certain density functions, which arise in the theory of stochastic point processes. In principle, the complete stochastic description requires the entire hierarchy of equations although a few of the leading equations may yield information sufficient from a practical standpoint. In the above analysis, the authors assumed the particle behavior to be independent of the continuous phase. The generalization to the situation, where continuous phase variables and particle behavior depend on each other has been presented by Ramkrishna⁵¹⁾. The concentrations of environmental substances, represented by the vector $\mathbf{C}(t)$, vary with time as a result of the biological activity of *all* the cells in the population. Any randomness in the rate of multiplication of the population should therefore produce a random variation in the environmental variables. Thus $\mathbf{C}(t)$ would be a vector-valued random process.

In this section, we will outline the theory of the stochastic formulation of segregated models. Indeed while their applications are only important in dealing with small populations, they also reveal the statistical foundation of the deterministic segregated models presented earlier. In outlining the theory, we will retain the vector description of the physiological state of the cell; furthermore we will deal with exactly the same functions of cell growth and cell division introduced in Section 2. The population and its environment are assumed to be uniformly distributed in space, which implies a well-stirred culture. In the following sections, we define the various density functions with which we must deal. The derivation of the equations satisfied by the density functions will be omitted because of the lengthy book-keeping procedures but the equations themselves possess a systematic structure suggestive of the significance of the constituent terms.

3.1 The Master Density Function

Since we must be concerned with the distribution of physiological states of all the cells in the population and the environmental variables, we define a master density function^a

$$J_\nu(\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_\nu; \mathbf{c}; t) d\mathbf{v}_1 d\mathbf{v}_2 \dots d\mathbf{v}_\nu d\mathbf{c} \quad (68)$$

which represents the probability that at time t , there are a total of ν cells in the population, comprising a cell in each of the infinitesimal volumes $d\mathbf{v}_i$ located about $\mathbf{z}_i$, $i = 1, 2, \dots, \nu$ and the environmental vector $\mathbf{C}$ is in a volume $d\mathbf{c}$ located at $\mathbf{c}$ somewhere in the m -dimensional volume $\mathcal{C}$. Aside from the physiological state, cells are assumed to be indistinguishable. The multivariate probability density function for the concentration vector $\mathbf{C}$, denoted $f_{\mathbf{C}}(\mathbf{c}, t)$ is given by

$$f_{\mathbf{C}}(\mathbf{c}; t) = \sum_{\nu=0}^{\infty} \frac{1}{\nu!} \prod_{i=1}^{\nu} \int_{\mathcal{V}} d\mathbf{v}_i J_\nu(\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_\nu; \mathbf{c}; t) \quad (69)^b$$

^a This is an extension of the density function introduced by Janossy⁵²⁾ in dealing with nucleon cascades. Here we assume that at most one cell can be of a given physiological state. This condition is not unreasonable. However, this constraint could be removed in a more general development. See for example⁵⁰⁾.

^b The product symbol is used to represent multiple integration in the physiological state space.

where we have integrated over all possible physiological states and accounted for the fact that the value of J_ν is insensitive to the permutation of its arguments. The probability distribution for the total number of cells in the population is

$$P_\nu(t) = \frac{1}{\nu!} \int_{\mathcal{C}} d\mathbf{c} \prod_{i=1}^{\nu} \int_{\mathcal{V}_i} d\mathbf{v}_i J_\nu(\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_\nu; \mathbf{c}; t) \quad (70)$$

Clearly

$$\int_{\mathcal{C}} f_{\mathcal{C}}(\mathbf{c}; t) d\mathbf{c} = 1, \quad \sum_{\nu=0}^{\infty} P_\nu(t) = 1 \quad (71)$$

so that the *normalization* condition on the master density function is given by

$$\int_{\mathcal{C}} d\mathbf{c} \sum_{\nu=0}^{\infty} \frac{1}{\nu!} \prod_{i=1}^{\nu} \int_{\mathcal{V}_i} d\mathbf{v}_i J_\nu(\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_\nu; \mathbf{c}; t) = 1 \quad (72)$$

Equation (72) lays down the means of calculating expectations of any quantity which depends on the population and its environment.

Mathematically, we write

$$E[\] = \int_{\mathcal{C}} d\mathbf{c} \sum_{\nu=0}^{\infty} \frac{1}{\nu!} \prod_{i=1}^{\nu} \int_{\mathcal{V}_i} d\mathbf{v}_i J_\nu(\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_\nu; \mathbf{c}; t) [\] \quad (73)$$

In the next section, we use (72) to calculate the expectations of certain important quantities associated with the population.

3.2 Expectations. Product Density Functions

The number density function is the quantity of central interest to the description of the population. If there are ν cells at time t with one cell in each of the infinitesimal volumes $d\mathbf{v}_i$ located about $\mathbf{z}_i$, $i = 1, 2, \dots, \nu$, the number density function $n(\mathbf{z}, t)$ is given by

$$n(\mathbf{z}, t) = \sum_{i=1}^{\nu} \delta(\mathbf{z} - \mathbf{z}_i) \quad (74)$$

where $\delta(\mathbf{z} - \mathbf{z}_i)$ is Dirac's function^a. The total number of cells in the population $N(t)$ is

$$N(t) = \int_{\mathcal{V}} n(\mathbf{z}, t) d\mathbf{v} \quad (75)$$

which has the value ν , when (74) holds. By changing the range of integration in (75) to any volume in the physiological state space the number of cells in that volume can be calculated. The expected value of $n(\mathbf{z}, t)$ is obtained by substituting (74) into (73). The properties of the delta function lead to the result

$$E[n(\mathbf{z}, t)] = \int_{\mathcal{C}} d\mathbf{c} \sum_{\nu=1}^{\infty} \frac{1}{(\nu-1)!} \prod_{i=1}^{\nu-1} \int_{\mathcal{V}_i} d\mathbf{v}_i J_\nu(\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_{\nu-1}, \mathbf{z}; \mathbf{c}; t) \quad (76)$$

The right hand side of (76), when multiplied by $d\mathbf{v}$, yields the probability that there is

^a The delta function has the properties, $\delta(\mathbf{z} - \mathbf{z}_i) = 0$ $\mathbf{z} \neq \mathbf{z}_i$.

For any function $f(\mathbf{z})$, $\int_{\mathcal{V}} f(\mathbf{z})\delta(\mathbf{z} - \mathbf{z}_i)d\mathbf{v} = f(\mathbf{z}_i)$. In particular $\int_{\mathcal{V}} \delta(\mathbf{z} - \mathbf{z}_i)d\mathbf{v} = 1$.

These results hold for any range of integration enclosing $\mathbf{z}_i$.

a cell at time t in the population with its physiological state in dv located about z^a . Thus the expected population density has also a probability interpretation. We introduce the notation

$$E[n(z, t)] \equiv f_1(z, t) \quad (77)$$

and call it *product density of order 1*, a term first used by A. Ramakrishnan who originated it⁵³. This product density is *not a probability density* function. Moreover, it has the property that

$$E[N(t)] = \int_{\mathcal{V}} f_1(z, t) dv \quad (78)$$

The expected number of cells in any region of the physiological state space is obtained by carrying out the integration in (78) over that region. One may expect that it is the function $f_1(z, t)$, which is featured in the deterministic segregated model equation (15)^b.

Higher moments of the population density, which are required for calculation of the fluctuations about the expected value, are obtained by taking expectations of products of the type $\prod_{k=1}^r n(z_k, t)$, $r = 2, 3, \dots$. In view of the prime significance of the second moment, we consider this in detail.

Thus we let

$$E[n(z_1, t) n(z_2, t)] \equiv f_2(z_1, z_2; t) \quad z_1 \neq z_2 \quad (79)$$

and call $f_2(z_1, z_2; t)$ *product density of order 2*. The left hand side of (79) is obtained by using (73), which yields

$$f_2(z_1, z_2; t) = \int_{\mathcal{C}} dc \sum_{\nu=2}^{\infty} \frac{1}{(\nu-2)!} \prod_{i=1}^{\nu-2} \int_{\mathcal{V}} dv_i' J_{\nu}(z_1', z_2', \dots, z_{\nu-2}', z_1, z_2; c; t) \quad (80)$$

The right hand side of (80), when multiplied by $dv_1 dv_2$ represents the joint probability that the population includes two cells at time t , one in dv_1 located about z_1 and the other in dv_2 located about z_2 . However, it is not a probability density function. When the constraint $z_1 \neq z_2$ is removed, then the procedure of taking expecting leads to

$$E[n(z_1, t)n(z_2, t)] = f_2(z_1, z_2; t) + f_1(z_1, t) \delta(z_1 - z_2) \quad (81)$$

The second moment of the total population is obtained by

$$\begin{aligned} E[N(t)^2] &\equiv \int_{\mathcal{V}} dv_1 \int_{\mathcal{V}} dv_2 E[n(z_1, t)n(z_2, t)] \\ &= \int_{\mathcal{V}} dv_1 \int_{\mathcal{V}} dv_2 f_2(z_1, z_2; t) + \int_{\mathcal{V}} f_1(z, t) dv \end{aligned} \quad (82)$$

^a This follows because the right hand side of (76) is the sum of the probabilities of *all* mutually exclusive and exhaustive situations under which a cell may be found in dv .

^b The subscript 1 was introduced earlier in the number density function to be in conformity with the notations in this section.

Equation (82) is particularly important in that it points to the strategy of analysis, viz., higher moments of the population, which are required for the calculation of fluctuations, are calculated from higher order product densities. The second moment requires the first *and* the second order product densities. Similarly the r^{th} moment of $N(t)$ is related to all the product densities of order ranging from 1 to r . For details we refer to ⁴⁸⁾. The r^{th} order product density is defined by

$$f_r(z_1, z_2, \dots, z_r; t) = E[n(z_1, t)n(z_2, t) \dots n(z_r, t)], \quad z_i \neq z_j \quad (83)$$

Using (73) one has

$$f_r(z_1, z_2, \dots, z_r; t) = \int d\mathbf{c} \sum_{\nu=r}^{\infty} \frac{1}{(\nu-r)!} \prod_{i=1}^{\nu-r} \int d\mathbf{v}_i J_{\nu}(z'_1, z'_2, \dots, z'_{\nu-r}, z_1, z_2, \dots, z_r; \mathbf{c}; t) \quad (84)$$

from which the r^{th} order product density inherits the interpretation that, when multiplied by $d\mathbf{v}_1 d\mathbf{v}_2 \dots d\mathbf{v}_r$, it represents the probability that the population at time t includes r cells, one in each of the volumes $d\mathbf{v}_i$ located about z_i , $i = 1, 2, \dots, r$. Again, it is important to recognize that f_r is not a probability density function. The r^{th} moment of $N(t)$ is given by

$$E[N(t)^r] = \sum_{k=1}^r C_k^r \prod_{i=1}^k \int d\mathbf{v}_i f_k(z_1, z_2, \dots, z_k; t) \quad (85)$$

where C_k^r are Stirling numbers of the *second kind*, which are combinatorial parameters and are tabulated in standard handbooks⁵⁴⁾.

Of special significance is the variance $V[N(t)]$ of the population which is obtained from

$$V[N(t)] = E[N^2(t)] - E[N(t)]^2 \quad (86)$$

The coefficient of variation (C. O. V.) is given by

$$\text{C. O. V.} = \frac{\sqrt{V[N(t)]}}{E[N(t)]} \quad (87)$$

Although the calculation of C. O. V. requires the first and second order product densities, an estimate of it is possible if based on the assumption of *statistical independence* of the physiological states of individual cells^a. Thus

$$f_2(z_1, z_2; t) = f_1(z_1, t) f_1(z_2, t), \quad z_1 \neq z_2 \quad (88)$$

^a The motivation for this assumption is that in a microbial culture, each cell has been assumed to grow and multiply independently of other cells. The disclaimer, however, is the fact that in view of the cells sharing and influencing a common environment, the physiological states of individual cells may become correlated in course of time.

The substitution of (88) into (82), (86) and (87) culminates in

$$\text{C. O. V.} = \frac{1}{\sqrt{E[N(t)]}} \quad (89)$$

which demonstrates how at high enough population levels, the fluctuations become negligible. Based on statistical independence, the r^{th} moment becomes

$$E[N(t)^r] = \sum_{k=1}^r C_k^r E[N(t)]^r \quad (90)$$

Equation (90) is a pointer to the *stochastic completeness* of the equation to be obtained for $f_1(\mathbf{z}, t)$, by which is implied that all information pertaining to the stochastic nature of the growing population is calculable from the first order product density.

The product density functions in the foregoing discussion had been freed from the concentration variables in the environment by integration over $\mathcal{C}$. It is clearly possible to define a product density function $f_r(\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_r; \mathbf{c}; t)$ such that on multiplication by $dv_1 dv_2 \dots dv_r d\mathbf{c}$ it will represent the joint probability at time t that the population includes r cells, one in each of dv_i located about $\mathbf{z}_i$, $i = 1, 2, \dots, r$, and that the concentration vector $\mathbf{C}(t)$ in the abiotic phase is in $d\mathbf{c}$ located about $\mathbf{c}$. Formula (84) may then be adapted to relate this product density function to the master density function J_ν by excluding the integration over $\mathcal{C}$. Thus

$$f_r(\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_r; \mathbf{c}; t) = \sum_{\nu=r}^{\infty} \frac{1}{(\nu-r)!} \prod_{i=1}^{\nu-r} \int_{\mathfrak{B}} dv'_i J_\nu(\mathbf{z}'_1, \mathbf{z}'_2, \dots, \mathbf{z}'_{\nu-r}, \mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_r; \mathbf{c}; t) \quad (91)^a$$

Evidently

$$f_r(\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_r; t) = \int_{\mathcal{C}} d\mathbf{c} f_r(\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_r; \mathbf{c}; t) \quad (92)$$

3.2.1 Expectation of Environmental Variables

It was observed earlier that a random change in the population (arising from random cell fission since growth has been regarded as deterministic) would introduce randomness in the concentration of the environmental substances. The rate of *consumption* of the abiotic phase variables $\bar{\mathbf{C}}(t)$ is given by

$$\bar{\mathbf{C}}(t) = - \int_{\mathfrak{B}} \boldsymbol{\gamma} \cdot \bar{\mathbf{R}}(\mathbf{z}, \mathbf{c}) n(\mathbf{z}, t) d\mathbf{v} \quad (93)$$

so that

$$E[\bar{\mathbf{C}}] = - \int_{\mathfrak{B}} E[\boldsymbol{\gamma} \cdot \bar{\mathbf{R}}(\mathbf{z}, \mathbf{c}) n(\mathbf{z}, t)] d\mathbf{v} \quad (94)$$

^a No change in notation is used in representing the $\mathbf{c}$ -dependent product densities from those that are independent of $\mathbf{c}$, beyond spelling out the arguments completely.

where the right hand side is of course obtained by using the integrand of (93) in combination with (74) into (73). It is readily shown by the above procedure that

$$E[\bar{C}] = - \int_{\mathcal{C}} d\mathbf{c} \int_{\mathcal{V}} d\mathbf{v} \gamma \cdot \bar{\mathbf{R}}(\mathbf{z}, \mathbf{c}) f_1(\mathbf{z}; \mathbf{c}; t) \quad (95)$$

where $f_1(\mathbf{z}; \mathbf{c}, t)$ is the first order $\mathbf{c}$ -dependent product density function. For the second moment of $\bar{C}$, there arise covariance terms of the type $E[\bar{C}_i \bar{C}_j]$, which may be shown to be

$$\begin{aligned} E[\bar{C}_i \bar{C}_j] = & \int_{\mathcal{C}} d\mathbf{c} \int_{\mathcal{V}} d\mathbf{v}_1 \int_{\mathcal{V}} d\mathbf{v}_2 \left[\sum_{k=1}^n \gamma_{ik} \bar{\mathbf{R}}_k(\mathbf{z}_1, \mathbf{c}) \right] \left[\sum_{k=1}^n \gamma_{jk} \bar{\mathbf{R}}_k(\mathbf{z}_2, \mathbf{c}) \right] f_2(\mathbf{z}_1, \mathbf{z}_2; \mathbf{c}; t) \\ & + \int_{\mathcal{C}} d\mathbf{c} \int_{\mathcal{V}} d\mathbf{v} \left[\sum_{k=1}^n \gamma_{ik} \bar{\mathbf{R}}_k(\mathbf{z}, \mathbf{c}) \right] \left[\sum_{k=1}^n \gamma_{jk} \bar{\mathbf{R}}_k(\mathbf{z}, \mathbf{c}) \right] f_1(\mathbf{z}; \mathbf{c}; t) \end{aligned} \quad (96)$$

Again as before, Eq. (96) reinforces the necessity for calculating higher order product densities if fluctuations about expected values are to be calculated. The expected value of $\mathbf{C}(t)$ is obtained from

$$E[\mathbf{C}(t)] = \int_{\mathcal{C}} \mathbf{c} f_{\mathbf{C}}(\mathbf{c}, t) d\mathbf{c} \quad (97)$$

and for the second moment, the following covariance terms are evaluated.

$$E[C_i C_j] = \int_{\mathcal{C}} c_i c_j f_{\mathbf{C}}(\mathbf{c}, t) d\mathbf{c} \quad (98)$$

Finally, we consider a function $F(\mathbf{C})$, which is analytic in $\mathbf{C}$ and examine its expected value $E[F(\mathbf{C})]$. We assume that $F(\mathbf{C})$ is expressible in terms of a convergent Taylor series about $E[\mathbf{C}]$. Thus

$$F(\mathbf{C}) = F(E[\mathbf{C}]) + \sum_{n=1}^{\infty} \{ (\mathbf{C} - E[\mathbf{C}]) \cdot \nabla \}^n F(E[\mathbf{C}]) \quad (99)^a$$

so that

$$E[F(\mathbf{C})] = F(E[\mathbf{C}]) + \sum_{n=1}^{\infty} \{ E[\mathbf{C} - E[\mathbf{C}]] \cdot \nabla \}^n F(E[\mathbf{C}]) \quad (100)$$

From (100), we may infer that when the concentration fluctuations are negligible

$$E[F(\mathbf{C})] \approx F(E[\mathbf{C}]) \quad (101)$$

Eq. (101) is useful in establishing the connection between the stochastic model equations to be presented in the next section and those in the deterministic formalism. The circumstances under which the fluctuations in the environmental variables may be neglected will be considered at a later stage.

^a For an explanation of this notation see for example "Advanced Calculus", by A. E. Taylor.

It is useful to note that any cell property, which depends on the physiological state of the cell and the environment, becomes a random quantity through its dependence on the randomly varying vector $\mathbf{C}(t)$. Thus $F(\mathbf{z}, \mathbf{C})$ is a random process. Its expectation is easily shown to be

$$E[F(\mathbf{z}, \mathbf{C})] = \int_{\mathfrak{V}} \frac{F(\mathbf{z}, \mathbf{c}) f_1(\mathbf{z}; \mathbf{c}; t) d\mathbf{v}}{f_1(\mathbf{z}, t)} \quad (102a)$$

Thus $F(\mathbf{z}, \mathbf{C})$ may be the growth rate vector $\dot{\mathbf{Z}}(\mathbf{z}, \mathbf{C})$, the transition probability function $\sigma(\mathbf{z}, \mathbf{C})$ and so on. Similarly, it is also possible to define expectations of quantities that depend on the physiological states of say r cells. If $F(\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_r, \mathbf{C})$ is a random process because of $\mathbf{C}(t)$, then

$$E[F(\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_r, \mathbf{C})] = \int_{\mathfrak{V}} \frac{F(\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_r, \mathbf{c}) f_r(\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_r; \mathbf{c}; t) d\mathbf{v}}{f_r(\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_r; t)} \quad (102b)$$

3.3 Stochastic Model Equations

The statistical foundation of the segregated model is contained in the equation for the master density function J_ν . The basis of its derivation is the application of the laws of probability to the projection of a specified state of the population and its environment at time $t + dt$ from all allowable states at time t . We do not provide the details here since they have been presented elsewhere⁵¹. The product density equations are then obtained from the master density equation. Finally, the conditions, under which the deterministic segregation model equations are valid, are elucidated.

3.3.1 The Master Density Equation

The equation for the master density function can be conveniently written with some additional notation. We denote the gradient operator in the physiological state space by ∇ as before and the gradient operator in the environment concentration space by ∇_C . The master density equation is given by

$$\begin{aligned} & \frac{\partial J_\nu}{\partial t} + \sum_{i=1}^{\nu} \nabla \cdot [\beta \cdot \bar{\mathbf{R}}(\mathbf{z}_i, \mathbf{c}) J_\nu] + \nabla_C \cdot [J_\nu \sum_{i=1}^{\nu} \gamma \cdot \bar{\mathbf{R}}(\mathbf{z}_i, \mathbf{c})] \\ &= -J_\nu \sum_{i=1}^{\nu} \sigma(\mathbf{z}_i, \mathbf{c}) + \sum_{i \neq j}^{\nu} \sum_{j=1}^{\nu} J_{\nu-1}(\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_{i-1}, \mathbf{z}_i + \mathbf{z}_j, \mathbf{z}_{i+1}, \dots, \\ & \quad \mathbf{z}_{j-1}, \mathbf{z}_{j+1}, \dots, \mathbf{z}; \mathbf{c}; t) \times \sigma(\mathbf{z}_i + \mathbf{z}_j, \mathbf{c}) p(\mathbf{z}_i + \mathbf{z}_j, \mathbf{c}) \end{aligned} \quad (103)$$

Equation (103) represents the most complete statistical equation for the segregated model. The left hand side of (103) is a “continuity” operator in the ν -fold direct sum^a of the physiological state space and the environmental concentration space. The master

^a By a ν -fold direct sum of the physiological state space $\mathfrak{V}$, denoted $\mathfrak{V} \oplus \mathfrak{V} \oplus \dots \oplus \mathfrak{V}$, we mean the collection of all vectors $[\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_\nu]$ where $\mathbf{z}_i \in \mathfrak{V}$ $i = 1, 2, \dots, \nu$.

density equation is thus a *difference, differential* equation; its solution must be considered together with an initial condition. If the initial number of cells is fixed at, say N_0 , Eq. (103) is, in principle, solvable because it represents a closed set of linear equations. Hence an analytical solution is possible for Eq. (103), using the method of characteristics. The characteristic curves are defined by

$$\frac{dz_i}{dt} = \beta \cdot \bar{\mathbf{R}}(z_i, c) \quad (104)$$

$$\frac{dc}{dt} = \sum_{i=1}^{\nu} \gamma \cdot \bar{\mathbf{R}}(z_i, c) \quad (105)$$

which lie in the ν -fold direct sum of the physiological state space and the environmental concentration space. Eq. (102) may then be rewritten as an ordinary differential equation in J_ν . However, the inherent combinatorial complexity makes such an approach utterly impractical. It is in this connection that the simulation procedure of Shah et al.³³⁾ becomes important, for in essence it automatically eliminates the less probable sample paths and provides for more efficient computation of the averages.

Eq. (103) provides the source of all other equations for the segregated model in density functions, which are derived from the master density function.

Thus the use of Eq. (103) in Eq. (69) leads to the following equation in $f_C(c, t)$

$$\frac{\partial}{\partial t} f_C(c, t) + \nabla_C \cdot \{f_C(c, t) E[\dot{\mathbf{C}} | \mathbf{C}(t) = c]\} = 0 \quad (106)$$

where the conditional expectation of $\dot{\mathbf{C}}$ is given by

$$E[\dot{\mathbf{C}} | \mathbf{C}(t) = c] = \frac{\int \gamma \cdot \bar{\mathbf{R}}(z, c) f_1(z; c; t) dz}{f_C(c, t)} \quad (107)$$

From Eq. (107), it is evident that the solution of Eq. (106) depends upon a knowledge of the first order product density function $f_1(z; c; t)$. Thus equations must be obtained for the product densities.

3.3.2 Product Density Equations

We first consider equations in the product densities defined by Eq. (91), which may be derived either by exploiting their probability interpretations or by using Eq. (103) for the master density function in conjunction with Eq. (91). The procedure would lead to the following equation for the *first order* product density $f_1(z; c; t)$.

$$\begin{aligned} & \frac{\partial}{\partial t} f_1(z; c; t) + \nabla \cdot [\beta \cdot \bar{\mathbf{R}}(z, c) f_1(z; c; t)] + \nabla_C \cdot [\gamma \cdot \bar{\mathbf{R}}(z, c) f_1(z; c; t)] \\ & = -\sigma(z, c) f_1(z; c; t) + 2 \int_{\mathbb{R}^n} \sigma(z', c) p(z', z, c) f_1(z'; c; t) dz' \end{aligned} \quad (108)$$

The higher order product density equation is similarly obtained. Using the abbreviation $f_r(\dots; c; t)$ for the r^{th} order product density $f_r(z_1, z_2, \dots, z_r; c, t)$ we have

$$\frac{\partial}{\partial t} f_r(\dots; c; t) + \sum_{i=1}^r \nabla \cdot [\beta \cdot \bar{\mathbf{R}}(z_i, c) f_r(\dots; c; t)] + \nabla_C \cdot [f_r(\dots; c; t) \sum_{i=1}^r \gamma \cdot \bar{\mathbf{R}}(z_i, c)]$$

$$\begin{aligned}
&= -\sum_{i=1}^r \sigma(z_i, c) f_r(\dots; c; t) + \sum_{i \neq j}^r \int_{\mathcal{V}} \sigma(z', c) p(z', z_j, c) f_r(z_1, z_2, \dots, z_{j-1}, z', z_{j+1}, \dots, \\
&\quad z_r; c; t) dv' + 2 f_{r-1}(z_1, z_2, \dots, z_{i-1}, z_i + z_j, \dots, z_{j-1}, z_{j+1}, \dots, z_{r-1}; c; t) \\
&\quad \sigma(z_i + z_j, c) p(z_i + z_j, z_i; c)
\end{aligned} \tag{109}$$

The product density functions in Eqs. (108) and (109) are those that depend on the environmental concentration variables. By integrating the above equations w.r.t c over $\mathcal{C}$, we obtain the equations in the c -independent product density functions (see for example Eq. (92)). Thus the expected population density $f_1(z, t)$ satisfies the following equation

$$\begin{aligned}
&\frac{\partial}{\partial t} f_1(z, t) + \nabla \cdot \{\beta \cdot E[\bar{\mathbf{R}}(z, C)] f_1(z, t)\} = -E[\sigma(z, C)] f_1(z, t) \\
&\quad + 2 \int_{\mathcal{V}} E[\sigma(z', C) p(z', z, C)] f_1(z', t) dv'
\end{aligned} \tag{110}$$

Equation (110) is thus the generalization of Eq. (15) for populations in which the population's environment varies randomly. This generalization could well have been anticipated. The expectations in Eq. (110) are defined by Eq. (102a), in view of which, the equation of interest in this situation is Eq. (108) for the c -dependent first order product density. Interestingly enough, it is not coupled to any other equation as is, for example, Eq. (15). Thus the expected value of the population density can be obtained if Eq. (108) can be solved, which must be done subject to an appropriate initial condition. If initially there are N_0 cells of physiological states, say $z_1, z_2, \dots, z_{N_0}$ in an environment of concentration c_0 , then

$$f_1(z; c; 0) = \delta(c - c_0) \sum_{i=1}^{N_0} \delta(z - z_i) \tag{111}$$

which is an example of how initial conditions are specified for Eq. (108).

3.3.3 Stochastic versus Deterministic Models

We had observed in the preceding section that the hierarchy of equations (108) and (109) are the segregated model equations for a stochastic treatment of cell populations. The expected value is obtained from Eq. (108), while the fluctuations about the expected value are calculated by solving the hierarchy (109) for $r = 2, 3, \dots$, and using Eqs. (92) and (85). We now explore the conditions under which the stochastic equations condense into the deterministic equations (13) and (15).

If the environment concentration fluctuations are negligible (for reasons to be examined presently), then from Eq. (101), we have

$$\begin{aligned}
E[\bar{\mathbf{R}}(z, C)] &\approx \bar{\mathbf{R}}(z, E[C]); E[\sigma(z, C)] \approx \sigma(z, E[C]) \\
E[\sigma(z', C) p(z', z, C)] &\approx \sigma(z', E[C]) p(z', z, E[C])
\end{aligned} \tag{112}$$

which converts Eq. (110) to

$$\begin{aligned} \frac{\partial}{\partial t} f_1(\mathbf{z}, t) + \nabla \cdot [\beta \cdot \bar{\mathbf{R}}(\mathbf{z}, E[\mathbf{C}]) f_1(\mathbf{z}, t)] = -\sigma(\mathbf{z}, E[\mathbf{C}]) f_1(\mathbf{z}, t) \\ + 2 \int_{\mathfrak{U}} \sigma(\mathbf{z}', E[\mathbf{C}]) p(\mathbf{z}', \mathbf{z}, E[\mathbf{C}]) f_1(\mathbf{z}', t) d\mathbf{v}' \end{aligned} \quad (113)$$

which is identical to Eq. (15) containing $E[\mathbf{C}]$ in place of $\mathbf{c}$. An equation must be obtained for $E[\mathbf{C}]$, which is defined by (97). When Eq. (106) is multiplied by $\mathbf{c}$ and integrated over $\mathfrak{C}$, the use of (95) and (97) yields

$$\frac{d}{dt} E[\mathbf{C}] = -E[\bar{\mathbf{C}}] \quad (114)^a$$

From Eq. (102) it is also possible to write

$$E[\bar{\mathbf{C}}] = -\int_{\mathfrak{U}} E[\gamma \cdot \mathbf{R}(\mathbf{z}, \mathbf{c})] f_1(\mathbf{z}, t) d\mathbf{v} \quad (115)$$

and for negligible concentration fluctuations, we have from the invocation of (101)

$$\frac{d}{dt} E[\mathbf{C}] = \int_{\mathfrak{U}} \gamma \cdot \bar{\mathbf{R}}(\mathbf{z}, E[\mathbf{C}]) f_1(\mathbf{z}, t) d\mathbf{v} \quad (116)$$

Eq. (116) is identical to the deterministic equation (13), in which $\mathbf{c}$ is replaced by $E[\mathbf{C}]$.

We have thus shown that the stochastic equations for the segregated model are the same as the deterministic equations when the environmental concentration fluctuations are negligible. It is important, however, to recognize that the absence of fluctuations in $\mathbf{C}$ does not necessarily imply that the population is free from fluctuations. There may persist substantial fluctuations in small populations. The expected population density is still obtained by the solution of Eqs. (113) and (116). The fluctuations are then to be calculated from the equations in the higher order $\mathbf{c}$ -independent product density equations, which are readily obtained from (109) by integration w.r.t $\mathbf{c}$. The result for negligible concentration fluctuations is

$$\begin{aligned} \frac{\partial}{\partial t} f_r(\dots; t) + \sum_{i=1}^r \nabla \cdot [\beta \cdot \bar{\mathbf{R}}(\mathbf{z}_i, E[\mathbf{C}]) f_r(\dots; t)] = -\sum_{i=1}^r \sigma(\mathbf{z}_i, E[\mathbf{C}]) f_r(\dots; t) \\ + \sum_{i \neq j} \sum_{\mathfrak{U}} \sigma(\mathbf{z}', E[\mathbf{C}]) p(\mathbf{z}', \mathbf{z}_j, E[\mathbf{C}]) f_r(\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_{j-1}, \mathbf{z}', \mathbf{z}_{j+1}, \dots, \mathbf{z}_r; t) \\ + 2 f_{r-1}(\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_{i-1}, \mathbf{z}_i + \mathbf{z}_j, \mathbf{z}_{i+1}, \dots, \mathbf{z}_{j-1}, \mathbf{z}_{j+1}, \dots, \mathbf{z}_{r-1}; t) \\ \sigma(\mathbf{z}_i + \mathbf{z}_j, E[\mathbf{C}]) p(\mathbf{z}_i + \mathbf{z}_j, E[\mathbf{C}]) \end{aligned} \quad (117)$$

where we have used Eq. (102b) and (101).

^a In obtaining Eq. (114), use has been made of the following properties. First, $\nabla_{\mathbf{C}} \mathbf{c} = \mathbf{I}$, a unit dyadic in the m -dimensional concentration vector space $\mathfrak{C}$. Second, the *regularity condition* (see for example ³) that the dyadic $\mathbf{c} \bar{\mathbf{R}}(\mathbf{z}, \mathbf{c}) f_{\mathbf{C}}(\mathbf{c}, t) = \mathbf{0}$ holds for $\mathbf{c}$ on the *boundary* of $\mathfrak{C}$.

Equations (113) and (117) may be suitable even for sizable fluctuations in C , if the dependence of $\bar{R}(z, C)$, $\sigma(z, C)$ and $p(z', z, C)$ on C in the range of prevailing concentrations, is such that the fluctuations in $\bar{R}$, σ and p are negligible. The circumstances, under which concentration fluctuations may be negligible deserve some elaboration. If the total amounts of all the environmental substances in the abiotic phase are sufficiently large to overcome the effects of random consumption (or production) by the population, then one may expect the concentration fluctuations to be small. A second possibility is that the population may be sufficiently large for fluctuations in $\bar{C}$ to be negligible (note that such fluctuations can be calculated from Eq. (96)) relative to $E[\bar{C}]$. If the initial concentration of C is known exactly, one has then a completely deterministic situation with Eqs. (113) and (116) as the only model equations to be considered. A summary of the relevant equations for various situations in population growth is provided in Table 1.

Table 1

Small Populations				Large Populations
Negligible fluctuations in Environment		Considerable fluctuations in Environment		Expected values
Expected values	Fluctuations	Expected values	Fluctuations	Eq. (113) & Eq. (116) or Eq. (15) & Eq. (13)
Eq. (113) & Eq. (116)	Eqs. (117) $r = 2, 3, \dots$ & Eq. (85)	Eq. (108)	Eqs. (109) $r = 2, 3, \dots$ Eq. (92) & Eq. (85)	

4 Correlated Behavior of Sister Cells

Powell^{15, 56)} has presented evidence that the life spans of sister cells are positively correlated, while those of the mother and the daughter are negatively correlated. The strong positive correlation between the life spans of sister cells has also been reported by Schaechter et al.⁴³⁾ The existence of such correlations has been the basis of criticism of age distribution models⁵⁷⁾. Fredrickson et al.³⁾ have pointed out that there is no machinery in Von Foerster's model⁷⁾ to account for such effects.

It was observed at the beginning of this article that the necessity to take *explicit* account of such correlations arises for *simple* indices of the physiological state because of their inability to probe into the cause of correlated behavior. Crump and Mode⁵⁸⁾ have analyzed an age dependent branching process in which they have accounted for correlations among sister cells. They consider an arbitrary number of sister cells and the problem of correlation between their life spans. Unfortunately, their treatment, which is cast in the mathematical language of branching processes^{1, 59)}, does not blend

with the methods used herein. A framework more suitable to us is available in the extension of the product density approach^{48, 50}.

In what follows we will regard the population as distributed according to their age since we are concerned with correlations between *life spans* of sister cells. It is of course conceivable that other indices of the physiological state may also be correlated for sister cells at the instant of division.

4.1 Statistical Framework

In dealing with the distribution of cells in the physiological state space in 3.1 we had allowed at most one cell in the population to be of any given physiological state. This assumption, although unessential for the development, is a reasonable and very useful simplification. However, in the distribution of cells along the age coordinate, sister cells are necessarily of identical age, so that the aforementioned simplification must be abandoned. We will assume instead that at most *two* cells can be identified of any given age in the population. Thus any two cells of a given age will be necessarily sister cells. Instead of defining a master density function^a as in Section 3.1, we directly define the product densities of interest to us. It is convenient to make a distinction between a “singlet”, which means a cell without its sister, and a “doublet”, which refers to a pair of sister cells. Since only binary division is considered, there can at most be two sister cells; thus “multiplets” with more than two sister cells need not be considered. There can at most be *one* singlet of a given age for if there are two cells of a given age, they would be deemed a doublet. Also there can be no more than one doublet of a given age. Now we define *two first order* product densities $f_1^1(a, t)$ and $f_1^2(a, t)$ as below.

$$f_1^1(a, t)da \equiv \{\text{Expected number of singlets in the age range } (a, a + da) \text{ at time } t\}$$

$$f_1^2(a, t)da \equiv \{\text{Expected number of doublets in the age range } (a, a + da) \text{ at time } t\}$$

These product densities have also probability interpretations. Thus $f_1^1(a, t)da$ represents the probability that there is a singlet at time t between a and $a + da$, while $f_1^2(a, t)da$ is the probability that there is a doublet at time t in $(a, a + da)$. If $f_1(a, t)da$ is the expected number of cells at time t with age between a and $a + da$, then

$$f_1(a, t) = f_1^1(a, t) + 2 f_1^2(a, t) \quad (118)$$

The function $f_1(a, t)$ has no probability interpretation. Denoting the actual total number of singlets in the population by r_1 and the doublets by r_2 , we have for their expected values.

^a We refer to ⁵⁰ for a more complete exposition of the statistical framework presented here, which shows how the product densities arise from the appropriate master density function.

$$E[r_i] = \int_0^{\infty} f_1^i(a, t) da \quad i = 1, 2 \quad (119)$$

The expected values in any age range $[a_1, a_2]$ are obtained by performing the integration in (119) over the interval $[a_1, a_2]$. The expected total number of cells in the population is

$$E[N(t)] = \int_0^{\infty} f_1(a, t) da \quad (120)$$

For the fluctuations about expected values, one must have the *second* order product densities $f_2^{ij}(a_1, a_2, t)$, $i, j = 1, 2$ defined by

$$f_2^{ij}(a_1, a_2, t) da_1 da_2 = \text{Expected [number of "i-lets" in } (a_1, a_1 + da_1) \times \text{number of "j-lets" in } (a_2, a_2 + da_2 \text{ at time } t)]$$

which is also the joint probability that at time t , there is an "i-let" in $(a_1, a_1 + da_1)$ and a "j-let" in $(a_2, a_2 + da_2)$. We may also define the product density

$$f_2(a_1, a_2, t) = \sum_{i=1}^2 \sum_{j=1}^2 ij f_2^{ij}(a_1, a_2, t) \quad (121)$$

which has no probability interpretation. The product density functions $f_1^i(a, t)$ and $f_2^{ij}(a_1, a_2, t)$ provide for the calculation of the second moment of the population⁵⁰. Thus

$$E[N(t)^2] = \sum_{i=1}^2 i^2 E[r_i] + \int_0^{\infty} \int_0^{\infty} f_2(a_1, a_2, t) da_1 da_2 \quad (122)$$

Hence equations must be obtained for the product density functions for a complete stochastic analysis of the population. For suitably large populations, however, the fluctuations would be negligible so that the product density functions $f_1^i(a, t)$ and $f_2^i(a, t)$ are of paramount interest. In the next section, we derive equations for these densities for a cell population in which there is high positive correlation between the life spans of sister cells. However we neglect any correlation between the life span of a parent cell with those of its offsprings.

4.2 A Simple Age Model

In a cell population, which multiplies by binary division, the singlets are produced from doublets (when one of the cells in a doublet divides, a singlet is formed) and the doublets are formed by binary division of individual cells, whether the dividing cell is a singlet or belongs to a doublet. The strong positive correlation between sister cells (siblings) indicates that if one of the siblings divides^a, it is highly likely that the other would divide soon after. Thus we define the two transition probabilities for cell division.

^a It is assumed that no two cells can divide exactly at the same instant. This statement also holds for siblings.

$$\Gamma_1(a)dt = \Pr \{ \text{singlet of age } a \text{ at time } t \text{ will divide in the next} \\ \text{time interval } (t, t + dt) \}$$

$$\Gamma_2(a)dt = \Pr \{ \text{cell belonging to a doublet of age } a \text{ at time } t \text{ will divide} \\ \text{in the next time interval } (t, t + dt) \}$$

Clearly, the transition probability functions have been taken to be time-independent. Furthermore, $\Gamma_1(a)$ must be substantially *larger than* $\Gamma_2(a)$, since the former refers to the fission probability for a cell whose sister has already divided. The higher the positive correlation between the life spans of siblings, the larger should be the magnitude of $\Gamma_1(a)$ relative to $\Gamma_2(a)$.

The basis for the derivation of the equations for $f_1^1(a, t)$ and $f_1^2(a, t)$ is their probability interpretations. Thus a singlet of age between a and $a + da$ at time $(t + dt)$ must have been a singlet of age $(a - dt)$ at time t and failed to divide during t to $t + dt$, or must have come from a doublet of age $(a - dt)$ and one of the siblings divided during t to $t + dt$. In mathematical terms

$$f_1^1(a, t + dt)da = f_1^1(a - dt, t) da[1 - \Gamma_1(a - dt)dt] + 2 f_1^2(a - dt, t) \Gamma_2(a - dt)dt \quad (123)$$

When Eq. (123) is suitably rearranged and divided by dt , then on letting $dt \rightarrow 0$, we have

$$\frac{\partial}{\partial t} f_1^1(a, t) + \frac{\partial}{\partial a} f_1^1(a, t) = -\Gamma_1(a) f_1^1(a, t) + 2 \Gamma_2(a) f_1^2(a, t) \quad (124)$$

Since every birth gives rise to a doublet, there are no singlets of age zero, so that Eq. (124) has the boundary condition

$$f_1^1(0, t) = 0 \quad (125)$$

Eq. (124) must also be subject to an initial condition. The equation for $f_1^2(a, t)$ is derived by recognizing that a doublet of a given age (a) at time $(t + dt)$ necessarily arises from a doublet of age $(a - dt)$ at time t , neither sibling dividing between t and $t + dt$. Thus

$$f_1^2(a, t + dt)da = f_1^2(a - dt, t)da[1 - 2 \Gamma_2(a - dt)dt] \quad (126)$$

from which one obtains

$$\frac{\partial}{\partial t} f_1^2(a, t) + \frac{\partial}{\partial a} f_1^2(a, t) = -2 \Gamma_2(a) f_1^2(a, t) \quad (127)$$

The fact that a doublet of age zero can arise from the division of any cell of arbitrary age leads to the following boundary condition for Eq. (127).

$$f_1^2(0, t) = \int_0^\infty \Gamma_1(a) f_1^1(a, t)da + 2 \int_0^\infty \Gamma_2(a) f_1^2(a, t)da \quad (128)$$

Equations (124) and (127) are thus coupled equations in the first order product densities.

Since boundary condition (128) is also coupled, one must solve simultaneously for the functions $f_1^1(a, t)$ and $f_1^2(a, t)$. Before we present a solution for this problem, it is useful to identify the differential equation in the function $f_1(a, t)$ defined by (118). Multiplying Eq. (127) by 2 and adding Eq. 124, one obtains

$$\frac{\partial}{\partial t} f_1(a, t) + \frac{\partial}{\partial a} f_1(a, t) = -\Gamma_1(a) f_1^1(a, t) - 2 \Gamma_2(a) f_1^2(a, t) \quad (129)$$

One may define a transition probability function $\Gamma(a, t)$ for the division of a cell of age a (without specification of whether it is a singlet or belongs to a doublet). The explicit time-dependence in this function would be clear from the following expression for $\Gamma(a, t)$ based on the total probability theorem.

$$\Gamma(a, t) = \Gamma_1(a) \frac{f_1^1(a, t)}{f_1(a, t)} + \Gamma_2(a) 2 \frac{f_1^2(a, t)}{f_1(a, t)} \quad (130)$$

In Eq. (130) the coefficient of $\Gamma_1(a)$ is the probability that a cell of age a is a singlet, while the coefficient of $\Gamma_2(a)$ is the probability that the said cell is one of a doublet. In general, these probabilities are clearly time-dependent. Eq. (129) may now be written as

$$\frac{\partial}{\partial t} f_1(a, t) + \frac{\partial}{\partial a} f_1(a, t) = -\Gamma(a, t) f_1(a, t) \quad (131)$$

which is the same as Von Foerster's equation (Eq. 126) without the concentration of the environment. The boundary condition (27) is similarly obtained by multiplying Eq. (128) by 2 and adding to (125); thus

$$f_1(0, t) = 2 \int_0^{\infty} \Gamma(a, t) f_1(a, t) da \quad (132)$$

This equivalence to Von Foerster's equation brings to further focus the observation of Fredrickson et al.³⁾ in regard to criticisms of age distribution models⁵⁷⁾ based on their "neglect" of correlation between the life spans of siblings. Fredrickson et al. have pointed out that age distribution models such as that of Von Foerster are not equipped to account for the aforementioned correlations. The model presently under discussion, in spite of its accounting for the correlated behavior of siblings, leads to Von Foerster's equation for the expected number density function $f_1(a, t)$. It must be recognized that the transition probability function $\Gamma(a, t)$ appearing in Von Foerster's equation, viewed as an empirically determined quantity, already has built into it, the effects of the correlated behavior of siblings. During repetitive growth with negligible effects of the environment, one must expect the above function to be time-independent and represented by $\Gamma(a)^a$. In such a case, it follows that

$$f_1^i(a, t) = g_i(a) f_1(a, t), \quad i = 1, 2 \quad (133)$$

^a Here, repetitive growth refers to the time-independence of the conditional probability density $f_{Z/A}$.

where $g_1(a)$ and $g_2(a)$ are time-independent probabilities (characteristic of repetitive growth), the former representing that for a cell of age a to be a singlet, and the latter referring to that for it being one of a doublet. Thus Eq. (130) would become

$$\Gamma(a) = \Gamma_1(a) g_1(a) + 2 \Gamma_2(a) g_2(a) \quad (134)$$

More generally, however, the time-dependence of $\Gamma(a, t)$ from Eq. (130) would seem to indicate a situation of non-repetitive growth. On the other hand, in view of the time-independence assumed for $\Gamma_1(a)$ and $\Gamma_2(a)$, one may have anticipated the growth situation to be repetitive. As Fredrickson et al.³⁾ have observed, repetitive growth (if at all attained) is attained only after the effects of the initial state of the population have become negligible, i.e. repetitive growth is an *asymptotic* growth situation. Thus during the initial stages, even if the functions, $\bar{R}(z, c)$, and $p(z, z', c)$ may be independent of c (the implication of which is that cellular processes repeat at identical rates in all cells), the conditional density $f_{Z/A}$ is time-dependent quantity.

The implications of the preceding paragraph is that the age model presented here may be able to describe *non-repetitive* growth situations, in which the functions $\bar{R}$, σ and p are independent of c . Thus explicit recognition of the correlated behavior of siblings enhances the potential of age distribution models to deal with populations in restricted cases of non-repetitive growth. The restriction appears in the time-independent forms assumed for the transition probabilities $\Gamma_1(a)$ and $\Gamma_2(a)$. Here, we do not address the problem of exactly what leads to such time-independence of the aforementioned transition probabilities. From the foregoing considerations, it is evident that the *description of growth situations more general than repetitive growth depends on models, which recognize the correlated behavior of microorganisms.*

For the situation of repetitive growth, the use of Eq. (133) in Eqs. (124), (127) and (129) yields the following differential equations for $g_1(a)$ and $g_2(a)$.

$$\frac{dg_1}{da} = -\Gamma_1 g_1(1 - g_1) + 2 \Gamma_2 g_2(1 + g_1) \quad (135)$$

$$\frac{dg_2}{da} = \Gamma_1 g_1 g_2 - 2 \Gamma_2 g_2(1 - g_2) \quad (136)$$

which must be solved subject to $g_1(0) = 0$ and $g_2(0) = 1$.

For non-repetitive growth with $\Gamma_1(a)$ and $\Gamma_2(a)$ time-independent, Eqs. (124) and (127) must be solved subject to appropriate initial conditions. Suppose one has initially n singlets of ages $a_1^1, a_2^1, \dots, a_n^1$, and m doublets (i.e., $2m$ cells) of ages $a_1^2, a_2^2, \dots, a_m^2$. The initial conditions are given by

$$f_1^1(a, 0) = \sum_{j=1}^n \delta(a - a_j^1); f_2^1(a, 0) = \sum_{i=1}^m \delta(a - a_i^2) \quad (137)$$

Eqs. (124) and (127) are readily solved for the case $\Gamma_1(a) \equiv \gamma_1$ and $\Gamma_2(a) \equiv \gamma_2$ to obtain

$$f_1^1(a, t) = \begin{cases} \sum_{j=1}^n \delta(a-t-a_j^1) e^{-\gamma_1 t} + \frac{2\gamma_2}{2\gamma_2 - \gamma_1} \sum_{k=1}^m \delta(a-t-a_k^2) (e^{-\gamma_1 t} - e^{-2\gamma_2 t}) & a > t \\ \frac{2\gamma_2}{2\gamma_1 - \gamma_2} h(t-a) (e^{-\gamma_1 a} - e^{-2\gamma_2 a}), & a < t \end{cases}$$

$$f_1^2(a, t) = \begin{cases} \sum_{k=1}^m \delta(a-t-a_k^2) e^{-2\gamma_2 t}, & a > t \\ h(t-a) e^{-2\gamma_2 a}, & a < t \end{cases} \quad (139)$$

where

$$h(t) \equiv \frac{1}{\alpha} e^{\frac{1}{2}\gamma_1 t} \left[\left(-\frac{1}{2}\gamma_1 \right) (n\gamma_1 + 2m\gamma_2) + 2\gamma_1\gamma_2(n+2m) \right] \sinh \alpha t$$

$$+ (n\gamma_1 + 2m\gamma_2) \alpha \cosh \alpha t$$

$$\alpha \equiv \frac{1}{2} \sqrt{\gamma_1^2 + 8\gamma_1\gamma_2}$$

We do not pursue the analysis of this model any further since its purpose has been for the sake of illustration only.

5 Conclusions

The behavior of microbial populations is a complex manifestation of the behavior of individual cells, whose characteristic diversity necessarily requires the framework of a statistical theory for a quantitative description of population dynamics. The problem of determining individual cell behavior in a population is also intimately connected with this statistical framework. Recent experimental techniques of microfluorometry, which have enabled biochemical engineers and microbiologists to quantitatively probe into the chemical composition of individual cells in a culture, have added considerable impetus to the development of structured, segregated models. There is sufficient evidence for such models to be mathematically tractable. The availability of simulation techniques for dealing with multivariate segregated models deserves special emphasis. The constraint of repetitive growth, which applies to segregated models using simple indices of the physiological state such as cell age or size, could possibly be relaxed to analyze more general situations of growth by determining and accounting for correlated behavior of cells. The statistical synthesis becomes more complete, when correlated behavior is accounted for. While the correlated behavior of sister cells can be accommodated by the methods presented here, the methodology for accounting for correlation between a parent and its offsprings is not clear at this stage.

The mathematical apparatus for analyzing the random behavior of small populations is available and is particularly applicable in situations, where the possibility exists that the population may become extinct. In this connection, it is well to observe that

the methods presented here are readily extended to ecosystems in which more than one species are usually present.

Population balance models are also useful in dealing with growth in hydrocarbon systems since the dynamics of droplet size distributions could have pronounced effects on these systems. We have had nothing to say about such models in this article because our concern has been with the statistics of *cellular populations*.

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6 Symbols

A	Age of a cell randomly selected from the population
a	Typical value of A
C	Environmental concentration vector (m-dimensional)
c	Typical point in environmental concentration space
$\dot{C}$	Rate of consumption of environmental substances
C_s	Substrate concentration
E	Expectation
f_r	Product density of r th order
f_C	Probability density for C
f_{Z/A}	Probability density for Z conditional on a knowledge of A
f_{Z/S}	Probability density for Z conditional on a knowledge of S
f₁ⁱ	Product density of order 1 for an "i-let", defined in 4.1
J_v	Janossy density or Master density
M	Mass of cell selected at random from the population
$\bar{M}$	Average mass-specific growth rate
m	Typical value of M
N	Total number of cells per unit volume of culture
n	Number density function
p	Partitioning function for physiological state
P_v	Probability distribution for N
$\bar{R}$	Biochemical reaction rate vector
r	Partitioning function for cell size
r_i	Number of "i-lets" per unit volume of culture
S	Size of a cell selected at random from the population
$\bar{S}$	Average size-specific growth rate
s	Typical value of S
t	Time
V	Variance
Z	Physiological state vector (n-dimensional)
z	Typical point in physiological state space
$\bar{Z}$	Average growth rate of cell

German Symbols

$\mathbb{C}$	Environmental concentration space
dc	Infinitesimal volume in $\mathbb{C}$
$\mathfrak{S}_s$	Hypersurface in physiological state space defined by Eq. (17)
$d\mathfrak{f}$	Infinitesimal surface on $\mathfrak{S}_s$
$\mathfrak{P}$	Physiological state space
$d\mathfrak{v}$	Infinitesimal volume in $\mathfrak{P}$

Greek Symbols

β	Stoichiometric matrix for biochemical constituents of the cell
γ	Stoichiometric matrix for environmental substances
Γ	Age-specific or size-specific transition probability

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Mass and Energy Balances for Microbial Growth Kinetics

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First, quantitative aspects on the problems of any sort of microbial growth are depicted starting from the most general term of growth yield, $Y_{X/S}$, followed by more meaningful parameters, i.e., growth yields based on total energy available in the medium, Y_{kcal} and based on catabolic activity, $Y_{X/C}$ involved physicochemical features, and in addition growth yield based on ATP generation, Y_{ATP} being connected with physiological features. Second, quantitative relationships with respect to stoichiometry, and mass and energy balances in the growth reactions are discussed to establish kinetic equations, including growth, substrate consumption, respiration, heat evolution and noncellular product formation applicable to process control in microbial cultivations.

1 Introduction

Mathematical models have long been used for analysis of the behavior of complex structures in reaction systems when complex structures cannot be fully or directly analyzed by means of experimental determinations. In bioreaction systems, on the other hand, mathematical models have been regarded as difficult to construct due to

the inherent complexity of the living system. However, recently, the applicability and necessity of mathematical models in microbial processes have been greatly emphasized in order to search for optimum operational conditions to achieve maximum productivity of the substance aimed for when these models were coupled with modern techniques of computer and automatic analyses^{1, 2, 3)}.

At the same time we must recognize that there is still a gap between the fundamental scientific theory and the experimental data determined in biological system. As a result, mathematical models must be constructed focussing limited amounts of biological information to enhance the productivity of an aiming substance. In this context, the selection of accurate characteristics of biological systems to incorporate in mathematical models is of great importance⁴⁾.

Many researchers in applied microbiology fields have doubted the value of simplified mathematical models which cannot fully reflect the complex features of biological system. However, tremendous development in instrumentation for detection and control of cultivation processes have reduced the gap between applied microbiologists and biochemical engineers, and proved the effective practicability of mathematical models when coupled with computer techniques to search out and establish the optimum production condition^{5, 6, 7)}. Thus, mathematical models have developed for use as a practical means to predict what goes on during cultivation and also improve the control processes so as to yield a high productivity of the substance aimed. From the present situation it is of great importance to construct appropriate mathematical models using limited experimental data on the basis of fundamental knowledge of biochemical reactions.

The mathematical models proposed so far could be divided into two groups: 'unstructured' and 'structured' models⁸⁾. Unstructured models scarcely imply the biological and physiological features of a living system and consequently they were mainly constructed using arbitrary or logistic equations to fit the experimental data^{9, 10)}. Structured models, however, can be further subdivided into two types, the one mainly consists of stoichiometric relationships on the basis of macroscopic features with respect to growth, substrate consumption, respiration and other components of the system^{11, 12)} and the other can be made up of microscopic features implying molecular kinetic theory and molecular biology^{13, 14, 15)}. As a result of microscopic consideration, the latter generally becomes difficult mathematically and consequently the number of rate constants involved in the models becomes so great that one cannot decide their values only by means of experimental analyses. This must be done with the aid of modern computer technology in order to establish kinetic constants.

This article deals with the construction of a structured model from macroscopic standpoints based on energetics, and stoichiometric and mass balances with respect to growth, substrate consumption, respiration, noncellular product formation, heat evolution and other components underlying the living system. Some of the control strategies during aerobic cultivation are also discussed on the basis of the structured model developed by automatically applying analyzed signals of oxygen and carbon dioxide in the air lines of a bioreactor.

2 Outline of Energetics

2.1 Growth Yields, $Y_{X/S}$, Y_{ave} , $Y_{X/O}$, and $Y_{X/C}$

2.1.1 $Y_{X/S}$

The efficiency of biomass produced to the amount of carbon source consumed by a microorganism was originally studied and determined by Monod¹⁶⁾ who used three bacteria growing on mineral media under anaerobic conditions. He found that the dry weight of an organism was in proportion to the amount of carbon source consumed as long as the substrate was the factor limiting growth. DeMoss et al.¹⁷⁾ studied the growth yields of *Streptococcus faecalis* and *Leuconostoc mesenteroides* where glucose in complex medium was almost completely used as the energy source. They observed that the amount of biomass produced was proportional to that of the energy source consumed just as found by Monod¹⁶⁾, however, the growth yield from glucose of *S. faecalis* was about 1.43 times higher than that of *L. mesenteroides*. As a result they suggested that *S. faecalis* obtained more energy than did *L. mesenteroides*. Further investigation by Bauchop et al.¹⁸⁾ concluded that the amount of growth of a microorganism was directly proportional to that of ATP produced through the anaerobic catabolism of carbohydrate when carbohydrate was used as the energy source.

On the other hand, practical interest in the growth yield from carbon sources has been accelerated since the commercial production of microbial protein as food and fodder has become a real necessity for mankind. For the production of single-cell protein we must take into account the cost of raw material. However, it is important practically to select a microorganism which is characterized by a high growth yield from substrate since this causes it to enhance biomass productivity and also to save cooling energy during cultivation.

Generally, growth yield from substrate is expressed either as grams of dry cells produced per gram substrate consumed or as grams of dry cells produced per mole substrate consumed.

$$Y_{X/S} = \Delta X / -\Delta S \quad (1)$$

where $Y_{X/S}$ = growth yield from substrate, $g \cdot g^{-1}$ or $g \cdot \text{mole}^{-1}$, X = biomass concentration, $g \cdot l^{-1}$, S = substrate concentration, $g \cdot l^{-1}$ or $\text{mole} \cdot l^{-1}$.

Growth yield from substrate defined by Eq. (1) cannot be used for the real evaluation of energy efficiency for the respective substrate because the denominator of Eq. (1) does not represent energy quantity from the substrate. In this context, other growth yields termed Y_{ave} and Y_{kcal} will be discussed later.

2.1.2 Y_{ave}

To evaluate growth yield from substrate on the same level, the dimension of dry cell produced per gram carbon of substrate consumed was sometimes used without taking into account the other constituent elements of the substrate¹⁹⁾. Corresponding to this,

another growth yield defined by the amount of dry cell per electron equivalent initially available from the substrate was proposed to settle the problem²⁰⁾. This can be written as:

$$Y_{\text{ave}} = \frac{Y_{X/S}}{Y_{\text{ave}/S}} \quad (2)$$

where Y_{ave} = growth yield based on electron available from substrate, $\text{g}(\text{ave})^{-1}$, $Y_{\text{ave}/S}$ = electron available from substrate, $\text{ave} \cdot \text{mole}^{-1}$.

Based on the number of moles of oxygen required for the perfect combustion of one mole of substrate, $Y_{\text{ave}/S}$ can be calculated by means of multiplying the amount of oxygen required for combustion by four, that is, the number of electrons required for the reduction of one molecule of oxygen. For an example, 6 moles of oxygen are required for the combustion of 1 mole of glucose, and 1 mole of oxygen corresponds to 4 equivalents of electron, i.e., 4 ave, thus, $Y_{\text{ave}/S}$ can be calculated as $6 \times 4 = 24$ ave per mole glucose.

Calculation examples of Y_{ave} based on $Y_{X/S}$ are taking $Y_{X/S}$ of *Penicillium chrysogenum* to be 0.43 g cell per g glucose consumed²¹⁾ and that of *Pseudomonas methanica* to be 0.56 g cell per g methane consumed²²⁾, the Y_{ave} values are

$$\begin{aligned} \text{for } P. \text{ chrysogenum, } Y_{\text{ave}} &= 0.43 \times 180/24 = 3.22 \text{ g}(\text{ave})^{-1}, \\ \text{for } P. \text{ methanica, } Y_{\text{ave}} &= 0.56 \times 16/8 = 1.12 \text{ g}(\text{ave})^{-1}. \end{aligned}$$

Growth yields from substrate in terms of $Y_{X/S}$ and Y_{ave} of various microorganisms growing aerobically in minimal media containing a sole carbon source are summarized in Table 1. The average value of Y_{ave} from 79 microbes calculated by Payne²³⁾ was 3.07 g cell produced per ave utilized, although carbon sources of 69 examples were C_4 to C_6 compounds. The values of Y_{ave} from C_1 to C_3 compounds in Table 1 are remarkably lower compared with those of C_6 compounds as well as the average value which was 3.07²³⁾. The presumed reasons for this are that firstly in some cases, when a low or high molecular substrate is transported into the cells, more energy (ATP) is expended for the entry of a low molecular weight substrate on the basis of the same mass of substrate consumed, for an example, one mole of ATP is required for the transport of 180 gram of glucose (= 1 mole) whereas 3 moles of ATP are required for 180 gram of acetate (= 3 moles). Thus, as a natural consequence, the transfer of acetate into cells requires three-times more ATP compared with that of the same mass of glucose²⁴⁾, and that secondly, as the energy for growth is ultimately provided in the form of ATP, in this context, if the efficiency of ATP formation in catabolic processes is lower, the lower Y_{ave} will be observed, and that thirdly, when one uses C_1 or C_2 compounds as substrate, extra energy-requiring processes are required in order to build up monomers as the components of macromolecules such as protein, RNA, DNA and so on compared with usual carbohydrate as substrate.

Table 1. Values of $Y_{X/S}$, Y_{ave} and $Y_{X/O}$ of various microorganisms growing aerobically in minimal media (presumably without producing noncellular products)

Organism	Substrate	$Y_{X/S}$			$Y_{X/O}$	Y_{ave}
		$\frac{g}{g}$	$\frac{g}{mole}$	$\frac{g}{g-C}$	$\frac{g}{g}$	$\frac{g}{ave}$
<i>Aerobacter aerogenes</i> ²⁵⁾	maltose	0.46	149.2	1.03	1.50	3.11
	mannitol	0.52	95.5	1.32	1.18	3.67
	fructose	0.42	76.1	1.05	1.46	3.17
	glucose	0.40	72.7	1.01	1.11	3.00
<i>Candida utilis</i> ²⁶⁾	glucose	0.51	91.8	1.28	1.32	3.82
<i>Penicillium chrysogenum</i> ²¹⁾	glucose	0.43	77.4	1.08	1.35	3.22
<i>Pseudomonas fluorescens</i> ²⁶⁾	glucose	0.38	68.4	0.95	0.85	2.85
<i>Rhodopseudomonas spheroides</i> ²⁷⁾	glucose	0.45	81.0	1.12	1.46	3.37
<i>Saccharomyces cerevisiae</i> ²⁸⁾	glucose	0.50	90.0	1.25	0.97	3.75
<i>Aerobacter aerogenes</i> ²⁵⁾	ribose	0.35	53.2	0.88	0.98	2.66
	succinate	0.25	29.7	0.62	0.62	2.12
	glycerol	0.45	41.8	1.16	0.97	2.99
	lactate	0.18	16.6	0.46	0.37	1.38
	pyruvate	0.20	17.9	0.49	0.48	1.78
	acetate	0.18	10.5	0.43	0.31	1.31
	acetate	0.36	21.0	0.90	0.70	2.62
	acetate	0.28	16.8	0.70	0.46	2.10
<i>Candida utilis</i> ²⁶⁾	ethanol	0.68	31.2	1.30	0.61	2.60
<i>Pseudomonas fluorescens</i> ²⁶⁾	ethanol	0.49	22.5	0.93	0.42	1.87
<i>Klebsiella</i> sp. ²⁹⁾	methanol	0.38	12.2	1.01	0.56	2.03
<i>Methylomonas</i> sp. ³⁰⁾	methanol	0.48	15.4	1.28	0.53	2.56
<i>Pseudomonas</i> sp. ³¹⁾	methanol	0.41	13.1	1.09	0.44	2.18
<i>Methylococcus</i> sp. ³²⁾	methane	1.01	16.2	1.34	0.29	2.02
<i>Pseudomonas</i> sp. ³³⁾	methane	0.80	12.8	1.06	0.20	1.60
<i>Pseudomonas</i> sp. ²²⁾	methane	0.60	9.6	0.80	0.19	1.20
<i>Pseudomonas methanica</i> ²²⁾	methane	0.56	9.0	0.75	0.17	1.12

2.1.3 $Y_{X/O}$ and $Y_{X/C}$

Growth yield based on oxygen, $Y_{X/O} = \Delta X / \Delta O_2$, gram cell produced per mole oxygen consumed means the efficiency of biomass produced to catabolic energy expended since the ΔO_2 corresponds to the representative value of overall catabolic activity when energy yielding reactions are mainly *via* the oxidative phosphorylation pathway without depending on the glycolytic pathway. These values are summarized in Table 1 in comparison with other growth yields. As there seems to be a proportional relationship between $Y_{X/O}$ and Y_{ave} in the table as previously observed by Minkevich et al.³⁸⁾, these values are arranged on $Y_{X/O}$ vs. Y_{ave} (Fig. 1). It is interesting in the figure to notice that, when a microorganism grows aerobically in minimal media utilizing a sole carbon source without producing any end-products, Y_{ave} remarkably depends on $Y_{X/O}$. In this respect, on the basis of the stoichiometry of the growth reaction, Minkevich et al.³⁸⁾ theoretically analyzed the relationship between $Y_{X/O}$ and Y , g-carbon/mole substrate, and derived the formula to express the tendency of $Y_{X/O}$ vs. Y_{ave} as observed in Fig. 1.

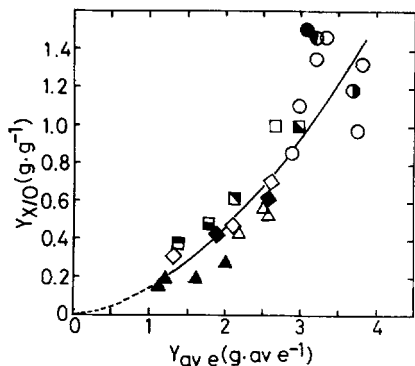


Fig. 1. Relationships between $Y_{X/O}$ and Y_{ave} of various microorganisms. Data are originated from Table 1. Marks: ● maltose, ○ glucose, ● mannitol, ● fructose, □ ribose, ■ succinate, ■ glycerol, ■ lactate, ■ pyruvate, ◇ acetate, ◆ ethanol, △ methanol, ▲ methane

Another growth yield based on catabolic activity can also be represented by taking account of the difference between the heat of combustion of carbon source consumed and that for the sum of end-products discharged. This has been little assessed hitherto perhaps due to experimental complexities, particularly for the quantitative determinations of end-products.

The heat generation by catabolism can be written by the following equation in the case of carbon source in complex media.

$$\Delta H_C = \Delta H_S \cdot (-\Delta S) - \sum \Delta H_P \cdot \Delta C_P \quad (3)$$

where ΔH_C = heat generation by catabolism, $\text{kcal} \cdot \text{l}^{-1}$, ΔH_S = heat of combustion of substrate, $\text{kcal} \cdot \text{mole}^{-1}$, ΔH_P = heat of combustion of end-product, $\text{kcal} \cdot \text{mole}^{-1}$, C_P = product concentration, $\text{mole} \cdot \text{l}^{-1}$.

It must be mentioned in Eq. (3) that, when a carbohydrate is used in complex media, very little of the carbon source is assimilated into cells, in other words, mostly dissimilated as the energy source^{18, 34)} and that, when one uses a carbon source in minimal media, part of the carbon source must be assimilated to the cellular substance, using the other for energy source. Thus, Eq. (3) can be applicable in the case of carbon source in complex media. When one uses minimal media, the amount of carbon source dissimilated must be corrected on the balance of the fate of carbon source consumed (see Section 2.2.3).

Thus, the growth yield based on catabolic activity, $Y_{X/C}$ can be written as:

$$\begin{aligned} Y_{X/C} &= \frac{\Delta X}{\Delta H_C} = \frac{\Delta X}{\Delta H_S \cdot (-\Delta S) - \sum \Delta H_P \cdot \Delta C_P} \\ &= \frac{Y_{X/S}}{\Delta H_S - \sum \Delta H_P \cdot Y_{P/S}} \end{aligned} \quad (4)$$

where $Y_{P/S}$ = product yield from substrate, $\text{mole} \cdot \text{mole}^{-1}$.

The values of heat of combustion on various substances are listed in Table 2 and compared with those theoretically calculated. The calculation of the heat of combustion of each substance can be made by means of assessing heat production either on the basis of available electrons from each molecule³⁵⁾ on the basis of oxygen consumed for the complete combustion of substrate³⁸⁾. According to Okunuki³⁵⁾, it is assumed that, first, the electrons in the C—C bond or C—H bond of a respective molecule produce heat energy of 26.05 kcal per equivalent of electron, and that, second, those electrons in C=O, CHOH and CH₂OH in the respective molecule also produce additional heat energy of 19.5, 13 and 13 kcal per equivalent of electrons respectively, one can calculate the value of the heat of combustion of the respective compound.

As an example, the heat of combustion of ethanol can be calculated as follows: one molecule of ethanol consists of 1 of C—C and 5 of C—H bonds, as a result, 12 (= (1 + 5) × 2) electrons in total relate to these bonds. Thus, heat of combustion based on 12 electrons can be calculated to be 312.6 (= 12 × 26.05) kcal per mole ethanol. Additional heat production based on CH₂OH of ethanol molecule can be taken to be 13 kcal per mole ethanol. As a result, the heat of combustion of ethanol can be calculated as the sum of 312.2 + 13, i.e., 325.6 kcal per mole ethanol. The value calculated for ethanol is in accord with the determined value of 326.5 (see Table 2). Most values predicted on the basis of the above assumption except those of formic acid and formaldehyde (Table 2) seem to be good correlation compared with the determined values³⁶⁾.

Table 2. The values of heat of combustion of various substances³⁶⁾ compared with those theoretically calculated (see in text)

Substance	Heat of combustion ³⁶⁾ kcal · mole ⁻¹	Heat of combustion predicted kcal · mole ⁻¹
methane	212.8	208.4 (26.05 × 8)
methanol	173.7	169.3 (26.05 × 6 + 13)
ethanol	326.5	325.6 (26.05 × 12 + 13)
glycerol	397.8	403.7 (26.05 × 14 + 13 × 3)
formaldehyde	134.1	123.7 (26.05 × 4 + 19.5)
acetaldehyde	278.8	280.0 (26.05 × 10 + 19.5)
acetone	436.3	437.3 (26.05 × 16 + 19.5)
formic acid	62.9	52.1 (26.05 × 2)
acetic acid	208.6	208.4 (26.05 × 8)
lactic acid	326.0	325.6 (26.05 × 12 + 13)
pyruvic acid	280.0	280.0 (26.05 × 10 + 19.5)
tartaric acid	275.1	286.5 (26.05 × 10 + 13 × 2)
maleic acid	320.1	325.6 (26.05 × 12 + 13)
succinic acid	357.1	364.7 (26.05 × 14)
fumaric acid	320.0	312.0 (26.05 × 12)
xylose	561.5	563.0 (26.05 × 20)
galactose	670.7	690.2 (26.05 × 24 + 13 × 5)
glucose	673.0	690.2 (26.05 × 24 + 13 × 5)
rhamnose	718.3	729.0 (26.05 × 26 + 13 × 4)
maltose	1350.2	1354.4 (26.05 × 48 + 13 × 8)

When a microorganism grows in a complex medium without producing products, $Y_{X/S}$ is identical with $Y_{X/C}$ (Eq. 4), however, when considerable extracellular products are discharged from the cells, the former becomes a pretended yield differing completely from the latter. Within this context, the evaluation either by $Y_{X/S}$ or $Y_{X/C}$ was conducted in the growth of *Streptococcus agalactiae*³⁷⁾ as an example (Table 3). From the table, $Y_{X/S}$ values in the aerobic cultivations are almost two-fold compared to those in the anaerobic cultivations. However, the discrepancies as to $Y_{X/S}$ could be dispensed with on the basis of $Y_{X/C}$ giving more or less the same value as can be seen in Table 3, although the case of pyruvate in the aerobic culture seems to be a little larger value compared to the others, probably because of an overestimation of end-products as could be seen in its carbon balance in Table 3. One point which is necessary for the prediction of $Y_{X/C}$ is that the carbon balance must be accurately determined to get to the extent of enough balance.

Table 3. Estimation of $Y_{X/C}$ and Y_{kcal} of *Streptococcus agalactiae* growing on either glucose or pyruvate as the energy source in a complex medium. Experimental data required for the calculation were from ³⁷⁾

	aerobic		anaerobic	
	glucose	pyruvate	glucose	pyruvate
$Y_{X/S} \text{ g} \cdot \text{mole}^{-1}$	51.60	12.45	21.40	6.87
$Y_{X/C} \text{ g} \cdot \text{kcal}^{-1}$	0.29	0.41	0.32	0.32
$Y_{X/O} \text{ g} \cdot \text{mole}^{-1}$	42.0	59.6	—	—
$Y_{kcal} \text{ g} \cdot \text{kcal}^{-1}$	0.114	0.132	0.120	0.114
$Y_{P/S}$ Ethanol	—	—	0.14	—
Lactic acid	0.79	0.31	1.50	0.23
mole Acetic acid	0.86	0.70	0.25	0.72
mole Formic acid	0.12	0.04	0.33	0.53
Acetoin	0.09	0.004	—	—
Carbon recovered %	96.8	103.8	93.3	96.4

For the calculation: $\Delta H_P = 553.5 \text{ kcal} \cdot \text{mole}^{-1}$ for acetoin (theoretically estimated, see in Section 2.1.3), other ΔH_P : see Table 2, $\Delta H_a = 5.3 \text{ kcal} \cdot \text{g}^{-1}$

2.2 Growth Yield Based on Total Energy, Y_{kcal}

2.2.1 General Considerations

A growth yield based on total energy available from the medium, Y_{kcal} can be written as²³⁾:

$$Y_{kcal} = \frac{\Delta X}{\Delta H_a \cdot \Delta X + \Delta H_C} \quad (5)$$

where Y_{kcal} = growth yield based on total energy available, $\text{g} \cdot \text{kcal}^{-1}$, ΔH_a = heat of combustion of dry cell, $\text{kcal} \cdot \text{g}^{-1}$, ΔH_C = heat generation by catabolism, $\text{kcal} \cdot \text{l}^{-1}$.

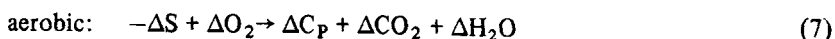
The denominator of Eq. (5) means that total energy available consists of two parts, i.e., energy incorporated biosynthetically into cellular materials and that expended by catabolism.

For the estimation of Y_{kcal} , a value of ΔH_a has been proposed to be 5.3 kcal per g cell based on a calorimetric analysis^{20, 23}. Another value of ΔH_a to be 4.2 can be taken on the basis of energy balance during the growth of *Saccharomyces cerevisiae* (see Section 3.5). Payne²³ proposed the two methods to estimate ΔH_C defined by Eq. (5). First, when a microorganism grows aerobically either in minimal media or complex media, it can be calculated on the basis of the amount of oxygen consumed multiplied by the energy quantity available for the reduction of oxygen, 106 kcal · mole⁻¹,²³. However, one must use this means with great care because this can be only be applied when no metabolic end-products are discharged extracellularly, in other words, when the growth is supported mainly by oxidizing carbon source with oxygen. If the two dissimilating pathways, that is, aerobic and anaerobic, operate simultaneously in any sort of growth as can be seen in aerobic degradation of sugar by *Saccharomyces*²⁸, the heat generation by catabolism should be assessed considering the sum of each value with respect to the two distinct pathways. This significant feature will be fully discussed on the thermodynamic basis in the case of aerobiosis (Section 3.5).

Second, it is true that the value of ΔH_C can be assessed on the basis of the difference between the heat of combustion of the energy source and that for the sum of end-products within the limits of cultivation in complex media. In fact, when one uses minimal media, dissimilated substrate should correspond to the difference between the total substrate consumed and the substrate incorporated into cellular materials (see Section 2.2.3).

2.2.2 Y_{kcal} in Complex Media

Since a carbon source in any sort of complex media almost completely dissimilated by catabolism, the overall fate of carbon source can be represented by:



Heat generation by catabolism can be expressed by Eq. (3), thus, the substitution of ΔH_C (Eq. 3) into Eq. (5) yields that

$$\begin{aligned} Y_{kcal} &= \frac{\Delta X}{\Delta H_a \cdot \Delta X + \Delta H_S \cdot (-\Delta S) - \sum \Delta H_P \cdot \Delta C_P} \\ &= \frac{Y_{X/S}}{\Delta H_a \cdot Y_{X/S} + \Delta H_S - \sum \Delta H_P \cdot Y_{P/S}} \end{aligned} \quad (8)$$

If any end-products are not produced ($\Delta C_P = 0$), Eq. (3) might be reduced as follows (discussed in Section 3.5):

$$\Delta H_C = \Delta H_S \cdot (-\Delta S) = \Delta H_O \cdot \Delta O_2 \quad (9)$$

where ΔH_O = heat generation based on oxygen consumed, 106²³⁾ and 108³⁸⁾, kcal · mole⁻¹.

Thus, Y_{kcal} provided that $\Delta C_P = 0$ in complex media can be written by the substitution of Eq. (9) into Eq. (5).

$$Y_{kcal} = \frac{1}{\Delta H_a + \Delta H_S/Y_{X/S}} \quad (10-1)$$

or

$$Y_{kcal} = \frac{1}{\Delta H_a + \Delta H_O/Y_{X/O}} \quad (10-2)$$

The values of Y_{kcal} of *Streptococcus agalactiae* growing in complex media can be assessed by Eq. (8) using the data of $Y_{X/S}$ and $Y_{P/S}$ in Table 3 and also the values of heat of combustion of respective substrate in Table 2. Assessed values of Y_{kcal} are described in Table 3.

2.2.3 Y_{kcal} in Minimal Media

The sole carbon source in minimal media is naturally metabolized partly via biosynthetic pathways and the other via catabolic pathways. Thus, the fate of carbon source can be written as:



The amount of substrate equivalent to that of cellular carbon produced from the sole carbon source will be approximated by the following equation^{45, 46)}, if there is no assimilation of carbon dioxide which would be brought from the air.

$$-\Delta S_C = \frac{\alpha_2}{\alpha_1} \Delta X \quad (12)$$

where $-\Delta S_C$ = amount of substrate equivalent to cellular carbon produced, mole · l⁻¹, α_1 = carbon content of substrate, g · mole⁻¹, α_2 = carbon content of cell, g · g⁻¹.

Thus, taking account of the difference between $-\Delta S$ and $-\Delta S_C$, one can predict the amount of substrate that might be dissimilated in energy yielding processes⁴³⁾.

$$-\Delta S - (-\Delta S_C) = -\Delta S - \frac{\alpha_2}{\alpha_1} \Delta X = -\Delta S (1 - \frac{\alpha_2}{\alpha_1} Y_{X/S}) \quad (13)$$

In this context, Eq. (3) applicable in the case of complex media, must be corrected by Eq. (13) when one uses minimal media. This can be written as:

$$\Delta H_C = \Delta H_S \cdot (-\Delta S) \cdot (1 - \frac{\alpha_2}{\alpha_1} Y_{X/S}) - \sum \Delta H_P \cdot \Delta C_P \quad (14)$$

Substitution of Eq. (14) into Eq. (5) yields:

$$Y_{kcal} = \frac{Y_{X/S}}{\Delta H_a \cdot Y_{X/S} + \Delta H_S (1 - \frac{\alpha_2}{\alpha_1} Y_{X/S}) - \sum \Delta H_P \cdot Y_{P/S}} \quad (15-1)$$

When $\Delta C_p = 0$, Eq. (15-1) reduces to

$$Y_{\text{kcal}} = \frac{Y_{X/S}}{\Delta H_a \cdot Y_{X/S} + \Delta H_s \left(1 - \frac{\alpha_2}{\alpha_1} Y_{X/S}\right)} \quad (15-2)$$

In conclusion, Y_{kcal} in any sort of aerobic growth without producing noncellular products in minimal media can be assessed either by Eq. (10-2) or Eq. (15-2).

An example of Y_{kcal} estimation is shown below when *Candida utilis* grows in minimal medium utilizing glucose as the sole carbon source (see Table 1).

By Eq. (10-2):

$$Y_{\text{kcal}} = \frac{1}{5.3 + 106/42.2} = 0.128 \text{ g} \cdot \text{kcal}^{-1}$$

By Eq. (15-2):

$$Y_{\text{kcal}} = \frac{91.8}{5.3 \times 91.8 + 673 \left(1 - \frac{0.5}{72} 91.8\right)} = 0.126 \text{ g} \cdot \text{kcal}^{-1}$$

provided that $\alpha_2 = 0.5 \text{ g} \cdot \text{g}^{-1}$

The values of Y_{kcal} of heterotrophs growing aerobically in minimal media without producing any particular products are summarized in Table 4. Judging from the values of Y_{kcal} in Table 4, the estimations by Eq. (10-2) would be appreciated on equal terms with those by Eq. (15-2). As a whole, the average value of Y_{kcal} in the case of glucose seems to be apparently larger than those of other substrates, particularly in the cases of C_1 and C_2 compounds. A gap observed in Y_{kcal} due to the difference of substrate might be explained by the same reasons as were discussed for the growth yield in terms of Y_{ave} (see 2.1.2). In addition, the values of Y_{kcal} for methane based on Eq. (15-2) were relatively larger than those based on Eq. (10-2). The discrepancy observed might be caused by the complicated features of methane metabolism involving the problem of whether or not Eq. (15-2) based on the approximative character of Eq. (12) is no longer applicable for methane-utilizing bacteria.

2.3 Growth Yield Based on ATP Generation, Y_{ATP}

2.3.1 Y_{ATP} in Energy Coupled Growth

Bauchop et al.¹⁸⁾ found that the amounts of growth of *Streptococcus faecalis*, *Saccharomyces cerevisiae* and *Zymomonas mobilis* growing anaerobically in complex media were directly proportional to the moles of ATP produced by catabolism.

The average value obtained from the three organisms was 10.5 g cell produced per 1 mole of ATP generated, ranging from 8.3 to 12.6.

Thus, Y_{ATP} can be defined by:

$$Y_{\text{ATP}} = \frac{\Delta X}{\Delta \text{ATP}} = \frac{Y_{X/S}}{Y_{A/S}} \text{ g} \cdot \text{mole}^{-1} \quad (16)$$

Table 4. Y_{kcal} values of heterotrophs growing aerobically in minimal media presumably without producing noncellular products. Values of $Y_{X/S}$, $Y_{X/O}$ and ΔH_S required for the estimations are from Tables 1 and 2, and $\alpha_2 = 0.5 \text{ g} \cdot \text{g}^{-1}$, $\Delta H_a = 5.3 \text{ kcal} \cdot \text{g}^{-1}$ and $\Delta H_O = 106 \text{ kcal} \cdot \text{mole}^{-1}$ are assumed

Organism	Substrate	Y_{kcal}	$\text{g} \cdot \text{kcal}^{-1}$
		Eq. 15-2	Eq. 10-2
<i>Aerobacter aerogenes</i>	maltose	0.104	0.133
<i>Candida utilis</i>	glucose	0.126	0.128
<i>Penicillium chrysogenum</i>	glucose	0.107	0.129
<i>Pseudomonas fluorescens</i>	glucose	0.096	0.108
<i>Rhodopseudomonas spheroides</i>	glucose	0.112	0.132
<i>Saccharomyces cerevisiae</i>	glucose	0.123	0.114
<i>Aerobacter aerogenes</i>	glucose	0.101	0.121
		average 0.116	
<i>Aerobacter aerogenes</i>	ribose	0.089	0.115
	succinate	0.073	0.093
	glycerol	0.108	0.114
	lactate	0.050	0.070
	pyruvate	0.059	0.082
		average 0.085	
<i>Aerobacter aerogenes</i>	acetate	0.048	0.063
<i>Candida utilis</i>	acetate	0.092	0.106
<i>Pseudomonas fluorescens</i>	acetate	0.075	0.080
		average 0.077	
<i>Candida utilis</i>	ethanol	0.112	0.093
<i>Pseudomonas fluorescens</i>	ethanol	0.077	0.076
		average 0.090	
<i>Methylobacterium methanolicum</i>	methanol	0.107	0.087
<i>Klebsiella</i> sp.	methanol	0.081	0.089
<i>Pseudomonas</i> sp.	methanol	0.088	0.078
		average 0.088	
<i>Pseudomonas</i> sp.	methane	0.077	0.046
<i>Methylococcus</i> sp.	methane	0.104	0.059
<i>Pseudomonas</i> sp.	methane	0.054	0.044
<i>Pseudomonas methanica</i>	methane	0.050	0.040
		average 0.066	

where Y_{ATP} = growth yield based on ATP generation, $\text{g} \cdot \text{mole}^{-1}$, $Y_{A/S}$ = ATP yield from energy source catabolized, $\text{mole} \cdot \text{mole}^{-1}$.

For the estimation of Y_{ATP} , $Y_{A/S}$ must be firstly assessed on the basis of the established pathways accompanied by ATP formation, and secondly calculated by the amount of ATP on the basis of the experimental data with respect to energy source utilized and/o end-products formed. In this context, many workers have long been using complex media in which monomer units such as amino acids are contained in sufficient concentration to

enable the substrate to be almost completely catabolized. Therefore, the average value of Y_{ATP} to be about 10 observed in many organisms in the above conditions (see Table 5) seems to come within the category of the so-called energy coupling growth. In other words, the rate of ATP synthesis on the catabolic pathways would be concerned with a rate-limiting step in the whole reaction sequence. Comparing the theoretical estimation of Y_{ATP} to be 33³⁹⁾, the average Y_{ATP} mentioned above is generally about one-third. This fact would suggest that the greater part of ATP synthesized might be expended for other purposes besides the ATP required for activation and polymerization purposes in the construction of a cellular body.

On the other hand, a large value of Y_{ATP} compared to 10 can be seen in some cases where first, a microorganism is capable of intracellular accumulation of some energy storage materials such as polysaccharides and poly- β -hydroxy butyrates as observed in *Ruminococcus albus* growing on cellobiose⁷⁸⁾, and second, a microorganism capable of utilizing other nutrients as an energy source such as amino acids; one sometimes fails to count the extra ATP formation. However, in fact, remarkably large Y_{ATP} values (18.7 to 23.5) were observed in *Lactobacillus casei* growing in glucose-limited chemostat cultures in complex media⁴⁰⁾. This large Y_{ATP} value could not be elucidated using the above mentioned reasons, and remains an interesting problem for the future.

Table 5. Growth yields based on ATP generation of microorganisms growing anaerobically in complex media

Organism	Substrate	Y_{ATP} , g · mole ⁻¹
<i>Aerobacter aerogenes</i>	glucose ²⁵⁾	10.3
	fructose ²⁵⁾	10.7
	mannitol ⁴⁸⁾	10.0
	gluconic acid ⁴³⁾	11.0
<i>Aerobacter cloacae</i>	glucose ⁴⁹⁾	11.5
<i>Actinomyces israeli</i>	glucose ⁵⁰⁾	12.5
<i>Bifidobacterium bifidum</i> ⁵¹⁾	glucose	13.1
	lactose	10.4
	galactose	9.9
	mannitol	11.8
<i>Clostridium bifidum</i>	glutamic acid ⁵²⁾	10.9
<i>Clostridium thermoaceticum</i>	glucose ⁵³⁾	10.0
<i>Desulfovibrio desulfuricans</i>	pyruvic acid ⁵⁴⁾	9.6
<i>Escherichia coli</i>	glucose ⁴⁹⁾	9.4
<i>Lactobacillus plantarum</i>	glucose ⁵⁵⁾	9.4
<i>Streptococcus faecalis</i>	glucose ¹⁸⁾	11.5
	arginine ¹⁸⁾	10.0
	ribose ¹⁸⁾	12.6
	glucose ³⁷⁾	11.1
<i>Streptococcus agalactiae</i>	pyruvic acid ³⁷⁾	10.0
	glucose ¹⁸⁾	10.5
<i>Saccharomyces cerevisiae</i>	glucose ¹⁸⁾	8.3
<i>Zymomonas mobilis</i>	glucose ¹⁸⁾	10.7
average:		10.7

Estimations of P/O ratio, the efficiency of ATP formation in relation to oxygen consumed via the oxidative phosphorylation pathway have always up to now been attempted by means of preparing crude-cell-free extract related to the system^{41,42}. The value measured *in vitro*, however, cannot represent the real value of P/O ratio *in vivo* due to a large gap between the two conditions⁴³. In this context, many attempts to predict the P/O ratio *in vivo* have been carried out on the basis of the Y_{ATP} concept^{28,37,44,45,46}.

A prediction of P/O ratio based on Y_{ATP} concept where *Aerobacter aerogenes* grows aerobically in minimal medium⁴⁷ is shown below as an example. The net gain of ATP per one mole glucose by this organism is: first, 2 moles ATP are produced when one mole glucose is catabolized via the glycolytic pathway, and second, one extra mole ATP is formed per one mole acetate produced, third, some ATP is formed via oxidative phosphorylation under aerobic condition. Data required for the estimation of P/O ratio are⁴⁷: for anaerobic condition, specific growth rate, $\mu = 0.4 \text{ h}^{-1}$, specific rate of glucose consumption, $\nu = 0.0154 \text{ mole} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$, specific rate of acetate production, $Q_P = 0.0102 \text{ mole} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$, and for aerobic condition, $\mu = 0.4$, $\nu = 0.0062$, $Q_P = 0$ and $Q_{O_2} = 0.01078 \text{ mole} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$.

Prior to the calculation of P/O ratio, one must evaluate Y_{ATP} value. As the glucose in the minimal medium serves both as carbon and energy sources, the amount of glucose dissimilated can be approximated by Eq. (13). Assuming that the glucose dissimilated is able to associate with ATP formation in glycolysis, specific rate of ATP formation, Q_{ATP} can be calculated by the sum of glucose dissimilated and acetate produced⁴⁷.

$$Q_{ATP} = \nu \left(1 - \frac{\alpha_2}{\alpha_1} Y_{X/S} \right) \times 2 + Q_P \times 1 \quad (17)$$

Thus,

$$Y_{ATP} = \frac{\mu}{Q_{ATP}} = \frac{\mu}{2 \nu \left(1 - \frac{\alpha_2}{\alpha_1} Y_{X/S} \right) + Q_P} \quad (18)$$

Substitution of $\mu = 0.4$, $\nu = 0.015$, $Q_P = 0.0102$, $Y_{X/S} = \mu/\nu = 25.97$, $\alpha_2 = 0.5 \text{ g} \cdot \text{g}^{-1}$, $\alpha_1 = 72 \text{ g} \cdot \text{mole}^{-1}$ into Eq. (18) gives that $Y_{ATP} = 11.3 \text{ g} \cdot \text{mole}^{-1}$.

Thus, the prediction of P/O ratio based on $Y_{ATP} = 11.3$ is as follows: Q_{ATP} in aerobic culture can be written as:

$$Q_{ATP} = 2 \nu \left(1 - \frac{\alpha_2}{\alpha_1} Y_{X/S} \right) + Q_P + 2 (P/O) Q_{O_2} \quad (19)$$

Arranging Eq. (19) we have

$$P/O = \frac{\mu - 2 \nu \left(1 - \frac{\alpha_2}{\alpha_1} Y_{X/S} \right) Y_{ATP}}{2 Y_{ATP} Q_{O_2}} \quad (20)$$

Substituting $Y_{ATP} = 11.3$, $\nu = 0.0062$, $Y_{X/S} = \mu/\nu = 64.5$, $\mu = 0.4$, $Q_P = 0$, $Q_{O_2} = 0.01078$ into Eq. (20), we have that $P/O = 1.32 \text{ mole ATP per g-atom oxygen}$.

2.3.2 Y_{ATP} in Energy Uncoupled Growth

As already mentioned, if the energy-yielding metabolism is fully coupled with macromolecular syntheses for a cellular body, that is, the limitation of growth is at the level of energy production, Y_{ATP} could be counted to be more or less 10 in many organisms (Table 5). However, this is the contrary if the formation rate of one or more essential compounds required for cell biosyntheses is rate-limiting rather than that of ATP in energy-yielding processes, energy would be excess and wasted as heat without coupling the growth, that is, the so-called energy uncoupled growth, unless some control system is able to regulate the rate of energy production⁵⁴).

Hernandez et al.⁴⁹) studied whether or not the energy uncoupled growth occurred when the growth of *Escherichia coli* is limited by factors other than ATP supply to biosyntheses (Table 6). In the table when the minimal medium contains excess EDTA (3×10^{-3} M) or citric acid (10^{-2} M) as chelating agent, the final amount of cell produced at the end of batch culture was apparently less compared with the case of basal medium in spite of more or less the same amount of ATP being produced, probably because some essential metals such as iron and magnesium might become a limiting factor for biosyntheses. As a result, the Y_{ATP} value was improved when the chelating agent was excluded from the medium and finally attained almost the general value of $Y_{ATP} = 10$ (Table 6) when the minimal medium was adjusted by the addition of amino acids and yeast extract.

Table 6. Y_{ATP} of *Escherichia coli* grown anaerobically in glucose-minimal media with the addition of some substances⁴⁹) (Courtesy of American Soc. for Microbiology)

Addition to minimal medium	Total cell yield, g	Total ATP yield, mole	Y_{ATP} , g · mole ⁻¹
EDTA	0.44	0.268	1.6
Citric acid	1.09	0.262	4.1
Amino acids + citric acid	1.48	0.284	5.2
None (minimal medium)	1.38	0.204	6.4
Amino Acids	1.89	0.239	7.9
Amino acids + vitamins			
+ nucleic acid precursors	1.72	0.238	7.2
Amino acids + yeast extract	2.40	0.255	9.4

Energy uncoupling growth due to pantothenate starvation during the growth of *Zymomonas mobilis* has been studied by means of adjusting the pantothenate concentration in the medium^{56, 57, 58}). Typical results are shown in Table 7⁵⁸). In the table it can be seen that the specific growth rate tends to decrease with the decrease in pantothenate concentration in spite of specific rate of glucose consumption being more or less constant. As a result, growth yield from glucose becomes lower in the pantothenate limited conditions suggesting that excess amount of ATP produced on the pathway of glucose dissimilation might be wasted as heat without coupling biosynthetic activity. Another interesting result observed in Table 7 is that ATP pool, that is. ATP content in

Table 7. Effects of pantothenate on the growth of *Zymomonas mobilis* in anaerobic culture⁵⁸⁾, (Courtesy of Cambridge Univ. Press)

Medium	Pantothenate concentration $\text{g} \cdot \text{l}^{-1}$	$Y_{X/S}$ $\text{g} \cdot \text{mole}^{-1}$	μ h^{-1}	ν $\text{mole} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$	ATP pool $\text{mg} \cdot \text{g}^{-1}$
Complex	—	7.0	0.37	0.054	1.54
Defined ^a	5×10^{-3}	6.4	0.39	0.061	1.55
Defined ^a	1×10^{-7}	2.8	0.20	0.067	3.15
Minimal	5×10^{-3}	4.5	0.28	0.064	3.55
Minimal	1×10^{-6}	2.9	0.16	0.057	4.52

^a minimal medium + 20 amino acids

the cell, seems to increase in the cases of less pantothenate concentration and which remains a problem to be solved in the future.

Other examples related to energy uncoupled growth are first, during the aerobic growth of *Aerobacter aerogenes*⁵⁴⁾, the growth yields from carbon sources in ammonia media have been observed to be double compared to those with nitrate as the nitrogen source, suggesting that nitrate consistently depressed the anabolic activities regardless of its lack of influence on the catabolic activities.

Second, the aerobic growth of *Azotobacter vinelandii*, a nitrogen-fixing bacterium, was extremely affected by the dissolved oxygen concentration in the culture medium, the growth yield from glucose in high oxygen concentration (i.e. 4 ppm) resulted in quite a low value of $Y_{X/S}$ (i.e. $5 \text{ g} \cdot \text{mole}^{-1}$). On the other hand, $43 \text{ g} \cdot \text{mole}^{-1}$ was observed in the oxygen-limited culture where the dissolved oxygen concentration was controlled less than 0.1 ppm⁵⁹⁾. The facts mentioned above suggest that the molecular oxygen in the culture media may cause either inhibition or repression of the nitrogenase system which is essential for nitrogen fixation and thereby leads to the energy uncoupled growth.

Third, in anaerobic carbohydrate metabolism, it often happens that the utilization of carbohydrates by a microorganism still continues even after the growth appears to have ceased. As an example, the time courses of alcohol formation from rice-starch in sake brewing are shown in Fig. 2. It is evident in the figure that ethanol accumulation

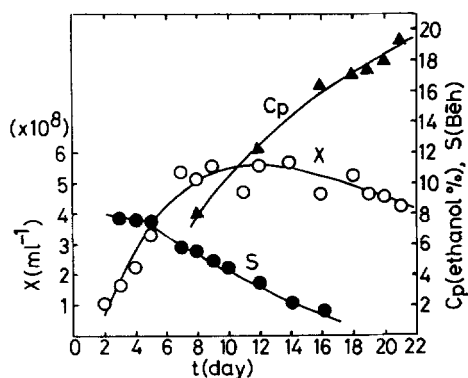
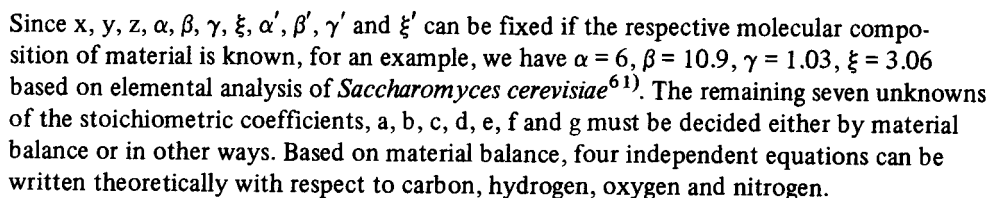


Fig. 2. Time courses during the growth of *Saccharomyces cerevisiae* in ethanol formation (sake brewing)⁷⁶⁾. raw material = starch of rice, temperature = 8 to 13 °C, C_p = ethanol concentration, S = carbohydrate concentration, X = biomass concentration

3 Mass and Energy Balances during Microbial Growth

A stoichiometric equation with respect to growth, carbon source utilization, noncellular product formation, oxygen uptake, carbon dioxide evolution and so on can be regarded as the basis of the law of conservation of substrate metabolized by a microorganism. The representative expression proposed at ⁶⁰⁾ was applied as the basic equation on the computer-aided cultivation for the production of baker's yeast. This can be written as:



Firstly, material balance for carbon is widely examined during the course of cultivation since this can be readily confirmed by actual proof on the basis of experimental data. Secondly, the other two elements, hydrogen and oxygen, have not been taken into consideration, particularly due to its being almost impossible to determine water production during growth. Thirdly, nitrogen balance also has not been practically examined, probably because some analytical difficulties in the nitrogen sources used, particularly in complex media, and some variation of nitrogen content of cells depending on culture conditions^{61, 62)}

Within this context, other rate equations such as carbon source, oxygen, carbon dioxide, noncellular product and so on must be established independently and collectively in the form of simultaneous equations so as to apply these equations for culture process controls.

Instead of stoichiometric equation (Eq. 21), the following overall expression on the growth reaction in minimal media will be written as:



where Q_{CO_2} = specific rate of carbon dioxide evolution, $\text{mole} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$, Q_{O_2} = specific rate of oxygen consumption, $\text{mole} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$.

Thus, the carbon balance in Eq. (22) is

$$\alpha_1 \nu = \alpha_2 \mu + \alpha_3 Q_{CO_2} + \alpha_4 Q_P \quad (23)$$

where α_1 = carbon content of substrate, $\text{g} \cdot \text{mole}^{-1}$, α_2 = carbon content of cells, $\text{g} \cdot \text{g}^{-1}$, α_3 = carbon content of carbon dioxide, $\text{g} \cdot \text{mole}^{-1}$, α_4 = carbon content of product, $\text{g} \cdot \text{mole}^{-1}$.

Another mass balance of oxygen on the growth reaction can be expressed⁶³⁾:

$$A\nu = B\mu + Q_{O_2} + CQ_P \quad (24)$$

where A = amount of oxygen required for the combustion of substrate to CO_2 , H_2O and NH_3 if the substrate contains nitrogen, $\text{mole} \cdot \text{mole}^{-1}$, B = amount of oxygen required for the combustion of dry cells to CO_2 , H_2O and $NH_3 = 41.7 \text{ mmole} \cdot \text{g}^{-1}$,⁶³⁾ (other values: see Section 3.5.3), C = amount of oxygen required for the combustion of noncellular product to CO_2 , H_2O and NH_3 if product contains nitrogen, $\text{mole} \cdot \text{mole}^{-1}$.

Eq. 24 can be usefully applied to the assessment of undetermined metabolic quotient for example, the rate of hydrocarbon utilization by microorganism can be assessed by Eq. (24) by substituting the experimental values of μ and Q_{O_2} when $Q_P = 0$. Mass balances with respect to carbon and oxygen during the growth of *Rhodopseudomonas spheroides* S when glucose was used as the carbon source are shown in Table 8²⁷⁾. Judging from the results in Table 8, carbon and oxygen balances based on Eqs. (23) and (24) are well accorded with the same accuracy, suggesting that the oxygen balance equation would be one valuable factor in these simultaneous equations that must be constructed in the cultivation system. In addition, the term μ described in Eqs. (23) and (24), can be eliminated from the equations when one takes the mass balances of energy source consumed in complex media.

Table 8. Carbon and oxygen balances by Eqs. (23) and (24) during the growth of *Rhodopseudomonas spheroides* S in the glucose media²⁷⁾

μ	ν	Q_{O_2}	Q_{CO_2}	Q_P	Carbon-balance	Oxygen-balance
0.051	0.70	1.44	1.66	0.094	1.04	0.98
0.097	1.09	2.28	2.36	0.132	1.09	1.09
0.143	1.77	2.64	2.72	0.367	1.02	1.02
0.167	1.89	3.57	3.48	0.179	1.02	1.02

Carbon balance = $(\alpha_2\mu + \alpha_3Q_{CO_2} + \alpha_4Q_P)/\alpha_1\nu$

Oxygen balance = $(B\mu + Q_{O_2} + CQ_P)/A\nu$

$\mu: \text{h}^{-1}$, $\nu: \text{mmole} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$, $Q_{O_2}: \text{mmole} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$, $Q_{CO_2}: \text{mmole} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$,

$Q_P: \text{mmole} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$ as glucose, $\alpha_1 = 72 \text{ g} \cdot \text{mole}^{-1}$, $\alpha_2 = 0.5 \text{ g} \cdot \text{g}^{-1}$, $\alpha_3 = 12 \text{ g} \cdot \text{mole}^{-1}$,

$\alpha_4 = 72 \text{ g} \cdot \text{mole}^{-1}$, $A = 6 \text{ mole} \cdot \text{mole}^{-1}$, $B = 0.0417 \text{ mole} \cdot \text{g}^{-1}$, $C = 6 \text{ mole} \cdot \text{mole}^{-1}$

3.3 ATP Generation during Growth

ATP generation during growth either in energy coupled or energy uncoupled was fully discussed before, here again the fundamental features of ATP generation in the cells will be firstly considered to set up kinetic equations for growth, substrate utilization and respiration. Here it is assumed that ATPs formed on energy yielding processes are immediately utilized mainly for cellular biosyntheses and partly for maintenance metabolism. Although the meaning of maintenance metabolism has not yet been clearly and accurately formulated, this metabolism would imply that some amounts of ATP must be essentially used for maintaining life such as for the mechanical movement of cells, the active transport of certain substances and for the organization of individual organs, for example, membrane and cell wall. The repair of macromolecular substances, considered dynamically, is also significant work in maintenance metabolism. Apart from these complex structures of maintenance metabolism, a metabolic quotient as a representative value of maintenance activity has since been assessed experimentally as the energy requirement when the growth rate corresponds to zero⁴⁰⁾.

For the assessment of a maintenance coefficient, first, mass balance on ATP formed (= utilized) during a small interval can be written as:

$$(\Delta \text{ATP})_F = (\Delta \text{ATP})_M + (\Delta \text{ATP})_G \quad (25)$$

where subscript: F: formed, M: maintenance and G: growth.

Second, assuming that the amount of ATP utilized for maintenance metabolism is proportional to the cell density at an arbitrarily given time, and that ATP requirement for the growth is proportional to the amount of biomass produced, the following equation will be established^{40,45,47)}.

$$(\Delta \text{ATP})_M = m_A X \Delta t \quad (26)$$

$$(\Delta \text{ATP})_G = \frac{\Delta X}{Y_{\text{ATP}}^{\text{MAX}}} \quad (27)$$

where m_A = maintenance coefficient for ATP, $\text{mole} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$, $Y_{\text{ATP}}^{\text{MAX}}$ = maximum growth yield for ATP, $\text{g} \cdot \text{mole}^{-1}$. Here, $Y_{\text{ATP}}^{\text{MAX}}$ differs from Y_{ATP} on the point in which Y_{ATP} is variable as a function of μ as defined by μ/Q_{ATP} whereas $Y_{\text{ATP}}^{\text{MAX}}$ is constant (see Eq. 28).

Substituting Eqs. (26) and (27) into Eq. (25) we obtain

$$Q_{\text{ATP}} = m_A + \frac{1}{Y_{\text{ATP}}^{\text{MAX}}} \mu \quad (28)$$

where Q_{ATP} = specific rate of ATP formation, $\text{mole} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$.

If it is possible to assess the rate of ATP formation as a function of growth rate during the course of cultivation, one can predict the values of m_A and $Y_{\text{ATP}}^{\text{MAX}}$ as far as a linear relationship between μ and Q_{ATP} is permissible. Estimations of m_A and $Y_{\text{ATP}}^{\text{MAX}}$ have been limited so far because the difficulties of the assessment of ATP formed during cultivation (Section 2.3). The assessments of m_A and $Y_{\text{ATP}}^{\text{MAX}}$ were

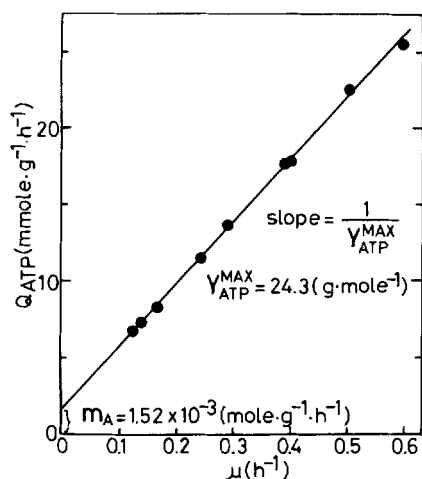


Fig. 3. Assessments of maintenance coefficient, m_A and maximum growth yield based on ATP generation, Y_{ATP}^{MAX} of *Lactobacillus casei* anaerobically growing in complex medium when glucose was used as the energy source⁴⁰⁾

successfully achieved⁴⁰⁾ during the anaerobic growth of *Lactobacillus casei* in glucose-limited chemostat cultures when a complex medium was used. The part of original data⁴⁰⁾ was rearranged by Q_{ATP} vs. μ (Fig. 3). A straight line could be observed in the figure and consequently m_A and Y_{ATP}^{MAX} can be assessed to be $1.52 \text{ mmole} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$ and $24.3 \text{ g} \cdot \text{mole}^{-1}$ respectively.

3.4 Relationships between Substrate Consumption, Growth, Respiration and Noncellular Products

3.4.1 Carbon Source in Minimal Media

When the carbon source consumed is mainly metabolized for cellular syntheses without discharging any noncellular products in the medium, the carbon source consumed might approximately be balanced by the following equation, based on the heat of combustion of the substrate.

$$\Delta H_S \cdot (-\Delta S) = \Delta H_S \cdot (-\Delta S_M) + \Delta H_S \cdot (-\Delta S_W) + \Delta H_S \cdot (-\Delta S_C) \quad (29)$$

where ΔH_S = heat of combustion of substrate, $\text{kcal} \cdot \text{mole}^{-1}$, $\Delta H_S \cdot (-\Delta S)$ = total substrate consumed, $\Delta H_S \cdot (-\Delta S_M)$ = substrate expended for maintenance metabolism, $\Delta H_S \cdot (-\Delta S_W)$ = substrate expended for biosynthetic activity, $\Delta H_S \cdot (-\Delta S_C)$ = substrate incorporated into cellular components.

In Eq. (29), first, the term of maintenance metabolism would be assumed on the same basis as already discussed in Eq. (26). This can be written as:

$$\Delta H_S (-\Delta S_M) = m' X \Delta t \quad (30)$$

where m' = maintenance coefficient based on heat of combustion, $\text{kcal} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$.

Second, the term of substrate expended for biosynthetic activity can be arranged as follows:

$$\Delta H_S \cdot (-\Delta S_W) = \Delta H_S \frac{-\Delta S_W}{\Delta X} \Delta X = \Delta H_S \cdot Y_W \cdot \Delta X \quad (31)$$

where Y_W = substrate catabolized for true biosynthetic activity to build up 1 gram cell, mole $\cdot g^{-1}$.

Third, if the substrate is incorporated into cellular components, it might be represented with the substitution of the heat of combustion of dry cell (Eq. 5), the term of $\Delta H_S \cdot (-\Delta S_C)$ in Eq. (29) can be written as:

$$\Delta H_S \cdot (-\Delta S_C) = \Delta H_a \cdot (\Delta X) \quad (32)$$

Substituting Eqs. (30) to (32) into Eq. (29) we have

$$\nu = \frac{m'}{\Delta H_S} + \left(Y_W + \frac{\Delta H_a}{\Delta H_S} \right) \mu \quad (33)$$

Eq. (33) corresponds to the representative form between μ and ν^{64} , that is

$$\nu = m + \frac{1}{Y_G} \mu \quad (34)$$

where m = maintenance coefficient for substrate = $m'/\Delta H_S$, mole $\cdot g^{-1} \cdot h^{-1}$, Y_G = true growth yield for substrate = $\Delta H_S/(Y_W \Delta H_S + \Delta H_a)$, g $\cdot$ mole $^{-1}$.

When considerable amounts of noncellular products are discharged from cells, the left-hand side of Eq. (29) must be modified by considering the difference between the heat of combustion of original substrate and that for the sum of the end-products, that is:

$$\begin{aligned} \Delta H_S \cdot (-\Delta S) - \sum \Delta H_P \Delta C_P \\ = \Delta H_S \cdot (-\Delta S_M) + \Delta H_S \cdot (-\Delta S_W) + \Delta H_S \cdot (-\Delta S_C) \end{aligned} \quad (35)$$

The substitution of Eqs. (30) to (32) in Eq. (35) gives

$$\nu - \frac{\sum(\Delta H_P)}{\Delta H_S} Q_P = \frac{m'}{\Delta H_S} + \left(Y_W + \frac{\Delta H_a}{\Delta H_S} \right) \mu = m + \frac{1}{Y_G} \mu \quad (36)$$

3.4.2 Carbon Source in Complex Media

As mentioned before in Eq. (3) carbon source in complex media would be considered almost completely dissimilated in energy yielding processes. Within this context, the energy balance in terms of carbohydrate (energy source) consumed can be represented by the sum of Eqs. (30) and (31).

$$\Delta H_S \cdot (-\Delta S) = m' X \Delta t + \Delta H_S Y_W \Delta X \quad (37)$$

Arranging Eq. (37) one obtains

$$\nu = \frac{m'}{\Delta H_S} + Y_W \mu \quad (38)$$

3.4.3 Respiration

Assuming that energy yielding processes are mostly dominated by the oxidative phosphorylation rather than the substrate level phosphorylation, and that most of the oxygen consumed is oxidized *via* oxidase systems, the amount of oxygen consumed would be generally regarded as a representative value in catabolism and consequently this could be subdivided in two parts as shown in Eq. (37)⁶⁴⁻⁶⁷.

$$Q_{O_2} = m_O + \frac{1}{Y_{GO}} \mu \quad (39)$$

where m_O = maintenance coefficient for oxygen, $\text{mole} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$, Y_{GO} = true growth yield for oxygen, $\text{g} \cdot \text{mole}^{-1}$.

3.5 Heat Evolution during Growth

3.5.1 General

Heat evolution during cultivation is an inevitable feature in any sort of microbial reactions. The estimation of heat evolution in industrial cultivation is a significant problem since the heat evolution must be removed during cultivation so as to maintain an optimum temperature for growth. Experimental determination of heat evolution has been conducted on a small scale so far, probably because of the practical difficulties of calorimetric analysis, particularly in aerobic systems. Direct measurements of heat evolution during the growth of *Saccharomyces cerevisiae*, either aerobic or anaerobic, have been carried out successfully by calorimetric analysis⁶⁸. The other method of heat balance based on heat losses and gains of the reactor has been carried out during aerobic cultivations of bacteria, yeast and fungi to establish the relation of heat evolution to oxygen consumption⁷³. A similar heat balance method was also used during cultivation when a semi-solid substrate was used for the production of hydrolases^{69, 70}. A calorimetric method gives actual data, however, this can be applicable under very limited culture conditions. Similarly, the heat balance methods mentioned above are only applicable in the case where a small reactor is used. In this context, it is necessary to establish an indirect means for the assessment of heat evolution during growth. Theoretically, the heat evolution during cultivation can be calculated on the basis of the difference between the heat of combustion of substrate consumed and that for the sum of the products formed during growth. However, there might be some troublesome problems due to the practical difficulties of rapid and accurate analyses of end-products in the culture broth.

When no particular products other than biomass and carbon dioxide are observed during the growth in minimal media, the heat evolution during cultivation was calculated from the difference between the heat of combustion of substrate and that for biomass produced by Guenther⁷¹ who took the heat of combustion of dried yeast to be $3.63 \text{ kcal} \cdot \text{g}^{-1}$. This method has also been used for the estimation of heat production where the production of single-cell proteins were concerned from various carbon sources such as carbohydrate, n-paraffin and methane⁷². However, few estimations

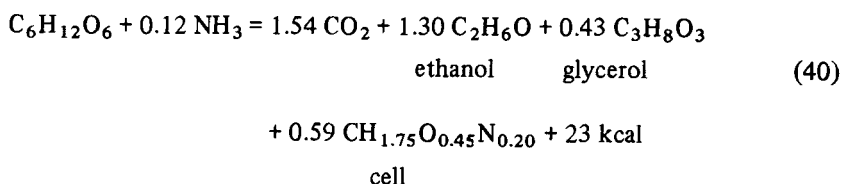
of heat production have been reported so far where noncellular products accompanied growth.

As to the other practical means for the estimation of heat production, it was found experimentally that one mole oxygen consumption was equal to 124 kcal heat production⁷³⁾. In this connection the means for the estimation of heat evolution based on the amount of oxygen consumption was further studied theoretically to elucidate the relationships between the two metabolic quotients^{38, 74)}. In this respect Minkevich et al.³⁸⁾ found a theoretical value to be 108 kcal heat generation per one mole oxygen consumed. Here, the problems of heat generation during microbial growth are discussed on the basis of the thermodynamic concepts using Battley's data⁶⁸⁾. It sets out to ascertain whether the heat evolution can be estimated practically, based on metabolic quotients such as substrate consumption, product formation and respiration.

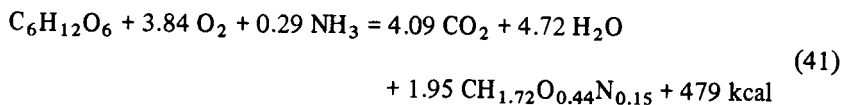
3.5.2 Estimation of Heat Generation Based on Substrate Consumed and Products Formed

During aerobic or anaerobic growth of *Saccharomyces cerevisiae*, stoichiometric equations including the quantity of heat production have been established as follows⁶⁸⁾

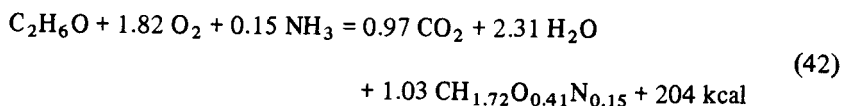
Anaerobic growth on glucose:



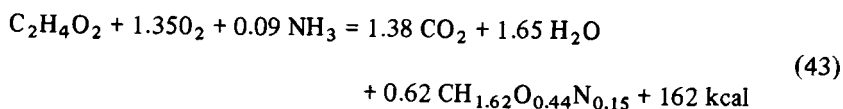
Aerobic growth on glucose:



Aerobic growth on ethanol:



Aerobic growth on acetic acid:



Based on Eqs. (40) to (43), indirect estimations of heat evolution accompanying growth can be calculated from the difference between the heat of combustion of substrate con-

sumed and that for the sum of products formed, that is

$$\Delta H_C = \Delta H_S \cdot (-\Delta S) - \Sigma \Delta H_P \cdot (\Delta C_P) - \Delta H_a \cdot \Delta X \quad (44)$$

where ΔH_C = heat production accompanying growth, $\text{kcal} \cdot \text{l}^{-1}$.

Although the values of ΔH_S and ΔH_P are quoted from a physicochemical book, the heat of combustion of dry cells, ΔH_a must be determined directly by a calorimetric method. A value of ΔH_a determined²³⁾ was $5.3 \text{ kcal} \cdot \text{g}^{-1}$ while another⁷¹⁾ was used as 3.63 for the estimation of heat evolution during growth. This discrepancy between the two values might be caused by the fact that, in the former, cellular nitrogen might be oxidized to nitrogen oxide and that, in the latter, cellular nitrogen might return to the ammonia originally used as the nitrogen source.

Within this context, an attempt was made to assess the values of ΔH_a , by using the experimental data of Eqs. (40) to (43). For an example, ΔH_a value during aerobic growth on glucose (Eq. 41) can be calculated as follows:

$$\Delta H_a \Delta X = \Delta H_S(-\Delta S) - \Delta H_C = 673 - 479 = 194 \text{ kcal} \cdot \text{l}^{-1}$$

due to $\Delta C_P = 0$, and as the amount of cell produced, ΔX is to be $1.95 \text{ CH}_{1.72}\text{O}_{0.44}\text{N}_{0.15} = 44.58 \text{ g} \cdot \text{l}^{-1}$, thus, ΔH_a can be taken to be $194/44.58 = 4.35 \text{ kcal} \cdot \text{g}^{-1}$. Estimated values of ΔH_a in the cases of Eqs. (40) to (43) are cited in Table 9. Although widely distributed in ΔH_a , values in Table 9, the mean value of $4.2 \text{ kcal} \cdot \text{g}^{-1}$ was used for the further estimation of heat evolution.

Table 9. Oxygen balance based on Eq. (45) and estimations of oxygen required for the oxidation of dry cell, B, heat of combustion of dry cell, ΔH_a and heat production per oxygen consumed, ΔH_O during the growth of *Saccharomyces cerevisiae* (see Eqs. 40 to 43)

	Oxygen balance	B <u>mmole</u> g cell	ΔH_O <u>kcal</u> mole	ΔH_a <u>kcal</u> g cell
Anaerobic:				
glucose	1.0 (6.03/6)	45.0	—	3.90
Aerobic:				
glucose	1.0 (5.98/6)	48.0	124.7	4.35
ethanol	0.98 (2.93/3)	49.7	112.7	5.31
acetic acid	1.0 (2.02/2)	47.4	120.0	3.30
average:	1.0	47.5	118.9	4.20

ΔH_S : 673 kcal/mole glucose, 326.5 kcal/mole ethanol, 398.7 kcal/mole glycerol, 208.6 kcal/mole acetic acid,

A: 6 mole O_2 /mole glucose, 3 mole O_2 /mole ethanol, 3.5 mole O_2 /mole acetic acid.

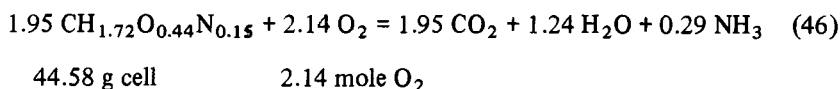
3.5.3 Estimation of Heat Evolution Based on Respiration

As already mentioned, the oxygen balance in any sort of microbial growth could be established on the basis of Eq. (24) (Table 8). Cooney et al.⁷³⁾ found a very useful method to estimate the quantity of heat production based on the amount of oxygen consumed during aerobic cultivations. Empirical constant observed was to be 124 kcal heat evolved per one mole oxygen consumed. Here, the empirical relationship observed⁷³⁾ will be discussed in terms of thermodynamics.

During the growth of *Saccharomyces cerevisiae* represented by Eqs. (40) to (43), the oxygen balance can be examined by the following equation (see Eq. 24).

$$\text{oxygen balance} = \frac{B\Delta X + \Delta O_2 + C\Delta C_P}{A(-\Delta S)} \quad (45)$$

Oxygen required for the combustion of dry cell, B in Eq. (45) can be assessed on the basis of the molecular composition of dry cells by assuming that the cells are oxidized to CO₂, H₂O and NH₃ since the nitrogen was originally supplied as ammonia. For example, in the case of Eq. (41), the following balance equation can be established.



Thus, we obtain that $B = 2140/44.58 = 48$ mmole oxygen per g cell. Assessed values of B (see Table 9) are a little larger than the value of 41.7 mmole · g⁻¹⁶³⁾. As might be expected in Table 9, it proves that the oxygen balance based on Eq. (45) is well established even during anaerobic growth. Therefore, in the cases of aerobic cultivation without producing non-cellular products, that is $\Delta C_P = 0$ (Eqs. 41–43), the following equation can be established based on Eq. (45).

$$\Delta O_2 = A \cdot (-\Delta S) - B \cdot \Delta X \quad (47)$$

On the other hand, the quantities of heat evolution accompanying growth represented by Eqs. (41) to (43) can also be written by Eq. (44).

$$\Delta H_C = \Delta H_S \cdot (-\Delta S) - \Delta H_a \cdot \Delta X \quad (44)$$

provided that: $\Delta C_P = 0$.

In this respect Minkevich et al.³⁸⁾ found that the heat of combustion of an organic substance and dried cells could be calculated by multiplying the proportional constant of 108 kcal/mole O₂ by the amount of oxygen required for the oxidation of each substance. The relation observed can be written as follows:

$$\begin{array}{l} \Delta H_S = \Delta H_{O_A} \\ \Delta H_a = \Delta H_{O_B} \end{array} \quad (48)$$

where ΔH_O = heat of combustion based on the amount of oxygen required for the combustion of organic substance, kcal/mole O_2 .

Thus the substitution of Eq. (48) into Eq. (47) gives:

$$\Delta H_O \Delta O_2 = \Delta H_S \cdot (-\Delta S) - \Delta H_a \Delta X \quad (49)$$

Comparing Eq. (49) with Eq. (44), we can derive the relation between heat generation and oxygen consumption which was observed experimentally⁷³⁾ or theoretically³⁸⁾.

$$\Delta H_C = \Delta H_O \cdot \Delta O_2 \quad (50)$$

The values of ΔH_O during the aerobic growth of *S. cerevisiae* were calculated by $\Delta H_C / \Delta O_2$ using the experimental values of ΔH_C and ΔO_2 observed in Eqs. (41) to (43). It is interesting to note that the average value of ΔH_O to be 118.9 (see Table 9) is almost the same as the determined value of 124⁷³⁾, and the theoretical values of 106²³⁾ and 108³⁸⁾.

It can be concluded that Eq. (50) might be an applicable means to estimate the quantity of heat production during any sort of cultivation, either of submerged or semi-solid state, when no particular noncellular products are discharged from the cells⁷⁵⁾.

3.5.4. Estimation of Heat Evolution under Glucose Repression

The so-called glucose effect can be seen when *Saccharomyces cerevisiae* grows aerobically in high-sugar concentration in which noncellular products such as ethanol are generally produced even in fully aerated conditions^{28,61,77)}. Here, the problems of the heat evolution during such aerobic repression are discussed. An overall growth reaction in any sort of aerobic repression can be written as:

$$-\Delta S + \Delta O_2 = \Delta CO_2 + \Delta H_2O + \Delta C_P + \Delta X + \Delta \text{heat}$$

Thus, the estimation of heat evolution by Eq. (44) is,

$$\Delta H_C = \Delta H_S \cdot (-\Delta S) - \sum \Delta H_P \cdot (\Delta C_P) - \Delta H_a \cdot \Delta X \quad (44)$$

Another means is also concerned. The carbon substrate consumed is mainly metabolized *via* two separate pathways; one is the glycolytic pathway to produce noncellular products such as ethanol, glycerol and carbon dioxide as final products and the other is the oxidative pathway in which water and carbon dioxide are the final products. Therefore, the overall equation for the fate of glucose consumed during aerobic repression can be written by subdividing the two distinct pathways as follows.

$$(-\Delta S)_F = (\Delta CO_2)_F + (\Delta X)_F + \Delta C_P + \Delta (\text{Heat})_F \quad (51)$$

$$(-\Delta S)_O + \Delta O_2 = (\Delta CO_2)_O + \Delta H_2O + (\Delta X)_O + \Delta (\text{Heat})_O \quad (52)$$

where subscripts: F = glycolytic pathway, O = oxidative pathway.

The following relationships are established between Eqs. (51) and (52).

$$\begin{aligned}
 (-\Delta S) &= (-\Delta S)_F + (-\Delta S)_O \\
 \Delta CO_2 &= (\Delta CO_2)_F + (\Delta CO_2)_O \\
 \Delta X &= (\Delta X)_F + (\Delta X)_O \\
 \Delta(\text{Heat}) &= \Delta(\text{Heat})_F + \Delta(\text{Heat})_O
 \end{aligned} \tag{53}$$

Heat evolution *via* glycolytic pathway (Eq. 51) can be balanced based on Eq. (44).

$$(\Delta H_C)_F = \Delta H_S \cdot (-\Delta S)_F - \Sigma \Delta H_P \cdot (\Delta C_P) - \Delta H_a \cdot (\Delta X)_F \tag{54}$$

And heat evolution *via* oxidative pathway (Eq. 52) can be balanced either by Eq. (44) or Eq. (50).

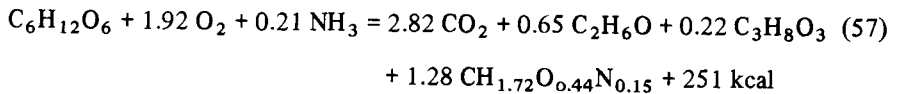
$$\begin{aligned}
 (\Delta H_C)_O &= \Delta H_S \cdot (-\Delta S)_O - \Delta H_a \cdot (\Delta X)_O \\
 &= \Delta H_O \cdot \Delta O_2
 \end{aligned} \tag{55}$$

The summation of Eqs. (54) and (55) based on Eq. (53) yields Eq. (44), however, the combination of $\Delta H_O \cdot \Delta O_2$ with Eq. (54) gives

$$\begin{aligned}
 \Delta H_C &= \Delta H_S \cdot (-\Delta S)_F - \Sigma \Delta H_P \cdot \Delta C_P - \Delta H_a \cdot (\Delta X)_F + \Delta H_O \cdot \Delta O_2 \\
 &= (\Delta H_C)_F + \Delta H_O \cdot \Delta O_2
 \end{aligned} \tag{56}$$

It should be emphasized when Eq. (56) is compared with Eq. (50) that the estimation of heat evolution accompanying aerobic repression would be of no use assessing only by Eq. (50) in the light of the thermodynamic considerations. However, when the quantity of heat evolution accompanying glycolytic pathway (see Eq. 40) is considerably less than that produced *via* oxidative pathway (see Eq. 41), it might be permitted to be able to estimate roughly the heat evolution on the basis of the amount of oxygen consumed. An example meeting the above-mentioned condition is shown below.

The summation of Eqs. (40) and (41) yields a growth reaction of *S. cerevisiae* growing under aerobic repression.



where $0.59 CH_{1.75}O_{0.45}N_{0.20}$ in Eq. (40) is adjusted to $0.61 CH_{1.72}O_{0.44}N_{0.15}$ by using the same molecular formula of Eq. (41) for sake of convenience.

Based on Eq. (44), the estimation of heat evolution can be calculated as follows:

$$\Delta H_C = 673 - (0.65 \times 326.5 + 0.22 \times 397.8) - 4.2 \times 29.26 = 250 \text{ kcal}$$

substrate noncellular products biomass

where $\Delta H_a = 4.2 \text{ kcal} \cdot \text{g}^{-1}$ (see Table 9).

Another estimation of heat evolution based on Eq. (50) is as follows: $\Delta H_C = 118.9 \times 1.92 = 228 \text{ kcal}$ where $\Delta H_O = 118.9 \text{ kcal} \cdot \text{mole}^{-1}$ (see Table 9). The estimation based on Eq. (44) to be 250 kcal was in good agreement with the calorimetric determination to be 251 kcal as shown in Eq. (57) whereas some difference was observed when the estimation was based on Eq. (50).

It can be concluded that in aerobic repression the heat evolution accompanying growth should be calculable by Eq. (44) especially when noncellular products cannot be disregarded in mass balance.

4 Establishment of Growth Kinetic Equations

Simultaneous equations with respect to growth, substrate consumption, respiration, noncellular product formation and so on, applicable for automatic process control in any sort of microbial cultivation can be set up on the grounds of mass balances and bioenergetic considerations discussed already. Here, the experimental data in chemostat cultures of *Saccharomyces cerevisiae* growing on ethanol as the carbon source⁶²⁾ were used for the establishment of kinetic equations. The original data are shown in Table 10. Based on the original data, mass balances with respect to carbon and oxygen (see Section 3.2) were examined so as to determine whether or not noncellular products were discharged from the cells. These results are shown in Tables 11 and 12. It is quite evident from Tables 11 and 12 that a considerable amount of noncellular products, that is, 40 to 50 percent of the amount of ethanol consumed, was discharged in the culture medium in all runs, although the original work did not identify noncellular products.

To examine a linear relationship between Q_{O_2} and μ based on Eq. (39), the data of Q_{O_2} described in Table 10 were plotted against μ ($= D$), (see Fig. 4). A linear relationship between Q_{O_2} and μ confirms Eq. (39) giving $m_O = 0.4 \text{ mmole} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$ and $Y_{GO} = 38.5 \text{ g} \cdot \text{mole}^{-1}$. The substitution of Eq. (39) into Eq. (24) gives

$$A\nu - CQ_P = m_O + \left(B + \frac{1}{Y_{GO}}\right)\mu \quad (58)$$

The relationship of Eq. (58) is reconfirmed by means of plotting $(A\nu - CQ_P)$ vs. μ by using the values of Table 12, (Fig. 4).

Recently, there have been attempts to use the respiratory quotient, RQ, as an operational factor to maintain sugar concentration at a moderate level during cultivation and as a result to avoid the so-called glucose effect during the growth of *Saccharomyces cerevisiae*^{61, 77)}. This control means is based on the fact that oxygen consump-

Table 10. Experimental results in ethanol-limited chemostat cultures of *Saccharomyces cerevisiae*⁶²⁾ (Courtesy of John Wiley & Sons, Inc.)

D	X	S	$Y_{X/S}$	Q_{O_2}	Q_{CO_2}	RQ	N	C	P	H
% dry basis										
0.020	3.93	0.10	0.414	19.5	11.1	0.56	6.72	50.05	0.64	7.41
0.055	4.70	0.02	0.491	41.5	23.8	0.54	7.79	49.29	0.50	7.83
0.095	4.75	0.06	0.498	63.9	32.9	0.51	8.69	49.70	0.65	7.22
0.115	4.14	0.64	0.462	77.8	38.2	0.49	8.73	48.98	0.64	7.41
0.119	2.92	2.69	0.422	80.7	38.3	0.47	9.24	49.91	0.66	7.71

S_0 , ethanol concentration of fresh medium = $10 \text{ g} \cdot \text{l}^{-1}$

D: h^{-1} , X: $\text{g} \cdot \text{l}^{-1}$, S: $\text{g} \cdot \text{l}^{-1}$, $Y_{X/S}$: $\text{g} \cdot \text{g}^{-1}$, Q_{O_2} : $\text{ml} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$,
 Q_{CO_2} : $\text{ml} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$

Table 11. Carbon balance during the growth of *S. cerevisiae* in ethanol-limited chemostat cultures (see Table 10)

D = μ	ν	$\alpha_1 \nu$ $\times 10^{-2}$	$\alpha_2 \mu$ $\times 10^{-2}$	$\alpha_3 Q_{CO_2}$ $\times 10^{-2}$	$\alpha_4 Q_P$ $\times 10^{-2}$	$\alpha_4 Q_P / \alpha_1 \nu$
0.020	1.10	2.6	6.1	0.6	1.1	0.39
0.055	2.55	6.1	2.7	1.3	2.1	0.34
0.095	4.34	10.4	4.7	1.8	3.9	0.38
0.115	5.65	13.6	5.7	2.1	5.8	0.43
0.119	6.43	15.5	5.9	2.1	7.5	0.48

$\alpha_4 Q_P$: assessed by Eq. 23 provided that $\alpha_1 = 24 \text{ g} \cdot \text{mole}^{-1}$, $\alpha_2 = 0.5 \text{ g} \cdot \text{g}^{-1}$, $\alpha_3 = 12 \text{ g} \cdot \text{mole}^{-1}$,
 $\nu = D(S_0 - S)/X$, $\text{mmole} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$, μ : h^{-1} , $\alpha_1 \nu$: $\text{g} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$, $\alpha_2 \mu$: $\text{g} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$, $\alpha_3 Q_{CO_2}$: $\text{g} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$,
 $\alpha_4 Q_P$: $\text{g} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$

Table 12. Oxygen balance during the growth of *Saccharomyces cerevisiae* in ethanol-limited chemostat cultures (see Table 10)

D = μ	$A\nu$	Q_{O_2}	$B\mu$	CQ_P	$CQ_P / A\nu$
0.020	3.30	0.87	0.95	1.48	0.45
0.055	7.65	1.84	2.61	3.19	0.42
0.095	13.02	2.84	4.51	5.66	0.44
0.115	16.95	3.46	5.46	8.02	0.47
0.119	19.29	3.60	5.65	10.04	0.52

CQ_P : assessed by Eq. 24 provided that $A = 3 \text{ mole} \cdot \text{mole}^{-1}$, $B = 47.5 \text{ mmole} \cdot \text{g}^{-1}$ (see Table 9).

D: h^{-1} , $A\nu$: $\text{mmole} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$, Q_{O_2} : $\text{mmole} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$, $B\mu$: $\text{mmole} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$, CQ_P : $\text{mmole} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$

tion and carbon dioxide evolution can be readily measured by means of a paramagnetic oxygen analyzer and an infrared carbon dioxide analyzer. In addition, these instruments can be readily interfaced with a computer. Thus, automatically calculated RQ values during cultivation can be readily available as a control parameter in any sort of microbial cultivation.

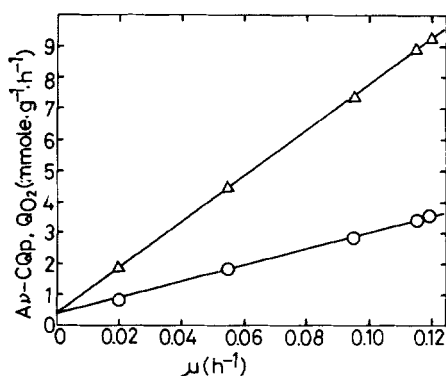


Fig. 4. Q_{O_2} vs. μ (Eq. 39) and $(Av - C_{qp})$ vs. μ (Eq. 58) in ethanol-limited chemostat cultures of *Saccharomyces cerevisiae* (data from Table 12)

The rates of oxygen consumption and carbon dioxide evolution are calculated from the mass balances of respective gas between inlet and outlet air lines of a reactor. The mass balance equations can be written⁶⁰.

$$\frac{dO_2}{dt} = I_{O_2} = XQ_{O_2} \quad (59)$$

$$= \frac{F_N}{V} \left(\frac{P_{O(in)}}{P - P_{O(in)} - P_{W(in)} - P_{C(in)}} - \frac{P_{O(out)}}{P - P_{O(out)} - P_{W(out)} - P_{C(out)}} \right)$$

$$\frac{dCO_2}{dt} = I_{CO_2} = XQ_{CO_2} \quad (60)$$

$$= \frac{F_N}{V} \left(\frac{P_{C(out)}}{P - P_{C(out)} - P_{W(out)} - P_{O(out)}} - \frac{P_{C(in)}}{P - P_{C(in)} - P_{W(in)} - P_{O(in)}} \right)$$

where I_{O_2} = oxygen consumption rate, mole $\cdot$ l⁻¹ $\cdot$ h⁻¹,

I_{CO_2} = carbon dioxide evolution rate, mole $\cdot$ l⁻¹ $\cdot$ h⁻¹,

F_N = flow rate of nitrogen gas, mole $\cdot$ h⁻¹, V = volume of culture medium, l,

P = total pressure, atm, P_O = partial pressure of oxygen, atm, P_W = partial pressure of water, atm, P_C = partial pressure of carbon dioxide, atm, subscripts: (out) = in outlet gas, (in) = in inlet gas.

Thus the respiratory quotient, RQ is readily calculated from Eqs. (59) and (60) as follows:

$$RQ = I_{CO_2}/I_{O_2} \quad (61)$$

Since the values of I_{O_2} , I_{CO_2} and RQ can be measured automatically and reliably in any sort of aerobic cultivation either on a laboratory scale or on an industrial scale, and in addition the RQ value sometimes can be used as a representative parameter to determine the overall state of the inside cell as can be seen in aerobic repression by *Saccharomyces cerevisiae*^{60, 77}, a quantitative relation between RQ and a desired product must be examined prior to using the RQ value as a control parameter.

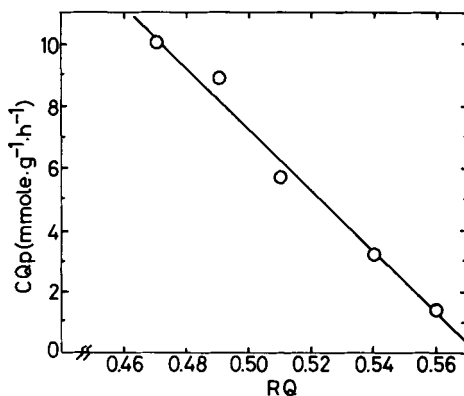


Fig. 5. Specific rate of noncellular product formation in terms of C_{Q_P} as a function of respiratory quotient, RQ (data from Tables 10 and 12)

In this context, RQ values in Table 10 and C_{Q_P} values in Table 12 were arranged so as to have an empirical equation between the two, (see Fig. 5). From the result observed in Fig. 5 the following equation can be written as:

$$C_{Q_P} = k_1 - k_2 RQ \quad (62)$$

where k_1, k_2 = empirical constants.

Now, kinetic equations required for the process control during the growth of *Saccharomyces cerevisiae* on ethanol as the carbon source would be summarized as follows:

Growth: based on Eq. (39) or Fig. 4,

$$\frac{dX}{dt} = Y_{GO} I_{O_2} - m_O Y_{GO} X \quad (a)$$

Noncellular product: based on Eq. (62),

$$\frac{dC_P}{dt} = \frac{1}{C} (k_1 - k_2 RQ) X \quad (b)$$

Substrate utilization: based on Eqs. (58) and (62),

$$-\frac{dS}{dt} = \frac{1}{A} \left[m_O X + \left(B + \frac{1}{Y_{GO}} \right) \frac{dX}{dt} + C \frac{dC_P}{dt} \right] \quad (c)$$

Heat evolution: based on Eq. (44),

$$\frac{dH_C}{dt} = \Delta H_S \cdot \left(-\frac{dS}{dt} \right) - \Delta H_P \cdot \frac{dC_P}{dt} - \Delta H_a \cdot \frac{dX}{dt} \quad (d)$$

Oxygen transfer: based on oxygen balance during culture,

$$\frac{dDO}{dt} = k_L a (DO^* - DO) - I_{O_2} \quad (e)$$

where DO = concentration of dissolved oxygen in culture medium, $\text{mole} \cdot \text{l}^{-1}$, DO^* = saturation concentration of DO , $\text{mole} \cdot \text{l}^{-1}$, $k_L a$ = volumetric oxygen-transfer coefficient, h^{-1} .

If simultaneous equations in differential form with respect to growth, noncellular product, substrate utilization, heat evolution and oxygen transfer as mentioned in Eqs. (a) to (e) could be programmed with a computer linked to the output signals of I_{O_2} , I_{CO_2} and RQ, one might be able to carry out automatic process control based on these equations and consequently discover the maximum productivity of aiming product in progressive cultivation. In addition the control method starting from $k_L a$ might be a novel means for the process control since $k_L a$ could be readily adjusted by means of changing conditions either by agitation speed or aeration, or both, and consequently I_{O_2} , I_{CO_2} and RQ would be inevitably affected by the change of $k_L a$. As a result, the physiological activities of the cells would be directly affected by $k_L a$ change, although one must undertake further research to find out whether or not this control method can effectively enhance the productivity of a desired product.

5 Conclusions

The objective of the review is to examine the macroscopic features of energetics and to establish the quantitative relationships, that is, kinetic equations, which must be materialized in any sort of microbial growth. Kinetic equations must be readily applicable to process control during cultivation. Within this context, then firstly, growth yields defined by total energy available in the medium, Y_{kcal} and defined by catabolic activity, $Y_{X/C}$, and based on the ATP generation, Y_{ATP} are more meaningful quotients in comparison with the most general term of growth yield, $Y_{X/S}$, which has been widely used so far. Secondly, based on growth yield concepts, the other quantitative features on the growth reactions, that is, stoichiometry, mass and energy balances, are analyzed and discussed to establish kinetic equations consisting of growth, substrate consumption, respiration, heat evolution and noncellular product formation. These simultaneous and differential equations might provide a modern technique applicable for use in fermentation industries so as to improve production methods, and ascertain the maximum productivity of the desired product at the same time providing a very promising future.

6 Nomenclature

A	amount of oxygen required for the combustion of substrate, mole · mole ⁻¹
B	amount of oxygen required for the combustion of dry cell, mole · g ⁻¹
C	amount of oxygen required for the combustion of noncellular product, mole · mole ⁻¹
C _p	concentration of noncellular product in culture medium, mole · l ⁻¹
D	dilution rate, h ⁻¹
DO	dissolved oxygen concentration in culture medium, mole · l ⁻¹
DO*	saturation concentration of DO, mole · l ⁻¹
ΔH _a	heat of combustion of dry cells, kcal · g ⁻¹
ΔH _C	heat generation by catabolism, kcal · l ⁻¹
ΔH _O	heat generation based on oxygen consumed, kcal · mole ⁻¹
ΔH _P	heat of combustion of noncellular product, kcal · mole ⁻¹
ΔH _S	heat of combustion of substrate, kcal · mole ⁻¹

I_{CO_2}	rate of carbon dioxide evolution, $\text{mole} \cdot \text{l}^{-1} \cdot \text{h}^{-1}$
I_{O_2}	rate of oxygen consumption, $\text{mole} \cdot \text{l}^{-1} \cdot \text{h}^{-1}$
k_{La}	volumetric oxygen-transfer coefficient, h^{-1}
m	maintenance coefficient for substrate, $\text{mole} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$
m'	maintenance coefficient based on heat generation, $\text{kcal} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$
m_A	maintenance coefficient for ATP generation, $\text{mole} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$
m_O	maintenance coefficient for oxygen, $\text{mole} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$
P	total pressure in gas phase, atm
P_C	partial pressure of carbon dioxide in gas phase, atm
P_O	partial pressure of oxygen in gas phase, atm
P_W	partial pressure of water in gas phase, atm
Q_{ATP}	specific rate of ATP generation, $\text{mole} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$
Q_{CO_2}	specific rate of carbon dioxide evolution, $\text{mole} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$
Q_{O_2}	specific rate of oxygen uptake, $\text{mole} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$
Q_P	specific rate of noncellular product formation, $\text{mole} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$
RQ	respiratory quotient = I_{CO_2}/I_{O_2} , $\text{mole} \cdot \text{mole}^{-1}$
S	substrate concentration in culture medium, $\text{mole} \cdot \text{l}^{-1}$
S_0	substrate concentration in fresh medium, $\text{mole} \cdot \text{l}^{-1}$
t	culture time, h
V	culture volume, l
X	biomass concentration in culture medium, $\text{g} \cdot \text{l}^{-1}$
Y_{ave}/S	total electron available from substrate, $\text{ave} \cdot \text{mole}^{-1}$
Y_{ave}	growth yield based on electron available, $\text{g} \cdot \text{ave}^{-1}$
Y_{ATP}	growth yield based on ATP generation, $\text{g} \cdot \text{mole}^{-1}$
Y_{ATP}^{MAX}	maximum growth yield based on ATP generation, $\text{g} \cdot \text{mole}^{-1}$
$Y_{A/S}$	ATP yield from substrate catabolized, $\text{mole} \cdot \text{mole}^{-1}$
Y_G	true growth yield from substrate, $\text{g} \cdot \text{mole}^{-1}$
Y_{GO}	true growth yield based on oxygen consumed, $\text{g} \cdot \text{mole}^{-1}$
Y_{kcal}	growth yield based on total energy available, $\text{g} \cdot \text{kcal}^{-1}$
$Y_{P/S}$	noncellular-product yield from substrate, $\text{mole} \cdot \text{mole}^{-1}$
$Y_{X/C}$	growth yield based on catabolic activity, $\text{g} \cdot \text{kcal}^{-1}$
$Y_{X/O}$	growth yield based on oxygen consumed, $\text{g} \cdot \text{mole}^{-1}$
$Y_{X/S}$	growth yield from substrate, $\text{g} \cdot \text{mole}^{-1}$
Y_W	substrate catabolized for true biosynthesis, $\text{mole} \cdot \text{g}^{-1}$
α_1	carbon content of substrate, $\text{g} \cdot \text{mole}^{-1}$
α_2	carbon content of cells, $\text{g} \cdot \text{g}^{-1}$
α_3	carbon content of carbon dioxide, $\text{g} \cdot \text{mole}^{-1}$
α_4	carbon content of noncellular product, $\text{g} \cdot \text{mole}^{-1}$
μ	specific growth rate, h^{-1}
ν	specific rate of substrate consumption, $\text{mole} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$

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Methane Generation by Anaerobic Digestion of Cellulose-Containing Wastes

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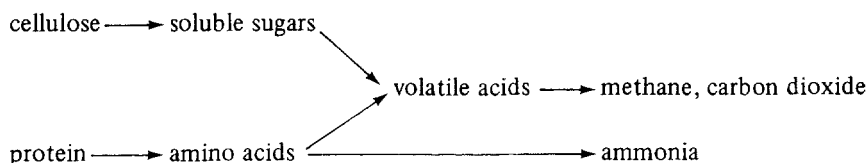
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The anaerobic digestion of cellulose-containing wastes, both municipal and agricultural, holds promise as a dual method of energy (methane) generation and waste treatment. Advances regarding the technical biochemistry and microbiology of this anaerobic-digestion process are reviewed. Experimental results concerning the kinetics of methane generation from various substances and the dependence of methanogenesis on operating parameters are assessed. Deficiencies in our present knowledge and areas of future research needs are identified. The feasibility of the process for farm-scale applications are examined.

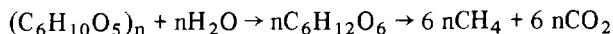
1 Introduction

Microbial conversion of organic matter to methane is indigenous to natural anaerobic ecosystems. This process, known as anaerobic digestion, has been widely adapted for stabilizing primary and secondary waste sludge in municipal wastewater pollution control plants. Indeed, much of the practical knowledge and most of the technology concerning methane generation by anaerobic digestion has been the outgrowth of this practice. It has become clear, however, that the methane generation capacity from wastewater sludge is not significant in alleviating national energy shortages. In North America, as elsewhere, the energy crisis (especially the exhaustion of natural gas reserves) has provided the recent impetus for research into methane generation by anaerobic digestion of other renewable resources, particularly cellulose-containing agricultural and municipal waste residues^{14, 17, 45, 46, 48, 55}. The ensuing discussion is an evaluation of the advances and unsolved problems pertaining to anaerobic digestion of cellulosic materials to methane.

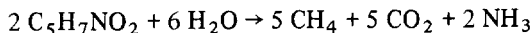
The basic route of methane generation from a complex cellulose-containing waste is illustrated by the following biochemical sequence:



Native celluloses consist of cellulose and hemicellulose fibers in a lignin matrix. Cellulose is a homopolymer of β -1, 4 linked glucose residues. It makes up about 10% of leaves, 40–50% of woody structures of plants and 98% of cotton, on a dry basis. Hemicellulose is a polymer of principally xylose and mannose subunits, but may contain a few molecules of arabinose and glucouronic acid as well. It comprises 10–30% of most plant tissue. Theoretically, methane and carbon dioxide are produced from cellulose in a 1:1 molar ratio as follows:



The protein content of woody plant tissue is normally low, seldom more than 3%. In some agricultural cellulosic wastes (manure, for example) the protein content can be considerably higher. The digestible proteinous material in manure is primarily bacterial cell mass. Stoichiometric yield of methane can be expressed as



It appears that the ratio of carbonaceous products from proteinous material is identical with that of cellulose. In practice, an average methane content of digester gas produced by these materials is higher than 50%, primarily due to the retention of carbon dioxide as bicarbonate in the liquid phase. Due to the added alkalinity of ammonia, more bicarbonate is retained in the case of protein digestion; thus, one may expect higher methane concentrations in the gas phase if all operating conditions are held constant.

It is generally accepted that the key biochemical events in converting cellulose to methane and carbon dioxide are cellulose hydrolysis, acidogenesis, and methanogenesis. These activities are assigned, in turn, to cellulolytic, acidogenic and methanogenic microbes. Anaerobic digestion, however, cannot be viewed as a sequence of independent phenomena. Rather, the process is characterized by complex interactions on both the biochemical and the cellular level. An interesting example is methane formation from ethanol, a major product of several cellulolytic anaerobic bacteria²⁵). The conversion of ethanol to methane in the presence of carbon dioxide by *Methanobacillus omelianskii* has been subject of numerous studies. More recently it has been shown that this species is, in fact, an association of two anaerobic bacteria⁹). One of the organisms, designated as "S" bacterium, produces acetate and hydrogen by dehydrogenation of ethanol. The second bacterium, *Methanobacterium M. O. H.*, utilizes the hydrogen for the reduction of carbon dioxide to methane. Bioconversion can take

place only at low partial pressures of hydrogen because hydrogen strongly inhibits the activity of the "S" organism. Synergistic association of anaerobic bacteria involving interspecies hydrogen transfer has been demonstrated for the cellulolytic mesophile, *Ruminococcus albus*²⁷⁾ and the cellulolytic thermophile, *Clostridium thermocellum*⁶²⁾. The interesting feature of co-culture is that hydrogen becomes the preferred "electron sink" rather than ethanol as in monoculture. The hydrogen is ultimately oxidized to methane by methanogenic bacteria.

2 Cellulose Hydrolysis and Acidogenesis

Undoubtedly, cellulolytic activity plays a key role in the anaerobic digestion of cellulosic wastes. Techniques for isolation and characterization of anaerobic cellulase-producing bacteria have been developed by Hungate²⁶⁾ who isolated 25 strains of anaerobic mesophilic organisms, primarily from rumen. Another 10 strains of cellulose digesting bacteria have been isolated from anaerobic sludge by Maki³⁷⁾. In later studies, the most actively cellulolytic species have been shown to be gram -, short rods (*Bacteroides* sp.) and cocci (*Ruminococcus* sp.)^{11,20,21,64)}. Many anaerobic cellulolytic species produce extracellular cellulases as indicated by clear zones developing at substantial distances around colonies in cellulose agar⁸⁾.

Only a few reports pertain to systematic study of anaerobic cellulases. Experimental data suggest that the cellulase complex of *R. albus* consists of several enzymes^{31,36)}. Smith et al.⁵¹⁾ showed that the anaerobic cellulase activity of *Ruminococcus* is partially inhibited by oxygen. Enzyme activity showed a broad pH profile with a maximum near pH 6.2. The products of cellulose hydrolysis, i.e. cellobiose and glucose, inhibited enzyme activity. As in the case of fungal cellulases, soluble cellulose (CMC), amorphous cellulose (pebble milled filter paper) and crystalline cellulose (cotton, "Sigma cell") were solubilized at lower rates, respectively.

The cellulose hydrolysis rates observed in some anaerobic cultures are given in Table 1. Generally, higher cellulolysis rates are observed at thermophilic temperatures than at mesophilic temperatures. In mixed cultures of sewage sludge organisms, the maximum rate of cellulose hydrolysis is reported to be at pH 7.5, while rumen microflora usually have maxima at more acidic conditions. Comparative studies of cellulase activities of anaerobic bacteria in pure culture are lacking. From available information, it can be inferred that organisms of rumen origin (*R. albus*, *Bacteriodes succinogenes*) are more actively cellulolytic than strains isolated from other environments^{11,60)}. Rumen cellulolytic bacteria, however, appear to have more exacting nutritional requirements¹¹⁾.

The effect of the cellulose hydrolysis products, particularly cellobiose, on cellulase synthesis is not yet resolved. Smith et al.⁵¹⁾, and Hammerstrom et al.¹⁸⁾ gave experimental evidence supporting the constitutive nature of anaerobic bacterial cellulases. By contrast, Leatherwood³⁶⁾ concluded that anaerobic cellulases are subject to catabolite repression. In mixed culture, as in anaerobic digestion, the repressibility of enzyme synthesis is a moot point, since no significant accumulation of either cellobiose or glucose is ever observed.

Table 1. Cellulase hydrolysis in anaerobic systems

Culture	Origin	Substrate	Temp. (°C)	pH	Rate of Cellulose Hydrolysis ($\mu\text{g ml}^{-1} \text{h}^{-1}$)
Mixed ¹⁹⁾	sewage	cellulose in sewage	25	7.4	5.90
Mixed ⁴⁵⁾	sewage	filter paper	40	7.5	58.30
Mixed ⁵⁵⁾	sewage	filter paper	60	7.5	83.30
Mixed ^{*)}	sewage	filter paper	40	6.5	33.30
Mixed ^{*)}	manure	filter paper	39	6.5	53.60
Mixed ³⁷⁾	sewage	filter paper	38	6.8	27.50
(2 pure cultures)					
Mixed ⁶⁵⁾	—	filter paper	60	7.5	200–250
(2 pure cultures)					
Pure (cell free)	rumen	filter paper	39	6.7	82.8

*) This work

The hydrolysis of lignocelluloses, both native and processed, is dependent on the lignin to cellulose ratio. According to Bryant¹¹⁾, the digestibility of grasses is highly correlated (correlation coefficient = 0.9) with the lignin content. Ranges of lignin to cellulose ratio for various cellulosic materials are given in Table 2. For each substrate, a low lignin to cellulose ratio implies a high degree of digestibility.

Despite their critical role, cellulolytic bacteria constitute a small fraction of the total acidogenic population in anaerobic digestors²¹⁾. In enrichment media with cellulose as the sole carbon source the number of cellulolytic bacteria, as counted by clear zones on cellulose agar, seldom rises above 1% of the acidogenic microflora. The distribution of the non-cellulolytic population is dependent on the type of waste. The predominant non-cellulolytic bacterium is *Streptococcus* in digesting piggery waste and *Enterobacter* in waste-water sludge²²⁾. Various genera of non-cellulolytic, anaerobic bacteria including *Peptostreptococcus*, *Propionibacterium*, *Bacteroides*, *Micrococcus*, and *Clostridium* can routinely be isolated from most digesting systems. Usually obligate anaerobes outnumber facultative organisms.

The interaction between cellulolytic and non-cellulolytic bacteria in anaerobic digestors is complex. On one hand, non-cellulolytic bacteria compete successfully with

Table 2. Effect of lignin on anaerobic digestibility by ruminant microflora¹¹⁾

Substance	Lignin/Cellulose Ratio	% Hydrolysis
Paper	0–0.5	20–99
Newsprint	0.34–0.43	23–37
Straw	0.10–0.46	40–60
Grasses	0.08–0.2	48–90
Wood	0.3–0.6	0–40

cellulolytic organisms for the soluble products of cellulose hydrolysis. Extracellular cellulases are destroyed by the proteolytic enzymes of non-cellulolytic bacteria. These effectively constrain both the biosynthesis and activity of cellulolytic enzymes. On the other hand, the bacteria lacking cellulolytic function may produce vitamins, growth factors and branched chain fatty acids, which are often essential for the growth of cellulolytic species¹¹). A synergistic association between the cellulolytic *R. albus* and a *Clostridium* has been shown to be beneficial for digesting cellulose³⁷).

The primary metabolic products of cellulolytic bacteria include aliphatic fatty acids (formic, acetic, propionic, butyric, valeric), lactic acid, succinic acid, ethanol, hydrogen, and carbon dioxide^{25, 26}). The product distribution and energy yields suggest that the Emden-Meyerhof pathway represents the principal route of carbohydrate metabolism²⁴). In some species, the Hexose Monophosphate pathway plays a supplementary but important role. Acetic and formic acids (and/or hydrogen and carbon dioxide) are often produced as major products by all known species, independent of source. Ethanol formation is commonly observed, but normally at lower concentrations. The production of other acids is more species specific. In mixed culture, some of the products of cellulolytic bacteria, particularly lactic, succinic acids, and ethanol, are rapidly metabolized by non-cellulolytic species. Lactic acid is converted primarily to propionic acid by virtually all species of *Propionibacterium*⁵). Propionic acid formation has been also observed via decarboxylation of succinic acid²⁹). The oxidation of ethanol to acetic acid, formic acid, hydrogen, and carbon dioxide regularly occurs in mixed cultures of acidogenic bacteria¹²).

Other than the ability to synthesize cellulases, cellulolytic bacteria are virtually biochemically indistinguishable from the majority of non-cellulolytic acidogens. The combined activity of the microflora results in rapid and efficient hydrolysis of solubilized saccharides from cellulose to a few simple aliphatic acids (formic, acetic, propionic, butyric), hydrogen and carbon dioxide. These then become substrates for methane synthesis.

3 Methanogenesis

Although scientific interest in methane production from decomposing vegetable and animal matter dates back to the 17th century, the techniques of isolation, nutrition, and systematic study of methanogenic bacteria have been developed only over the past 30 years. Early work on microbial methane formation has been reviewed by Barker⁵) and progress through 1966 by Stadtman⁵³). More recent developments have been reviewed and assessed by Wolfe⁶³). Altogether about a dozen species of methanogenic bacteria have been isolated and maintained in pure culture. They include short rods and curved rods (*Methanobacterium*), cocci (*Methanococcus*), spiral organisms (*Methanospirillum*), and a sarcina (*Methanosarcina*). All known species can utilize carbon dioxide and hydrogen as substrates for methane generation. The majority of species either require or benefit from various growth factors, particularly B vitamins. Generally, organisms isolated from the rumen have more exacting nutritional requirements than bacteria from mud or sludge. It has become evident that methanogenic bacteria diverged early from other prokaryotes and have unique metabolic characteristics³⁸).

The biochemical mechanism of methane formation from organic acids and its coupling to ATP synthesis is virtually unknown. Most workers have focused their attention on methanogenesis from either carbon dioxide or formate. The reduction of carbon dioxide to methane requires molecular hydrogen as an electron donor. However, neither the electron transport mechanism nor the intermediates of the postulated sequential reduction steps have been proven experimentally^{40, 53}. Recently, several species of *Methanobacterium* have been shown to possess a low molecular weight, fluorescent, electron transfer coenzyme (coenzyme F₄₂₀). This coenzyme appears to be involved in NADP linked, reversible hydrogenase-hydrogenlyase systems^{57, 58}. Thus, hydrogen and formate are essentially equivalent sources of electrons for the reduction of carbon dioxide to methane. The immediate precursor to methane is most likely methylcobalamine as methane forms rapidly from methylcobalamine and related compounds in cell-free extracts^{7, 40, 63}. Methyl group transfer involves a small coenzyme, 2-mercaptoethane-sulphonic acid, which occurs exclusively in methanogens³⁸.

In anaerobic digestors the major source of methane appears to be acetate rather than carbon dioxide or formate. Estimations based on carbon isotope labelling indicate that 67% to 75% of the methane is derived from acetate^{28, 49}. Methane formation probably involves the decarboxylation of a coenzyme-bound intermediate by an unknown mechanism. It has been shown that the methyl group of acetate is the precursor for methanogenesis as virtually all the methane comes from the methyl residue^{7, 56}. However, only a few methanogenic bacteria can utilize acetate in pure culture. Positively identified methane formers from acetate are *Methanosarcina barkeri*, a *Methanobacterium* sp. and an extreme thermophile, *Methanobacterium thermoautotrophicum*. Two or three other species, notably *M. ruminantium*, *M. formicum*, and an unidentified *Methanobacterium* are probable utilizers of acetate, at least in synergistic association with other bacteria. The latter three are common, perhaps predominant strains in anaerobic systems at mesophilic temperatures^{10, 22, 23, 59}.

The formation of methane from propionate and butyrate has not been demonstrated in pure culture. One can infer that they are converted to methane as both are known products of acidogenic activity, yet no accumulation of either occurs in stable operational digestors. Carbon isotope studies suggest that propionate utilization proceeds by oxidative conversion to pyruvate either via succinyl- or acrylyl-SCoA since all three carbon atoms can give rise to methane. Nelson et al.⁴⁴ postulated that butyrate is oxidized to acetate and methane by a mechanism resembling β -oxidation. However, experimental evidence is insufficient to positively prove that either propionate or butyrate are direct precursors of methane. In fact, the degradation of fatty acids to shorter chain length products can be accomplished by non-methanogenic species.

Several researchers have reported kinetic constants for methanogenic bacteria. Often, the data have been obtained under uncontrolled, ill-defined environmental and nutritional conditions, and objective assessment of the kinetic constants is a nearly impossible task. Growth models are either direct adaption or some variation of the conventional Monod-type model. At 35 °C, Lawrence and McCarty³⁵ reported maximum specific growth rates ($\mu_{\max}$) of 0.32, 0.31 and 0.37 day⁻¹ and saturation constants (K_s) of 154 mg l⁻¹, 32 mg l⁻¹ and 5 mg l⁻¹ in continuous cultures of mixed methanogenic population with acetate, propionate, and butyrate as limiting substrates, respectively.

Andrews and Graef³⁾ have estimated a similar maximum specific growth rate ($\mu_{\max} = 0.4 \text{ d}^{-1}$) but a considerably smaller saturation constant ($K_s = 2.0 \text{ mg l}^{-1}$) for acetate in mixed cultures in the mesophilic temperature range. The authors incorporated substrate inhibition by un-ionized acids in their model. Since the concentrations of un-ionized acids depend both on pH and total acid concentration, kinetic profiles differ markedly at constant pH values below pH 7. The authors gave no experimental evidence concerning the validity of their model. Our own observations gave kinetic profiles similar to those predicted by Andrews²⁾ when the instantaneous gas generation rate (lh^{-1}) was plotted against the organic acid concentration (mM) obtained from continuous digestors operating at stationary states at 39°C , pH 6.5. In these studies, the source of methane was either glucose or cellulose (filter paper) fed at the rate of $0.5\text{--}1.0 \text{ g l}^{-1} \text{ d}^{-1}$ on a reducing sugar basis. The kinetic parameters including the saturation constant ($K_s = 9.2 \text{ mM}$ as acetate) and the inhibition constant ($K_i = 103 \text{ mM}^{-1}$ as acetate) were established by standard regression analysis. The best model (in the least square sense) could explain only 50% of variability of the reaction rates with respect to volatile acid concentrations. In other words, the statistical variance was reduced by one half when the instantaneous acid concentration was taken into account in addition to the loading rate, implying that other important parameters are more significant than allowed for by Andrews and Graef³⁾.

Pohland and Ghosh⁴⁷⁾ have obtained a maximum specific growth rate of 3.4 d^{-1} and a corresponding saturation constant of 600 mg l^{-1} (as acetic acid) at 37°C , pH 6.1–6.8. The discrepancy in the rate constants between results of various investigators cannot be explained by chance selection of some particular methane forming species. Rather, it reflects the difficulty in establishing specific growth rate constants and saturation constants independently by using conventional methods of analysis such as the Lineweaver-Burk plot. The problem is compounded by the considerable scatter of experimental data points and poor reproducibility inherent in these heterogeneous populations.

In our studies, the kinetic information obtained using single substrates could not be used for predicting methane generation from a mixture of acids. For example, at retention times of 14 days, 39°C , pH 7 to 7.3, the utilization of acetic, propionic, and butyric acids was 92% to 98% when fed individually at $9.0 \text{ mM l}^{-1} \text{ d}^{-1}$. Propionate utilization efficiency fell to 74.5% and butyrate to 62.7% when the substrate consisted of 3:2 molar mixtures of acetate/propionate or acetate/butyrate, respectively. Acetic acid utilization remained unchanged at 97.3%. Acid mixtures of butyrate and propionate never reached steady state. Both acids accumulated daily and the experiment was abandoned after methane generation had ceased. When equimolar mixtures of all three acids were used propionic acid was not utilized at all. The disappearance of propionic acid was due solely to volumetric dilution if propionic acid was eliminated from the feed. The apparent lack of propionate utilization in the presence of other acids has also been observed previously⁵⁸⁾.

These observations suggest that propionate utilization is more complex than that of acetate or butyrate. It is likely that propionate utilization commences by the conversion of carbon dioxide to succinate. This reaction is envisioned as the reversal of propionate formation from succinyl-CoA. The eventual conversion of succinate to

acetate, which is the major precursor of methane from organic wastes, would require a reduction step. It is possible that the reaction sequence is blocked either by the absence of a suitable reducing agent (hydrogen, for example) or by the presence of acetate.

Of the various microflora in anaerobic digestors, methanogenic bacteria are the most sensitive to changes in environmental conditions. Methanogenic activity in mixed culture is only observed in the relatively narrow pH range of 6.2 to 8.0. In enriched culture, the pH profile is even narrower. Van den Berg et al.⁵⁹⁾ reported that at 40 °C acetate conversion to methane is optimal between pH 6.5 and pH 7.1. Outside this range, the reaction rate dropped rapidly at pH 6.2 and pH 7.3 and the reaction rates were about half the maximum. Acetate concentrations between 0.2 mM and 100 mM had little effect on the rate. The question of inhibition due either to the effect of pH on some critical enzyme conformation as implied by the above results, or to un-ionized acid inhibition as suggested by Andrews²⁾ is difficult to resolve. The concentration of undissociated aliphatic acids is pH dependent. At constant pH the undissociated acid concentration increases with total acid concentration. Thus at lower pH values the pH and concentration effects are virtually inseparable. Inhibition by protonated acids cannot account for the reaction rate decrease above pH 7.1. In anaerobic digestion processes, pH control can be difficult and pH changes occur more frequently than many investigators would like to admit. With cellulosic substrates the problem is compounded by continual acid accumulation because cellulolytic and acidogenic activities are not affected until the pH decreases well below pH 6.

The inhibition of methanogenesis by metal ions, particularly heavy metals, has been studied by a number of workers. Cation toxicity is a problem in sewage sludge digestors^{6, 34, 41, 42)}. According to Kugelman and Chin³⁴⁾, much of the data collected is erroneous and more research is needed for rational evaluation of toxic metal ion concentrations. Metal toxicity is seldom a serious problem during the digestion of cellulosic wastes. Heavy metals can be eliminated as insoluble sulfides by natural production of hydrogen sulphide from proteinous material (for example, manure), which precipitate as insoluble sulphides.

4 Substrate-Microbe Interactions

A distinguishing feature of anaerobic processes, in particular the conversion of cellulose to methane, is the low cell yield in comparison to aerobiosis. Generally, overall yields of 0.1 to 0.2 g biomass synthesized/g substrate degraded are observed with heterogeneous substrates. Using cellulose as the carbon source and ammoniacal nitrogen as the primary nitrogen source, we have obtained overall yield coefficients of 0.2 g biomass/g cellulose utilized at 14 days and 0.09 g biomass/g cellulose utilized at 28 days retention time in functioning anaerobic digesters at 39 °C and average pH 6.7. A comparison of carbon and nitrogen utilization with three different substrates obtained in our laboratory are presented in Table 3. The data represent the product distribution from 100 g of the respective substrates in semicontinuous anaerobic digesters operating at a stationary state. The casein and cellulose containing media were supplemented

Table 3. Carbon and nitrogen anaerobic conversions for three substrate types
 Temp. = 39 °C, Retention = 14 d, pH = 6.5–6.8, Loading = 1–4 g l⁻¹ d⁻¹

Component	100 g Protein (Casein)	100 g Carbohydrate (Cellulose)	100 g Cow Manure
Solids reduction (%)	75	71	49
Biomass synthesis (%)	21	12	6
Residual material (%)	4	17	45
<i>Carbon (as C, g)</i>			
Methane	19.1	17.4	6.4
Carbon Dioxide	13.4	12.3	4.3
Organic acids	1.5	1.5	1.0
Bicarbonate	5.0	4.0	2.0
Cell mass	11.3	5.5	3.5
<i>Nitrogen (as N, g)</i>			
Ammonia Nitrogen	12	-0.8	1.6
Dissolved Organic Nitrogen	1.1	—	0.3
Particulate Organic Nitrogen	1.6	0.8	0.6

with 100 ml l⁻¹ filtered effluent from a municipal anaerobic digester to satisfy the requirement for trace nutrients. In addition, ammonium hydroxide was supplied to the cellulose (filter paper) medium to give a carbon to nitrogen ratio of 30:1 in the feed.

The carbon metabolism from highly digestible substrates such as pure protein (casein) and cellulose (pulped filter paper) are comparable. Only the nitrogen conversion shows marked differences. Approximately 76% of the organic nitrogen of protein is mineralized to ammonia. In contrast, cellulose digestion results in nitrogen deficit inasmuch as ammonia nitrogen is converted to proteinous nitrogen at a total carbon utilized to nitrogen uptake ratio of about 40:1. Our experience has been that ammonia nitrogen cannot be replaced by nitrates. Nitrate has inhibited methane generation at concentrations exceeding 0.15 g l⁻¹ nitrate N. The results with manure reflect the more recalcitrant nature of native lignocelluloses. The nitrogen balance indicates that the protein content of the manure is a significant source of methane since much of the organic nitrogen is degraded to ammonia. Carbon to ammonia nitrogen ratios of 25:1 to 35:1 are generally considered optimal. Significantly lower C/N ratios, as in poultry manure, may cause ammonia toxicity particularly if the pH becomes alkaline^{22,41}. With some cellulosic substrates (paper, grass, wood, straw) nitrogen deficiency may become a problem. According to our observations, the development of the methanogenic microflora is impaired if C/N ratios exceed 50:1. The nitrogen could be supplied in an organic form (proteins, urea) or as ammoniacal nitrogen. Both cellulolytic and methane forming bacteria are remarkably poor utilizers of organic nitrogen^{10, 11, 23}. However, proteinous material is degraded to organic acids and ammonium ion at rates exceeding cellulose degradation.

Generally, protein co-utilization depresses the cellulose hydrolysis rate, presumably because of enhanced proteolytic activity against extracellular cellulases.

The most important parameters for evaluating the performance of anaerobic digestion systems are the volumetric gas productivity and solids gasification efficiency. The former expresses the specific rate of gas production (l gas/day/l reactor volume), while the latter expresses the extent of solids gasification (l gas/g solids added). The gas produced usually contains 50–75% methane, 20–45% carbon dioxide, 1–5% hydrogen 1–3% nitrogen and traces of hydrogen sulphide. Performance characteristics of some digesters are given in Table 4. The observations refer to digesters operated in a cyclic batch manner. Each day a defined fraction of the fluid is removed from the vessel and replaced with equal volume of liquid containing the substrate of choice. The fraction of material exchange is numerically equal to the inverse of the designated retention time. In these studies the data represent average values from functioning digesters operated under constant conditions for several months. In the thermophilic temperature range (55–60 °C) retention time of 5–7 days can be realized while at mesophilic temperatures (35–40 °C), 14 days or longer retention times are preferable. Our results at 39 °C showed that the minimum retention time attainable with a cellulosic substrate is 9–10 days. However, solids conversion efficiency drops sharply below a 14 day retention time. One of the cellulosic materials used, 90% pure cellulose, gave the highest gas yield. The conversion efficiency of processed (newspaper) and native (manure, grass) lignocellulosics are progressively lower. The high rates and conversion efficiencies observed with swine manure are probably due to methane formation from starches, which are inherently more easily digested than lignocellulosic substrates. Slightly alkaline conditions (pH of 8) enhance the rate of gas production. However, the pH tends to drop below 7 with substrates possessing high carbon to nitrogen ratios.

Our results obtained for the co-digestion of cow manure and grass at 39 °C are given in Figure 1. The volumetric productivity and the solids conversion efficiency are

Table 4. Typical performance of anaerobic digesters using cellulose-containing wastes

Waste Material	Retention time (days)	Temp. (°C)	pH	Gas volumetric productivity 1 gas 1 reactor volume-day	Solids conversion efficiency (l gas/g solids added)
Municipal refuse ¹⁴⁾	30	37	6.5–7.0	0.21	0.47
Municipal refuse ¹⁴⁾	30	65	6.5–7.0	0.31	0.69
Municipal refuse ¹³⁾	5	55	6.5–7.0	0.39	0.56
Municipal refuse ¹³⁾	6	60	6.8	–	0.42
Cellulose (paper)*	14	39	6.6–6.9	0.54	0.52
Cow manure/grass*	20	39	6.6–6.9	0.52	0.17
Cow manure*	20	39	6.6–6.9	0.42	0.21
Swine manure ³³⁾	15	35	8.0	1.24	0.65
Swine manure ³³⁾	30	35	8.0	0.80	0.76
Cow manure ⁶¹⁾	6	55	7.1–7.2	1.45	0.22
Cow manure ⁶¹⁾	6	60	7.3–7.4	1.51	0.23
Cow manure ⁶¹⁾	6	65	7.1–7.2	1.24	0.19

* This work

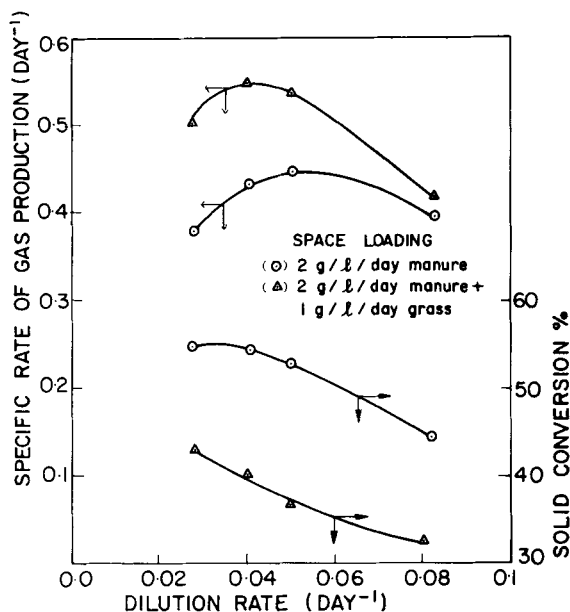


Fig. 1. Effect of codigestion of manure and grass on gas production at 39 °C, pH 6.4

plotted against the inverse of the retention time (dilution rate). The enrichment of the manure with native cellulose (grass) is beneficial at retention times of about 20 days. At lower retention times a significant fraction of the gas is generated by protein breakdown. Approximately 60% of the organic nitrogen in the manure was converted to ammonia. The manure feed contained 1.2% organic nitrogen (7.6% crude protein). Presumably, proteolysis depressed cellulose conversion. Solids conversion efficiency was poor at high dilution rates but improved as the retention time increased. Codigestion of animal wastes with other cellulose sources is an attractive practical method of combined enhanced methane generation and pollution abatement. The potential fertilizer or animal feed value of the manure is not impaired thus land disposal of the digested sludge is not a problem. Manure is probably the most economical source of nitrogen for the anaerobic digestion of nitrogen-deficient cellulosic substrates.

If co-digestion is practiced, the ratio of manure to cellulose must be carefully controlled to minimize the nitrogenous residue in the digester effluent. Poor utilization of nitrogen from the manure could result in an inadvertent discharge of ammonia and other soluble nitrogenous compounds. Such a discharge would negate the pollution abatement aspect of the anaerobic process.

Our results obtained for the effect of cellulose hydrolysis on gas generation are shown on Figure 2. The cellulose hydrolysis rates were determined from daily material balances while feeding 0.25, 0.5, 0.75, and 1.0 g l⁻¹ of cellulose to digesters operating at 39 °C, pH 6.6 to pH 6.9. The rate of gas generation is a linear function of the cellulose hydrolysis rate up to 1.2 g l⁻¹ d⁻¹. Under the experimental conditions the upper limit of cellulose hydrolysis was 1.3 g l⁻¹ d⁻¹ and did not seem to be a strong function of retention time between the practical range of 14 to 30 days. In our opinion, the improvement

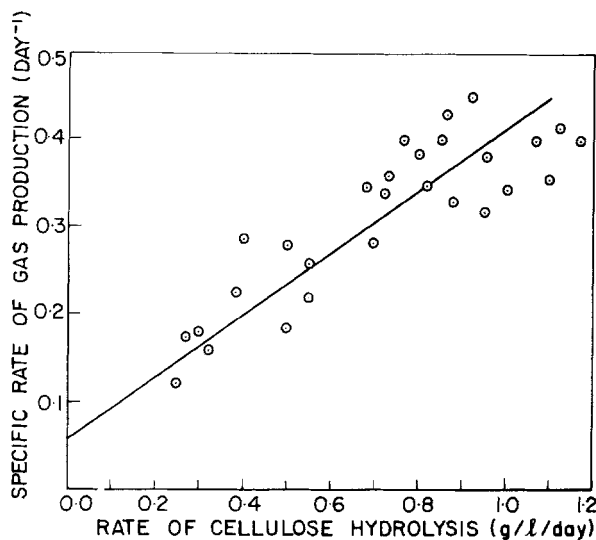


Fig. 2. Dependence of gas generation on cellulose hydrolysis at 39 °C

upon the rate of enzymatic cellulolysis, particularly at the mesophilic temperature ranges, is an unsolved but major problem. Reported rates⁶⁰⁾ of up to 5 $\text{gl}^{-1}\text{d}^{-1}$ of cellulose solubilization in some pure cultures is not feasible in conventional digestion systems at present. Several workers have suggested and developed models for multistage processes in which cellulolytic and acidogenic activity is separated from methane synthesis^{4, 16, 47)}. However, experimental results obtained so far have demonstrated only slight improvements³⁰⁾.

In practice, the anaerobic digestion process is subject to instabilities in performance. Variable feeding is likely to produce fluctuations in gas production rates. In cellulosic digestion systems, overloading occurs when the cellulose addition rate exceeds the maximum attainable rate of cellulolytic activity. Solids build-up can be corrected by a programmed feeding schedule. A more serious problem is the loss of methanogenic activity and the concomittant rise in acid levels. Recovery from "souring" is a slow process and may take several days after pH re-adjustment. Even under constant loading conditions, the performance of digestors is highly oscillatory. Typical data obtained in well-mixed digestors are shown in Figures 3 and 4. The residual cellulose represents mainly the recalcitrant fractions of the cellulosic feed (paper) which, however, cannot be differentiated from the biologically available cellulose substrate.

The methane content of the gas also showed day to day variation of 51% to 84% as determined by either gas chromatography or carbon dioxide absorption methods. The operating parameters were expected to have an effect on the gas composition. However, we found no statistically significant differences in mean gas compositions (58% to 60% methane) when either the loading rate or the retention time (14 to 28 days) was varied. The process does not appear to reach a steady in the traditional sense. At best, the process becomes stationary, i.e. the parameter values do not possess any significant time-dependent drift detectable by classical regression analysis. Oscillations

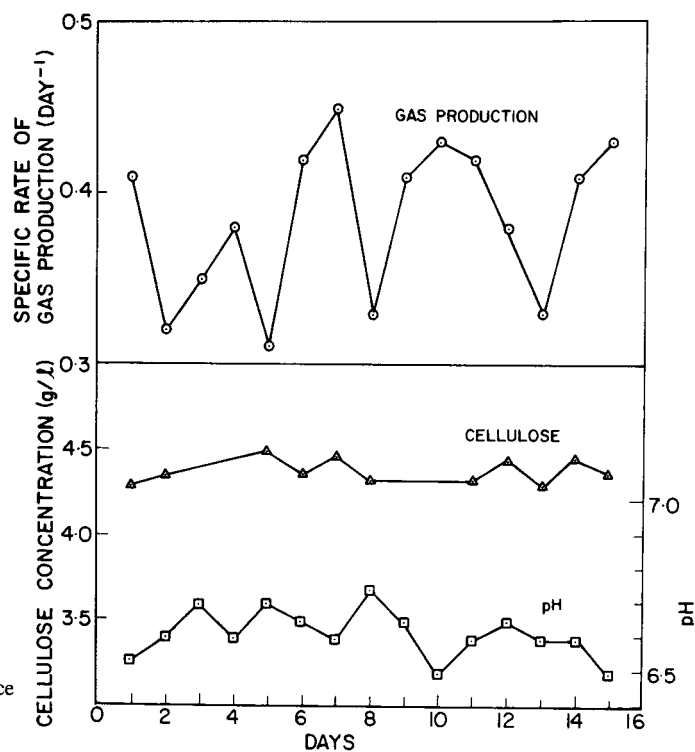


Fig. 3. Digester performance with 1 g l⁻¹ d⁻¹ cellulose at 39 °C

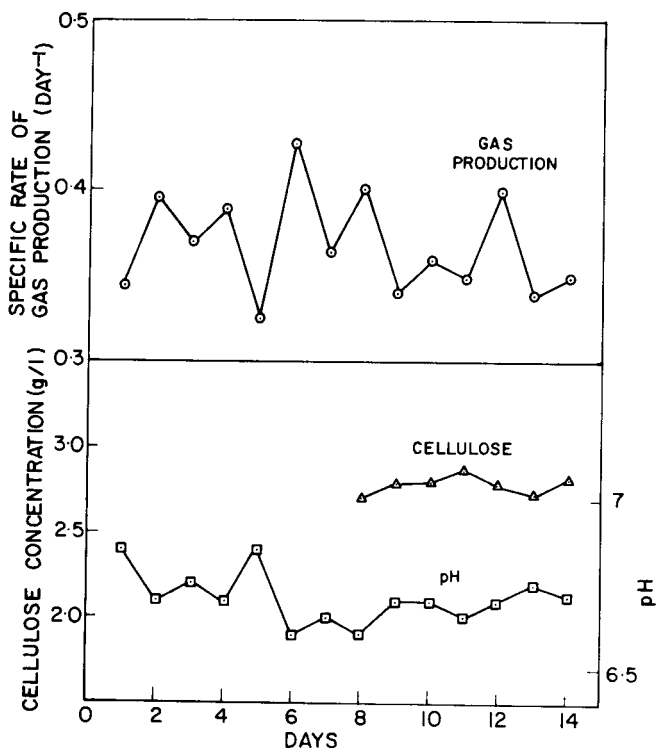


Fig. 4. Digester performance with 0.75 g l⁻¹ d⁻¹ cellulose at 39 °C

around the stationary point do not appear to be random and cannot be fully accounted for by analytical error.

Hobson et al.²³⁾ proposed that the oscillations may be due to periodic fluctuations in the key bacterial populations. We observed that the magnitude of fluctuations increases with the loading rate and is inversely related to the retention time. To our knowledge, no detailed analysis based on experimental data has ever been directed toward the study of these oscillations using cellulose as the primary substrate. Since both the density and the spectra of bacterial species may vary, this type of study may be a complex undertaking.

The choice of temperature is one of the few independent variables in the design of anaerobic processes. The rate of gas generation is higher in the thermophilic temperature range. In simple, small-scale digesters, for example farm units, a significant proportion of the gas must be utilized to maintain a constant temperature in cold climates³³⁾. This point is illustrated in Figure 5. The gas generation rate data, derived from batch studies with cow manure, is plotted against the digester temperature. The dotted lines indicate the estimated fuel requirement to increase the sensible heat content of the feed slurry to the operating temperature during external winter conditions of 10 °C. The net gain in fuel gas is the difference between the two values. Although the gas generation rates may not represent true limits due to insufficient acclimatization, the net gain in fuel gas must depend on the heat requirement, the operating temperature, and the retention time. In a practical design, these factors should be taken into account if optimization of the gas production rate is the goal. Even with more sophisticated design, selection of temperature or the retention time may not be unequivocal.

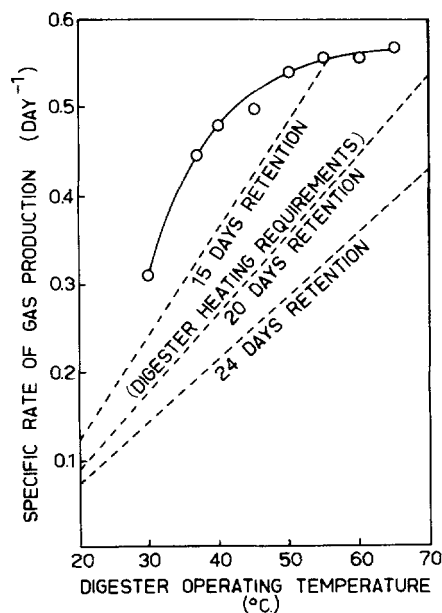


Fig. 5. Effect of temperature on gas production from cattle manure

5 Process Economics

Undoubtedly, the future of anaerobic digestion for energy generation depends on its impact on local as well as national energy budgets. Conventional sewage sludge digestion scarcely produces excess gas beyond the immediate need of the water pollution control plant. Our estimates indicate that a well designed anaerobic digestion system producing a digester gas of 60% methane content may satisfy 3–5% of the energy requirement of a city of population 100 000. This may seem small, but is by no means insignificant in view of natural gas shortages. On the farm, the situation is more promising. According to detailed calculations on actual agricultural energy budgets, the methane gas generated from the manure of a herd of 100 to 200 cattle (or 400 sows) could generate sufficient gas to satisfy all the farmstead heating requirements¹⁵. The major problem is long-term gas shortage, since energy utilization in agriculture (barn heating, corn drying, etc.) is highly seasonal. Large-scale feedlot operations will give excess gas. At present the most economical use of this gas appears to be generation of electricity in stationary gas engines. The added cost for scrubbing of the gas (to eliminate carbon dioxide and hydrogen sulphide) and transport to urban distribution centers is formidable. The national impact is somewhat uncertain. The bioconversion of urban solid wastes, agricultural wastes, and potential energy crops specifically used for methane generation, may give as much as 8% of the U. S. energy requirement by the year 2000¹.

The economic feasibility is subject to a number of variables. Klass and Ghosh³² estimated a net profit of \$ 4.13/metric ton of solid municipal waste by assuming a drop charge of \$ 3.75/ton and the pipeline gas charge of \$ 0.94/1000 A³ can be applied (1973 analysis) in bioconversion plants with a capacity of about 1000 tons/solids/day. Calculations indicate that small scale plants on farms (100 cows or 200 sows) may not become profitable for 10 or so years if one compares the cost of the digester, the value of the gas, and the price of commercial Liquid Petroleum Gas¹⁵. However, natural gas, a much cheaper fuel than LPG, is generally not available to farmers in remote, rural areas. At present, profitable operation requires a population of the equivalent of about 10 000 cattle, which effectively limits the process to large feedlot operations. However, the added benefits of waste stabilization for pollution control and animal-feed production in the form of a nutritious, digestible sludge, has not been included in assessment of the potential profitability.

On smaller farms, the economics favor combined anaerobic-aerobic systems. Thomas and Evison⁵⁴ have shown, that the nutrients of anaerobically digested piggery wastes can be recovered for producing feedgrade protein (*Candida utilis*) in aerobic reactors. In our laboratory, a multi-stage process has been developed in which the anaerobic effluent is enriched with chemically hydrolyzed hemicellulose and cellulose feedstock for single cell protein production. The digester gas generated under anaerobic conditions from crop residue-manure mixtures can be utilized to satisfy the energy requirement of the combined aerobic-anaerobic process.

6 Concluding Remarks

Anaerobic digestion already has an important role as a method of waste disposal. It now appears to have an important future role as a supplementary energy source. We have attempted to identify and analyse the major problems associated with methane fuel-gas generation by the anaerobic digestion of cellulose-containing wastes; these wastes which occur universally in vast quantities, would probably be the main sources of the required raw materials to ensure viability of any future process technology. More basic research aimed at improving the efficiencies of cellulose hydrolysis and methanogenesis is indicated. Equally important, well-documented reports on basic data are required to enable more rational design and operation of commercially-important plant-scale anaerobic digestion systems.

The additional benefits of anaerobic digestion to the above should not be overlooked. Environmental pollution abatement is achieved and the process sludge by-product is useful as soil fertilizer or animal feed. In addition, non-aseptic anaerobic digestion offers a more attractive method than the current efforts of aseptic aerobic prunes, to produce cellulases for use in the enzymatic hydrolysis of cellulose, our most abundant renewable resource, to glucose, which can be used as feedstock for the production of SCP, alcohol and chemicals. Finally, it may be profitable to investigate the feasibility of carrying out anaerobic digestions by solid-state processes so that higher gas productivity per unit volume of reactor capacity can be obtained. In this regard, prior art knowledge may be gleaned from industrial experience with solid-state conversions such as the Koji-type processes and composting.

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The Rheology of Mould Suspensions

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A survey is given of existing theories and measurements of the rheology of mould suspensions. The Casson equation gives a good description of rheology and the impeller method is very suitable for viscosity measurements. The parameters in the Casson equation can be correlated with morphology and volume fraction. Osmotic pressure has a pronounced effect on viscosity, while the pellet growth form can decrease viscosity significantly. The implications of mould rheology for the design of reactors are discussed.

1 Influence of Viscosity on Transport Phenomena in Stirred Tank Reactors (STR)

1.1 Introduction

There are a number of economically important industrial processes using moulds (Table 1). Although mushroom cultivation is carried out on solid media, most processes involve submerged culture in aerated and agitated reactors.

The physical properties of the suspensions involved in these processes are rather different from bacteria or yeast cultures. This is caused by the form in which the mould is growing, i.e. its morphology. The form of the mycelium can be divided into two categories: the filamentous form and the pellet form. In the filamentous form the mould forms hyphae, usually with one or more branches. The mycelial hyphae have a length of the order of 50–500 microns and a diameter of 2–10 microns.

This particular morphology, to which will be referred to as (mycelial) particles, causes entanglement and together with a high concentration of biomass in the reactor (20–50 kgm⁻³ as dry matter, 100–250 kgm⁻³ as wet biomass) leads to a suspension that can be very viscous, in the order of several thousand of centipoise. The rheological

Table 1. Economically important fermentation products from moulds

Product	Class	Organism
Penicillin	antibiotic	<i>Penicillium chrysogenum</i>
Griseofulvin	antibiotic	<i>Penicillium patulum</i>
Cephalosporin	antibiotic	<i>Cephalosporium acremonium</i>
Citric acid	bulk chemical	<i>Aspergillus niger</i>
Amylase	enzyme	<i>Aspergillus oryzae</i>
Pectinase	enzyme	
Glucose-oxidase	enzyme	
Modified steroids	drug	<i>Rhizopus nigricans</i>
Giberillic acid	plant growth hormone	<i>Giberella fujikuroi</i>
Ergot alkaloids	drug	<i>Claviceps sp.</i>
Kojic acid	food flavour	<i>Aspergillus sp.</i>
Single cell protein	food	<i>Trichoderma viride</i>
		<i>Penicillium sp.</i>
		<i>Fusarium sp.</i>
Mushroom	food	<i>Morchella sp.</i>
Tempeh and other	food	<i>Rhizopus sp. a.o.</i>
Oriental products		

behaviour is usually very non-Newtonian, leading to relatively low viscosities in regions of high shear rate (near the impeller) and very high viscosities in regions with low shear rate (near the wall).

In the pellet form the mycelium develops as spherical stable aggregates, consisting of a branched and partially intertwined network of hyphae. These pellets can have diameters of several millimeters. Suspensions of pellets are usually less viscous than suspensions of the filamentous growth form.

In most processes the dispersed or filamentous growth form is present, although in some processes pellets are present also, sometimes in considerable numbers. For the production of citric acid the pellet growth form is used widely. A review on the growth of moulds in the form of pellets has been given by Metz and Kossen (1977).

1.2 Viscosity as an Engineering Problem

The high viscosity of suspensions of filamentous mycelia leads to a number of important problems:

1.2.1 Mass Transport

The high viscosity of the suspension causes problems with the transfer of oxygen from the gas to the liquid phase. Coalescence of air bubbles in the vessel is very pronounced, which means that oxygen transfer is effectively taking place only in the impeller region. A second problem is that the mycelia have a tendency to form agglomerates. Nutrients, of which oxygen is often present in the lowest concentration due to its limited solubility in water, must diffuse into these agglomerates or be brought into them during break-up in the region of the impeller. Partial depletion of nutrients in the agglomerates can lead

to a loss of productivity. A possible further effect of poor oxygen transfer could be that the resulting low oxygen tension in the vessel increases the viscosity due to changes in the mycelial properties. Such a positive feed-back loop, leading to a rapid fall in oxygen tension could have disastrous effects upon the microbial process.

1.2.2 Momentum Transport

The high viscosity and the non-Newtonian character of mycelial suspensions leads to difficulties with mixing, which in turn strongly affects oxygen transfer. Because oxygen transfer is mainly concentrated in the region of the impeller the oxygen supply to the rest of the vessel must be achieved by mixing. A higher viscosity causes longer mixing times and can also cause so-called "dead corners" in the vessel, where the suspension can be stagnant for a considerable period of time (Fig. 2).

Dissolved oxygen and sugar (in continuous culture or fed-batch culture) are nutrients, that are exhausted within a period of the order of a minute. Therefore stagnant zones in the bioreactor can easily decrease the productivity of the mould.

Mixing is important also for control processes, e.g. pH control by means of alkali or acid addition. Bad mixing can cause considerable instabilities in the control system due to a time lag between acid/alkali addition and the resulting signal from the pH electrode.

1.2.3 Heat Transport

Heat transfer to cooling coils, which are used in most large scale vessels, is difficult in viscous suspensions due to the fact that the viscosity is near maximum in such regions of low shear rate. Even a complete standstill of the fluid in certain parts of the cooling zone is not impossible. The cooling problem is aggravated because the

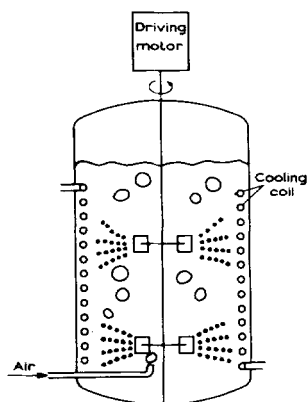


Fig. 1. Schematic drawing of an agitated aerated vessel

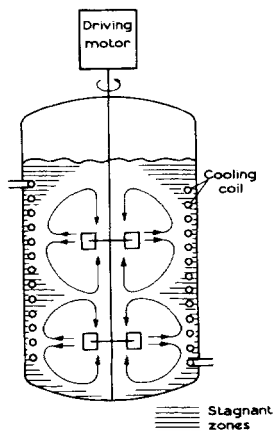


Fig. 2. Mixing of a viscous suspension in a bioreactor

temperature difference between the vessel contents (operating temperatures ranging from 25–30 °C) and the cooling water is small, with heat production of the order of 5 kWm^{-3} , with about 50% from the mould and 50% by stirring.

1.2.4 Inter-Relations Between Processes in a STR

The mass transfer, mixing and heat transfer processes influence the conditions under which a microbial process is working. These conditions influence growth, morphogenesis (development of morphology) and product formation. It can be concluded therefore that a number of feed-back mechanisms exist, making the behaviour of the system rather complicated and difficult to predict. The interrelations of the processes in a vessel containing a viscous mycelial broth are shown schematically in Fig. 3.

Use of the pellet growth form has the advantage of eliminating the problem of suspension viscosity to a great extent. However, another problem arises, that is the possible diffusion limitation on the transport of nutrients into the pellets.

The mechanistic influence of the viscosity is illustrated in Fig. 4. Unfortunately, for each of the processes the optimum flow conditions are different. This means, due to the non-Newtonian character of the suspension that for every process a different rheological aspect will be the most relevant.

1.3 Rheological Aspects

1.3.1 Distribution of Rate of Shear

The shear rate varies in a stirred vessel from very high values in the region close to the impeller to low values some distance away.

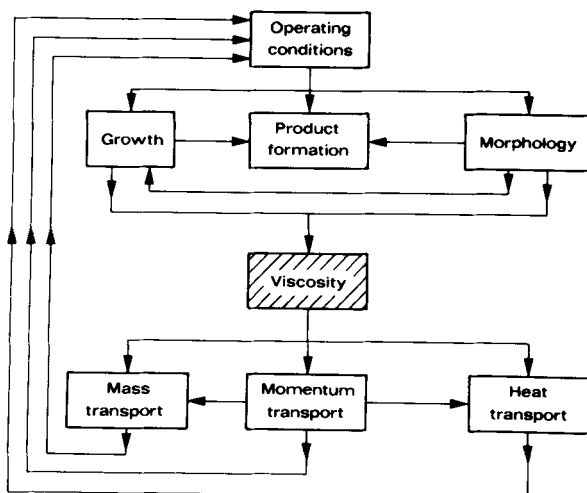


Fig. 3. Schematic representation of the interrelations between processes in a viscous mould process

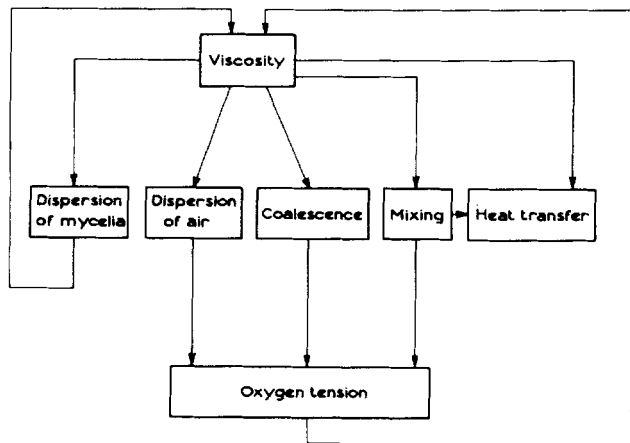


Fig. 4. Block diagram for the relations between suspension viscosity and several processes in stirred reactors

1.3.2 Shear and Elongational Flow Fields

Both shear flow fields (velocity gradient normal to the direction of flow) and elongational flow fields (velocity gradient in the direction of the flow) exist in bioreactors (Fig. 5).

It has been established (Batchelor (1971), Mewis and Metzner (1974) and Van den Tempel (1977)) that there can be considerable differences in viscosity for the same fluid under these two flow conditions. Elongational flow fields can play a role in vessels, especially near the tip of turbine blades (Van 't Riet (1975)).

1.3.3 Elastic Phenomena

For processes where shearing forces are applied for short periods of time, for instance near impeller blades, it is very possible that the elastic properties of the suspension exert an influence. For processes where the forces are applied more or less continuously, elasticity will probably not play a role.

1.3.4 Time Dependence

Suspensions often show a distinct time dependent rheological behaviour due to the breaking and building up of structures. The forces between particles and the local flow conditions determine whether this time dependence is of importance.

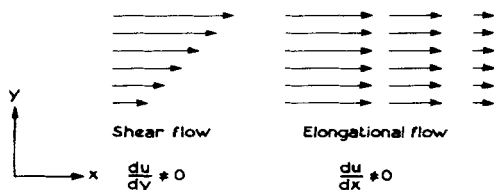


Fig. 5. Shear and elongational flow fields

The time dependence identified here, must be distinguished from the time dependence of the viscosity on a longer term, as a result of the changing conditions during a process (i.e. morphology, nutrient concentration).

The use of the correct rheological aspects for the processes in a reactor would imply complete knowledge of the shear stress/shear rate relations and elastic properties in several different flow fields. With the present state of the art of the rheology of suspensions this is virtually impossible. The only way to estimate the importance of the different aspects is by means of an educated (engineering) guess of ratios of time constants or forces.

2 Rheological Models for the Viscosity of Mould Suspensions

2.1 Survey of Existing Rheological Models for Filamentous Mould Suspensions

So far the rheology of mould suspensions has been described in terms of a time independent, non-viscoelastic fluid. This implies that there is a distinct relationship between the shear stress and the shear rate in the fluid. Mycelial broths generally have pronounced non-Newtonian rheological characteristics. A number of different models have been proposed and a review has been given by Blanch and Bhavaraju (1976).

2.1.1 Bingham Model

Several authors (Karow et al. (1953), Deindorfer and Gaden (1955), Solomons and Perkin (1958), Solomons and Weston (1961) and Deindorfer and West (1960a) have described the rheological behaviour of mould suspensions in terms of so-called Bingham plastics. This means that the model includes the often observed yield stress, which has to be exceeded before the fluid will flow, while the behaviour at higher shear stresses is Newtonian. In mathematical terms:

$$\tau = \tau_o + \eta \dot{\gamma} \quad (1)$$

where τ = shear stress (Nm^{-2})
 τ_o = yield stress (Nm^{-2})
 η = viscosity (Nsm^{-2})
 $\dot{\gamma}$ = shear rate (ls^{-1})

2.1.2 Pseudoplastic Model

Other authors (Richards (1961), Sato (1966), Virgilio et al. (1964), Taguchi and Miyamoto (1966) and Deindorfer and West (1960a)) used a pseudoplastic model to correlate the rheological behaviour. This means that the rate of shear increases more than in proportion to the shearing stress.

In mathematical terms:

$$\tau = K \dot{\gamma}^n \quad (2)$$

where K = consistency index (Ns^nm^{-2})

n = power law index ($n < 1$).

This model fails to include a yield stress, which is the reason that certain authors (Sato 1966), Deindorfer and West (1960a)) used the following adaptation:

$$\tau = \tau_o + K \dot{\gamma}^n \quad (3)$$

2.1.3 Casson Model

Bongenaar et al. (1973) and Roels et al. (1974) have applied the Casson model to mycelial suspensions. This model, originally derived by Casson (1959) for pigment-oil suspensions has been applied successfully to a large variety of suspensions (Kooyman (1971), Charm (1963)). The mathematical expression is:

$$\sqrt{\tau} = \sqrt{\tau_o} + K_c \sqrt{\dot{\gamma}} \quad (4)$$

where K_c = "Casson" viscosity (Nsm^{-2})^{1/2}

Because these models do not obey Newton's law, i.e. the constant ratio between the shearing stress and the rate of shear, it is usual to deal with an apparent viscosity. This apparent viscosity is defined as the ratio of the total shearing stress to total rate of shear at a given value of the rate of shear. So for a pseudoplastic fluid or a Casson fluid the apparent viscosity decreases with increased shear rate.

2.2 Viscosity as a Function of Suspension Characteristics

The viscosity of suspensions is dependent upon a number of parameters (Myers (1962)):

- shape, size and mass of particles
- flexibility and deformability of particles
- surface forces on particles
- concentration of particles
- rate of shear.

For bioreactions, these parameters can depend upon:

- age of the broth

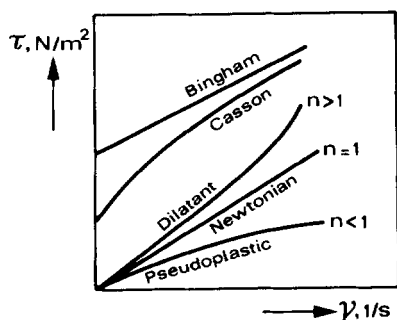


Fig. 6. Rheological models

- the presence or production of extracellular substances (especially important with polysaccharide processes)
- surface structure of the particles (pellets).

For concentrated mycelial suspensions only very few of these influences have been investigated.

2.2.1 Influence of the Morphology

Filamentous and pellet suspensions. While Deindoerfer and West (1960b) mentioned the relation between mould morphology and rheological behaviour, Carilli et al. (1961) have shown quantitatively that morphology influences viscosity. Two strains of *Penicillium chrysogenum* with marked morphological differences were compared yielding viscosity differences of the order of a factor 2. For a strain of *Aspergillus niger* three morphological types were compared: large pellets, small pellets and a filamentous suspension. A difference in viscosity of a factor 3 was found between the filamentous suspension and the large pellets, the small pellet suspension lying somewhere in between. Takahashi and Yamada (1960) obtained similar results for pulp-like and pellet suspension of *Aspergillus niger*. Their data show a difference of a factor 10.

Age of the broth. Several workers have shown the influence of culture age upon rheological properties e.g. during a batch process considerable changes can take place (Virgilio et al. (1964), Deindoerfer and West (1960a)). Morris et al. (1973) found that the viscosity varied considerably (Fig. 7) during a mould process due to changes of the morphology of the organisms.

Shape and size of the particles. Roels et al. (1974) developed a theoretical model for the rheological behaviour of filamentous mould suspensions. This model includes

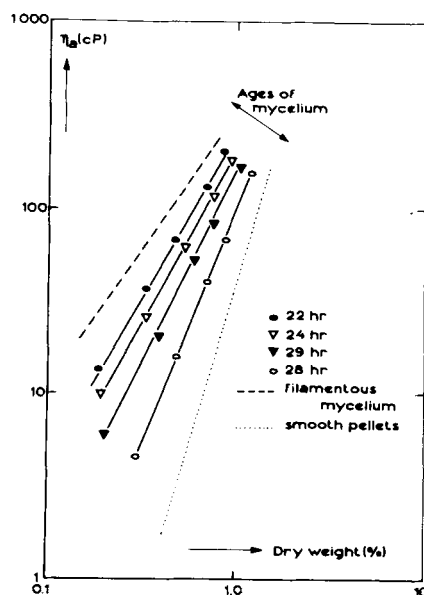


Fig. 7. Changes in the viscosity of *A. niger* pellet suspensions during growth in a tower reactor (Morris et al., 1973)

the influence of the morphology of the mould and the influence of biomass concentration, and is based upon the Casson model. These workers treated the filamentous mycelial particles like polymer molecules and assumed that each particle was coiled to give a spherical shape. By using the "excluded volume" concept a relation was established between the total volume fraction of the mycelial particles (behaving like spheres) and the particle properties.

$$\phi_c = \sqrt{\delta} \cdot C_M \quad (5)$$

where ϕ_c = volume fraction of mycelial particles as spheres (—)

δ = morphology factor dependent on length-diameter ratio of mycelial particles ($\text{m}^6 \text{kg}^{-2}$)

C_M = biomass concentration (kgm^{-3}).

The assumption was made that the spherical coils aggregate into linear agglomerates to which the development which led to the Casson model can be applied. Both K_c and $\sqrt{\tau_o}$ are then a function of the volume fraction of the mycelial particles and thus of the biomass concentration and a morphology factor.

$$\sqrt{\tau_o} = \sqrt{\delta_1} C_M \quad (6)$$

where $\delta_1 = f(\delta)$ (7)

and

$$\frac{K_c}{\sqrt{\tau_o}} = \eta_o^{1/2} \left[\frac{1}{\sqrt{\delta_1} C_M} + C_1 \right] \quad (8)$$

where η_o = viscosity of suspending medium (Nsm^{-2})

C_1 = constant ($\text{mN}^{-1/2}$).

When the Casson equation is written in the form

$$\sqrt{\tau} = \sqrt{\tau_o} \left(1 + \frac{K_c}{\sqrt{\tau_o}} \sqrt{\dot{\gamma}} \right) \quad (9)$$

then a relation can be given between shear stress, shear rate, biomass concentration and morphology factor by combining equation (6), (8) and (9):

$$\sqrt{\tau} = \sqrt{\delta_1} \cdot C_M [1 + f(C_M) \sqrt{\dot{\gamma}}] \quad (10)$$

where $f(C_M) = \eta_o^{1/2} \left(\frac{1}{C_M \sqrt{\delta_1}} + C_1 \right)$. (11)

Quantitative relations between the morphology factor and particle properties were not given.

The main objections against this theoretical model are:

- network interaction between the branched hyphae is considered to be negligible, which is very unlikely due to the character of the particles.

- The hyphae are treated as flexible chains and are supposed to form a spherical coil, which is again very unlikely and in conflict with many microscopic observations.
 - The assumptions of Casson that the coils arrange in the form of rouleaux with an orientation (45 degrees with the direction of flow) is questionable.
 - The randomising effect of the Brownian motion, essential for the validity of the polymer rheology theories, is virtually absent in mould suspensions.
- Consequently it must be concluded that the theoretical model of Roels et al. (1974) has too many objections against its assumptions to be a useful basis for further work. However, the introduction of a morphology factor is of great value.

2.2.2 Influence of the Biomass Concentration

Data on the influence of mycelial concentrations are also rather scarce. The data of Takahashi and Yamada (1960), interpreting mould suspensions as Bingham plastics, led to the following relation for pulp-like suspensions: ($C_M \leq 5 \text{ kgm}^{-3}$):

$$\eta \sim C_M^{1.1} \quad (12)$$

where η = viscosity (Nsm^{-2})

C_M = biomass concentration of mould as dry weight (kgm^{-3}).

Deindoerfer and Gaden (1955) found, when using *Penicillium chrysogenum*, the following relation between yield stress and mycelial concentration:

$$\tau_o \sim C_M^{2.3-2.5} \quad (13)$$

The data of Solomons and Weston (1961) for *Aspergillus niger* led to a relation between viscosity and mycelial dry weight:

$$\eta \sim C_M^{2.65} \quad (14)$$

Carilli et al. (1961) also measured the viscosity during the batch process of *Aspergillus niger* of different morphologies. The influence of the mycelial dry weight upon the viscosity was very strong for the filamentous form: the viscosity was roughly proportional to the square of the mycelial dry weight.

$$\eta \sim C_M^2 \quad (15)$$

As mentioned above, Roels et al. (1974) also considered the influence of the biomass concentration.

2.2.3 Influence of Hyphal Flexibility

Very little has been said about the influence of the flexibility of the hyphae, apart from the observation of Forgacs et al. (1957) for cellulose fibers, that the more flexible fibers give a lower suspension viscosity.

Flexibility of hyphae can vary due to:

- changes in cell wall composition
- changes in branching pattern
- variation of turgor pressure.

The cell wall composition is influenced by the culture conditions and changes during batch growth are possible. No relations are available, however, between culture conditions, cell wall compositions and hyphal flexibility.

Branching of hyphae makes them less flexible. A higher branching frequency can therefore be expected to increase viscosity. Turgor pressure is easily influenced by changes in the osmotic pressure of the culture fluid or the intracellular fluid. Higher osmotic pressures of the suspending medium give a lower turgor pressure, thereby making the hyphae more flexible.

2.3 A Rheological Model for Filamentous Mould Suspension Viscosity

2.3.1 Introduction

A model has been developed for the rheological behaviour of filamentous mycelial suspensions on the basis of the literature on other types of suspensions.

One of the most characteristic properties of mould suspensions is the entanglement of the hyphae. As has been shown for paper pulp suspensions by Forgacs et al. (1957) the particles form a network structure, which is continuously being built up and broken down. With increasing shear rate the network structure disappears until at sufficiently high shear rates the system flows like a Newtonian fluid. Visco-elastic properties often play a role at very low shear rates as a consequence of the network structure.

No fundamental model has been proposed that explains the non-Newtonian rheological behaviour of this kind of suspension satisfactorily. Most attempts to formulate a model have been based upon the kinetics of the flocculation–deflocculation reaction which proceeds under the influence of the rate of shear. The effect of the Brownian motion can be considered to be negligible for mycelial suspensions.

Models have been presented by Goodeve (1939), Gillespie (1960), Casson (1959), Cross (1965), Van den Tempel (1963), De Vries (1963), Ree and Eyring (1955), Kim et al. (1960), and Michaels and Bolger (1962). A review of these models is given by Sherman (1970). The model presented by Michaels and Bolger (1962) (see also Hunter and Nicol (1968), gives an approach that seems useful for mould suspensions. In this model the existence of aggregates is assumed, i.e. relatively weak, more or less spherical clusters of particles. When a suspension consisting of such aggregates is sheared, it is reasonable to assume that these aggregates break into smaller fragments. All the aggregates together form a network structure. At high shear rates the aggregates are assumed to be broken down completely e.g. there is no continuous structural network. Flocculation under the influence of shear rate is still taking place, but the shear stresses are pulling the clusters apart as soon as they are formed.

The stress necessary to produce deformation of such a system at a constant rate can be found from an energy balance:

$$E_{\text{tot}} = E_n + E_{\text{cr}} + E_v \quad (16)$$

where E_{tot} = total energy dissipation rate to produce deformation at constant rate (Wm^{-3})

E_n = energy dissipation rate required to break the aggregate network at given rate of shear (Wm^{-3})

E_{cr} = energy dissipation rate required to break the bonds formed due to collisions at a rate corresponding to the deformation rate (Wm^{-3})

E_v = energy dissipation rate due to purely viscous drag. (Wm^{-3})

As generally

$$E = \tau \cdot \dot{\gamma} \quad (17)$$

Eq. 16 becomes:

$$\tau = \tau_n + \tau_{\text{cr}} + \tau_v \quad (18)$$

where τ , τ_n , τ_{cr} and τ_v have analogous meanings to E_{tot} , E_n , E_{cr} and E_v .

The contribution to τ of the different terms in Eq. (18) is shown schematically in Fig. 8.

2.3.2 Network Energy Dissipation Rate (E_n , τ_n)

The network energy dissipation term is of the greatest importance at low rates of shear. At zero shear rate the network yield stress τ_n equals τ_o , the conventional yield stress. At very high shear rates the network yield stress is negligible.

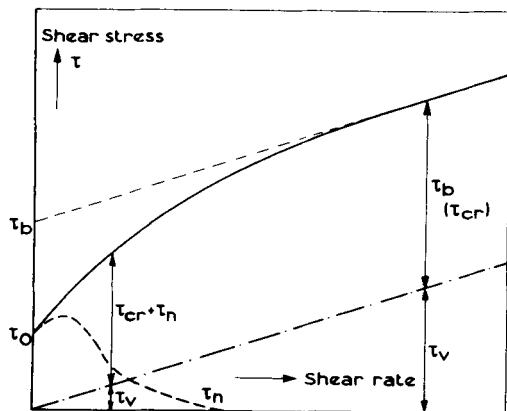


Fig. 8. Contribution of the various shear stress terms for a flocculating suspension (Eq. 18).

$$\lim_{\dot{\gamma} \rightarrow 0} \tau_n = \tau_0 \quad (19)$$

$$\lim_{\dot{\gamma} \rightarrow \infty} \tau_n = 0 \quad (20)$$

There is no derivation available for the network yield stress term at other values of $\dot{\gamma}$. However, the most important factors influencing the network yield stress will be:

- number of particles
- length of the particles
- flexibility of the particles.

The relations between network yield stress and these parameters are so complicated that the development of a more quantitative model has not been attempted.

From the literature it is known that the yield stress of clay suspensions is proportional to the cube of the volume fraction of clay particles (Norton et al. (1944), Michaels and Bolger (1962) and for red blood cell suspensions several workers have found that the yield stress was proportional to the cube of the volume fraction of cells (see Koojyman (1971). The literature provides data on floc strength: Kaiser (see Michaels and Bolger (1962) found that the mechanical strength of compacts of colloidal zinc oxide particles (and so the network strength) was described by the relation:

$$\tau_n \sim \frac{H \cdot \phi_A^3}{d^2} \quad (21)$$

where H = interaggregate binding force (N)

ϕ_A = volume fraction aggregates (–)

d = mean aggregate diameter (m)

Forgacs et al. (1958) determined floc strength of paper pulp suspensions and found that yield stress is proportional to the fiber concentration (measurements at fiber concentrations < 1% by weight). They also found that increases in the fiber length and decreases in fiber flexibility, increased the network strength.

2.3.3 Creep Energy Dissipation Rate (E_{cr} , τ_{cr})

The creep energy represents the energy used to break network bonds. Therefore, according to Michaels and Bolger (1962)

$$E_{cr} = \left(\frac{\text{No of collisions}}{m^3 \cdot s} \right) \left(\frac{\text{No of bonds formed}}{\text{collision}} \right) \left(\frac{\text{work}}{\text{bond}} \right) \quad (22)$$

The collision frequency is given by Von Smoluchowski (1917) as:

$$f = k_o \cdot N_A^2 \cdot V_A \cdot \dot{\gamma} \quad (23)$$

where f = collision frequency (ls^{-1})

N_A = number of aggregates/volume (lm^{-3})

V_A = volume of aggregate (m^3)

k_o = constant, depending on shape of aggregate (m^3)

as

$$N_A = \phi_A / V_A \quad (24)$$

where ϕ_A = volume fraction aggregates Equation (23) can be written as

$$f = k_o \frac{\phi_A^2}{V_A} \cdot \dot{\gamma} \quad (25)$$

Assuming

$$C_2 = \phi_A / \phi_s, \quad (26)$$

in which C_2 can be considered as a concentration factor and where

$$\phi_s = \text{volume fraction particles}$$

then Eq. (25) becomes:

$$f = k_o C_2^2 \frac{\phi_s^2}{V_A} \cdot \dot{\gamma} \quad (27)$$

At very high rates of shear the aggregates consist of a single particle and the constant C_2 becomes:

$$\lim_{\dot{\gamma} \rightarrow \infty} C_2 = 1 \quad (28)$$

If the number of bonds formed per collision is assumed to be proportional to the particle volume, then the overall equation for E_{cr} at high rates of shear becomes:

$$E_{cr} = \text{const. } \phi_s^2 \dot{\gamma} \cdot W \quad (29)$$

where W = work required to break a bond (J)

Using equation (17) for τ_{cr} at high shear rates:

$$\tau_{cr} = \text{const. } \phi_s^2 \cdot W \quad (30)$$

At low rates of shear the relationship for τ_{cr} is much more complicated. The number of bonds formed due to collision is much higher than at high shear rates and depends on the aggregate size, the number of particles per aggregate and several particle properties. No attempt has been made to quantify these relationships.

2.3.4 Viscous Flow Energy Dissipation Rate (E_v , τ_v)

The viscous flow energy dissipation rate can be calculated from an Einstein-like analysis. For very dilute suspensions of spherical particles the Einstein relation is:

$$\eta_{\infty} / \eta_o = 1 + 2.5 \phi_s \quad (31)$$

where η_o = viscosity of suspending medium (Nsm^{-2}).

η_∞ = viscosity at infinite shear rate (Nsm^{-2})

Together with Eq. (17) this gives:

$$E_v = \tau_v \cdot \dot{\gamma} = \eta_\infty \dot{\gamma}^2 = \eta_o (1 + a\phi_s) \dot{\gamma}^2 \quad (32)$$

For concentrated suspensions of particles with different shapes Eq. (31) has to be modified (see also Section 2.4.2)

$$\eta_\infty = \eta_o (1 + a\phi_s + b\phi_s^2 + c\phi_s^3 + \dots) \quad (33)$$

where a, b, c, etc. are parameters, related to particle properties.

Sherman (1970) gives a thorough review of the values that have been found for the constants b, c etc. for a variety of systems. No indication can be given, however, about the values for mould suspensions.

At high rates of shear the total shear stress τ will be equal to:

$$\tau = \tau_{cr} + \tau_v \quad (34)$$

where τ_{cr} is equal to the Bingham yield stress (τ_b) (see Fig. 8).

Thus by extrapolation of the Newtonian part of the shear stress versus shear rate diagram to zero shear-rate, a Bingham yield stress can be obtained, which can be expressed as [see Eq. (30)]:

$$\tau_b = \text{const. } \phi_s^2 \cdot W \quad (35)$$

Only very few data have been found in the literature for suspensions which can contribute to the validity of Eq. (35). Michaels and Bolger (1962) and Hunter and Nicol (1968) both found that for clay suspensions the Bingham yield stress is proportional to the square of the volume fraction solids. However, Thomas (1961, 1962) found a proportionality with the cube of the volume fraction for a variety of small particle suspensions. He also showed that particles with a high shape factor (length/thickness ratio) have higher Bingham yield stress values.

It may be concluded that no complete model has been developed for mould suspensions that describes the rheological behaviour as a function of concentration and particle properties. However, three rheological parameters have been identified namely the yield stress τ_o , the Bingham yield stress τ_b and the viscosity at infinite shear rate η_∞ for which dependency on concentration and particle properties are available.

2.4 Rheological Models for the Viscosity of Suspensions of Mycelial Pellets

2.4.1 Viscosity of Mycelial Pellet Suspensions

As mentioned in section 2.1 very few data on the rheological behaviour of mycelial suspensions have been published. This is even more the case for pellet suspensions.

Takahashi and Yamada (1960) gave the following relation for the influence of the biomass concentration on the viscosity of pellet-like suspensions (for $C_M \leq 10 \text{ kgm}^{-3}$):

$$\eta \sim C_M^{1.1} \quad (36)$$

Carilli et al. (1961) found that for the pellet form the influence of the biomass concentration was much less than for the filamentous form [Eq. (15)]:

$$\eta \sim C_M^{0.2-0.5} \quad (37)$$

The results of Morris et al. (1973), Fig. 7, can be interpreted as:

$$\eta_a \sim C_M^n \quad (38)$$

where $2.3 < n < 3$, depending upon the structure as a function of the age of the mycelium.

2.4.2 General Relations

Many relations have been given, most of them empirical, to describe the relative viscosity η_r of suspensions (Sherman (1970), Jeffrey and Acrivos (1976)). The relative viscosity η_r is defined as:

$$\eta_r = \frac{\eta_s}{\eta_o} \quad (39)$$

where η_s = viscosity of the suspension (Nsm^{-2})
 η_o = viscosity of suspending medium (Nsm^{-2}).

The viscosity of a suspension is made up of a hydrodynamic and a structural viscosity. The hydrodynamic viscosity represents the contribution which is caused by the flow of the suspended particles through the medium. This viscosity is independent of the shear rate and is the limiting viscosity at infinite shear-rate (η_∞). The structural viscosity is the contribution by a permanent interaction (flocculation–deflocculation) of the suspended particles represented in Eq. (18) by τ_n and τ_{cr} . This viscosity depends strongly on the shear-rate.

If there is no interaction between particles ($\phi_s < 0.03$), the relation of Einstein [Eq. (31)] can be used. For more concentrated suspensions [Eq. (33)] is useful. The higher terms in this equation bring into account interaction due to collisions between the particles in the absence of flocculation. However, these types of relations are not completely satisfactory, because the influence of the particle diameter is neglected; this can cause large errors.

Frequently used semi-empirical relations which contain parameters which are a function of the particle diameter are as follows:

$$\text{Mooney (1951): } \ln \eta_r = \frac{a \phi_s}{(1 - \phi_s/\phi_{\max})} \quad (40)$$

for $\phi_s < 0.2$, $a = 2.5$

where ϕ = volume fraction particles (—)
 a = Einstein-constant (—)
 $\phi_{\max}$ = maximum packing density (—)

Krieger/Dougherty (1959):

$$\eta_r = (1 - \phi_s/\phi_{\max})^{-2.5 \phi_{\max}} \quad (41)$$

Eilers (1941):

$$\eta_r = \left(1 + \frac{2.5 \phi_s}{2 (1 - \phi_s/\phi_{\max})}\right)^2 \quad (42)$$

for $d_p < 0.1$ mm

Vand (1948):

$$\eta_r = 1 + 2.5 \phi_s + 7.349 \phi_s^2 \quad (43)$$

Nicodema et al. (1973), Chong et al. (1971), Jinescu (1974) and Quemada (1977) have investigated the above relations and found that for smooth rigid spheres with $d_p < 0.1$ mm only the relation of Eilers was applicable.

The particle size distribution can be expected to play an important role (Chong et al. (1971)). So the use of the above relations (which are all derived from data on monodisperse systems) for a system with a particle size distribution can lead to large errors. Because of all these restrictions, it may be anticipated that the relations quoted above, have limited applicability for pellet suspensions.

2.5 Conclusions

From the foregoing it must be concluded that there are no reliable existing equations to describe the rheological behaviour of pellet suspensions in general. Neither is it possible to make use of existing equations for suspensions of other particles.

3 The Measurement of the Viscosity of (Mycelial) Suspensions

3.1 Impeller Viscometer

Most viscosity measurements reported on mycelial suspensions have been obtained with conventional viscometers, usually of the concentric cylinder or cone and plate types. Use of these viscometers can cause large errors, because of problems arising due to the nature of the suspension. The main problems encountered with both concentric cylinder or cone and plate viscometers are (Bongenaar et al. (1973)):

- particles, especially pellets, are often of the same order of magnitude as the annulus of the apparatus, resulting in the destruction of pellets and flocs during the experiment

- formation of less dense layers next to the surface of the measuring bodies
- a tendency of the suspension to become inhomogeneous because of settling and particle interaction.

Recently a study by Cheng and Le Grys (1975) confirmed these factors when using mould suspensions.

Tube viscometers could cope with the first problem by use of large diameter tubes, but the other problems may still be significant. In fact it must be concluded that most experimental results on the rheological behaviour of mould suspensions obtained with conventional concentric cylinders or cone and plate viscometers, are of limited value.

An alternative method was developed by Bongenaar et al. (1973). Instead of a concentric cylinder a standard 6-blade Rushton-turbine or other type of impeller was used. This solved the problem of settling and water layers, because the stirrer achieved mixing and no water layer could develop adjacent to the stirrer blades.

3.2 Theory

The use of the impeller viscometer system is based upon the following theory.

The power consumption of an impeller can be written as:

$$P = P_o \rho N^3 D_R^5 \quad (44)$$

where P = power consumption (W)

P_o = power number (—)

ρ = density of fluid (kgm^{-3})

N = impeller speed (ls^{-1})

D_R = impeller diameter (m).

In the laminar flow regime ($Re < 10$), the power number for an impeller is given by:

$$P_o = \frac{c}{Re} \quad (45)$$

where c = constant (—), depending on impeller dimensions.

For a non-Newtonian fluid the Reynolds number can be written as

$$Re = \frac{\rho N D_R^2}{\eta_a} \quad (46)$$

where η_a = apparent viscosity (Nsm^{-2}).

The power consumption can be related to the torque on the impeller:

$$P = 2 \pi N \cdot M \quad (47)$$

where M = torque (N m).

Combining Eq. (47) with (44), (45) and (46) gives a relation between torque and impeller speed:

$$M = \frac{c}{2 \pi} \eta_a N D_R^3 \quad (48)$$

So by measuring the moment M as a function of the impeller speed N for a *Newtonian* liquid ($\eta_a = \text{constant}$), the constant c can be determined.

According to Calderbank and Moo-Young (1959) the average shear rate $\dot{\gamma}_{av}$ around the impeller in the laminar flow regime is equal to:

$$\dot{\gamma}_{av} = k \cdot N \quad (49)$$

the constant k being about 10. This relation is independent of the rheological characteristics of the fluid, but is dependent on the measuring system used.

The shear stress can be written:

$$\tau = \eta_a \cdot \dot{\gamma}_{av} \quad (50)$$

For a liquid that follows the power-law:

$$\tau = K \dot{\gamma}^n \quad (51)$$

where K = consistency index ($\text{Ns}^n \text{m}^{-2}$)

n = flow index (—).

Combination of Eqs. (49), (50) and (51) then gives:

$$\frac{\tau}{\dot{\gamma}} = \eta_a = K \dot{\gamma}^{n-1} = K (kN)^{n-1} \quad (52)$$

For a given *non-Newtonian* power-law liquid of known rheological behaviour, η_a can be calculated from equation (48), and when the moment M is measured as a function of the impeller speed N , then k can be found from Eq. (52).

Eqs. (48), (49) and (50) result in:

$$\tau = \frac{2 \pi M k}{c D_R^3} \quad (53)$$

Now since the constants c and k are known, $\dot{\gamma}_{av}$ can be calculated using Eq. (49) and τ using Eq. (53).

3.3 Apparatus

3.3.1 Haake Rotovisco

In order to measure the torque on the impeller of the viscometer system, a Haake Rotovisco RV 11 (Gebrüder Haake, Berlin) was used. This rotational viscometer consists of a measuring head connected to a control panel (Fig. 9). The cup is a glass beaker. Five different combinations of glass beaker and impeller were used. The dimensions are given in Table 2. Systems 1–4 are standard 6-blade Rushton turbine impellers, system 5 is a Haake ‘Messeinrichtung’ FL 10.

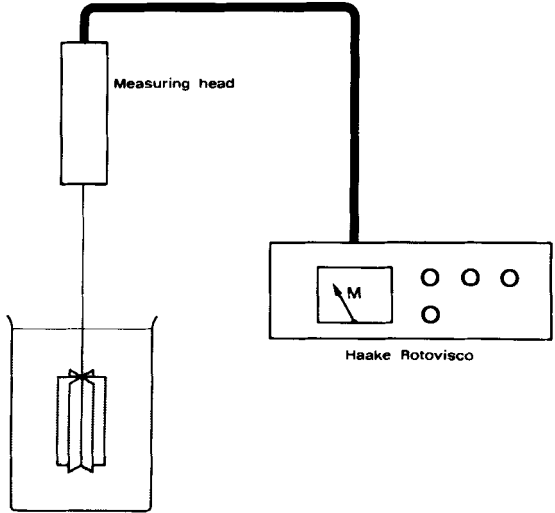


Fig. 9. Impeller viscometer

Table 2. Dimensions of the impeller viscometer (systems 1–4 standard 6-blade Rushton turbine impellers; system 5 Haake ‘Messeinrichtung’ FL 10)

	System 1	System 2	System 3	System 4	System 5
D_R (mm)	96.0	56.8	50.0	35.0	40.0
H (mm)	20.0	11.5	20.6	9.0	60.0
W (mm)	23.3	14.3	12.5	9.0	38.5
S (mm)	74.5	42.7	33	26.0	—
V (l)	2.5	0.600	0.450	0.300	2.5
D_T (mm)	146	93	75	70	146
D_T/D_R	1.52	1.64	1.50	2.00	3.65

Constant temperature precautions are not necessary, because the viscosity of this kind of suspension is only slightly dependent on temperature. The temperature dependence is mainly that of the suspending medium, but the variations in this viscosity are negligible compared to the viscosity of the suspension.

3.3.2 Torsion Wire Viscometer

In practice, only rather viscous suspensions can be measured with the Haake Rotovisco in the low shear rate region. This is due to the minimum torque that can be determined with this viscometer.

To overcome this problem, another system has been developed to measure the less viscous suspensions, especially in the low shear rate region. This apparatus, the torsion wire viscometer, was developed by Bongenaar et al. (1973), and consists of a measuring cup attached to a floating body in a larger vessel filled with water (Fig. 10).

The floating body is attached to the bottom of the vessel by a torsion wire. The torque applied by the impeller is measured as the angular rotation of the floating body.

The torsion wire viscometer is based upon the equation:

$$M = \frac{\pi}{32} d_w^4 \cdot T_c \cdot \frac{\alpha}{L} \quad (54)$$

where d_w = wire diameter (m)

T_c = torsion constant of the wire (Nm^{-2})

α = angle of rotation (rad)

L = length of the wire (m).

As can be seen, there is a linear relationship between the moment M of the impeller and the angle of rotation α of the floating body. The rotation was related to the moment

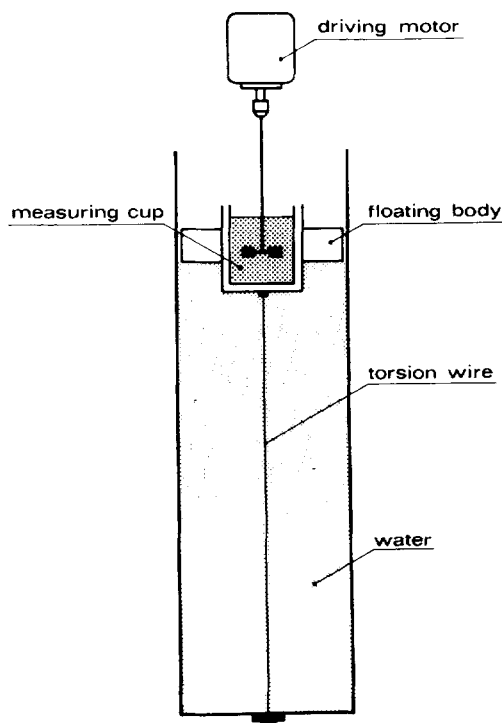


Fig. 10. Torsion wire viscometer

using the Haake Rotovisco (Fig. 11). Because of the fact that a minimal torque gives a rather large angle of rotation, this apparatus has proved to be very useful, especially when dealing with low viscosity suspensions and/or low shear rates.

3.3.3 Operating Range of the Viscometer Systems

The range of shear rates and apparent viscosities that can be covered by the viscometers are restricted by three bounds:

- Reynolds number must be < 10
- a minimum torque that can be measured
- a maximum torque that can be measured.

The Reynolds number restriction leads to a relation between $\dot{\gamma}$ and η_a

$$\frac{\rho N D_R^2}{\eta_a} \leq 10 \quad (55)$$

$$\dot{\gamma} = k \cdot N \quad (49)$$

From Eqs. (49) and (55) follows:

$$\dot{\gamma} \leq \frac{k \cdot 10 \eta_a}{\rho D_R^2} \quad (56)$$

The torque limitations lead also to a relation between η_a and $\dot{\gamma}$.

From Eq. (48)

$$\eta_a = \frac{2\pi M}{c \cdot N D_R^3} \quad (57)$$

Haake Rotovisco. The Haake Rotovisco has a minimum torque of:

$$M_{\min} = 1.6 \cdot 10^{-4} \text{ Nm}$$

So Eq. (57) becomes:

$$\eta_a \geq \frac{3.2 \pi \cdot 10^{-4}}{c \cdot N D_R^3} \quad (58)$$

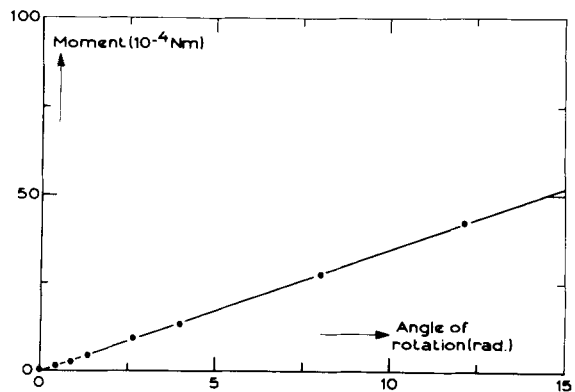


Fig. 11. Calibration of the torsion wire viscometer

The maximum torque is given by: $M_{\max} = 5 \cdot 10^{-2}$ Nm, so

$$\eta_a \leq \frac{0.1 \pi}{c \cdot ND_R^3} \quad (59)$$

Torsion wire viscometer. The same restriction for Re is valid for this viscometer. The minimum torque is: $M_{\min} = 0.3 \cdot 10^{-4}$ Nm, so

$$\eta_a \geq \frac{0.6 \pi \cdot 10^{-4}}{c \cdot ND_R^3} \quad (60)$$

A distinct maximum torque for this system is not given, because this depends on the properties of the torsion wire. After too many rotations, the wire distorts permanently.

With the values for k , c and D_R for the different systems (Table 2, 4 and 6) and by using Eq. (49) a plot can be made of the area in which the different impeller viscometer systems can be used (Fig. 12).

3.4 Procedure

To ensure a fair degree of reproducibility a number of precautions have to be taken:

1. The sample has to be de-aerated. This is achieved either by increasing the impeller speed into the early transition region or by lowering the pressure.
2. The torque has to be measured at different impeller speeds in the laminar flow regime, either from high to low speed with the readings taken quickly after each other or alternatively with mixing at high speed between each measurement for a short period. These procedures minimise settling.
3. The readings should be taken immediately after the start of the impeller (about 5 s) to prevent the influence of any time dependent behaviour.

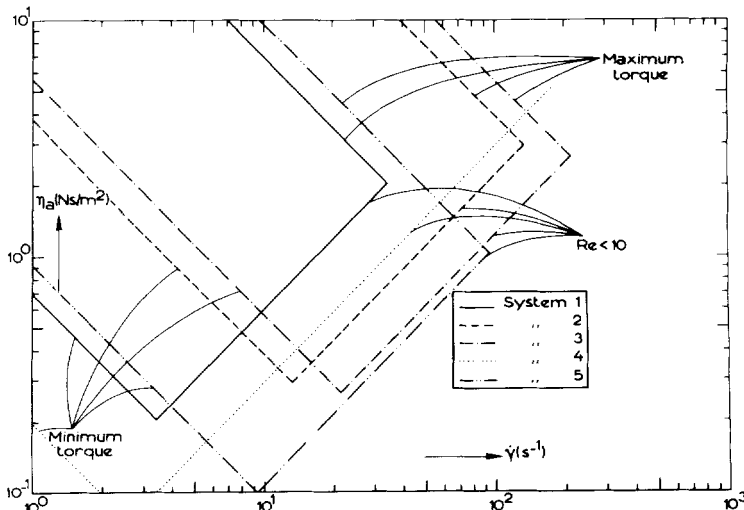


Fig. 12. Operating ranges of the impeller viscometer (system 4 in combination with the torsion wire viscometer)

3.5 Calibration

To be able to convert the torque readings at different stirrer speeds into a shear stress/shear rate relationship, the impeller system has to be calibrated in two ways i.e. with Newtonian and non-Newtonian fluids of known rheological properties.

Calibration with Newtonian fluids. The various impeller systems (Table 2) were calibrated with three Newtonian fluids to find the value of the constant c in Eq. (45) i.e. with glycerol, a solution of polyvinylpyrrolidone (PVP) in water and a silicone fluid. The viscosities are given in Table 3 and were determined with a Ferranti-Shirley (MK II) cone and plate viscometer.

Care has to be exercised when using these calibration fluids e.g. glycerol is very hygroscopic, and at 20 °C the viscosity falls down from 1.412 Ns/m² to 0.523 Ns/m² if 5% water is present (Hodgman et al. (1959)), while PVP can break down at high shear rates, so that it is difficult to get a reproducible plot of τ vs $\dot{\gamma}$. The best fluid to use is silicone oil, it is non-hygroscopic and very stable.

By taking the logarithm of Eq. (45):

$$\log P_o = \log c - \log Re \quad (61)$$

The resulting power number – Reynolds plots are given in Fig. 13.

The slope of the lines is -1 as predicted by Eq. (61). The values of the constant c were determined from the intercepts. The data for each type of stirrer are summarized in Table 4.

Calibration with non-Newtonian fluids. The calibration fluids used were: a solution of 2% carboxypolymethylene (Carbopol 940, Goodyear) in water and a solution of 1% carboxymethyl-cellulose (CMC) in water. The rheological properties (Table 5), according to the power-law equation, were determined in a concentric cylinder viscometer (Contraves, Zürich, combination Bb) and a cone and plate viscometer (Ferranti-Shirley MK II).

The apparent viscosity of the fluids in the impeller systems was evaluated by means of Eq. (57) and measurements of torque *versus* stirrer speed.

$$\eta_a = \frac{2\pi M}{c \cdot ND_R^3} \quad (57)$$

The average shear rate $\dot{\gamma}_{av}$, Eq. (50), was defined for the impeller viscometer, as the shear rate for which the apparent viscosity was equal in both measuring systems (Contraves/Ferranti-Shirley and the impeller viscosimeter).

Table 3. Viscosities of Newtonian calibration fluids (temp. 20 °C)

Solution	Viscosity (Nsm ⁻²)	Density (kgm ⁻³)
A: glycerol	1.863	1261
B: PVP (7.5%)	1.139	1025
C: PVP (15%)	1.394	1027
D: silicone fluid	1.063	977

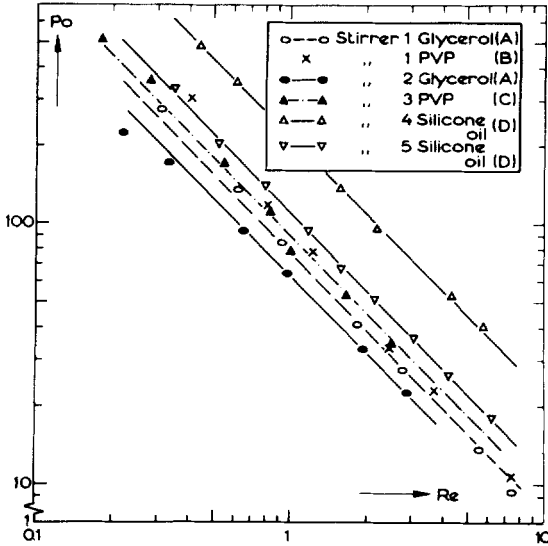


Fig. 13. Power number–Reynolds number relation for impeller viscometer with various calibration fluids

Table 4. Values of the constant c in Eq. (45) from calibration of the impeller viscometer with Newtonian fluids

System	Calibration fluid	c
1	A, B	80
2	A	62
3	C	90
4	D	81
5	D	222

Table 5. Rheological properties of non-Newtonian calibration fluids (temp. 20 °C)

Calibration fluid	Power law index n (–)	Consistency index K (Ns ⁿ /m ²)
2% carbopol (E)	0.27	6.1
1% CMC (F)	0.61	3.7

Combination of Eqs. (52) and (57) gives:

$$\frac{2 \pi M}{c \cdot D_R^3 K N} = (kN)^{n-1} \quad (62)$$

Taking the logarithm of both sides and rearranging yields:

$$\log M = \log (A \cdot k^{(n-1)}) + n \log N \quad (63)$$

$$\text{in which the constant } A = \frac{c D_R^3 \cdot K}{2 \pi} \quad (64)$$

The proportionality constant k between the average shear rate and the stirrer speed can now be determined from a plot of $\log M$ vs $\log N$, the slope being n and the intercept $A \cdot k^{(n-1)}$. The results are given in Table 6.

As can be expected, the value of k differs for the different stirrer geometries, but is constant for one system with different fluids. A log-log plot of the torque versus stirrer speed for the calibration fluids in the turbine system gives the same slope as that for the Contraves data (Fig. 14). This is further support for the impeller method.

3.6 Behaviour of Viscometers with Mould Suspensions

A number of experiments were performed to investigate the behaviour of the impeller viscometer with mould suspensions using *Penicillium chrysogenum* grown in laboratory reactors (details given in Section 4.2.1).

3.6.1 Wall Effects

To investigate possible slip effects at the wall due to the formation of water layers, 4 baffles were placed in the vessel with a width of 0.1 of the vessel diameter. As can be seen from Fig. 15 the baffles had no influence upon the measured torque values, so the conclusion can be drawn that wall effects are unimportant.

Table 6. Values of k from the relation $\dot{\gamma} = k \cdot N$ for impeller viscometer systems

Calibration fluid	k				
	system 1	system 2	system 3	system 4	system 5
C	15.6	14.5	20.0	3.6	37.0
D	16.2				

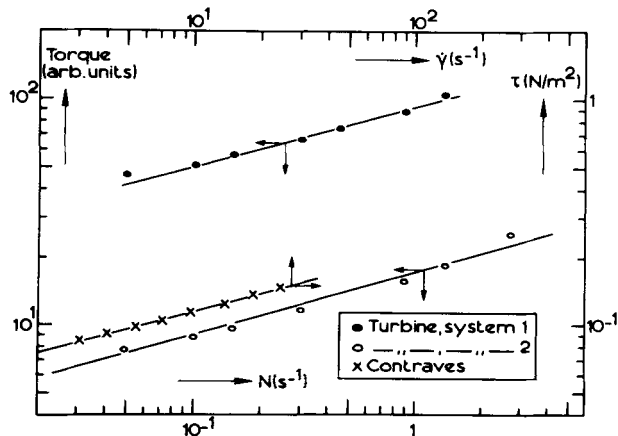


Fig. 14. Torque vs impeller speed in impeller viscometers compared with shear stress-shear rate plot for calibration fluid (C)

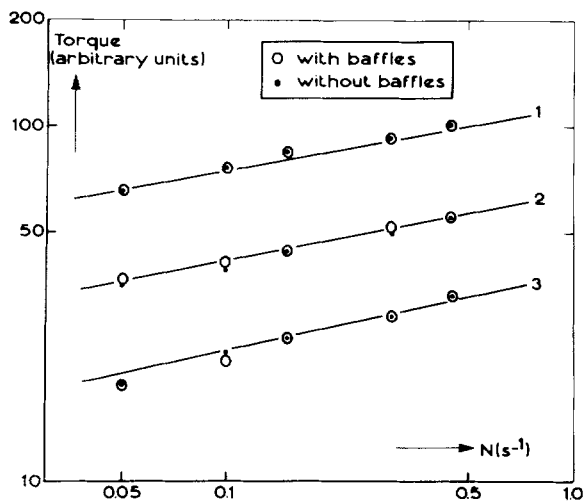


Fig. 15. Influence of baffles upon rheology measurements. Line 1 and 2: filamentous suspensions, 14.8 and 8.2 kg m^{-3} resp. Line 3: pellet suspension, $\phi_s = 60\%$

3.6.2 Time Dependence

The time dependent behaviour of a filamentous mould suspension was investigated with stirrer combination 3 at stirrer speeds of 0.9 and $0.45 \text{ (s}^{-1}\text{)}$. The torque was measured during a period of 7 minutes and as can be seen from Fig. 16 the time dependence was very pronounced.

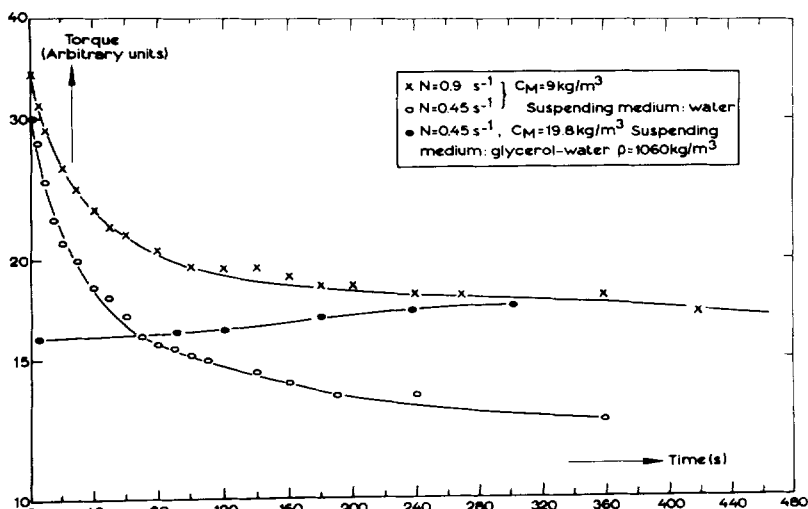


Fig. 16. Time dependence of torque in impeller viscometer with filamentous suspension

The reasons for this phenomenon could include:

1. flocculation/deflocculation equilibrium
2. the orientation of the particles in the laminar flow field
3. centrifugation of the suspension, so that a region with lower concentration develops around the impeller
4. a separation effect of the suspension behind the stirrer blades, resulting in the measurement of a lower viscosity
5. sedimentation of the suspension.

Taking these points in order the following counterarguments are possible:

1. As the suspension was stirred at high impeller speeds for a short period before measurement, only flocculation could happen. This would, however, lead to a higher viscosity with increasing time.
2. As long thin particles in a laminar flow field tend to form coils (see e.g. Mason (1950)), orientation of the mycelial particles is improbable. An effect would be expected in elongational flow fields in the vicinity of the impeller blades. In this kind of flow field rod-like particles do tend to align (Goldsmith and Mason (1967)). However, for the laminar flow regime used neither elongational flow nor alignment of the particles should occur.
3. The centripetal acceleration driving the particles away from the impeller is equal to v^2/r , where v = tangential velocity (ms^{-1}) and r = impeller radius (m).

For the turbine impeller:

$$\frac{v^2}{r} = \frac{2(\pi N D_R)^2}{D_R} = 2\pi^2 N^2 D_R \quad (65)$$

With $N = 1 \text{ s}^{-1}$ and $D_R = 0.05 \text{ m}$, the value of the centripetal acceleration becomes: $v^2/r = 1 \text{ ms}^{-2}$, which is small compared to the gravitational acceleration. This makes it very unlikely that centrifugation plays a role.

5. The possibility of sedimentation effects was investigated using the following experiment: mycelium recovered by filtration, was washed and redispersed in a glycerol-water mixture with a density of 1060 kgm^{-3} , which is about equal to the density of the wet mycelium. Time dependence was investigated with impeller system 1 at two different dry weight concentrations (8.6 and 19.8 kgm^{-3}) over a period of 5 minutes. No time effect could be detected, as can be seen from Fig. 16. *Thus the conclusion can be drawn that sedimentation of the mycelial mass is responsible for the observed time effects in Fig. 16.*

The same time dependent behaviour was found with pellet suspensions although less pronounced (Fig. 17).

3.6.3 Turbulence

Turbulent flow conditions for which the relation [Eq. (48)] between torque and stirrer speed is no longer applicable, are easily detected by a sharp rise in the power consumed. The power number then becomes more or less constant and the torque on the impeller is proportional to the square of the impeller speed (Fig. 18).

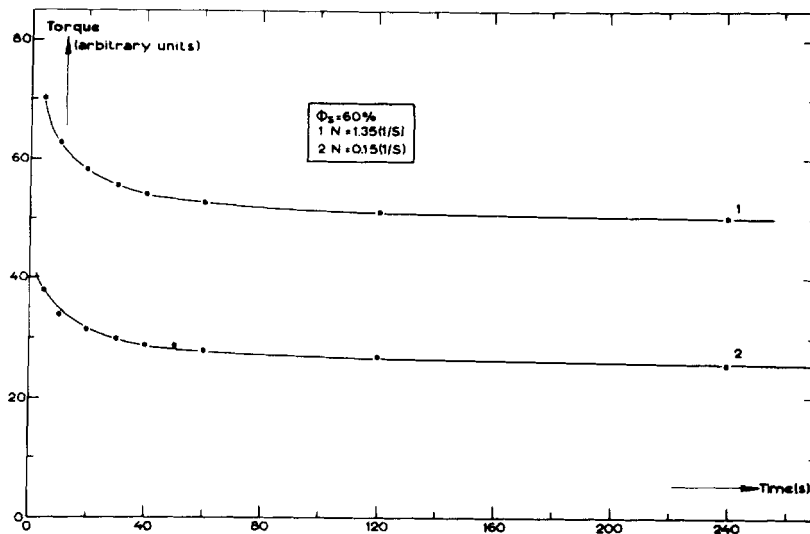


Fig. 17. Time dependence of torque in impeller viscometer for a pellet suspension

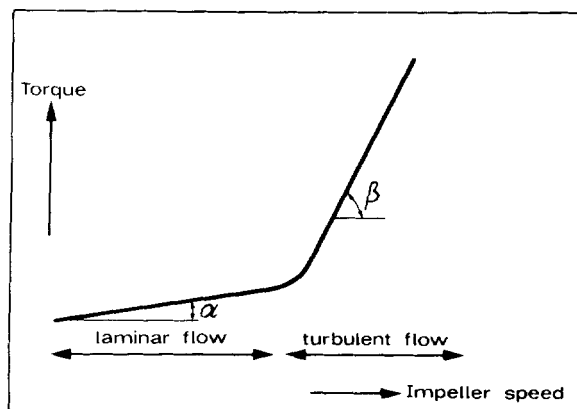


Fig. 18. Plot of torque vs impeller speed for the impeller viscometer

3.6.4 Influence of the Procedure

The way measurements are taken influences the results. When increasing stirrer speeds are used, the measured torque is systematically lower than with decreasing stirrer speeds (Fig. 19). This confirms that sedimentation takes place at low stirrer speeds. Mixing of the suspension at high stirrer speeds for a short period before each measurement gives the same results as when using decreasing stirrer speeds.

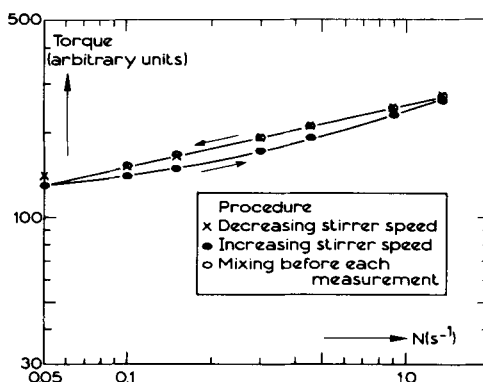


Fig. 19. Influence of procedure upon torque reading of impeller viscometer

Another important factor is the time at which the torque reading is made, due to the time dependence discussed above. Taking the reading directly after adjustment of the stirrer speed, that is after about 5 seconds, gives very good reproducibility. Extrapolation from a half logarithmic plot of torque versus time to $t = 0$ also gives very good reproducibility, but leads to slightly higher values (Fig. 20). The method of instantaneous readings was usually followed because of its relative simplicity.

A third important aspect is the time that has elapsed after the sample is taken from the reactor. To investigate this, the viscosity of a sample of filamentous mycelium was measured directly after taking the sample and again after 24 hours standing at room temperature without aeration. Only small differences were detected in the rheological properties (Table 7), although the deviation is a systematic one.

For pellet suspensions the time after which the sample is measured has a considerable influence (Table 8).

The explanation for this phenomenon is probably that due to autolysis during the 24 hours the pellets were collapsed. The collapse was accompanied by a decrease of the volume fraction of pellets (Table 8) and because of this the torque measured was much lower.

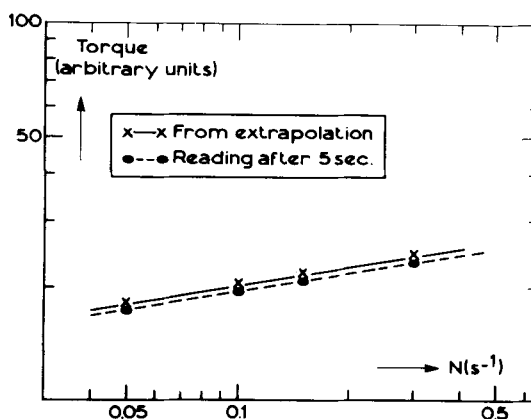


Fig. 20. Comparison of different procedures for the determination of the torque.

Table 7. Effect of storage of a filamentous mould suspension (at room temperature) upon rheological behaviour

N (ls^{-1})	Torque (arb. units)	
	directly	after 24 h
0.05	6.6	6.2
0.10	7.6	6.6
0.15	8.2	7.8
0.30	9.2	8.7
0.45	10.1	9.5
0.90	12.5	11.7
1.35	15.0	14.0

Table 8. Effect of storage of a pellet suspension upon rheological behaviour

N (s^{-1})	Torque (arbitrary units)			
	Suspension 1		Suspension 2	
	directly $\phi_s = 40\%$	after 24 h $\phi_s \approx 26\%$	directly $\phi_s = 60\%$	after 24 h $\phi_s = 39\%$
0.1	3	—	22	4.5
0.15	3.5	1	25	5
0.3	4	1.5	29	5.5
0.45	4.5	1.5	33	7
0.9	6.5	turb.	42	9.5
1.35	9	turb.	49	11.5

3.6.5 Influence of Pellets in a Filamentous Suspension

The influence upon the rheological behaviour of pellets in the suspension was investigated by adding pellets to a pulp-like suspension. Pellets with a diameter greater than 0.55 mm obtained by sieving from a suspension, were added in quantities up to 12 volume percent of the original suspension. The results of the rheology measurements are given in Table 9.

Addition of pellets slightly decreased the values of the yield stress and the Casson viscosity. If the pellets had no effect, the decrease of the rheological parameters could be predicted from the results of the dry weight experiments, the only effect being due to dilution. Full influence on a dry weight basis would mean that a sharp increase of yield stress and Casson viscosity could be expected (addition of 12 volume percent pellets means an increase in dry weight of 50%). A comparison of the effect of the pellet addition with the two hypothetical cases described suggests that the contribution of the pellets per unit dry weight is less than 10% of that of the filamentous mycelium. This is illustrated in Fig. 21. These results are a justification of the calculation procedure follow-

Table 9. Influence of pellets upon rheological behaviour of filamentous mould suspensions

Volume percent pellets added	Yield value (arb. units)	Casson viscosity (arb. units)
0	13.24	2.48
1	13.07	2.52
2	13.31	2.45
3	13.08	2.49
4	13.00	2.50
6	12.64	2.50
8	12.83	2.47
10	12.68	2.36
12	11.38	2.43

ed; the dry weight of the pellets, when present in a suspension, was subtracted from the total dry weight. Roels et al. (1974) have also successfully used the impeller method, obtaining very reproducible results with suspensions of *Penicillium chrysogenum*.

3.6.6 Resume

The impeller viscometer method has proven to be very useful for rheology measurements on mould suspensions. Reproducible results can be obtained with this method providing certain precautions are taken as part of the experimental procedure. When dealing with filamentous mould suspensions a simple correction can be made to account for the presence of mycelial pellets.

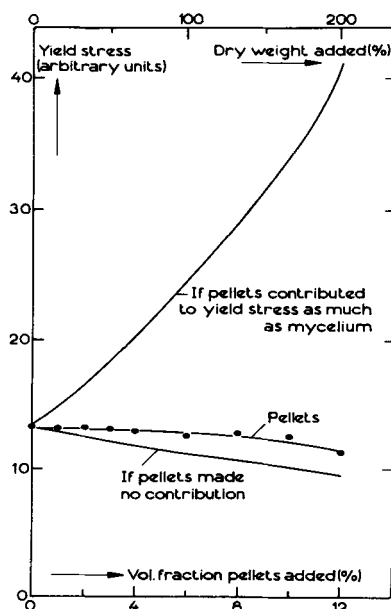


Fig. 21. Effect of pellets upon yield stress of a filamentous suspension

4 Survey of Experimental Results

4.1 Measurements of Roels et al.(1974)

The experimental results of Roels et al. showed that the relation between yield stress and biomass concentration is similar from that expected theoretically. The correlation found was [compare with Eq. (6)]:

$$\tau_o = \delta_1 \cdot C_M^{2.5} \quad (66)$$

Thus the morphology index δ_1 is given by:

$$\delta_1 = \frac{\tau_o}{C_M^{2.5}} \quad (67)$$

Roels et al. (1974) found that the morphology index decreased during batch reactions a fact which is in agreement with observations of fragmentation of mycelial particles in that type of process.

They also found that in batch reactions of *Penicillium chrysogenum* the yield stress first increased and then became more or less constant towards the end of the process (Fig. 22). The consistency index $K_c/\sqrt{\tau_o}$ had a tendency to decrease near the end of the process after being constant in the early stages (Fig. 23).

4.2 Measurements Presented in this Study

4.2.1 Materials and Methods

Organisms. The experiments were done with strains of *Penicillium chrysogenum* obtained from the Royal Dutch Fermentation Industries Ltd., Delft, called strain A and strain B. Another organism used was *Sporotrichum pulverulentum*, Novobranova CBS

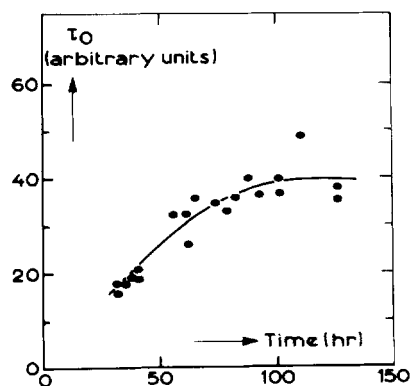


Fig. 22. Yield torque as a function of reaction time

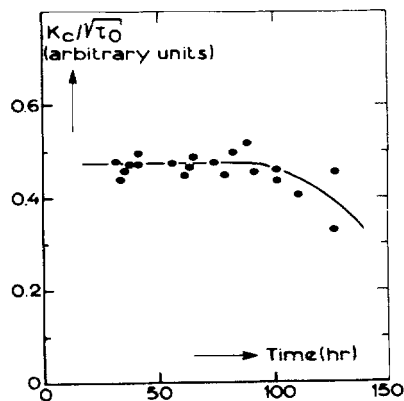


Fig. 23. Consistency index as a function of reaction time

67171, obtained from Centraal Bureau voor Schimmelcultures. The organisms were kept on malt-agar slants (1% malt-agar, Oxoid) and were transferred to fresh slants every 10 days. Every two months new spores from a dry sand stock were used.

Growth media. Synthetic growth media were used for the culture of the organisms, i.e. Medium M12 and M13 for *Penicillium chrysogenum* and M20 for *Sporotrichum pulverulentum*. For the culture of *Penicillium chrysogenum* in the pellet form, medium M11 was used. The compositions are given in Table 10. Before use the media were sterilized at 120 °C for 20 minutes with the exception of glucose which was sterilized separately at 110 °C for 30 minutes. Polypropylene glycol (P2000) was pre-added as anti-foaming agent in the quantity 0.1 vol.%.

Apparatus. The batch cultures were carried out in a 14-liter reactor (New Brunswick Scientific Co. type MA 114). The continuous culture was carried out in a 5-liter reactor (New Brunswick Scientific Co. type MA 105). A schematic drawing of the vessels is given in Fig. 24. More details are given by Metz (1976).

Culture conditions. A summary of the culture conditions for the growth of *Penicillium chrysogenum* is given in Table 11. Experiment no. 201 was carried out in a commercial reactor at the Royal Dutch Fermentation Industries.

Dry weight determination. Dry weight of mycelium was determined by vacuum filtration of a 50 ml sample of suspension over filter paper, washing with 50 ml distilled water and drying at 100 °C until constant weight.

Determination of the morphology of mycelial particles. A semi-automatic method has been developed for the quantitative representation of hyphal morphology. Photographs of mycelial particles were made through a microscope. Much attention was paid to obtaining unbiased samples. Mycelial suspension was diluted about 100x and shaken thoroughly to disintegrate any aggregates; this prevented the selective photography of not aggregated and thus possibly shorter and more compact particles. A drop of this

Table 10. Medium composition for the culture of *Penicillium chrysogenum* and *Sporotrichum pulverulentum*

Constituent	Concentration (kgm ⁻³)			
	M11	M12	M13	M20
Glucose	40	10	10	20
Lactose	—	25	—	—
Yeast-extract (Oxoid Ltd.)	—	—	—	5
(NH ₄) ₂ SO ₄	15	10	10	—
KH ₂ PO ₄	3.0	7.5	7.5	—
MgSO ₄ · 7 aq.	1.6	1.6	1.6	—
FeSO ₄ · 7 aq.	0.10	0.18	0.18	—
CuSO ₄ · 0 aq.	0.004	0.008	0.008	—
ZnSO ₄ · 7 aq.	0.02	0.05	0.05	—
MnSO ₄ · 4 aq.	0.02	0.05	0.05	—
CaCl ₂ · 0 aq.	—	0.05	0.05	—
NaEDTA	—	0.60	0.60	—
CaCO ₃	3.0	—	—	—

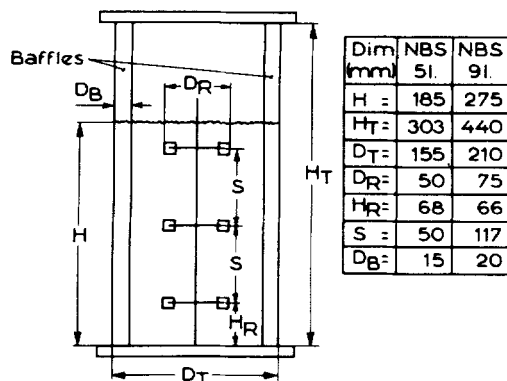


Fig. 24. Dimensions of the 5 and 9 l reactor

suspension was brought under the microscope. Particles to be photographed were selected by a random process of moving the microscope table step-wise and photographing the particle that was visible, unless this was unsuitable due to cluster formation.

Table 11. Culture conditions of the *Penicillium chrysogenum* suspensions used for rheology measurements

Exp. No.	Age (h)	Reactor	Medium	Stirrer speed (rpm)	pH	Dry weight (kgm ⁻³)	Remarks
70	96	NBS-9	M12	400-600	7.0	14.3	continuous culture
73-3		NBS-3	M13	1000	5.5	3.6	continuous culture
73-5-1						12.7	continuous culture
73-5-2						6.3	continuous culture
86-1	48	NBS-9	M12	1000	7.0		
86-2						9.6	diluted
86-3						8.0	diluted
86-4						6.8	diluted
86-5						6.0	diluted
88-0	67	NBS-9	M12	1000	7.0	8.0	
88-1						19.0	concentrated
88-2						15.2	diluted
88-3						12.7	diluted
88-4						10.8	diluted
88-5						9.5	diluted
90	48	NBS-9	M12	700	7.0	9.3	
201-2	83	commercial (150 m ³)	comm.	ca. 60	-	22.8	strain B
201-3	106					25.0	strain B
201-4	130					28.4	strain B
202-3	84					18.9	strain B
202-4	108					23.9	strain B
202-6	132					30.0	strain B

The resulting photographs were projected by means of an adapted microfilm reader upon an electronic digitizer attached to an IBM 1130 computing system. The digitizing table determined the X- and Y-coordinates of a point that is touched with a special pen, with an accuracy of about .25 mm.

A program was written for the IBM 1130 computer that calculated the distances between points. When the photograph of a mycelial particle was projected upon the digitizing table and a number of points along the main hypha were touched with the pen, the program calculated the length of the main hypha (Fig. 25). Calibration by means of a standard millimeter photographed with the microscope, enabled the determination of absolute values of hyphal lengths.

To determine the hyphal morphology properly, at least 100 mycelial particles from every sample were analyzed in this way.

4.2.2 Representation of the Data

When the data from the rheology measurements are plotted in a shear stress—shear rate diagram, curved lines can be drawn through the data points. A characteristic set of curves is given in Fig. 26.

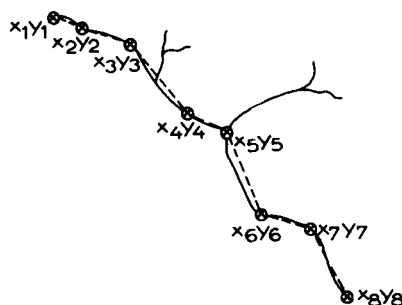


Fig. 25. Calculation of particle parameters by means of a digitizing table

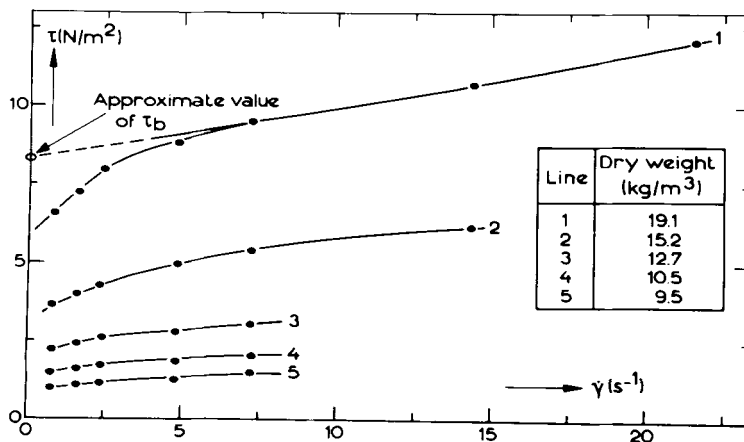


Fig. 26. Characteristic shear stress—shear rate diagram for filamentous mould suspensions (exp. 88)

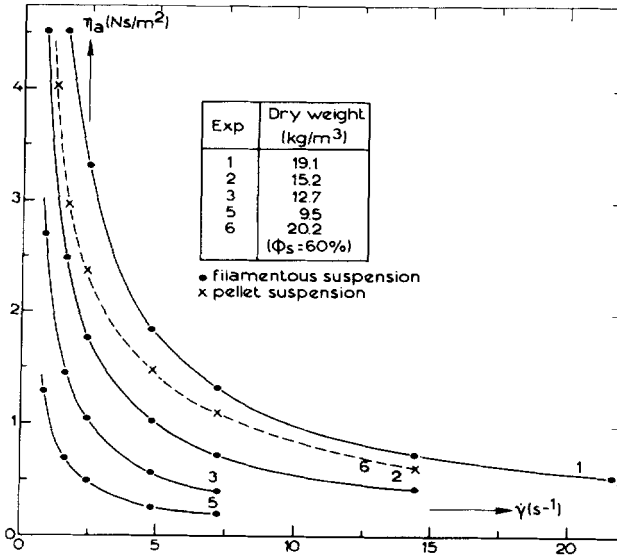


Fig. 27. Apparent viscosity as a function of the shear rate for mould suspensions

To give an idea of the apparent viscosities involved, the apparent viscosity, defined by Eq. (50) is plotted *versus* the shear rate in Fig. 27.

As can be seen from Fig. 26 the suspension behaves as predicted by the qualitative model given in Section 2.3, i.e. the behaviour at low shear rates is clearly non-Newtonian, and at higher shear rates of a more Newtonian character. The non-Newtonian properties are stronger at higher particle concentrations, as expected.

To correlate the rheological behaviour with the particle concentration and the particle properties, the yield stress τ_o , the viscosity at infinite shear rate η_∞ and the Bingham yield stress τ_b must be calculated.

To find the yield stress τ_o , the curves must be extrapolated to zero shear rate. Therefore the data were fitted with Eq. (4).

$$\sqrt{\tau} - \sqrt{\tau_o} = K_c \cdot \sqrt{\dot{\gamma}} \quad (4)$$

A plot of $\sqrt{\tau}$ versus $\sqrt{\dot{\gamma}}$ gives a straight line with slope K_c and intersection with the ordinate at $\sqrt{\tau_o}$. A set of characteristic data is given in Fig. 28.

The value of the yield stress τ_o and the "Casson viscosity" K_c were obtained by linear regression according to the least-squares method.

To find a value for the limiting viscosity η_∞ the slope of the Casson-curve is determined at infinite shear rate. For this slope can be written:

$$\frac{d\tau}{d\dot{\gamma}} \Big|_{\dot{\gamma} \rightarrow \infty} = \left(\frac{d\sqrt{\tau}}{d\dot{\gamma}} \frac{d\tau}{d(\sqrt{\tau})} \right) \Big|_{\dot{\gamma} \rightarrow \infty} \quad (68)$$

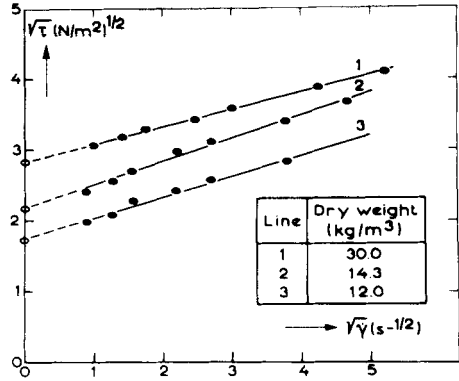


Fig. 28. A plot of $\sqrt{\tau}$ vs $\sqrt{\dot{\gamma}}$ according to the Casson model for filamentous mould suspensions

From Eq. (4)

$$\frac{d(\sqrt{\tau})}{d\dot{\gamma}} = \frac{K_c}{2\sqrt{\dot{\gamma}}} \quad (69)$$

and

$$\frac{d\tau}{d(\sqrt{\tau})} = 2\sqrt{\tau} \quad (70)$$

Equations (69) and (70) give:

$$\left(\frac{d\tau}{d\dot{\gamma}}\right)_{\dot{\gamma} \rightarrow \infty} = \left(\frac{K_c}{\sqrt{\dot{\gamma}}} (\sqrt{\tau_0} + K_c \sqrt{\dot{\gamma}})\right)_{\dot{\gamma} \rightarrow \infty} = K_c^2, \quad (71)$$

Thus the limiting viscosity can be written:

$$\eta_{\infty} = K_c^2 \quad (72)$$

The Bingham yield stress τ_b is found by extrapolation of a straight line through the 3 data points in the shear stress vs shear rate diagram at the highest shear rate (Fig. 26).

The values of τ_0 , η_{∞} and τ_b obtained for the various experimental runs are given in Table 12.

4.2.3 Influence of Dry Weight and Particle Properties

The particle properties investigated are:

- size
- branching frequency
- flexibility.

For the size of the hyphal particles three parameters are available (Fig. 29):

- the effective length or the length of the main hypha L_e

Table 12. Data of rheology measurements with filamentous suspensions of *Penicillium chrysogenum* strain A (exp. 70, 86, 88, 90, 73, 74) and B (exp. 201/202)

Exp. No.	K_c ($N\frac{1}{3}s^{-1}m^{-1}$)	$\sqrt{\tau_0}$ ($N\frac{1}{3}s^{-1}m^{-1}$)	K_c^2 (Nsm^{-2})	τ_0 (Nm^{-2})	Corr. coeff.	τ_b (N/m^2)	C_M (kgm^{-3})	L_e ($10^{-6}m$)	L_t ($10^{-6}m$)	L_{ngu} ($10^{-6}m$)	L_g^* (-)	$\delta_1 \cdot 10^3$ [Eq. (66)]
70	0.325	2.18	0.106	4.75	0.99	7.81	14.3	513 ± 137	890 ± 298	153 ± 19	92 ± 34	6.14
86-1	0.290	1.76	0.084	3.10	0.99	5.00	12.0	302 ± 94	417 ± 132	112 ± 43	55 ± 23	6.21
86-2	0.245	1.32	0.060	1.74	1.00	2.74	9.6					6.09
86-3	0.209	1.01	0.044	1.03	1.00	1.56	8.0					5.69
86-4	0.164	0.83	0.027	0.69	0.99	0.94	6.8					5.72
86-5	0.159	0.54	0.025	0.29	0.96	0.33	6.0					3.29
88-0	0.140	0.60	0.020	0.36	0.94	0.40	8.0	195 ± 40	260 ± 71	69 ± 13	27 ± 7	1.99
88-1	0.233	2.41	0.054	5.83	0.99	8.22	19.0					3.70
88-2	0.201	1.75	0.040	3.05	0.99	4.38	15.2					3.39
88-3	0.139	1.36	0.019	1.84	0.99	2.32	12.7					3.20
88-4	0.115	1.11	0.013	1.23	0.99	1.53	10.8					3.21
88-5	0.110	0.91	0.012	0.83	0.98	1.00	9.5					2.98
90	0.135	0.91	0.018	0.83	1.00	1.10	9.3	330 ± 150	452 ± 213	88 ± 29	49 ± 23	3.15
201-2	0.506	2.07	0.256	4.30	1.00	8.44	22.8	144 ± 50	159 ± 50	67 ± 16	18 ± 6	1.73
201-3	0.525	2.54	0.276	6.47	1.00	12.99	25.0	165 ± 75	202 ± 91	70 ± 35	21 ± 11	2.07
201-4	0.228	2.48	0.052	6.16	0.99	9.42	28.4	132 ± 70	143 ± 72	58 ± 36	18 ± 11	1.43
202-3	0.303	1.78	0.109	3.16	0.99	5.96	18.9	103 ± 33	119 ± 43	41 ± 14	24 ± 10	2.03
202-4	0.361	2.30	0.130	5.28	1.00	8.94	23.9	88 ± 32	99 ± 43	36 ± 12	21 ± 10	1.89
202-5	0.321	2.39	0.103	5.73	1.00	9.42	24.5	—	—	—	—	—
202-6	0.241	2.85	0.058	8.14	1.00	11.10	30.0	64 ± 28	72 ± 36	27 ± 12	15 ± 7	1.65
73-3	0.196	0.55	0.038	0.30	0.95	0.39	3.6	188 ± 109	271 ± 184	55 ± 30	49 ± 35	12.2
73-5a	0.330	2.93	0.109	8.57	0.99	13.02	12.7	190 ± 110	264 ± 172	63 ± 38	55 ± 51	14.9
74-5b	0.199	0.92	0.040	0.84	1.00	1.17	6.3					8.4

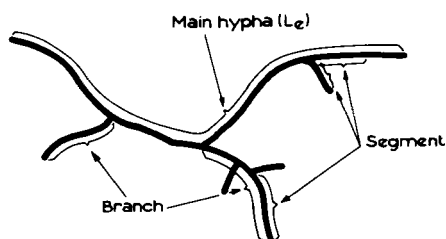


Fig. 29. Schematic representation of a mycelial particle

- the total length of the hyphae L_t
- the dimensionless length L_e^* (= effective length/diameter)
The branching frequency can be characterized by:
- the length of the hyphal growth unit L_{hgu} = total length per number of growth tips (Trinci (1973)).^a

4.2.4 Results

Filamentous suspensions. The particle parameters of the suspensions are given in Table 12.

To describe the influence of the dry weight and the dimensions of the particles upon viscosity, the following equation has been used:

$$\text{rheological parameter (i.e. } \tau_o, K_c, \eta_\infty \text{ or } \tau_b) = \text{const. } (C_M)^\alpha (L_e^*)^\beta (L_{hgu})^\gamma (L_t)^\delta (L_e)^\epsilon \quad (73)$$

The parameters ($\alpha, \beta, \gamma, \delta, \epsilon$) in this equation were estimated from the experimental data for *Penicillium chrysogenum* strains A and B by means of a multiple regression method after linearization of the equation by means of a logarithmic transformation. A standard SPSS regression program was used. For further details of this method see Kim and Kohout (1975).

In all cases the dimensionless length L_e^* proved to give a better correlation with the rheological parameters than the effective or total length as the size parameter. Eq. (73) then becomes:

$$\text{rheological parameter (i.e. } \tau_o, K_c, \eta_\infty \text{ or } \tau_b) = \text{const. } (C_M)^\alpha \cdot (L_e^*)^\beta \cdot (L_{hgu})^\gamma \quad (74)$$

The values of α, β and γ , obtained by means of the multiple regression method using all the results of Table 12, are given in Table 13. A value of zero in Table 13 means that the power was not significantly different from zero (significance level of 95%).

^a In the semi-automatic method for the quantitative representation of hyphal morphology this length of the hyphal growth unit was calculated by $L_{hgu} = L_t / (n + 2)$, where n is the number of branches of the main hypha

Table 13. Parameters in Eq. (74) estimated by means of a multiple regression method

Rheological parameter	α	β	γ
τ_o	2.5 ± 0.2	0.8 ± 0.2	0
K_c	1.0 ± 0.2	0	0.6 ± 0.2
$\eta_\infty (= K_c^2)$	1.9 ± 0.4	0	1.2 ± 0.4
τ_b	2.8 ± 0.2	1.0 ± 0.2	0

The influence of the dry weight was investigated separately by means of dilution experiments. Two samples from experiments 86 and 88 were concentrated by filtration and diluted again with the filtrate (series 86-1 to 86-5 and 88-0 to 88-5), so that a range of mycelial concentrations was obtained. The results of the corresponding rheology measurements are given in Table 12. The values of α for the different rheological parameters were calculated by means of linear regression after logarithmic transformation. The results are summarized in Table 14.

Slight differences occur in the values of α compared with those obtained by means of multiple regression on all experiments (Table 13). This could mean that under the experimental conditions factors other than those accounted for in Eq. (77) have influenced the rheological parameters, although this influence is limited. Flexibility of the mycelial hyphae could be such a factor. It is worthwhile noting that the data of Roels et al. (1974) yield a value of 2.7 for the α corresponding to τ_o .

A comparison of the values of α , β and γ obtained from the experimental data with the correlations predicted by the models presented in Section 2.3 show that they only correspond to the order of magnitude. The model equations are therefore of very limited value. It is clear from Table 13 that the influence of the dry weight upon viscosity is by far the most important and that the branching frequency only makes a significant contribution towards the limiting viscosity.

A characterization of the morphology of the mycelial particles can be achieved by eliminating the dry weight influence. From the correlation for the yield stress τ_o [Eq. (74) and Table 13]:

$$\frac{\tau_o}{C_M^{2.5}} = \text{const.} (L_e^*)^{0.8} \quad (75)$$

Table 14. Values for α in Eq. (74) from dilution experiments (exp. 86 and 88)

Rheological parameter	α
τ_o	2.7 ± 0.1
K_c	0.9 ± 0.1
$\eta_\infty (= K_c^2)$	1.7 ± 0.2
τ_b	3.5 ± 0.2

This group $\tau_0/C_M^{2.5}$ is the same as the morphology factor δ_1 , introduced by Roels et al. (1974) (see Eq. (66)). In Fig. 30 the morphology factor δ_1 ($=\tau_0/C_M^{2.5}$) is plotted versus the dimensionless length L_e^* .

From a regression analysis of $\log \delta_1$ versus $\log L_e^*$ and a statistical evaluation, it must be concluded, however, that no linear correlation can be assumed between $\log \delta_1$ and $\log L_e^*$ (see Table 15). Therefore the correlation given in Eq. (75) is of limited value and can only be used as an indication of the positive correlation between δ_1 and L_e^* .

Thus for mould suspensions the Casson equation parameters τ_0 and K_c can be estimated from the dry weight, the dimensionless hyphal length and the length of the hyphal growth unit by means of the correlations:

$$\tau_0 = 1.67 \cdot 10^{-4} \cdot C_M^{2.5} \cdot (L_e^*)^{0.8} \quad (\text{Nm}^{-2}) \quad (76)$$

$$K_c = 5.454 C_M^{1.0} L_{\text{hgu}}^{0.6} \quad (\text{Nsm}^{-2})^{1/2} \quad (77)$$

The accuracy of these correlations is adequate, i.e. 95% of the values of τ_0 and K_c , predicted by these empiric models have a deviation from the experimental values of less

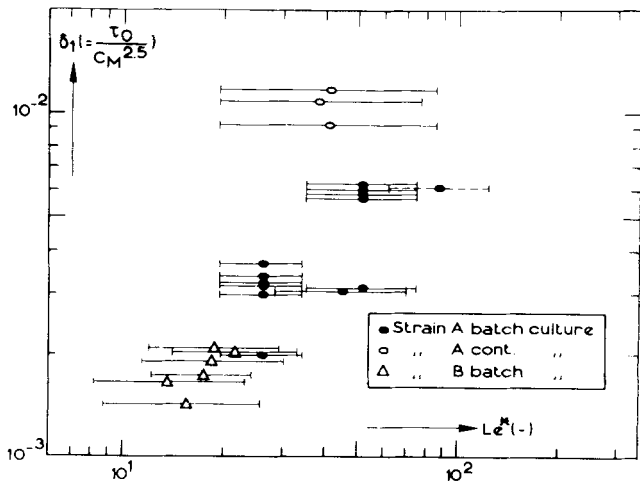


Fig. 30. Morphology factor δ_1 , as a function of the dimensionless length L_e^*

Table 15. Statistical evaluation of regression analysis for morphology factor as function of dimensionless length (L_e^*) (linearized by means of logarithmic transformation), according to the procedures described by Himmelblau (1970)

Morphol. index	s_F^2	s_e^2	Degrees of freedom ν_1 ν_2	F (s_F^2/s_e^2)	$F_{0.95}$ (ν_1, ν_2)	s_3^2	s_y^2	Degrees of freedom ν_1 ν_2	F (s_3^2/s_y^2)	$F_{0.95}$ (ν_1, ν_2)
L_e^*	0.3736	0.0290	17 655	12.87	1.7	24.91	0.0378	1 672	659.8	3.84

than 20%. For mycelial suspensions grown in continuous culture (exp. 73) the correlations do not apply at all, as can be seen in Fig. 30 where the δ_1 -values for the continuous culture suspensions are included. Apparently there is some other factor influencing the viscosity. This could very well be the flexibility of the particles, as the conditions are quite different from those in batch culture.

Influence of hyphal flexibility. The effect of the osmotic pressure upon hyphal flexibility was investigated by addition of sodium chloride or distilled water to a filamentous mould suspension (*Penicillium chrysogenum* strain B).

The osmotic pressure of the suspending medium was determined by a cryoscopic method. This method is based upon the drop of the freezing point. The osmometer (Knauer, Berlin) was calibrated with distilled water and sodium chloride solutions. The yield stress of the suspensions is plotted *versus* the osmotic pressure of the suspending fluid in Fig. 31.

The influence of the osmotic pressure upon the yield stress is considerable, between the highest and the lowest value a factor 2.5 exists. Below a certain osmotic pressure the yield stress is at a maximum. This can be explained by the state of water saturation of the cells. Above a certain osmotic pressure the yield stress reaches a minimum which is probably caused by plasmolysis of the cells.

There are strong indications that osmotic pressure exerts considerable influence on hyphal flexibility through changes in turgor pressure of the cells. Osmotic pressure changes therefore can account for variations in the rheological behaviour.

Pellet suspensions. When the rheology measurements of pellet suspensions are plotted in a shear stress *vs* shear rate diagram, the same non-Newtonian behaviour appears as with filamentous mould suspensions (Fig. 32).

By plotting $\sqrt{\tau}$ *versus* $\sqrt{\dot{\gamma}}$ it can be seen that pellet suspensions can also be described very well with a Casson model (Fig. 33).

Generally, pellet suspensions were found to be less viscous than filamentous suspensions with the same biomass concentration. This is illustrated in Fig. 27, where the apparent viscosity of a pellet suspension of *Penicillium chrysogenum* is given as a function of the shear rate, together with the results of a filamentous suspension with about the same biomass concentration.

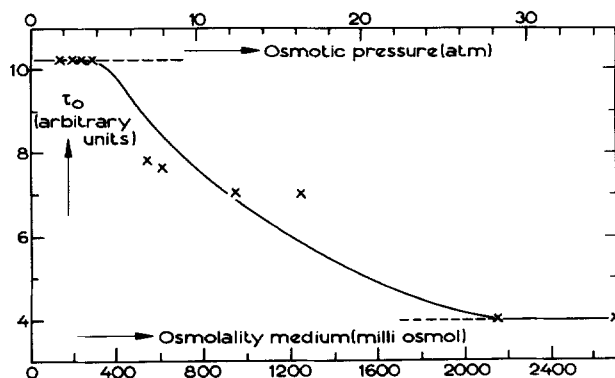


Fig. 31. Influence of the osmotic pressure of suspending medium upon the yield stress of a mould suspension

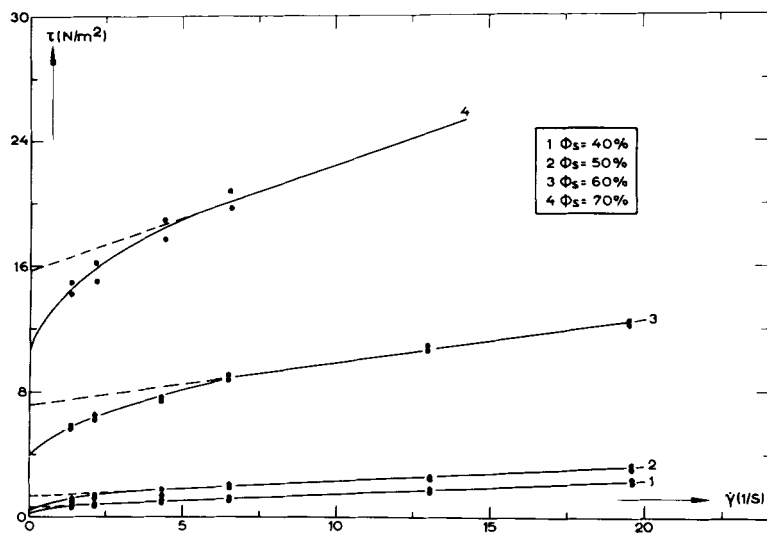


Fig. 32. Characteristic shear stress-shear rate diagram for pellet suspensions

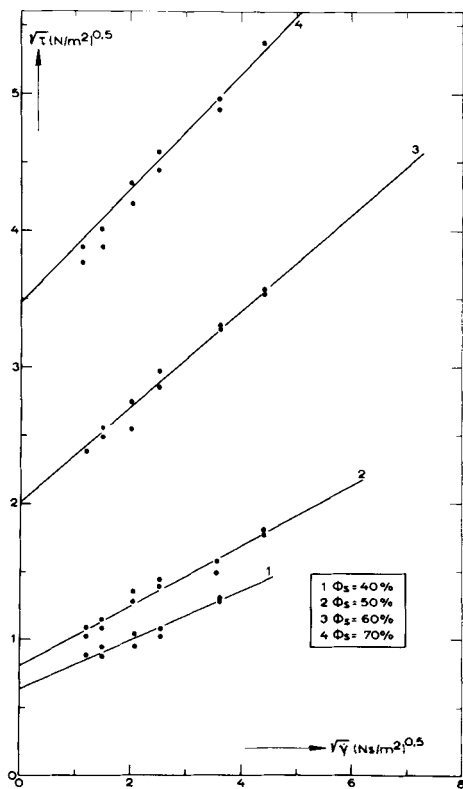


Fig. 33. A plot of $\sqrt{\tau}$ vs $\sqrt{\dot{\gamma}}$ according to the Casson model for pellet suspensions

Volume fraction of pellets. As could be expected it was not possible to correlate the rheological behaviour of pellet suspensions with biomass concentration. Although the theories discussed in Section 2.4 predicted a correlation between the viscosity of a pellet suspension with the volume fraction of pellets (ϕ_s), it was not possible to find such a distinct correlation. In Fig. 34 the results are given of some experiments with *Penicillium chrysogenum* and *Sporotrichum pulverulentum*. As can be seen, there is not a distinct relation between the apparent viscosity and the volume fraction of pellets for different suspensions.

The outer structure of the pellets seems to be of importance, since suspension A consisted of loose 'hairy' pellets of *Sporotrichum pulverulentum* (Fig. 35), while suspension B consists of very smooth dense pellets of the same organism (Fig. 36).

Experiments were carried out with spheres of polystyrene to study the effect of the volume fraction. These spheres were taken as a model for the 'ideal' smooth pellet and had about the same size and density as mycelial pellets ($d_m = 1.35$ mm, $\rho = 1030$ kgm⁻³). The results are given in Fig. 34, and show that the influence of the volume fraction alone can be neglected. So the conclusion can be drawn that the viscosity of pellet suspensions is mainly caused by the interaction of the particles and is dependent upon their outer structure. In future work, this structure needs to be defined quantitatively.

4.3 Flocculation Experiments

A set of experiments have been performed by Metz (1976) to investigate the flocculation behaviour of filamentous mycelial suspensions. The experiments were carried out in a flat plate column, constructed of two perspex plates, one centimeter from each other (Fig. 37).

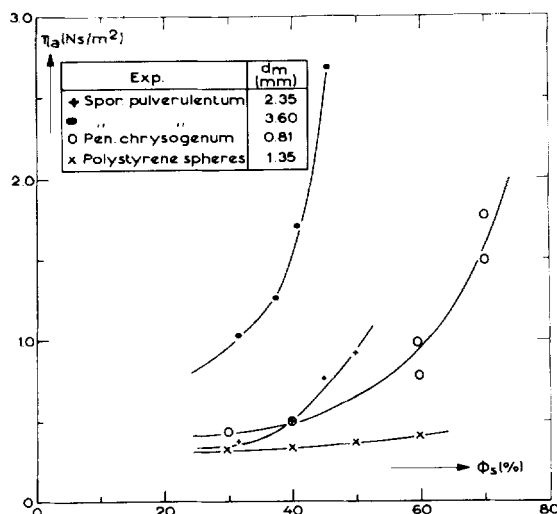


Fig. 34. Apparent viscosity as a function of the volume fraction of pellets (ϕ_s) for $N = 0.9$ 1/s. A (●): suspension of loose 'hairy' pellets of *Sporotrichum pulverulentum*, B (+): suspension of smooth dense pellets of *Sporotrichum pulverulentum*, C (○): suspension of pellets of *Penicillium chrysogenum*, D (×): suspension of polystyrene spheres

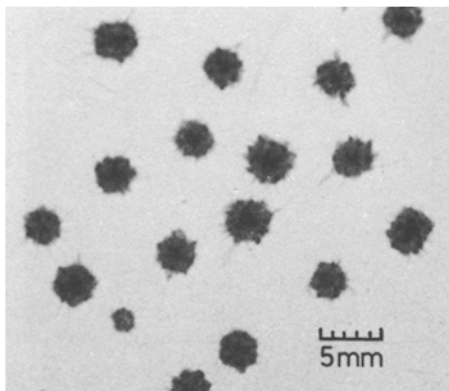


Fig. 35. Photograph of pellets of *Sporotrichum pulverulentum* (loose 'hairy' type)

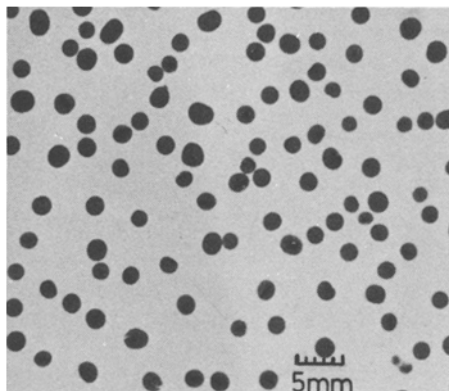


Fig. 36. Photograph of pellets of *Sporotrichum pulverulentum* (smooth dense type)

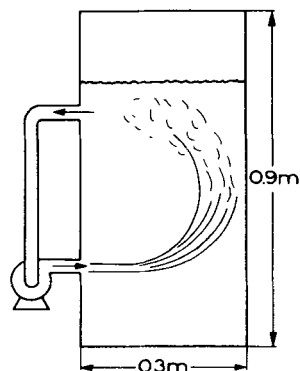


Fig. 37. Flat column for flocculation studies

A centrifugal pump (Stewart, 2500 r.p.m.) was connected to the column by means of openings along the side. Mould suspensions were pumped around thus simulating two dimensionally the situation in a stirred reactor. The distance between pump outlet and column inlet was very short and the suspension could thereby be studied almost immediately after the deflocculation in the pump had taken place by taking photographs of the flat plate column. Indirect lighting at the back of the column was used together with an electronic flash of $1/50000$ s to enhance contrast. A picture of a mould suspension being circulated through the column is given in Fig. 38.

As can be seen the floc size in the jet-stream coming from the centrifugal pump, where deflocculation takes place, is small near the entrance and increases with increasing residence time, until near the opposite wall the floc size has become nearly equal to that in the rest of the vessel. Calibration of the volumetric flow rate from the pump allowed an estimation of the time needed for complete flocculation. This appeared to be of the order of 0.5 s. This value was only slightly influenced when the dry weight concentration was varied from about 13 to 20 kgm^{-3} .

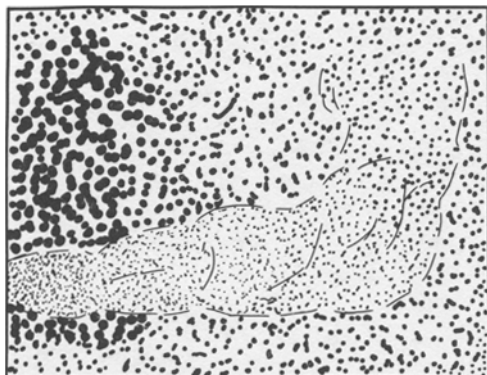


Fig. 38. Picture of a mould suspension in the flat plate column; flow from inlet (left) into column. Flocculation is nearly completed at opposite wall (right)

The conclusion can be drawn that the flocculation process in concentrated mycelial suspensions is very rapid and that the flocculation time is of the same order of magnitude as that for deflocculation. So a dynamic flocculation–deflocculation process does exist in mould suspensions as was assumed in the theoretical discussions. This is in agreement with the fact that no time dependent rheological behaviour has been found that can be attributed to structural effects. This conclusion is not valid for very dilute suspensions (dry weight $< 0.1 \text{ kgm}^{-3}$) where the flocculation time can be considerable ($\approx 5 \text{ min}$) due to the low collision frequency of particles.

5 Consequences for Reactor Design

As a typical example a reactor of 100 m^3 with one turbine impeller ($D_R = 1/3 T$) and a height (H) over tank diameter (T) ratio of 2:1, can be considered i.e.

$$\begin{aligned} T &= 4 \text{ m} \\ H &= 8 \text{ m} \\ D_R &= 1.33 \text{ m} \\ N &= 2 \text{ s}^{-1}. \end{aligned}$$

The Casson equation in combination with equations (76) and (77) for a typical *Penicillium* broth) $L_e^* = 55$; $L_{\text{hgu}} = 80 \text{ }\mu\text{m}$; $C_M = 30 \text{ kgm}^{-3}$) leads to:

$$\sqrt{\tau} = 4.51 + 0.57 \sqrt{\dot{\gamma}}. \quad (78)$$

The value of the viscosity near the impeller can be found by calculating the viscosity at infinite shear rate η_∞ by means of Eq. (72):

$$\eta_\infty = K_c^2 = 0.32 \text{ Nsm}^{-2} (= 320 \text{ cP}).$$

The Reynolds number is:

$$Re = \frac{\rho \cdot ND_R^2}{\eta} = 1.1 \times 10^4.$$

With these values in mind some observations can be made about mixing time and mass transfer in the reactor

5.1 Mixing Time

Methods to calculate the mixing time for stirred vessels with pseudoplastic fluids are rather scarce. One of the older theories is that due to Norwood and Metzner (1960). The theory is commented upon by Skelland (1967) and Blanch and Bhavaraju (1976). With this theory a typical mixing time of 16 s has been calculated. This is much lower than observed in reactors of similar size with high viscosities, where mixing times are of the order of several minutes (see Bylinkina et al. (1973)). For a Newtonian low viscosity fluid a mixing time of 85 s can be expected (Beek and Muttzall (1975)). In fact mixing times ten to fifty times larger than those calculated by the Norwood and Metzner method have been found by Godleski and Smith (1962) for pseudoplastic fluids.

The conclusion has to be that, although we now have the correlations to calculate viscosity as a function of mycelial properties, an adequate theory to relate viscosity to mixing in stirred vessels is lacking. Bryant (1977) comes to the same conclusion. Further work would be required to achieve this objective, perhaps with particular emphasis on local velocities near cooling coils and on the existence of secondary flow regions far away from the impeller.

5.2 Mass Transfer

Because little is known about the flow of pseudoplastic fluids in large vessels and because flow has a tremendous influence on transport phenomena (see Fig. 3) no mechanistic theories are available to give the relation between mass transfer and flow conditions for viscous fluids. However, the following method of calculation provides at least the possibility for an 'order of magnitude estimate' of the mass transfer.

In a mycelial broth with a viscosity of 320 cP the rate of coalescence of gas bubbles is very high e.g. in a flat plate column, starting with bubbles of 5 mm, this size has increased to approximately 5 cm after the bubbles have travelled over a distance of less than 1 m. This means that mass transfer in industrial reactor with highly viscous fluids occurs mainly in the impeller region. If one assumes that all the mass transfer occurs in the impeller region and that the liquid/air mixture leaving the impeller zone is in equilibrium, the mass transfer can easily be calculated (see Fig. 39).

Under steady state conditions:

$$\phi_p (C^* - C_1) = r_{O_2} V = k_1 a V (C^* - C_1) \quad (79)$$

or

$$k_1 a = \frac{\phi_p}{V} \quad (80)$$

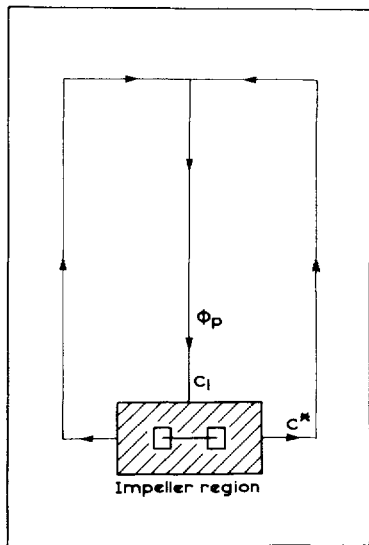


Fig. 39. Simplified model for the mass transfer in a reactor with a highly viscous broth

where ϕ_p = pumping capacity of the impeller ($\text{m}^3 \text{s}^{-1}$)
 C^* = oxygen concentration by saturation with air (kgm^{-3})
 C_l = oxygen concentration in the liquid phase (kgm^{-3})
 r_{o_2} = the oxygen consumption rate ($\text{kgm}^{-3} \text{s}^{-1}$)
 V = volume of the vessel (m^3)
 $k_1 a$ = mass transfer coefficient (ls^{-1}).

The pumping capacity ϕ_p is given by:

$$\phi_p = C_3 N D^3 \text{ with } 0.6 < C_3 < 1.3 \text{ (Re} > 10^4 \text{)} \quad (81)$$

Thus

$$k_1 a \cong \frac{N D^3}{V} \quad (82)$$

For the 100 m^3 vessel the $k_1 a$ would be $\cong 4.7 \times 10^{-2} \text{ s}^{-1}$. This equation gives the right order of magnitude of $k_1 a$ for large reactors with viscous fluids. However, it is not sufficiently accurate for design purposes.

Since $\frac{\phi_p}{V}$ is the reciprocal of the circulation time (t_c) so:

$$k_1 a = \frac{1}{t_c} \quad (83)$$

This shows the close connection between mass transfer and flow.

A final point of interest is the explanation for the shift of oxygen uptake rate vs dissolved oxygen curves with increasing stirrer speed (Fig. 40).

Results of this kind have been given by Wang and Fewkes (1977). Two explanations are possible:

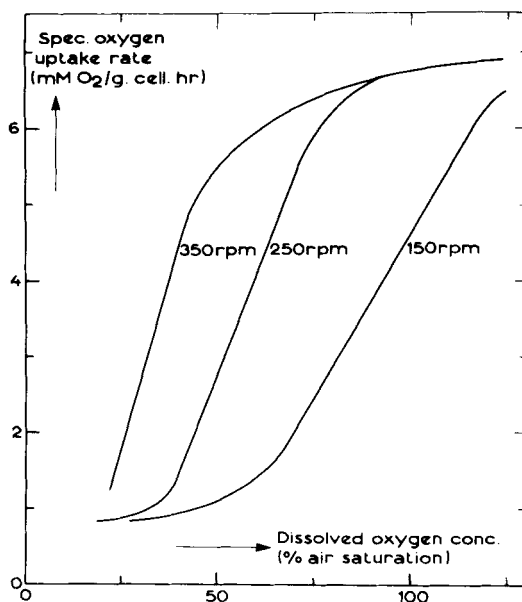


Fig. 40. Effect of impeller speed on dissolved oxygen and specific oxygen uptake rate of *Streptomyces niveus*

- oxygen transfer is limited by diffusion into flocs. An increase of N results in smaller flocs and so in an increased oxygen uptake rate.
- The reactor contains secondary flow regions that exchange slowly with the bulk of the reactor contents. An increase in N decreases the size of these regions and increases the exchange.

As shown in Section 4.3 the break up and formation of flocs takes place within a second. Therefore the flocs are homogeneously saturated in the impeller region, where the main part of the mass transfer from air to fluid takes place. Furthermore the flocs are so close to each other (at $C_M = 30 \text{ kgm}^{-3}$) that practically no free liquid is left between the flocs. This leads to the conclusion that floc size is not important for mass transfer: all flocs are homogeneously saturated with oxygen near the impeller, and this oxygen is homogeneously consumed in the rest of the reactor. If pellets are present to a large extent the situation is different of course. Thus it can be concluded that the shift of the curves in Fig. 40 is caused by imperfect mixing (if there are no pellets present).

This conclusion is confirmed by the fact that in a highly viscous broth the mixing time is several minutes and the oxygen consumption time (without further supply) is of the order of 10–20 s. How organisms behave in a situation where they regularly spend minutes in regions with oxygen depletion is a subject that needs thorough research. The efficiency of biomass synthesis will certainly decrease, due to a lower yield (Stouthamer), but much needs to be known.

The final conclusion has to be that mixing is an extremely important phenomenon in high viscosity reactions and that much has to be learned about its causes and effects.

6 Nomenclature

a, b, c ...	parameters related to particle properties [Eq. (33)]	—
A	constant [Eq. (64)]	Nms^n
c	constant [Eq. (45)]	—
C_1	constant [Eq. (8)]	$mN^{-1/2}$
C_2	concentration factor	—
C_3	constant [Eq. (81)]	—
C^*	oxygen concentration by saturation with air	kgm^{-3}
C_l	oxygen concentration in the liquid phase	kgm^{-3}
C_M	biomass concentration	kgm^{-3}
d_m	mean particle diameter	m
d_p	particle diameter	m
d_w	wire diameter	m
D_R	impeller diameter	m
E_{cr}	creep energy dissipation rate	W
E_n	network energy dissipation rate	W
E_{tot}	total energy dissipation rate	W
E_v	viscous flow energy dissipation rate	W
f	collision frequency	1/s
H	tank height	m
k	constant [Eq. (49)]	—
k_O	constant [Eq. (23)]	m^3
k_{1a}	mass transfer coefficient	ls^{-1}
K	consistency index	Ns^nm^{-2}
K_C	'Casson' viscosity	$(Nsm^{-2})^{1/2}$
L	length	m
L_e	effective length	m
L_g^*	dimensionless length	—
L_{hgu}	length of the hyphal growth unit	m
L_t	total length	m
M	torque	Nm
n	number of branches of the main hypha	—
n	power law index	—
N	impeller speed	ls^{-1}
N_A	number of aggregates per volume	lm^{-3}
P	power consumption	W
P_O	power number	—
r	impeller radius	m
r_{O_2}	oxygen consumption rate	$kgm^{-3}s^{-1}$
Re	Reynolds number	—
s_e^2	error mean square in regression analysis	—
s_r^2	residual mean square in regression analysis	—
s_y^2	estimated variance of the mean of a sample	—
s_3^2	estimated variance between the mean of a sample and the regression line	—
t_c	circulation time	s
T	tank diameter	m
T_c	torsion constant	Nm^{-2}
u	fluid velocity	ms^{-1}
v	tangential velocity	ms^{-1}
V	vessel volume	m^3
V_A	volume of aggregate	m^3
W	work required to break a bond	J
$\alpha, \beta, \gamma, \delta, \epsilon$	constants [Eq. (73)]	—

α	angle of rotation	rad
$\dot{\gamma}$	shear rate	ls^{-1}
$\dot{\gamma}_{\text{av}}$	average shear rate	ls^{-1}
δ	morphology factor [Eq. (5)]	$\text{m}^6 \text{kg}^{-2}$
δ_1	morphology factor [Eq. (7)]	$\text{Nm}^{5.5} \text{kg}^{-2.5}$
η	viscosity	Nsm^{-2}
η_0	viscosity of suspending medium	Nsm^{-2}
η_∞	viscosity at infinite shear rate	Nsm^{-2}
η_a	apparent viscosity	Nsm^{-2}
η_r	relative viscosity	—
η_s	viscosity of the suspension	Nsm^{-2}
ν	degrees of freedom	—
ρ	density of fluid	kgm^{-3}
τ	shear stress	Nm^{-2}
τ_0	yield stress	Nm^{-2}
τ_b	Bingham yield stress	Nm^{-2}
τ_{cr}	creep shear stress	Nm^{-2}
τ_n	network yield stress	Nm^{-2}
τ_v	viscous shear stress	Nm^{-2}
ϕ_A	volume fraction aggregates	—
ϕ_c	volume fraction of mycelial particles as spheres [Eq. (5)]	—
ϕ_{max}	maximum packing density	—
ϕ_p	pumping capacity of the impeller	$\text{m}^3 \text{s}^{-1}$
ϕ_s	volume fraction particles, solids, pellets	—

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Scale-up of Surface Aerators for Waste Water Treatment

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Laboratory-scale experiments using flat-blade turbines of different sizes (scale 1:3) working on the surface of the liquid have shown that the sorption number Y and the power number Ne are only dependent on the Froude number Fr . Using this information, the sorption correlation $Y(Fr)$ and the power correlation $Ne(Fr)$ for different types of surface aerators were measured on the laboratory scale. Some additional geometric parameters, namely the form and number of blades, the degree of blade enclosure and the ratio of liquid height/aerator diameter were also varied. The validity of these scale-up correlations has been proved by measurements on one large surface aerator of 4.3 m diameter, this being exactly 10 times larger than its geometrically similar model used in laboratory tests.

The conclusion that the sorption number Y and the power number Ne are functions of the Froude number Fr implies that the dimensionless number for the efficiency E [kg O_2 /kWh], which is derived from these three numbers, is also a function of Fr . From the definition of this number it is apparent that the efficiency of surface aerators is inversely proportional to the square root of the aerator diameter, i.e. the efficiency decreases as the aerator size increases. This implies that only relatively small surface aerators with diameters of approx. 1 m can be recommended.

The paper ends with a critical comparison of the findings of the recently published pertinent literature of the years 1977/78 with the conclusions drawn from this work.

Introduction

Surface aerators nowadays represent a commonly used device for the intensification of oxygen input into biological waste water treatment plants. It is, therefore, all the more amazing that there are still no reliable correlations for the scale-up and the design of these devices that would also allow an objective comparison between them and other types of apparatus (e.g. roll aerators, submerged stirrers with gas supply from below, perforated pipes, porous plates, nozzles, injectors, etc.)

In this paper, scale-up criteria for a range of different types of surface aerators will be presented.

Problem Formulation Using the Theory of Similarity

Mass transfer in gas-liquid systems is given, according to the two-film theory, by the equation

$$G = k_L \cdot A \cdot \Delta c \quad (1)$$

where

G is the mass transfer rate through the gas-liquid interface,

k_L is the liquid-phase mass transfer coefficient,

A is the interfacial area between gas and liquid and

Δc is the concentration difference between the saturation concentration c_s in the interface and the concentration c of oxygen dissolved in the bulk of the liquid:

$$\Delta c = c_s - c.$$

The quantities k_L and A can be measured separately only with difficulty. Besides this, in a general consideration of the absorption process the main parameter is the absorption rate coefficient $k_L A$, which is defined by Eq. (1) as

$$k_L A \equiv G/\Delta c \quad (2)$$

Therefore the ratio $G/\Delta c$ is the main object of our investigations. The term $G/\Delta c$ depends upon many different parameters which, for the purpose of clarity, may be divided into three classes: geometric, material and process related parameters.

With regard to *geometric* parameters, it must be remembered that for a given surface aerator arranged under given conditions in a vessel or a pool, only the diameter d of the surface aerator (as the characteristic length) is required to transform all the other geometric parameters into (dimensionless) geometric ratios.

The relevant *material* parameters are density ρ , kinematic viscosity ν and surface tension σ of the liquid, whilst the material parameters of the gas—with the sole exception of diffusivity D of the gas in the liquid—may be neglected.

At first glance the only relevant *process* parameter appears to be the rotational speed n of the aerator. However, as the undulating flow patterns on the liquid surface are highly dependent on the gravitational constant g (e.g. under the same process

conditions the liquid wave formation on the moon would be very different to that on the earth), it must also be considered an important process parameter.

Thus, for the *main* parameter $G/\Delta c$ the following functional dependency results:

$$G/\Delta c = f_1(d; \rho, \nu, \sigma, D; n, g) \quad (3)$$

Using dimensional analysis, this dependency of eight parameters reduces to one of only five dimensionless numbers:

$$\frac{G}{\Delta c \cdot d^3} \left(\frac{\nu}{g^2} \right)^{1/3} = f_2 \left\{ \frac{n^2 d}{g}, \frac{nd^2}{\nu}, \frac{\nu}{D}, \frac{\sigma}{\rho(\nu^4 \cdot g)^{1/3}} \right\}. \quad (4)$$

To simplify this, the following symbols will be introduced:

$$\begin{aligned} Y &\equiv \frac{G}{\Delta c \cdot d^3} \left(\frac{\nu}{g^2} \right)^{1/3} && \text{Sorption number}^a, \\ Fr &\equiv n^2 \cdot d/g && \text{Froude number,} \\ Re &\equiv nd^2/\nu && \text{Reynolds number,} \\ Sc &\equiv \nu/D && \text{Schmidt number,} \\ \sigma^* &\equiv \sigma/[\rho(\nu^4 \cdot g)^{1/3}] && \text{material number.} \end{aligned}$$

Hence

$$Y = f_2(Fr, Re, Sc, \sigma^*). \quad (4 a)$$

Any information concerning the form of the function f_2 can be given only by experiment

Since the main application of these stirrers is the aeration of waste water, the laboratory tests will be performed on a practically identical system (pure water-air). Thus the numerical values of the material numbers Sc and σ^* remain constant.

The analogy with gas-liquid contacting using submerged stirrers¹⁴⁾, would lead one to expect that in the turbulent flow region ($Re > 10^4$) the Reynolds number is irrelevant and hence the Eq. (4 a) reduces to:

$$Y = f_3(Fr) ; \quad Sc, \sigma^* = \text{idem} \quad (5)$$

The function f_3 is the expected scale-up correlation for the mass transfer rate achieved by a surface aerator and will be called the *sorption correlation*.

The power consumption P of a given surface aerator is, in general, dependent upon the same parameters as the term $G/\Delta c$ (except for the diffusivity D). Therefore, by way of dimensional analysis we obtain the relationship

$$Ne = f_4(Fr, Re, \sigma^*) \quad (6)$$

^a Due to the basic equality between *absorption* and *desorption* processes, the term "*sorption*" will be used in the following text.

where

$$Ne \equiv P/(\rho \cdot n^3 \cdot d^3)$$

is the power or Newton number.

The analogy with respect to power consumption of a stirrer submerged in a liquid and supplied with gas from below¹⁶⁾ leads to the assumption that, in this case, the Reynolds number and the material number are irrelevant in the turbulent flow region and thus the *power correlation* also reduces to

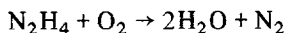
$$Ne = f_5(Fr). \quad (7)$$

Experimental Techniques and Verification of the Assumed Scale-up Criteria for Surface Aerators

The laboratory experiments were performed on models of different types of surface aerators. Their diameters ranged between 200 and 400 mm and, therefore, between 1/10 and 1/20 of the full-scale. The aerators were installed concentrically in open shallow cylindrical vessels (tubs), their diameters D being 4 to 6 times the aerator diameter and the liquid height H usually equal to the aerator diameter. To avoid the rotation of the liquid and hence vortex formation, the vessels were fitted with four baffles (width $D/10$) attached to the wall.

The power consumption was measured by a mechanical method involving the rotational speed and the torque of the motor. The accuracy obtained was $\pm 5\%$.

The mass transfer rate G was measured in the water-air system at $20 \pm 1^\circ \text{C}$ under steady-state conditions using aqueous solutions of hydrazine to remove the dissolved oxygen. The oxidation of hydrazine according to the over-all equation



at $\text{pH} = 12$ and catalysed by $0.01 \text{ M CuSO}_4/1$, proceeds relatively quickly at room temperature and leaves no reaction products that would change the chemical or physical properties of the system. Comparative measurements under non-steady-state conditions (i.e. absorption of O_2 from air followed by its desorption by nitrogen) gave exactly the same results. The concentration of dissolved oxygen was measured in both cases by an oxygen membrane electrode and was recorded on a pen recorder (For further practical details see¹⁸⁾). The accuracy of these measurements was approx. $\pm 10\%$.

To verify the expected scale-up criteria, Eq. (5) and Eq. (7), it is necessary to carry out experiments on at least two geometrically similar but differently sized sets of equipment. The rather complicated construction and hence high cost of surface aerator models proved to be prohibitive and, therefore, the verification experiments were performed with three flat-blade turbines (6 blades, blade height $d/5$, blade width $d/4$) of diameters $d = 90$, 180 and 270 mm. The stirrers were so positioned in the liquid that the supporting discs of

the blades rested on the surface and the lower sections of the blades were submerged, thus enabling the stirrer to act as a surface aerator (see Fig. 1). The three steel vessels (tubs) used had diameters of $D = 5d$ and liquid heights of $H = d$.

For these geometrical conditions Fig. 1 shows the power correlation $Ne(Fr)$ and Fig. 2 the sorption correlation $Y(Fr)$. These results prove conclusively that the expected scale-up criteria obtained by assuming the irrelevance of the Reynolds number and the dominant role of the Froude number are indeed correct.

The sorption correlation was measured for values of $D/d = 3.3 - 9.0$ and the results in Fig. 2 show that this geometric number does not influence the mass transfer within the measured range.

Fig. 1. Power correlation $Ne(Fr)$ or a flat-blade turbine positioned at the liquid surface

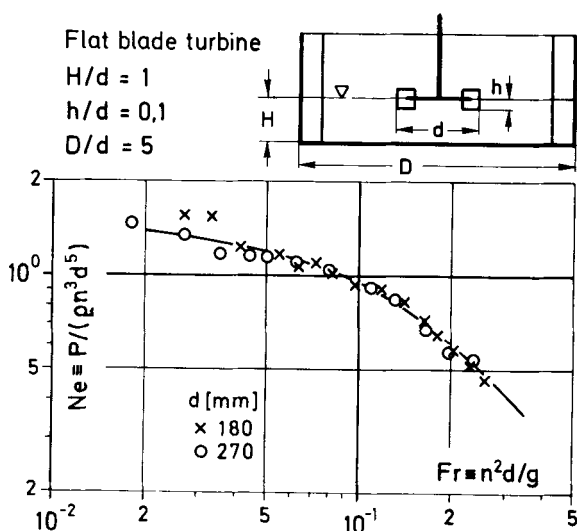
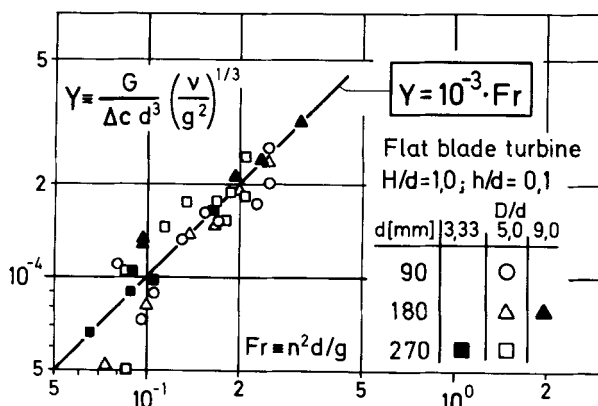


Fig. 2. Sorption correlation $Y(Fr)$ for a flat-blade turbine positioned at the liquid surface



Power and Sorption Correlations for Different Types of Surface Aerators

After the verification of the assumed scale-up criteria with flat-blade turbines, the power and sorption correlations for different types of surface aerators can be evaluated using only one model of each type of surface aerator. The seven models of surface aerators examined in this work were more or less similar to those employed industrially. They differed from each other with respect to the *shape* of their blades (bent, slightly curved or straight but radially inclined), the degree of *blade enclosure* (fully enclosed top and bottom, partially or completely opened below) and the *number* z of blades (between 8 and 36). Table 1 and Fig. 3 review their typical design. In all experiments the stirrer was positioned so that it was just submerged in the liquid.

Despite the relatively great differences in shape between the types A and B, their power correlations (Fig. 4) and the sorption correlations (Fig. 5) are surprisingly similar. The common feature in both models is that their blades are bent or slightly curved. The curving of the blades seems to have a greater influence on the aerator performance than all the other geometric parameters. This results in a high dependancy on the direction of rotation: when the stirrer is "pushing" the liquid, this doubles its power consumption and also enhances the absorption rate compared with "dragging". (The deviation from the expected values of $Ne(Fr)$ in the range of $Fr = 0.1 - 0.2$ in the case of aerator type B for dragging is probably due to resonance between the stirrer and the liquid waves which increases both power and mass transfer.)

Figures 6 and 7 show the power and the sorption correlations for the aerator type C, proving that the number of blades has a minor effect: Tripling z from 12 to 36 increases the power by only 30–50% and enhances the absorption rate by approximately the same amount but only for $Fr > 0.1$. This type of aerator has straight, radially inclined blades, which results in the surprising discovery that there is no difference between "pushing" and "dragging".

Figures 8 and 9 represent the power and the sorption correlations for the aerator type D. Since the blades are also only inclined, no influence is found by changing the direction of rotation. Whereas the change in H/d does not influence the power correlation, it has a relatively large effect on the sorption correlation (especially at $Fr < 0.1$).

Table 1. Characteristics of the models used

Type	$d[\text{mm}]$	z	Blade arrangement
A	206	24	Blades bent at the outer end, fully enclosed top and bottom
B	245	8	Blades slightly curved along their whole length, fully enclosed top and bottom
C 1	215	12	Blades straight but inclined at 30° to the radial; partially opened below (degree of blade enclosure 0.74, in C 3: 0.88)
C 2	215	36	
C 3	430	36 + 36	
D 1	285	8	Blades straight but inclined at 6° to the radial; completely opened below
D 2	506	8	

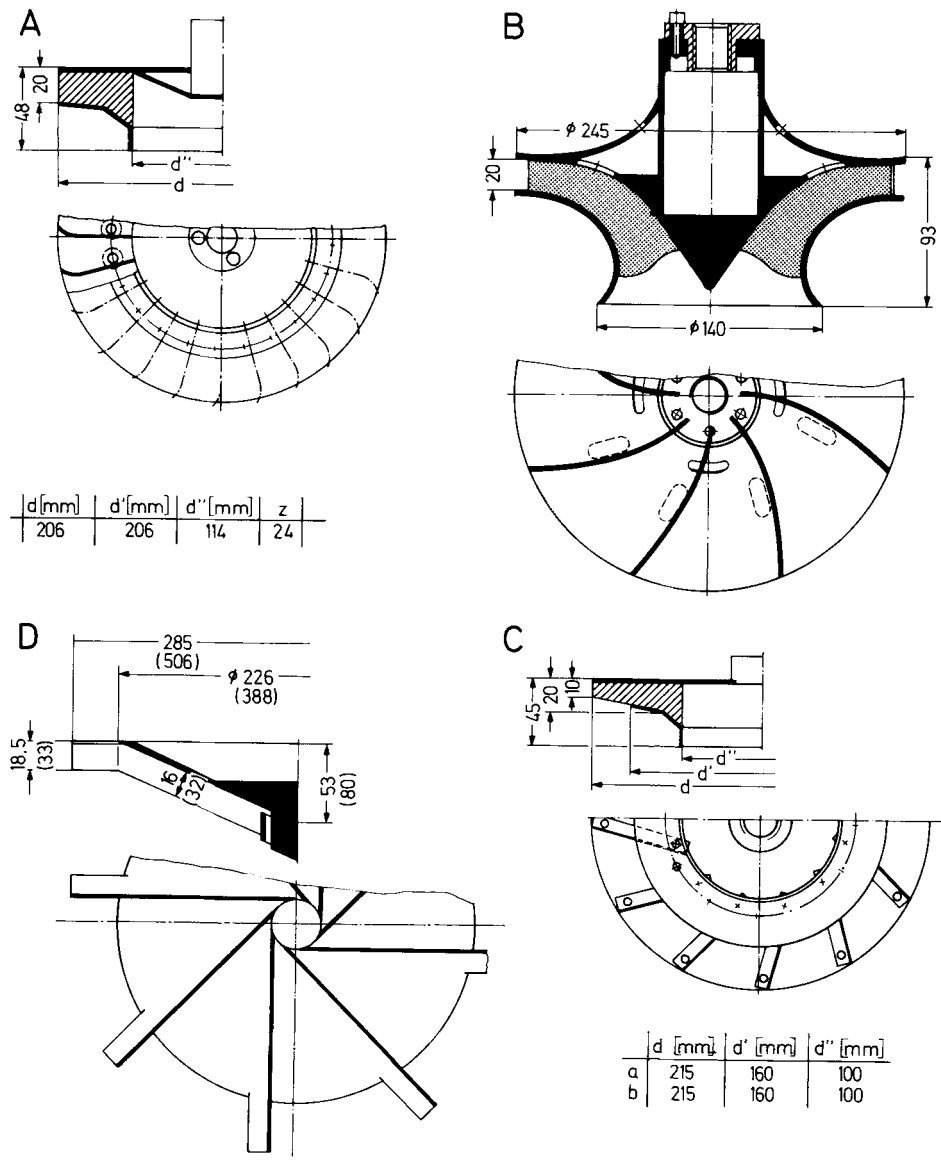


Fig. 3. Schematic diagrams of aerator types used

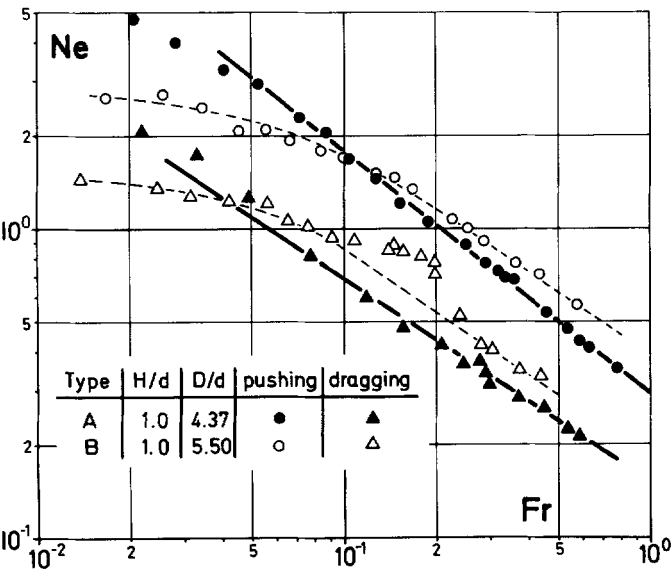


Fig. 4. Power correlations $Ne(Fr)$ for surface aerator types A and B

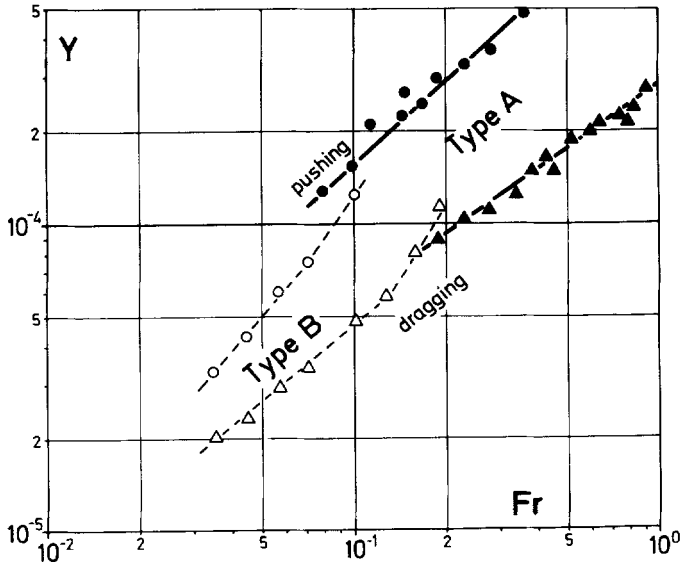


Fig. 5. Suction correlations $Y(Fr)$ for surface aerator types A and B

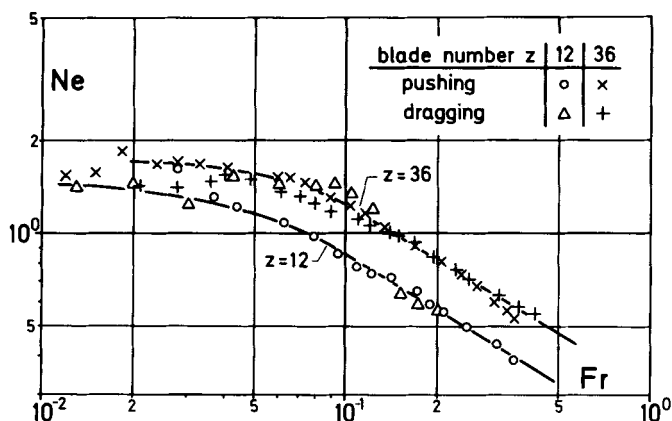


Fig. 6. Power correlations $Ne(Fr)$ for surface aerators type C ($H/d = 1.0$; $D/d = 4.2$; $d'/d = 0.74$)

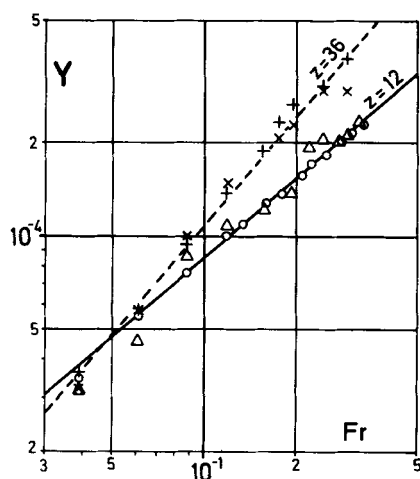


Fig. 7. Sorption correlations $Y(Fr)$ for surface aerator type C. (Legend and geometrical conditions see Fig. 6)

The explanation for this is that the liquid circulation produced by the aerator in deeper basins is less affected by the base and walls of the basin than in shallow ones. Improved circulation enhances the turbulence on the liquid surface, thus increasing the interfacial area.

In order to compare the aerator types C 1 and D 1 (which were measured at approx. equal H/d values), the data for C 1 from Fig. 7 is represented in Fig. 9 by a dotted line. The agreement between the stirrers is excellent, although there is little similarity in their shapes (also see $Y(Fr)$ for types A and B in Fig. 5). To verify the relevance of the ratio H/d , additional measurements were made with aerator type C 1 at $H/d = 0.63$. The full circles in Fig. 9 confirm this.

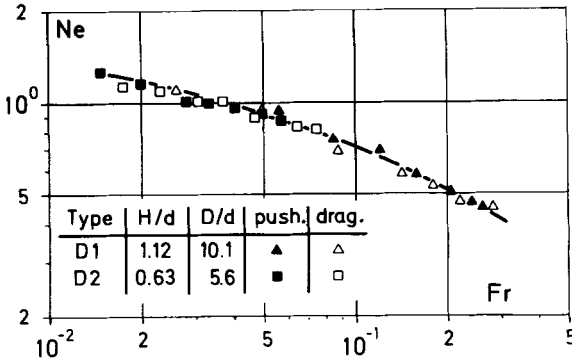


Fig. 8. Power correlations $Ne (Fr)$ for surface aerators type D

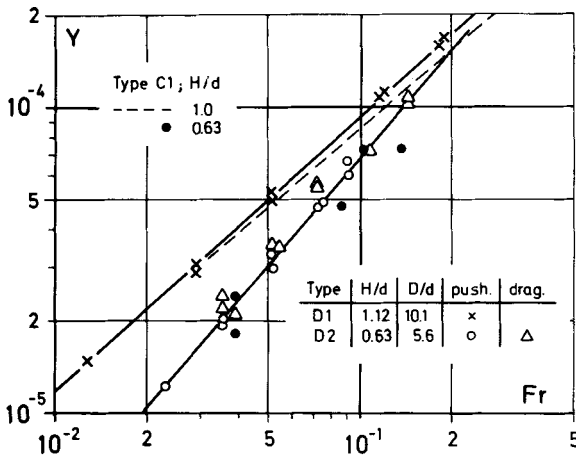


Fig. 9. Sorption correlations $Y (Fr)$ for surface aerators type D and their comparison with those for type C1 under equal geometric conditions

Comparative Measurements of Power and Absorption Rate of an Industrial Scale Surface Aerator ($d = 4.3$ m) and its Geometrically Similar Laboratory Scale Model ($d = 0.43$ m)

It must be stressed that the described experiments with flat-blade turbines and differently shaped surface aerator types were all obtained using laboratory scale models which were approx. 1/10 of the usual industrial scale.

To ensure the validity of the proposed criteria, additional measurements were performed on a full-scale surface aerator ($d = 4.3$ m), which is currently operating in the biological waste water treatment plant of Bayer AG in Leverkusen-Bürrig. None of the previously tested surface aerator models were exactly geometrically similar to this particular full-scale aerator and, therefore, it was necessary to construct a 1/10 full-scale model.

This aerator model is already included in Table 1 as Type C 3. It was similar to C 1 and C 2 but was different in two respects: firstly, a higher degree of blade enclosure ($d'/d = 0.88$) and secondly, the number of blades was double that of C 2, the additional ones being positioned between the existing blades and having a rectangular shape (height $0.028 d$ and length $0.046 d$).

The full-scale aerator chosen for measurements was one of 8 aerators which are arranged in the rectangular waste water treatment pool in two rows. Each aerator is $4.3 m$ in diameter, the distance between their shafts is $4.65 d$ and between the shaft and the wall $2.9 d$. The liquid height is $3.0 m$ (Installation conditions: $H/d = 0.70$; $D/d = 1.8 - 2.4$.) The rotational speed was $n = 21 \text{ min}^{-1}$ and therefore $Fr = 0.054$.

The power measurements were performed when none of the other aerators in the pool were in operation. It was measured by an electrical method and calculated using the formula:

$$\begin{aligned}
 P[W] &= \sqrt{3} \cdot U[V] \cdot I[A] \cdot \cos \varphi \cdot \eta_{\text{Motor}} \cdot \eta_{\text{Gear}} = \\
 &= 1.73 \cdot 518 \cdot 88 \cdot 0.82 \cdot 0.93 \cdot 0.93 = \\
 &= 55900
 \end{aligned}$$

The power number gives a value of $Ne = 0.89$.

Additional measurements were made to ascertain the influence of the degree of submergence of the blades on the power consumption. The water level in the pool was raised or lowered, so that the blades were either completely in the water or partly out of it. The parameter h was defined as the height of the blades taken at the tip, that were under the water. h was varied between 0.155 and $0.280 m$. The correlation obtained is $Ne (d/h)^{0.4} = 3.05$.

The absorption rate G in the plant with all aerators in operation was determined from the oxygen demand of the activated sludge. Under steady-state conditions the oxygen input is equal to the oxygen demand of the activated sludge and this results

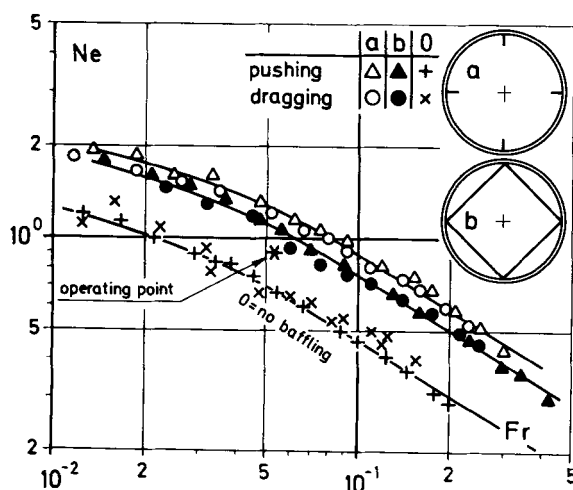


Fig. 10. Power correlations $Ne (Fr)$ for the surface aerator type C 3 under different baffling conditions. Operating point of full-scale aerator: $Fr = 0.054$; $Ne = 0.89$

in a constant level of dissolved oxygen (DO) in the aeration pool. The oxygen demand can be estimated from the respiration curve of the bacteria in the sludge. The respiration measurements gave a value for G of between 54 and 60 kg O₂/h for each aerator at a Δc level of between 4.5 and 6.5 mgO₂/l. This results in a sorption number of Y between 6×10^{-5} and 1×10^{-4} , see the range indicated in Fig. 11.

The respiration measurements were performed by taking liquid samples from different locations and immediately saturating them with oxygen. The oxygen consumption by the activated sludge was then measured by an oxygen membrane electrode.

To obtain Δc values, the temperature of the liquid (and hence saturation concentration) and concentration of dissolved oxygen (DO concentration) were measured on the liquid surface. Although both measurements are accurate, their results do not necessarily represent the actual conditions under which the absorption occurs.

It has to be remembered that surface aerators are relatively inefficient liquid agitators and that the mixing time (which is an expression for surface renewal and back-mixing) increases with the square of the tank diameter¹⁵. From this, it is obvious that the bulk of the water in large basins is not, as a rule, ideally mixed and that, therefore, the temperature and the DO value measured at the surface do not represent the mean values. However, the consumption of dissolved oxygen by bacteria takes place throughout the whole volume. Therefore the equilibrium between the O₂ consumption and O₂ supply occurs under different conditions. We can expect that the mean values of Δc have to be greater than those measured at the surface (lower temperature at the bottom of the pool resulting in a higher c_s value and a lower DO concentration); with $\Delta c \approx 8.0$ mg/l and $G = 54$ kgO₂/h the sorption number is $Y \approx 5 \cdot 10^{-5}$, see the "operating point" in Fig. 11.

Laboratory measurements with the model (C3) of the full-scale aerator were performed in a circular pool with $D = 2.86$ m ($D/d = 6.65$ and $H/d = 0.7$). It cannot be expected that the kind of baffling that was used up to this point (4 vertical baffles of width $D/10$) results in the same liquid pattern as produced in the rectangular pool

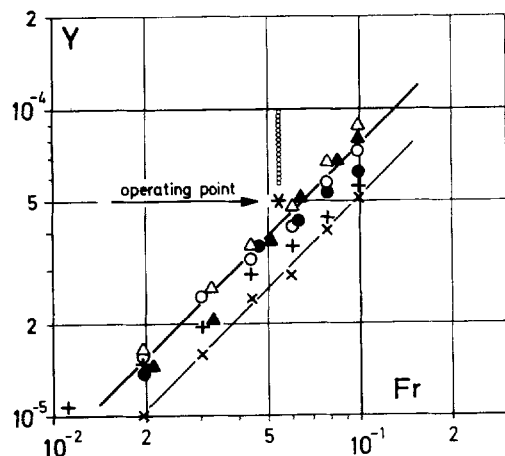


Fig. 11. Sorption correlations $Y(Fr)$ for the surface aerator type C 3 under different baffling conditions. For the explanation of the measured range and the defined "operating point" of the full-scale aerator see the text (Legend as in Fig. 10)

with 8 aerators arranged in two rows. For this reason, these measurements were carried out under different baffling conditions: without baffles; with a square frame which transformed the circular pool to a square one; 4 vertical baffles of width $D/10$ as had been used before.

Figure 10 shows the power correlation $Ne (Fr)$ for C 3 under these three different baffling conditions. We see that this has a considerable influence on the power consumption of the aerator. In Fig. 10 the operating point of the full-scale aerator is also plotted. Assuming that the square frame best represents the actual baffling in the rectangular pool, the deviation of the corresponding correlation is -20% .

A comparison of $Ne (Fr)$ between C 2 (Fig. 6) and C 3 (Fig. 10) shows that C 3 uses 35% less power under identical operating conditions. Provided that the difference in the number of blades (36 as compared to $36 + 36$ very small blades) has no influence on this result, this finding proves the large effect of the degree of blade enclosure (C 2 : 0.74; C 3 : 0.88) on the power.

Figure 11 shows the sorption correlation for C 3 under different baffling conditions. It is evident that this has a smaller effect on the mass transfer compared with that on the power consumption (practically no difference between the two types of baffles). It is surprising that the operating point of the full-scale aerator lies approx. 20% above the laboratory results. The only possible explanation for this is that the Δc value taken ($\Delta c \approx 8 \text{ mg/l}$) is still too low (see ¹³).

When one considers that the scale factor between both devices is 10, the deviation in results of $\pm 20\%$ is generally acceptable and the proposed scale-up criteria are proved to be valid.

A Comparison Between Different Types of Surface Aerators and the Determination of the Optimum Operating Conditions

The previously discussed power and sorption correlations for different surface aerator types permit an exact design of a full-scale aerator of the same type for a given oxygen uptake G . They do not allow, in this form, an objective comparison between different aerator types or a determination of optimum operation conditions. For this purpose it was necessary to formulate a new dimensionless number for the aerator efficiency $E \equiv G/P [\text{kg O}_2/\text{kWh}]$. For the dimensionless formulation of this term, the numbers Y , Ne and Fr were combined in the following way:

$$E^* \equiv Y \cdot Ne^{-1} \cdot Fr^{-3/2} = \frac{G \cdot d^{0.5}}{P \cdot \Delta c} \cdot \rho \cdot (\nu^2 g^5)^{1/6} \quad (8)$$

Since all three numbers are functions of the Froude number, the efficiency number E^* must also be a function of this number.

In Fig. 12 E^* is plotted against Fr for all surface aerator types tested at $H/d = 1$. It can be seen that the influence of the Froude number on E^* is not pronounced. This shows that there are no significant optimum operating conditions for surface aerators.

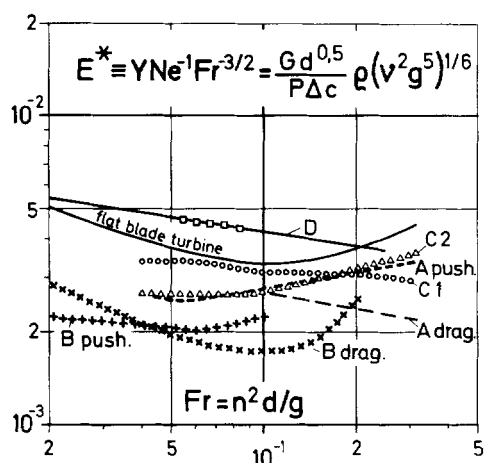


Fig. 12. Comparison of aerator efficiencies between the types tested ($H/d = 1.0$)

The previous comparisons of power and sorption correlations for different surface aerator types showed that the difference between them was less than could be expected from their shapes. The presentation in Fig. 12 confirms this. The difference in efficiencies between the tested types is relatively small and does not exceed 50%. It appears that the aerators with blades open from below are more effective than those with blades fully enclosed top and bottom.

To estimate the efficiency $E \equiv G/P$ the relationship $E^*(Fr)$ has to be expressed in a dimensional form valid for water only [$\rho(\nu^2 \cdot g^5)^{1/6} = 67.05$]. Then we have to assume values for E^* and Δc . We will consider $E^* = 4 \cdot 10^{-3}$ (which is a generous value according to our findings, see Fig. 12) and an optimistic value for the concentration difference of $\Delta c = 8$ ppm (e.g. water temperature 20 °C and thus $c_s = 8.8$ ppm, DO conc. 0.8 ppm). With these values we can write

$$E [\text{kg O}_2/\text{kWh}] = 1.72/\sqrt{d} \quad (9)$$

From this we obtain:

Table 2

d [m]	E [kg O ₂ /kWh]
0.5	2.4
1	1.7
2	1.2
3	1.0
4	0.85

To obtain the efficiencies under so-called standard conditions, the E values in Table 2 have to be multiplied by the factors

$$1.14 \text{ (20 } ^\circ\text{C, } c = 0 \text{ mg O}_2\text{/l) and}$$

$$1.41 \text{ (10 } ^\circ\text{C, } c = 0 \text{ mg O}_2\text{/l)}$$

It is apparent that the efficiency decreases with the square root of the aerator diameter. This is by far the most important finding of this work. However, this conclusion can be directly deduced from the theory of similarity, assuming that the power and sorption correlations are only functions of the Froude number (see the structure of E^* , Eq. (8)). This result proves the power of the theory of similarity and shows the limitations of surface aerators. It is clear from Table 2 that surface aerators with $d \geq 1$ m can not be recommended.

Influence of Salts and Surfactants on the Oxygen Uptake of Surface Aerators

It is known that certain additives to water influence the mass transfer rate in gas-liquid contacting¹⁷⁾. In general, there are two types: salts of strong electrolytes and miscible organic liquids like alcohols, ketones, etc., which transform a so-called coalescent system into a non-coalescent one, i.e., they prevent the coalescence of primarily produced fine bubbles to larger ones and thus increase the interfacial area. The second type is represented by organic compounds which are surface active in such a way that they form films at the interface, thus reducing the absorption rate.

Due to the fact that the materials which prevent coalescence can only be effective when the primarily produced gas bubbles in the liquid (or water droplets in the air) are of such a size that they would otherwise be forced to coalesce, measurements were performed on a water/air system with the addition of sodium chloride and a detergent (PRIL) using aerator types C 1 (Fig. 13) and A (Fig. 14). It can be seen from these figures that these additives have no influence on mass transfer. From this, it is clear that the surface aerators do not produce fine bubbles or droplets.

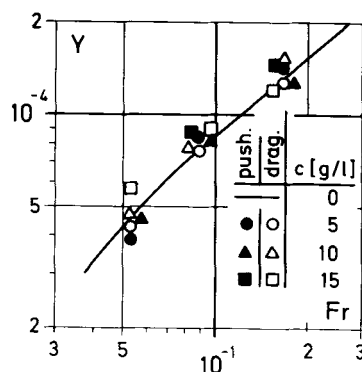


Fig. 13. Sorption correlation $Y(Fr)$ for surface aerator type C 1 measured at different salt concentrations

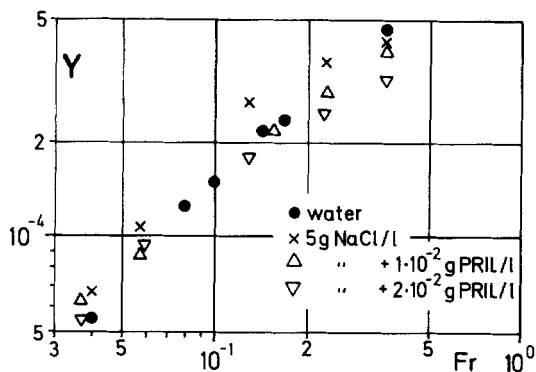


Fig. 14. Sorption correlation $Y(Fr)$ for surface aerator type A measured at different concentrations of salt and detergent

Review of Relevant Literature with Respect to the Conclusions Drawn from this Work

As has already been mentioned in the introduction, only a little data can be found in the literature where the scale-up of surface aerators for a required oxygen uptake is concerned. In contrast to the power consumption of this class of aerators (which was only once treated using the theory of similarity¹²⁾ and found to be predictable from the power correlation as in Eq. (7)), none of the relevant publications dealt with the absorption process using this technique. It is the prevailing opinion that the efficiency E of a surface aerator depends on its volume-related power P/V , as if the absorption process would not primarily take place at the liquid surface but in the bulk of it. Consequently, many papers and industrial brochures consider P/V as the scale-up criterion for the determination of the oxygen uptake G .

From the individual types of surface aerators discussed in the literature, power and absorption measurements suitable for evaluation and comparison were found only for the Simplex rotor^{1, 2, 3)} and for the BSK turbine^{2, 4, 5, 6)}, whereas the papers on the Simcar aerator⁸⁾ and the Hamburg rotor⁹⁾ do not include any information about power and absorption measurements. The literature data for the Simplex rotor is given in Table 3 and that for the BSK turbine in Table 4. The dependency $E^*(Fr)$ based on this data is presented in Fig. 15.

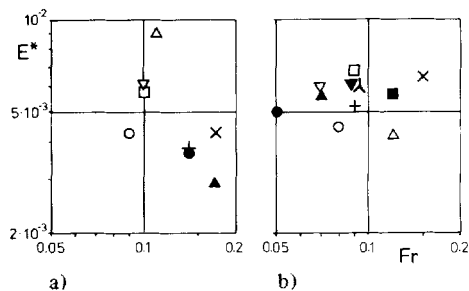


Fig. 15. Literature data evaluated according to the conclusions drawn from this work.

a) Simplex rotor; b) BSK-turbine

Table 3. Simplex-Rotor

Type	d [m]	n [min ⁻¹]	G ₁₀ [kg O ₂ /h]	P [kW]	V [m ³]	Fr	10 ⁴ Y	Ne	10 ³ E*	Ref.	Symbol
6	1.83	58	26.8	14	162	0.17	2.35	0.76	4.3	1)	×
	1.83	51	22.5	13.2	162	0.14	1.98	1.05	3.8	1)	+
5	1.52	41	8.3		120	0.07	1.27			1)	
	1.52	49	16.4		120	0.10	2.51			1)	
6	1.83	51	17.0	10.3	162	0.14	1.50	0.82	3.7	3)	●
6	1.83	57	18.6	14.3	630	0.17	1.63	0.81	2.9	3)	▲
10 HL	3.05	32.5	42.3	28.8	1200	0.09	0.80	0.69	4.3	2)	○
10 HL	3.05	35	100.1	32	680	0.11	1.90	0.61	9.0	2)	△
10 SL	3.05	34.3	136.0	62	1200	0.10	2.58	1.26	6.2	2)	▽
5 HL	1.52	49	18.1	6.4	120	0.10	2.78	1.44	5.8	2)	□

Note: Ref. 2); only the data with the highest efficiencies were considered.

Ref. 3); measurements under actual water treatment conditions. Sludge concentration 6 g/l.

G₁₀ and Y: these values refer to Δc = 11.25 ppm under so-called standard conditions (10 °C and c = 0 ppm).

Table 4. BSK-turbine, pushing

Type	d [m]	n [min ⁻¹]	G ₁₀ [kg O ₂ /h]	P [kW]	V [m ³]	Fr	10 ⁴ Y	Ne	10 ³ E*	Ref.	Symbol
	2.0	37	15.75	8.2	1200	0.08	1.09	1.06	4.5	2)	○
	2.0	45	64.8	36	1200	0.12	4.36	2.67	4.2	2)	△
	3.0	28	110.9	52.8	1200	0.07	2.21	2.13	6.0	2)	▽
	3.0	32	179.2	75.6	1200	0.09	3.57	2.05	6.8	2)	□
	1.25	66	14.2	4.1	54.5	0.15	3.91	1.00	6.5	4)	×
66/67	3.0	32.4	112.8	62.5	2220	0.09	2.25	1.63	5.2	5)	+
Favorit	3.0	32.4	146.2	68.6	2220	0.09	2.91	1.79	6.1	5)	⋈
	3.0	24.5	40	23	1200	0.05	0.80	1.39	5.0	6)	●
	3.0	28	72	37	1200	0.07	1.44	1.49	5.6	6)	▲
	3.0	32.5	123	58	1200	0.09	2.45	1.50	6.1	6)	▼
	3.0	37	173	87	1200	0.12	3.45	1.53	5.7	6)	■

Note: Ref. 2); only the data with the highest efficiencies were considered.

Ref. 5, 6); data for "just completely submerged" aerators were evaluated.

Ref. 6); the direction of rotation not indicated.

G₁₀ and Y: these values refer to Δc = 11.25 ppm under the so-called standard conditions (10 °C and c = 0 ppm).

Before beginning the discussion of these findings, it must be mentioned that it is not known whether the aerators used for these measurements were geometrically similar to each other. There is no information referring to this important aspect in the cited literature. The second difficulty arises in the fact that the parameter H/d , which has been found to be the important one in this work, was not kept constant in the tests. The third, equally important parameter, which was also varied throughout the measurements, is the degree of submergence of the aerator blades. Whenever possible, the data was taken for the case in which the blades were just completely submerged^{5, 6)}. When the degree of submergence was not indicated, the data was evaluated for a value at which the efficiency had a maximum²⁾.

Figure 15 a shows eight E^* values, four of them are scattered around $E^* = 4 \cdot 10^{-3}$ (a value which was found to be favorable in this work) but three of them lie 50% or higher. The majority of the E^* values found with BSK turbines (Fig. 15 b) lie at $E^* = 6 \cdot 10^{-3}$ and are therefore also 50% higher than expected according to this report. The data taken from²⁾, which was measured with BSK turbines of $d = 2$ and 3 m (see Table 3), suggest that the efficiency E would rise with $\sqrt{d}$ instead of falling with it according to the findings in this work.

Due to the non-observance of the geometrical similarity these findings cannot be interpreted unequivocally. Besides this, difficulties have to be considered when the oxygen uptake measurements are performed in large basins. In contrast to the laboratory scale, where the measured $c(t)$ values are position-independent due to the complete back-mixing of the aerated liquid, this is not true in large basins. Measurements¹³⁾ in waste treatment plants have revealed a dramatic drop of the horizontal velocity at the surface to only 1/3 of its value at a depth of 1 m!

Tests¹³⁾ in the activated sludge system under steady-state conditions have furthermore demonstrated that despite a high DO value at the surface of $4 \text{ mg O}_2/\text{l}$, zero DO levels were observed at lower depths ($H \geq 4 \text{ m}$).

Critical Discussion of the Pertinent Literature Published After the Conclusion of this Work

The most important research work on surface aerators was published by Schmidtke and Horváth¹⁹⁾ in 1977. They reported the $k_L a$ values obtained by experiments with four flat-blade turbines of the Rushton type (similar to those used in this work) with diameters $d = 0.054; 0.084; 0.152$ and 0.267 m operating under geometrically similar conditions ($D/d = 4.0; H/d = 2.0$) in square vessels.

The turbines were so positioned in the liquid, that the impeller blades were entirely flooded while at rest (approx. double the blade submergence as compared with that in our work). In all 215 tests were carried out under non-steady-state conditions in tap water at 20°C and evaluated in the form $k_L a = f(\text{Fr})$. A linear relationship of the form $k_L a = A \text{Fr} - B$ with different constants A and B was found for each impeller diameter.

These findings are represented in Fig. 16 in the form of a sorption correlation $Y(\text{Fr})$. The fact that the $Y(\text{Fr})$ curves lie higher than the straight line from Fig. 2

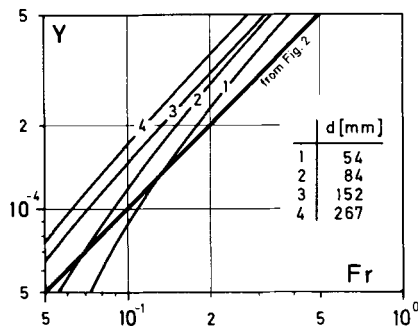


Fig. 16. Results of Schmidtke and Horváth¹⁹⁾ on flat-blade turbines represented in the form $Y(Fr)$ and compared with our measurements (Fig. 2)

representing our own measurements is due to the different blade submergences applied in both studies. It is surely of the greatest importance to learn that the dimensionless representation of the form $Y(Fr)$ does not satisfactorily correlate the absorption measurements on impellers of different diameters. This exemplifies the influence of at least one of the pertinent material parameters ν and σ . According to our own measurements in pure water, in an aqueous salt solution and in water with detergent (see Fig. 13 and 14), the effect of the surface tension σ on mass transfer seems to be negligible. From this point of view the results of Schmidtke and Horváth¹⁹⁾ suggest that the previous assumption of the irrelevance of the Reynolds number on mass transfer in the turbulent flow region was false. Correcting our standpoint and assuming the irrelevance of σ instead of ν , the dimensionless dependency Eq. (4) loses the expression $\sigma^* \equiv (Re^4 \cdot Fr)^{1/3}/We$ which is the only number representing σ , and thus instead of the sorption correlation Eq. (5), we now obtain the dependency

$$Y = f_3(Fr, Re, Sc),$$

which reduces, due to $Sc = \text{const.}$ in water at constant temperature, to

$$Y = f_4(Fr, Re); \quad Sc = \text{const.}$$

In order to correlate absorption measurements more easily it is advantageous to transform Re into the Galilei number $Ga \equiv Re^2/Fr = d^3 \cdot g/\nu^2$, because this number does not contain the rotational speed and in a given liquid is therefore altered only by the aerator diameter d :

$$Y = f_4(Fr, Ga); \quad Sc = \text{const.}$$

Using this set of dimensionless numbers, the final correlation formula given in the paper of Schmidtke and Horváth¹⁹⁾

$$k_L a = 0.98 \cdot 10^{-8} \cdot n^{2.41} \cdot d^{1.55}; \quad k_L a [\text{min}^{-1}]; n [\text{min}^{-1}]; d [\text{cm}]$$

can be converted into the following sorption characteristic:

$$Y = 1.41 \cdot 10^{-4} \cdot Fr^{1.205} \cdot Ga^{0.115}; Fr = 0.02 - 0.34 \\ Ga = 1.5 \cdot 10^9 - 2 \cdot 10^{11}$$

To verify this new sorption correlation $Y \cdot Ga^{-0.115} = f(Fr)$ on the basis of our own measurements on surface aerators, type C 3, we have to take the absorption number Y for the model aerator ($d = 0.43$ m) from the compensating straight line in Fig. 11 at $Fr = 0.054$ and for the full-scale aerator ($d = 4.3$ m) the mean Y value from the measured range:

d [m]	Y	Ga	$Y \cdot Ga^{-0.115}$
0.43	$4.2 \cdot 10^{-5}$	$7.8 \cdot 10^{11}$	$1.80 \cdot 10^{-6}$
4.30	$8.0 \cdot 10^{-5}$	$7.8 \cdot 10^{14}$	$1.55 \cdot 10^{-6}$

The agreement is within 20%.

Taking this sorption correlation for granted, it follows that the efficiency number E^* becomes:

$$E^* \equiv Y \cdot Ga^{-0.115} \cdot Ne^{-1} \cdot Fr^{-1.5} = \frac{G \cdot d^{0.155}}{P \cdot \Delta c} \cdot f(\nu, g, \sigma),$$

instead of Eq. (8).

This expression implies that for scale-up of surface aerators with the same Froude number, the efficiency would diminish by $d^{0.155}$ instead of $d^{0.5}$ as is stated in this work, which means that the efficiency would diminish very little as the diameter of the aerator increases.

J. Groot Wassink et al.²⁰⁾ studied a new type of surface aerator ("Ramix aerator") which consists of a pitched blade turbine with three blades ($\alpha = 30^\circ$), supported on their tips is a ring with 8 vertical blades. This construction enables the aerator to be always flooded, which prevents flow instabilities and spray formation. The laboratory scale measurements were carried out in square vessels with cross-shaped baffles on the bottom. The diameters of the three aerators were $d = 0.185; 0.30$ and 0.40 . The kinematic viscosity of water was altered in the range $\nu = (1 - 18) \cdot 10^{-6} \text{ m}^2/\text{s}$ by adding "Luviskol" (trade name for polyvinylpyrrolidone). The change in the diffusivity was achieved by variation of the temperature.

The power correlation was found to be

$$Ne = \text{const.} \cdot Fr^{-0.25} \cdot Re^{-0.10},$$

which is in respect to Re in contrast to our findings (see Fig. 1 and 8) and to the measurements of Bruxelmane²²⁾, which will be discussed later on. We can also check the above correlation by applying it to our results obtained on surface aerators of type C 3 (see Fig. 10), where a disagreement of -20% was found when the scale factor was 10. For this purpose the above correlation will be preferentially transformed to $Ne = \text{const.} \cdot Fr^{-0.30} \cdot Ga^{-0.05}$.

d [m]	Ne	Ga	NeGa ^{0.05}
0.43	1.10	$7.8 \cdot 10^{11}$	4.33
4.30	0.90	$7.8 \cdot 10^{14}$	4.99

The disagreement is now + 15%.

The sorption correlation of this aerator type was found to be

$$G/(\text{nd}^3 c_s) = \text{const.} \cdot \text{Fr}^{0.15} \cdot \text{We}^{0.30} \cdot \text{Sc}^{-0.25}.$$

For given geometric and material conditions it can be transformed into

$$k_L a = \text{const.} \cdot n^{1.9} \cdot d^{1.05},$$

which applied to the results of Schmidtke and Horváth leads to the expression

$$Y = \text{const.} \cdot \text{Fr}^{0.95} \cdot \text{Ga}^{0.033}.$$

For the scale-up of this aerator type results in

$$E^* \equiv Y \cdot \text{Ga}^{-0.033} \cdot \text{Ne}^{-1} \cdot \text{Fr}^{-1.5} = \frac{G d^{0.4}}{P \Delta c} \cdot f(\nu, g, \sigma),$$

which means that the efficiency would diminish by $d^{0.40}$. This value is similar to the $d^{0.50}$ found in our investigations.

M. Roustan²¹⁾ reported his measurements with vaned-disk stirrers, flat-blade turbine and pitched blade turbine impellers of different diameters working under different installation conditions as surface aerators. Besides the $k_L a$ values and the power requirement he also measured the pumping capacity, q and the mixing time, θ . He found that the pumping head, $H = P/q \cdot \rho \cdot g$ (derived from the equation $P = q \cdot \Delta p \hat{=} q \cdot \rho \cdot g \cdot H$) and the dimensionless expression $k_L a \cdot \theta$ are both functions of the surface-related power P/A and do not depend on the aerator type. To evaluate the dependence of the installation conditions (h/d , D/d) on the absorption rate he proposed the number $X \equiv k_L a \cdot H/\nu$, where ν is the tip speed of the aerator, and found two dependencies for $X(h/d)$, one valid for $d/D \leq 0.15$ and one for $d/D > 0.15$.

Due to the favorable circumstance that Roustan's paper contains a tabulation of all the measurements, we were able to transform them into the form used in this paper. Figures 17 and 18 show the dependencies $E^*(\text{Fr})$ for flat-blade turbine and pitched-blade turbine impellers, respectively. We learn from these graphs that in the range $\text{Fr} > 0.2$ the Froude number exerts a strong influence on the efficiency number E^* , whereas the installation conditions D/d , H/d and h/d have no significance. The compensating straight lines obey the following analytical expressions:

$$\begin{aligned} \text{flat-blade turbine:} \quad & E^* = 2.2 \cdot 10^{-3} \cdot \text{Fr}^{-0.37}; \text{Fr} = 0.2 - 2.0 \\ \text{pitched blade turbine:} \quad & E^* = 2.2 \cdot 10^{-3} \cdot \text{Fr}^{-0.43}; \text{Fr} = 0.2 - 3.0 \end{aligned}$$

	d [mm]	D/d	H/d	h/d
○	150	6,7	4,0	0,00
△				0,066
▽				0,13
●	200	5,0	3,0	0,00
▲				0,10
▼				0,15
■				0,20
*	900	8,1	1,06	0,00

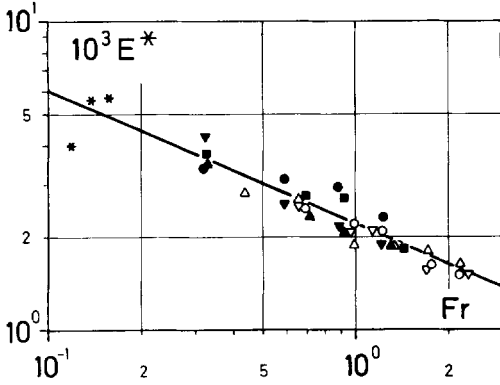


Fig. 17. The dependency $E^*(Fr)$ for flat-blade turbines according to Roustan²¹⁾. The straight line obeys the analytical expression $E^* = 2.2 \cdot 10^{-3} Fr^{-0.37}$

	d [mm]	D/d	H/d	h/d
○	131	7,6	4,6	0,15
○				0,23
●				0,30
●				0,38
△	167	6,0	3,6	0,06
△				0,12
▲				0,18
□	200	5,0	3,0	0,10
□				0,20
■				0,30
x	740	9,9	1,2	0,05
+				0,08

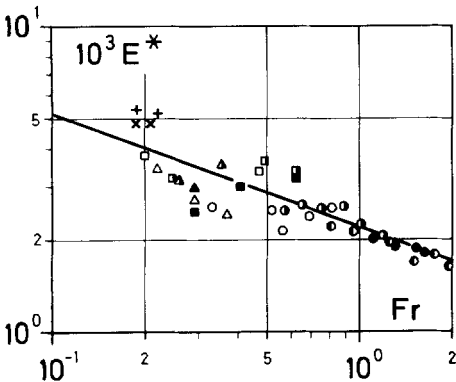


Fig. 18. The dependency $E^*(Fr)$ for pitched blade turbines according to Roustan²¹⁾. The straight line obeys the analytical expression $E^* = 2.2 \cdot 10^{-3} Fr^{-0.43}$

We see from these expressions that the aerator design also has little influence on the efficiency. We learn further that in this Froude range the efficiency is inversely proportional to $n^{0.8}$, which means that doubling the rotational speed would reduce the efficiency to about 57%! The accuracy of these measurements does not permit any definite statement concerning the influence of the scale on E . Considering the data for $d = 0.74$ m in Fig. 17 ($E^* \approx 5 \cdot 10^{-3}$) and comparing them with the value for $d = 0.13$ m indicated by the straight line at the same Froude number ($E^* = 4 \cdot 10^{-3}$), one finds that E is inversely proportional to $d^{0.37}$.

M. Bruxelmane²²⁾ investigated the hydrodynamic behavior of flat blade turbines (Rushton type) working as surface aerators in a cylindrical vessel of $D = 1.15$ m. He found with three turbines ($d = 0.20$; 0.245 and 0.348 m) that the Newton number Ne is solely a function of the Froude number Fr . In the range $Fr < 0.13$ the stirrer spreads the liquid over the surface in the form of a film. Here the dependency $Ne(Fr)$ is given by $Ne \propto Fr^{-0.37}$. The transition range $0.13 < Fr < 0.35$ is signified by a strong drop in the power number from $Ne = 1.0$ to $Ne = 0.57$ due to the formation of cavities behind the stirrer blades. In the range $Fr > 0.35$ the stirrer disperses the liquid into droplets. Here the power correlation is given by $Ne = 0.35 Fr^{-0.5}$. It is also to be expected that in each of these regions a different absorption characteristic prevails.

From the discussion of the results of the recent literature and comparison with the conclusions of our work, it can be seen that a certain inconsistency in the proposed scale-up criteria still exists. This is mainly due to the fact that the absorption rate measurements on the laboratory scale are scarcely more accurate than $\pm 10\%$. This is not sufficient to satisfactorily predict the performance of a large-scale aerator which normally is 5 to 10 times greater than the largest laboratory equipment. On the other hand the absorption measurements on the industrial scale are far less reliable than those done in the laboratory so that they can contribute only little to the reinforcement of the scale-up criteria found in the laboratory.

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Nomenclature

A	[m ²]	gas-liquid interfacial area
c	[ppm or kg m ⁻³]	conc. of gas dissolved in the liquid (DO)
c _s	[ppm or kg m ⁻³]	saturation conc. of the gas in the liquid
Δc	[ppm or kg m ⁻³]	concentration difference
d	[mm or m]	aerator diameter
d'	[mm or m]	diameter of blade enclosure
D	[mm or m]	vessel or pool diameter
E	[kg O ₂ /kWh]	aerator efficiency
g	[ms ⁻²]	gravitational constant
G	[kg O ₂ h ⁻¹ or kg O ₂ s ⁻¹]	oxygen uptake
h	[mm or m]	blade submergence
H	[mm or m]	liquid depth
k _L	[ms ⁻¹]	liquid-phase mass transfer coefficient
k _{LA}	[m ³ s ⁻¹]	sorption rate coefficient
n	[min ⁻¹ or s ⁻¹]	rotational velocity of the aerator
P	[W or kW]	power consumption of the aerator
z	[—]	number of blades
D	[m ² s ⁻¹]	diffusivity of the gas in the liquid
ρ	[kg m ⁻³]	liquid density
ν	[m ² s ⁻¹]	liquid kinematic viscosity
σ	[kg s ⁻²]	liquid surface tension

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